

THE 1,000 MEDICAL PROMPTS · VOLUME 1 OF 3

THESIS, RESEARCH & PUBLICATION.

400 clinician-grade AI prompts for every stage of your thesis and first papers, from finding a feasible topic to the published manuscript, case report and conference poster.

400

prompts · #1–400

13

sections

48 · 229 · 123

Master · Standard · Quick

1.1

Topic selection & research question

1.2

Literature search strategy

1.3

Writing the Introduction

1.4

Review of Literature

1.5

Aims, objectives & hypotheses

1.6

Study design & methodology

1.7

Sample size & statistics

1.8

Results, tables & figures

1.9

Discussion, conclusion & limitations

1.10

Protocol, ethics & synopsis

1.11

Thesis to manuscript & publication

1.12

Case reports & case series

1.13

Conference abstracts, posters & oral papers

Read this first.

These prompts are written the way a research supervisor thinks, not the way a chatbot answers. Many are long on purpose: a vague prompt gets you a vague, confident, wrong answer. Before you paste anything, five non-negotiables.

01 The AI does not know your patients.

Every prompt has a context block. Fill it honestly. If you say you have 200 cases a year when you have 40, the model will design a study you cannot finish.

02 Never accept a citation you have not opened.

Language models fabricate references that look flawless: real journal, real authors, wrong or non-existent paper. Verify every reference on PubMed by PMID or DOI before it enters your work.

03 Never paste identifiable patient data.

No names, hospital IDs, MRNs or scans with burned-in identifiers. De-identify before the data touches a chatbot. This is an ethics committee issue, not a preference.

04 The AI drafts. You decide.

Use it to speed up structure, phrasing and pre-checking. The scientific judgement, the assumptions and the interpretation stay yours. Your name goes on the paper, not the model's.

05 Declare AI use where required.

ICMJE and most journals require disclosure of generative AI use. AI cannot be an author. Check your journal's policy and your university's thesis rules before submission.

How to use each prompt

- › Copy the full prompt. The length is what makes the output specific.
- › Replace every [SQUARE BRACKET] with your real details. Options inside a bracket are separated by "/"; keep the one that applies.
- › Put anything you paste (abstracts, drafts, results) where the prompt shows [PASTE ...], above the task.
- › Use a model with web search or document upload switched on wherever the prompt asks for evidence.
- › Run each stage in a fresh chat so the model doesn't carry assumptions from earlier steps.
- › Push back. If the output is generic, reply: "Be more specific. Tie every point to my context."

Every prompt is built on **P-R-O-M-P-T**.

One formula, six layers. Each layer matches what OpenAI, Anthropic, Google and Microsoft recommend in their own prompting guides.

P Persona Who the AI should be: a senior expert with a trait	R Request One clear task, as numbered steps	O Output Shape, length, tone, tables	M Material Your context and anything you paste	P Parameters Rules, guidelines, accuracy guards	T Tweak Self-critique and the next step
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See it in action

BAD	"Tell me about pneumonia."
GOOD	"Explain community-acquired pneumonia management for an MBBS exam."
BETTER	"Act as a pulmonologist. Explain CAP management for an MBBS final-year student as a 6-point answer with empirical antibiotic choices per Indian guidelines. Keep it under 150 words, exam-oriented."

Three kinds of prompt

QUICK 40–80 words One focused output in a few sentences. Each has a TRY IT WITH example.	STANDARD 100–200 words One real task done properly: context, numbered task, rules, output, a pro tip and the next prompt to run.	MASTER 250–450 words A full workflow on its own page, with when to use it, why it exists and how to get more out of it.
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Tags you will see in the outputs

TAG	WHAT TO DO
[CITE] · [CITATION NEEDED]	Find a real source on PubMed and insert it yourself.
[VERIFY] · [ASSUMPTION — VERIFY FROM PUBLISHED DATA]	Replace the number with a sourced figure before use.
[UNVERIFIED BACKGROUND]	Treat as a lead to check, not a fact.
[ASK ME] · [NEED INFO] · [NOT REPORTED]	The model is missing information. Supply it, or leave it out.

Every prompt works in ChatGPT, Claude, Gemini and Copilot. → **NEXT** points to the prompt to run after this one.

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1.1

Topic selection & research question

Most thesis disasters are topic failures decided in a ten-minute corridor conversation. Build feasibility into the decision from day one.

35

prompts

4

Master

20

Standard

11

Quick

#1 Feasibility-first topic shortlist

#2 Five Indian abstracts → six thesis topics

#3 Socratic research-question workshop

#4 Guide-meeting pitch pack for your top 3 topics

#5 India gap finder: 15 topics from scratch

#6 Clinical hunch → PICOT question

#7 FINER stress-test of my research question

#8 Is this topic already done to death?

#9 Turn gaps into Consensus yes/no questions

#10 Topics I can do from existing hospital records

#11 Topics that fit a 12-month data window

#12 Topics needing only routine investigations

#13 Validate an existing clinical score in my population

#14 Quality-improvement and audit thesis topics

#15 Questionnaire-based topics without the KAP trap

#16 Head-to-head decision matrix for 2–3 topics

#17 Thesis title that passes the committee

#18 What could kill this topic? A risk register

#19 Choose a measurable primary outcome

#20 The questions my guide will ask about my topic

#21 Topics from the gaps in a clinical guideline

#22 Four-week feasibility check before you commit

#23 Five original angles on a crowded topic

#24 Two-month project for an MBBS student (ICMR-STs style)

#25 Ten quick topic seeds in my specialty

#26 Sharpen my research question to one line

#27 PICO or PECO? Frame my question correctly

#28 Check ongoing trials before you commit

#29 The follow-up that surfaces Indian gaps

#30 Identify the variables in my question

#31 My topic is too broad: narrow it three ways

#32 What infrastructure does this topic need?

#33 Does this Western topic apply in India?

#34 Hypothesis-generating or hypothesis-testing?

#35 Summarise my topic on one slide

Feasibility—first topic shortlist

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT (YEAR 1)

USE IT

Months 1 to 3 of your postgraduate course, before you commit to anything.

WHY

Most theses fail at the topic, not the analysis. This prompt checks every idea against your real patient numbers, equipment and time before you fall in love with it.

COPY THE PROMPT BELOW

PERSONA	You are a senior clinical research methodologist with over 20 years of experience supervising postgraduate theses in [YOUR SPECIALTY]. You have sat on an institutional ethics committee and you peer review for specialty journals. You are blunt about feasibility rather than encouraging.
MY CONTEXT	Specialty: [e.g. Respiratory Medicine] Programme and year: [MD / MS / DNB / DM / MCh], year [1 / 2 / 3] Institution type: [government medical college / tertiary private / district hospital] Eligible patients I can realistically access: about [N] per month Months available for recruitment and data collection: [X] Funding: [none / departmental consumables / grant of amount Y] Investigations and equipment I genuinely have: [list] What I do NOT have: [e.g. flow cytometry, PET-CT, research nurse] Statistical support: [none / department statistician / paid] Clinical interests: [2 to 4] Diseases with high burden in my setting: [list]
TASK	Generate 8 candidate research questions I could design, complete, defend and publish inside my timeline. For each, give exactly: 1. Working title as it would appear on a thesis cover page. 2. PICOT: Population, Intervention or Exposure, Comparator, Outcome, Timeframe. 3. The gap in two sentences: what is established, what remains unanswered (describe evidence generically). 4. Study design, and one line on why not the obvious alternative. 5. Feasibility score out of 5, justified against my context above. 6. Indicative sample size with every assumption stated. 7. Recruitment timeline in months, assuming 30% of eligible patients are lost. 8. Primary outcome and the exact instrument, scale or cut-off. 9. The three most likely reasons it fails. 10. Honest publication ceiling: local, national indexed, or international.
RULES	Mark every number you need as [ASSUMPTION — VERIFY FROM PUBLISHED DATA] instead of inventing prevalence, effect sizes, trial names or citations, because my guide will ask where each came from. Drop any question needing resources I said I do not have. At least 2 of the 8 must survive if recruitment halves.
OUTPUT	A ranked list from most to least feasible, then your top pick with reasoning.
TWEAK	Finish with AVOID THESE: three topic directions in my field that look attractive to postgraduates but routinely fail, and why.

How to get more out of it

- Run it twice, once with your real patient numbers and once with half. Any topic that survives both is the safe one.
- Take the top three to your guide as a one-page comparison, not a chatbot transcript.
- Read AVOID THESE before you fall in love with an idea.
- → NEXT #4 Guide-meeting pitch pack for your top 3 topics

Five Indian abstracts → six thesis topics

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT

USE IT

After you have saved 5 recent abstracts from Indian centres in your field (Semantic Scholar or PubMed, last 3 years).

WHY

Topics generated from memory repeat what already exists. Topics generated from real recent Indian abstracts start from what your peers have actually published, and from what they left out.

COPY THE PROMPT BELOW

PERSONA	You are a research mentor at an Indian medical college who has guided over 50 postgraduate theses to completion. You read abstracts like a reviewer: you look for what was not measured, not studied, or not studied here.
MY CONTEXT	Specialty and programme: [e.g. MD Pulmonary Medicine, year 1] Setting: [government medical college / private tertiary / district] Limits: 2-year thesis window, [limited / moderate] lab infrastructure, about [N] eligible patients per month. [PASTE ABSTRACT 1] [PASTE ABSTRACT 2] [PASTE ABSTRACT 3] [PASTE ABSTRACT 4] [PASTE ABSTRACT 5]
TASK	1. For each abstract, write one line on what it studied and one line on what it did not address (population, outcome, setting, follow-up or design). 2. From these, identify 3 research gaps relevant to Indian patients. 3. For each gap, propose 2 thesis topics, each with a PICO question and a measurable clinical outcome. 4. For all 6 topics, give: best study design, rough sample size range, primary outcome measure, the main feasibility challenge in an Indian government hospital, and publication potential in an indexed journal. 5. Rank the 6 and recommend the top 3 to present to my guide.
RULES	Work only from the abstracts I pasted. If a gap needs knowledge beyond them, label it [UNVERIFIED BACKGROUND — CONFIRM ON PUBMED] rather than stating it as fact. Name no additional papers unless you can give a verified PMID or DOI.
OUTPUT	A short table per gap, then the ranked shortlist with one-paragraph reasoning for each of the top 3.
TWEAK	End with the single question I should search on Consensus to check whether my top topic is already saturated.

How to get more out of it

- If an abstract is from a different population (Western, paediatric), say so in the context line. It changes which gaps are real.
- Keep the 5 abstracts. They become the first five studies in your Review of Literature.
- If fewer than 3 real gaps appear, your search was too narrow. Search the gap, not the disease name.
- → **NEXT #8** Is this topic already done to death?

Socratic research-question workshop

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT / MBBS INTERN

USE IT

When you have a vague idea ("something on sepsis in the ICU") but no question.

WHY

A good question is refined, not generated. This prompt makes the AI interview you, one question at a time, until the question is sharp enough to defend.

COPY THE PROMPT BELOW

PERSONA	You are a clinical epidemiologist who runs research-methods workshops for postgraduates. You never hand over answers early. You ask probing questions, one at a time, and you push until the question is specific, measurable and feasible.
MY CONTEXT	My rough idea in one or two sentences: [e.g. something about delirium in elderly ICU patients] Specialty and level: [e.g. MD Anaesthesiology, year 1] Setting: [e.g. 20-bed mixed ICU, government medical college] Time for data collection: [X months]
TASK	Run a workshop with me in rounds: 1. Ask me one question at a time about population, exposure or intervention, comparator, outcome, timeframe and setting. Wait for my answer before asking the next. 2. After every 3 answers, show the current version of my research question and score it on FINER (Feasible, Interesting, Novel, Ethical, Relevant), 1 to 5 each, with one line per score. 3. Point out the weakest FINER element and ask the question that fixes it. 4. Stop when every FINER score is 4 or more, or after 5 rounds. 5. Then give the final question in PICOT format, a one-line hypothesis, and a draft working title.
RULES	Do not suggest a finished question before round 3, because the point is to make me think. If an answer I give makes the study unethical or impossible, say so directly and explain why. Keep every question you ask under 30 words.
OUTPUT	Conversational, one question per turn. Use a short FINER scorecard table at each checkpoint.
TWEAK	At the end, list the 3 assumptions in my question that I must verify in the literature before my guide meeting.

How to get more out of it

- › Answer honestly, including "I don't know". The workshop is only as good as your answers.
- › Screenshot the final scorecard. It becomes slide 2 of your topic presentation.
- › Re-run the workshop for your second-choice idea and compare the final scores.
- › → **NEXT #7** FINER stress-test of my research question

Guide-meeting pitch pack for your top 3 topics

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT

USE IT

The day before you meet your guide or research committee to choose a topic.

WHY

Guides respond far better to a student who arrives with three shortlisted topics, the evidence gap behind each, and an honest feasibility view. This prompt builds that one-page pitch.

COPY THE PROMPT BELOW

PERSONA	You are a head of department who has chaired thesis-topic committees for 15 years. You know which questions guides ask, which topics get approved, and which get sent back.
MY CONTEXT	Specialty and year: [e.g. MS General Surgery, year 1] My guide's known interests: [list, or "unknown"] Department case load relevant to these topics: [numbers per month] Topic 1: [title + PICOT + why it matters] Topic 2: [title + PICOT + why it matters] Topic 3: [title + PICOT + why it matters] Evidence I have for each gap (searches, Consensus result, abstracts): [PASTE NOTES]
TASK	1. Build a one-page comparison table: title, design, sample size needed vs available, time to complete, cost, ethics complexity, novelty, publication ceiling. 2. Give each topic an overall recommendation: strong, possible, or weak, with one reason. 3. For each topic, write the 3 questions my guide is most likely to ask, with a short answer from my notes. 4. Write a 60-second verbal pitch for my first-choice topic. 5. List what I should bring to the meeting (search strings, hit counts, a feasibility estimate).
RULES	Base every feasibility judgement on the numbers I gave, not on general assumptions. Where my notes do not support a claim of novelty, write [GAP NOT YET VERIFIED] instead of asserting it, because a guide will test it. Keep my voice: a first-year resident, not a professor.
OUTPUT	The comparison table first, then the per-topic questions, the pitch script, and a checklist. Fits on two printed pages.
TWEAK	Finish with the one weakness in my first-choice topic that I should raise before my guide does, and how to present it.

How to get more out of it

- › Print the table. A one-page comparison beats scrolling a phone in a guide's office.
- › If your guide rejects all three, ask which constraint mattered most, then re-run topic-feasibility with that constraint added.
- › Practise the 60-second pitch out loud twice.
- › → **NEXT #17** Thesis title that passes the committee

5 India gap finder: 15 topics from scratch

STANDARD

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT (YEAR 1)

USE IT When you have no abstracts or ideas yet and need a starting list today.

PERSONA	You are a senior researcher in [SPECIALTY] at an Indian medical college, known for steering residents towards questions that matter to Indian patients rather than copying Western topics.
MY CONTEXT	I am doing my [MD / MS / DNB] in [SPECIALTY] at a [government medical college / private college] in India. I have limited lab infrastructure and a 2-year thesis window.
TASK	Generate 15 potential thesis topics that: 1. Address conditions with a high burden in India (e.g. TB, diabetes, COPD, anaemia, dengue, snakebite, hypertension) relevant to my specialty. 2. Have limited published data from Indian populations. 3. Can be done as observational, cross-sectional or simple interventional studies. 4. Need no advanced laboratory infrastructure. For each, state the gap it addresses in one line and rate feasibility out of 10.
RULES	Describe gaps generically. Do not name papers or quote prevalence figures from memory; write [VERIFY] where a number would help, because every gap must be checked on PubMed before I pitch it.
OUTPUT	A numbered table: topic, gap, design, feasibility /10. Then the 3 you would pick and why.

PRO TIP Paste your 3 favourites into Consensus as yes/no questions. Keep only the ones with mixed or insufficient evidence.

TRY IT WITH MD General Medicine, government medical college in Bihar.

→ **NEXT #9** Turn gaps into Consensus yes/no questions

6 Clinical hunch → PICOT question

STANDARD

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT / MBBS INTERN

USE IT When you have noticed something on the ward ("these patients seem to do worse") and want to turn it into a question.

PERSONA	You are a clinical research mentor who helps residents turn bedside observations into answerable research questions.
MY CONTEXT	What I noticed on the ward or in OPD: [describe the observation in 2–3 lines] Specialty and setting: [e.g. Paediatrics, tertiary PICU] Rough patient numbers per month: [N]
TASK	1. Restate my observation as a clear clinical question. 2. Convert it into 3 alternative PICOT (or PECO) questions: one descriptive, one comparative, one prognostic. 3. For each, name the study design that answers it and the primary outcome. 4. Say which is most feasible for a postgraduate thesis and why.
RULES	Keep each question to one sentence. If my observation is more likely explained by bias or chance (e.g. referral pattern), say so first, because that changes the question.
OUTPUT	A table of the 3 PICOT versions, then a recommendation in 3 sentences.

PRO TIP Write your observation down the day you notice it. Specific details ("three patients this month on drug X") make much better questions than vague impressions.

TRY IT WITH Elderly patients with hip fracture on our ward seem to develop delirium more often when surgery is delayed beyond 48 hours.

→ **NEXT #7** FINER stress-test of my research question

7 FINER stress-test of my research question

STANDARD

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT

USE IT Before you submit your topic to the department or research committee.

PERSONA	You are a research committee member at an Indian medical college who scores postgraduate topics every term. You are fair but you do not pass weak questions.
MY CONTEXT	My research question: [paste in PICOT form] Study design: [design] Resources and patient numbers: [brief] Time available: [months]
TASK	1. Score my question on each FINER criterion (Feasible, Interesting, Novel, Ethical, Relevant) from 1 to 5, with two lines of justification each. 2. Identify the single weakest criterion. 3. Rewrite the question twice: once to fix the weakest criterion, once to make it simpler. 4. Say which of the three versions you would approve and why.
RULES	Judge novelty only on what you can reason from the question itself; where novelty depends on the literature, mark it [CHECK ON PUBMED] instead of guessing, because false novelty claims are caught at the viva.
OUTPUT	A FINER scorecard table, the two rewrites, and a verdict of 3 sentences or fewer.

PRO TIP A score of 3 on "Novel" is fine for a thesis. A score below 3 on "Feasible" is not.

TRY IT WITH Is serum vitamin D level associated with exacerbation frequency in COPD patients at a tertiary hospital in north India?

→ **NEXT #8** Is this topic already done to death?

8 Is this topic already done to death?

STANDARD

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT

USE IT After shortlisting, when you suspect your topic may already be heavily studied.

PERSONA	You are a systematic reviewer who can tell within minutes whether a research question is saturated, contested or open.
MY CONTEXT	My proposed question: [PICOT] Search results I found (titles or abstracts from PubMed, Consensus or Perplexity): [PASTE TITLES / ABSTRACTS HERE]
TASK	1. Sort the studies I pasted by design and setting (Indian vs international). 2. Judge the evidence state: saturated (consistent findings, many studies), contested (mixed findings), or open (few or no studies). 3. If saturated, suggest 3 angles that would still be original (new population, setting, outcome, follow-up, or measurement method). 4. If contested, identify what explains the disagreement and whether my study could resolve it.
RULES	Judge only from what I pasted. If the sample is too small to judge, say so and tell me which search to run next, because a verdict based on six titles is not a verdict.
OUTPUT	One-line verdict, then a short table of studies, then the angles or next search.

PRO TIP Consensus's meter is a quick first pass. This prompt is the second pass on what it returns.

TRY IT WITH Paste 10 PubMed titles for "neutrophil-lymphocyte ratio sepsis mortality".

→ **NEXT #23** Five original angles on a crowded topic

9 Turn gaps into Consensus yes/no questions

STANDARD

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT

USE IT Between gap-finding (Perplexity) and topic selection, before you check evidence density.

PERSONA	You are a medical librarian who trains residents to query evidence engines such as Consensus effectively.
MY CONTEXT	Research gaps or problem areas I have noted: [PASTE 5–8 GAP AREAS HERE] Population of interest: [e.g. Indian adults with type 2 diabetes]
TASK	1. Convert each gap into one binary research question in the form "Does [factor] affect [outcome] in [population]?" 2. Give a second, narrower version of each question. 3. Tell me how to read the result: what a strong-agreement, mixed or insufficient-evidence result would mean for choosing my thesis.
RULES	Keep each question under 20 words and answerable with yes or no. Use my population exactly, because broad populations return Western data that does not apply to my setting.
OUTPUT	A numbered list: broad question, then narrow question, for each gap. Then a 3-row interpretation guide.

PRO TIP Keep only the questions whose result is mixed or insufficient. Strong agreement means that topic is done.

TRY IT WITH Gaps: sarcopenia in COPD, vitamin D in TB outcomes, early mobilisation in ICU.

→ **NEXT #8** Is this topic already done to death?

10 Topics I can do from existing hospital records

STANDARD

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT / DNB TRAINEE

USE IT When prospective recruitment looks slow, or your thesis window is short.

PERSONA	You are a clinical researcher who has published multiple retrospective and record-based studies from Indian hospitals, and you know their pitfalls.
MY CONTEXT	Specialty: [SPECIALTY] Records available: [e.g. ICU admission registers 2019–2024, discharge summaries, lab information system, OPD registers] How complete and reliable these records are, in my honest view: [brief] Approximate number of records: [N]
TASK	1. Suggest 8 thesis questions answerable from these records. 2. For each, list the variables needed and whether they are routinely recorded. 3. Flag the main bias for each (missing data, misclassification, selection). 4. Say which questions most need an ethics waiver of consent, and what the IEC will want.
RULES	Prefer outcomes that are hard to misrecord (death, readmission, length of stay) over soft outcomes. Mark any question that needs data my records probably lack as [CHECK DATA AVAILABILITY] rather than assuming it is there.
OUTPUT	A table: question, variables, bias risk (low / medium / high), IEC note.

PRO TIP Pull 20 random files before committing. If more than a quarter are missing your key variable, choose another question.

TRY IT WITH Medicine ward records of acute kidney injury, 2020–2024, about 900 admissions.

→ **NEXT #18** What could kill this topic? A risk register

11 Topics that fit a 12-month data window

STANDARD

1.1 TOPIC SELECTION & RESEARCH QUESTION · DNB TRAINEE / PG RESIDENT (LATE START)

USE IT When your approved data-collection window is 12 months or less.

PERSONA	You are a thesis coordinator who has rescued many late-starting postgraduate theses by right-sizing the question.
MY CONTEXT	Specialty: [SPECIALTY] Months left for data collection: [X] Eligible patients per month: [N] Current (too ambitious) topic idea, if any: [describe]
TASK	1. Tell me honestly whether my current idea fits the window, with the arithmetic. 2. Suggest 5 alternative or scaled-down questions with outcomes measurable within days or weeks, not months. 3. For each, estimate the patients needed and the months to recruit at my numbers.
RULES	Prefer cross-sectional, diagnostic-accuracy or short-follow-up designs. Show every calculation with its assumptions marked [ASSUMPTION], because I will have to defend the timeline to my guide.
OUTPUT	A verdict paragraph, then a table: question, design, patients needed, months to recruit.

PRO TIP Build in a 2-month buffer. Theses almost never finish recruitment on schedule.

TRY IT WITH DNB Orthopaedics, 10 months left, about 15 eligible patients per month.

→ **NEXT #22** Four-week feasibility check before you commit

12 Topics needing only routine investigations

STANDARD

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT

USE IT When there is no funding and you can only use tests the hospital already does free or routinely.

PERSONA	You are a researcher who has built a publication record entirely on routine clinical data in low-resource Indian hospitals.
MY CONTEXT	Specialty: [SPECIALTY] Routine tests available at no cost to patients: [e.g. CBC, LFT, RFT, ABG, chest X-ray, ECG, bedside USG] Tests I cannot afford: [list] Patient groups I see often: [list]
TASK	1. Suggest 10 thesis topics that use only the routine tests I listed. 2. For each, give the research question, the key routine variable, the outcome, and why the question still matters clinically. 3. Mark the 3 with the best publication potential.
RULES	Exclude any topic needing a test from my "cannot afford" list, and do not suggest workarounds that shift cost to patients, because the IEC will reject that.
OUTPUT	A numbered table with a one-line clinical-value note for each, then the top 3.

PRO TIP Ratios and indices built from routine tests are popular but crowded. Pair them with a hard outcome to stand out.

TRY IT WITH MD Medicine; routine CBC, RFT, LFT, ABG, ECG, chest X-ray; lots of sepsis, CKD and snakebite.

→ **NEXT #7** FINER stress-test of my research question

13 Validate an existing clinical score in my population

STANDARD

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT

USE IT When you want a solid, publishable design with a clear outcome, and a well-known score exists in your field.

PERSONA	You are a clinical epidemiologist experienced in external validation of prognostic and diagnostic scores in Indian populations.
MY CONTEXT	Candidate score or rule: [e.g. qSOFA, CURB-65, Alvarado, Wells] Population where it was developed (if known): [describe or "unsure"] My population and setting: [describe] Outcome the score predicts: [e.g. 28-day mortality]
TASK	1. Explain why validating this score in my population is (or is not) a genuine gap. 2. Outline the validation design: inclusion criteria, timing of scoring, outcome ascertainment, blinding of assessors. 3. Name the performance measures to report (discrimination, calibration) and what each means clinically. 4. Give the sample-size logic for validation studies (events needed, not just total patients).
RULES	Do not claim the score has never been validated in India; write [CHECK EXISTING INDIAN VALIDATIONS ON PUBMED], because that search decides whether this topic is novel.
OUTPUT	Four short headed sections, under 400 words in total.

PRO TIP Validation studies are strongest when the score is calculated before the outcome is known. Build that into your protocol.

TRY IT WITH CURB-65 for 30-day mortality in community-acquired pneumonia at a district hospital.

→ **NEXT #19** Choose a measurable primary outcome

14 Quality-improvement and audit thesis topics

STANDARD

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT / FACULTY

USE IT When your department wants a thesis that also improves care, or clinical trials are impractical.

PERSONA	You are a quality-improvement lead trained in PDSA cycles and clinical audit, who has supervised QI projects in Indian hospitals.
MY CONTEXT	Specialty and unit: [e.g. Obstetrics labour room] Care processes that often go wrong or vary: [list 3–5] Existing standards we could measure against: [e.g. WHO, national guideline, hospital protocol, or "unsure"]
TASK	1. Suggest 6 audit or QI topics, each with the standard, the measurable indicator and the target. 2. For the best one, outline a before-and-after design with one PDSA cycle. 3. Explain how to report it (SQUIRE 2.0 headings) and whether it counts as research needing full IEC review.
RULES	Name a standard only if it is a real, named guideline; otherwise write [STANDARD TO BE CONFIRMED], because an audit without a verifiable standard is not an audit.
OUTPUT	A table of 6 topics, then the outline for the best one, under 350 words.

PRO TIP Ask your IEC early whether QI counts as research at your institution. The answer changes consent requirements.

TRY IT WITH Time from admission to first antibiotic in suspected sepsis in the medical emergency.

→ **NEXT #7** FINER stress-test of my research question

15 Questionnaire-based topics without the KAP trap

STANDARD

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT / MBBS INTERN

USE IT When your topic will rely on a questionnaire or scale.

PERSONA	You are a psychometrics-trained researcher who reviews questionnaire studies and is tired of weak knowledge-attitude-practice (KAP) surveys.
MY CONTEXT	Area of interest: [e.g. inhaler adherence, depression in dialysis patients] Population: [describe] Languages my patients speak: [list]
TASK	1. Suggest 5 questionnaire-based questions that go beyond a descriptive KAP survey (e.g. linking a validated score to a clinical outcome). 2. For each, name the kind of validated instrument needed and whether a validated version in my patients' language is likely to exist. 3. Explain what translation and validation steps I need if it does not.
RULES	Do not name a specific instrument's Indian-language validation unless you are sure; write [CHECK FOR VALIDATED TRANSLATION], because using an unvalidated translation is a classic viva question.
OUTPUT	A table of 5 topics with instrument type and translation status, then the validation steps as a numbered list.

PRO TIP A validated instrument plus a hard clinical outcome makes a questionnaire study publishable.

TRY IT WITH Adherence to inhaled therapy and exacerbations in asthma, Hindi-speaking OPD patients.

→ **NEXT #7** FINER stress-test of my research question

16 Head-to-head decision matrix for 2–3 topics

STANDARD

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT

USE IT When you are stuck choosing between two or three final topics.

PERSONA	You are a pragmatic research supervisor who helps residents decide with criteria, not gut feel.
MY CONTEXT	Topic A: [title + design + patient numbers] Topic B: [title + design + patient numbers] Topic C (optional): [title + design + patient numbers] What matters most to me (rank): [e.g. finishing on time, publication, learning a skill, guide's interest]
TASK	1. Build a decision matrix with my priorities as weighted criteria. 2. Score each topic 1 to 5 per criterion, with one line of reasoning. 3. Calculate weighted totals and give a clear recommendation. 4. State what new information would change the recommendation.
RULES	Use only the facts I gave. Where a score needs information I did not supply, mark it [NEED INFO] rather than guessing, because a guessed matrix decides nothing.
OUTPUT	The matrix table, then a 3-sentence recommendation.

PRO TIP If two topics tie, pick the one with the more reliable primary outcome measurement.

TRY IT WITH A: CPAP adherence in OSA (cohort, 8/month). B: Lung ultrasound vs X-ray for pneumonia (diagnostic, 25/month).

→ **NEXT #4** Guide-meeting pitch pack for your top 3 topics

17 Thesis title that passes the committee

STANDARD

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT

USE IT When your topic is fixed and you need the formal thesis title.

PERSONA	You are a university thesis committee member who has seen hundreds of titles rejected for being vague, too long or overclaiming.
MY CONTEXT	Research question (PICOT): [paste] Study design: [design] Setting: [hospital, city] My university's title rules, if any: [e.g. word limit, must mention design]
TASK	1. Write 6 title options: descriptive, design-first, PICO-style, question-form, short, and conservative. 2. Check each against the common committee objections (vague population, missing design, causal words, excessive length). 3. Recommend one and explain why.
RULES	Use associational wording (association, correlation, comparison) unless my design is randomised, because causal wording in an observational title is the first thing examiners question.
OUTPUT	A numbered list of 6 titles with word counts, a one-line check result for each, and the recommendation.

PRO TIP Most universities like the design and setting in the title: "A prospective observational study at a tertiary care centre in north India".

TRY IT WITH Association of serum albumin with 28-day mortality in adult sepsis, prospective cohort, AIIMS Rishikesh.

→ **NEXT #19** Choose a measurable primary outcome

18 What could kill this topic? A risk register

STANDARD

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT

USE IT Right after your topic is approved, before protocol writing.

PERSONA	You are a research operations manager who has seen theses fail for practical reasons: recruitment, equipment breakdowns, lost records and ethics delays.
MY CONTEXT	Approved topic and design: [paste] Recruitment source and expected numbers: [brief] Key equipment or tests: [list] People I depend on (lab, radiology, statistician): [list]
TASK	1. List the 10 most likely risks to completing this thesis on time. 2. For each, rate likelihood and impact (low / medium / high). 3. Give one prevention step and one fallback plan for each. 4. Highlight the 3 risks to discuss with my guide now.
RULES	Be specific to my topic, not generic. Where a risk depends on local facts I did not give, write [CHECK LOCALLY], because generic risk lists get ignored.
OUTPUT	A risk-register table: risk, likelihood, impact, prevention, fallback. Then the top 3.

PRO TIP Review this register every 3 months. Add risks as they appear.

TRY IT WITH Prospective cohort of 150 stroke patients needing MRI within 72 hours.

→ **NEXT #22** Four-week feasibility check before you commit

19 Choose a measurable primary outcome

STANDARD

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT

USE IT When your topic is agreed but the primary outcome is still vague ("better outcome", "improvement").

PERSONA	You are a trial methodologist who insists every primary outcome is defined precisely enough for two observers to agree.
MY CONTEXT	Research question: [PICOT] Candidate outcomes I am considering: [list] Follow-up I can actually do: [e.g. in-hospital only, 30-day phone call, 3-month OPD visit]
TASK	1. Evaluate each candidate outcome for clinical relevance, measurability, reliability and feasibility with my follow-up. 2. Recommend one primary outcome and up to 3 secondary outcomes. 3. For the primary, write an operational definition: instrument, unit, cut-off and time point.
RULES	Prefer validated scales and objective measures. If an outcome needs a scale, say whether a validated version exists or write [VERIFY VALIDATED TOOL], because unvalidated outcomes weaken the whole thesis.
OUTPUT	An evaluation table, then the recommended outcomes, then the operational definition in 3–4 lines.

PRO TIP Your sample size depends on the primary outcome. Fix this before you calculate anything.

TRY IT WITH Candidates for ARDS: mortality, ventilator-free days, P/F ratio at day 3, length of ICU stay.

→ **NEXT #22** Four-week feasibility check before you commit

20 The questions my guide will ask about my topic

STANDARD

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT

USE IT Before your first topic discussion with your guide, or before a departmental topic presentation.

PERSONA	You are an experienced thesis guide in [SPECIALTY] who probes every new resident's topic for weaknesses.
MY CONTEXT	My topic and design: [paste] What I know about the background: [brief notes] My expected patient numbers and time: [brief]
TASK	1. List the 12 questions you would ask me about this topic, from basic to tough. 2. For each, show what a weak answer and a strong answer look like. 3. Mark the 3 questions I am least prepared for, based on my notes.
RULES	Ground the model answers in my notes. Where a strong answer needs facts I have not given, write [FIND THIS BEFORE THE MEETING] instead of supplying numbers, because my guide will ask where they came from.
OUTPUT	A numbered list of questions with weak and strong answer pairs, then the 3 danger questions.

PRO TIP Answer the 3 danger questions in writing before the meeting.

TRY IT WITH Topic: point-of-care lung ultrasound vs chest X-ray for pneumonia in children under 5.

→ **NEXT #4** Guide-meeting pitch pack for your top 3 topics

21 Topics from the gaps in a clinical guideline

STANDARD

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT / FACULTY

USE IT When a recent guideline in your field lists recommendations based on weak evidence or states "further research needed".

PERSONA	You are a guideline methodologist familiar with GRADE and with how evidence gaps turn into research agendas.
MY CONTEXT	Guideline document or recommendations section: [PASTE OR UPLOAD THE GUIDELINE / SECTION] My specialty and setting: [brief]
TASK	1. Extract every recommendation graded low or very low certainty, conditional, or expert opinion, and every explicit "research needed" statement. 2. For each, say whether it could become a feasible postgraduate thesis question in my setting. 3. Turn the best 5 into PICOT questions with a suggested design.
RULES	Quote the guideline wording for each gap in quotation marks so I can verify it. Do not describe recommendations that are not in the text I provided, because guideline misquotes are easy to catch.
OUTPUT	A table: quoted recommendation, certainty grade, feasible (yes / no), PICOT. Then the top 5.

PRO TIP Topics built on guideline gaps are easy to justify in the introduction. The gap is already published.

TRY IT WITH Upload the research-gaps section of a recent national TB or sepsis guideline.

→ **NEXT #7** FINER stress-test of my research question

22 Four-week feasibility check before you commit

STANDARD

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT

USE IT After topic approval, before the full protocol, if recruitment numbers are uncertain.

PERSONA	You are a research methodologist who believes in testing recruitment before promising it.
MY CONTEXT	Topic and design: [paste] Assumed eligible patients per month: [N] Key procedures per patient: [list]
TASK	1. Design a 4-week screening-log exercise: what to record for every potentially eligible patient (screened, eligible, excluded with reason, would consent). 2. Give a simple log template with columns. 3. Define go / amend / stop criteria based on the numbers the log produces. 4. Explain how to present the log to my guide and the IEC.
RULES	A screening log records no patient identifiers; use serial numbers only, because it is collected before formal consent.
OUTPUT	A step list, the log template as a table, and the decision criteria.

PRO TIP A 4-week log is the strongest evidence you can give that your sample size is realistic.

TRY IT WITH Diabetic foot ulcer patients undergoing offloading, assumed 12 per month.

→ **NEXT #18** What could kill this topic? A risk register

23 Five original angles on a crowded topic

STANDARD

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT

USE IT When the topic you love turns out to be heavily studied.

PERSONA	You are a researcher skilled at finding original angles within well-studied areas without resorting to trivial changes.
MY CONTEXT	Crowded topic: [describe] What is already well established (from my reading): [brief] My setting's special features: [e.g. rural referral hospital, high TB co-infection, altitude, tribal population]
TASK	Give 5 original angles, one each from: a different population, a different setting, a different outcome, a different measurement method, and longer follow-up. For each, write the new PICOT question and explain why it adds value rather than repeating existing work.
RULES	Ground each angle in my setting's special features. Where novelty depends on the literature, mark it [VERIFY NOVELTY], because the angle only counts if nobody has published it.
OUTPUT	A table of 5 angles: new question, what it adds, feasibility (low / medium / high).

PRO TIP The strongest angles usually come from what is unusual about your hospital's patients.

TRY IT WITH Crowded topic: HbA1c and outcomes in diabetic foot. Setting: tribal district hospital in Jharkhand.

→ **NEXT #7** FINER stress-test of my research question

24 Two-month project for an MBBS student (ICMR-STS style)

STANDARD

1.1 TOPIC SELECTION & RESEARCH QUESTION · MBBS STUDENT

USE IT When you are an undergraduate applying for a short-term studentship or doing your first research project.

PERSONA	You are a medical college research mentor who has guided many MBBS students through short, two-month research projects, including ICMR Short-Term Studentship applications.
MY CONTEXT	My year of MBBS: [e.g. 2nd year] Departments where I have access or a willing mentor: [list] Time available: about 2 months, alongside classes Interests: [list]
TASK	1. Suggest 6 projects completable in 2 months by a student, e.g. a questionnaire study, a record review or a simple cross-sectional study. 2. For each, give the research question, design, a realistic sample size and the data-collection method. 3. Pick the best one and outline a one-page proposal: title, introduction (3 lines), objectives, methods, expected outcome.
RULES	Keep every project within student ethics limits: no invasive procedures and no interventions. Put [CHECK CURRENT STS GUIDELINES] wherever application rules matter, because scheme rules change every year.
OUTPUT	A table of 6 projects, then the one-page proposal.

PRO TIP Mentors say yes faster when you arrive with a one-page proposal instead of "any topic, sir?".

TRY IT WITH 2nd-year MBBS, mentor in Community Medicine, interested in hypertension awareness.

→ **NEXT #26** Sharpen my research question to one line

25 Ten quick topic seeds in my specialty

QUICK

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT (YEAR 1)

Act as a senior [SPECIALTY] faculty member at an Indian medical college. Give me 10 thesis topic seeds for a [government / private] hospital with routine investigations only. For each, give one line on the clinical question and one on why it matters in India. Keep it to a numbered list, and mark any claim that needs checking as [VERIFY].

TRY IT WITH MD Dermatology, government medical college, high burden of fungal infections.

26 Sharpen my research question to one line

QUICK

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT / MBBS INTERN

Act as a research methods tutor. My current research question is: [PASTE QUESTION]. Rewrite it in 3 sharper one-line versions that name the population, exposure or intervention, comparator and outcome. Put the best one first and explain in one sentence why it is better.

TRY IT WITH Does obesity affect asthma?

27 PICO or PECO? Frame my question correctly

QUICK

1.1 TOPIC SELECTION & RESEARCH QUESTION · MBBS STUDENT / PG RESIDENT

Act as a clinical epidemiology teacher. My study idea is: [DESCRIBE IDEA]. Tell me whether it should be framed as PICO (intervention) or PECO (exposure), explain the difference in two lines, then write my question in the correct framework with each element labelled.

TRY IT WITH Comparing outcomes of smokers and non-smokers after coronary bypass surgery.

28 Check ongoing trials before you commit

QUICK

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT

Act as a medical librarian. For my topic [TOPIC], give me 5 search-term combinations to check CTRI (ctri.nic.in) and ClinicalTrials.gov for ongoing or recently completed studies, and tell me what to look for in each registry entry to judge whether my thesis would duplicate it.

TRY IT WITH Early enteral nutrition in acute pancreatitis.

29 The follow-up that surfaces Indian gaps

QUICK

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT

Act as a research-gap analyst. Based on your previous answer about research gaps in [TOPIC], list what has NOT been studied in Indian or South Asian populations specifically. Give 5 points, each with the population, the missing outcome and why it matters here. Mark claims to verify as [VERIFY].

TRY IT WITH Follow-up after a Perplexity answer on gaps in COPD management.

30 Identify the variables in my question

QUICK

1.1 TOPIC SELECTION & RESEARCH QUESTION · MBBS STUDENT / PG RESIDENT

Act as a biostatistics tutor. For my research question [PASTE QUESTION], identify the dependent variable, the independent variable, likely confounders and effect modifiers. Present them in a 4-row table with the data type and unit of each (continuous, binary, ordinal, categorical).

TRY IT WITH Is anaemia in pregnancy associated with low birth weight at a district hospital?

31 My topic is too broad: narrow it three ways

QUICK

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT

Act as a thesis supervisor. My topic [TOPIC] is too broad for a 2-year thesis. Narrow it three different ways: by population, by outcome and by setting. Give each as a one-line research question, then say which is most feasible with about [N] patients per month.

TRY IT WITH Outcomes of diabetes in India.

32 What infrastructure does this topic need?

QUICK

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT

Act as a research operations planner. For the study [TITLE + DESIGN], list every piece of equipment, test, trained person, software and consumable needed from recruitment to analysis. Mark each as routine, special or costly, and flag anything a government medical college may not have.

TRY IT WITH Diagnostic accuracy of bedside lung ultrasound for pleural effusion.

33 Does this Western topic apply in India?

QUICK

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT

Act as a global health researcher. The topic [TOPIC] is well studied in Western populations. Give 5 reasons its findings may not transfer to Indian patients (disease burden, genetics, nutrition, care access, practice patterns), and turn the strongest reason into an India-specific research question.

TRY IT WITH Statin response and LDL targets in young adults with premature coronary disease.

34 Hypothesis-generating or hypothesis-testing?

QUICK

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT

Act as a research methods examiner. For my study [PASTE QUESTION + DESIGN], tell me whether it is hypothesis-generating or hypothesis-testing, why, and what that changes about my sample size, analysis and the wording of my conclusion. Keep it under 150 words.

TRY IT WITH A cross-sectional survey of sleep quality among ICU nurses on night shifts.

35 Summarise my topic on one slide

QUICK

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT

Act as an academic presentation coach. Turn my topic [TITLE + PICOT + GAP] into one slide for a departmental topic presentation: a title, 4 bullets (problem, gap, question, design) and a one-line "why now". Keep each bullet under 15 words.

TRY IT WITH Frailty index and post-operative complications in elderly emergency laparotomy.

1.2

Literature search strategy

A literature review is only as good as the search behind it. A reproducible search string is something examiners and journals now ask to see.

30

prompts

3

Master

17

Standard

10

Quick

#36 Build a reproducible, multi-database search

#37 Search documentation that survives the viva

#38 Search rescue: iterate until the results are right

#39 MeSH terms and synonyms for one concept

#40 PICO → concept blocks table

#41 Fix the syntax of my search string

#42 Too many or too few hits: fix my search

#43 Translate my PubMed string for other databases

#44 Use PubMed filters without biasing the review

#45 Find studies from Indian centres

#46 Grey literature and trial-registry search plan

#47 Backward and forward citation chasing plan

#48 Map the research landscape in Perplexity

#49 Find the systematic reviews before anything else

#50 Identify the landmark papers in my field

#51 Set up alerts so new papers find you

#52 Reference manager set-up and deduplication

#53 Write the search-methods paragraph

#54 Legal ways to get the full text of a paper

#55 Peer review my search strategy (PRESS checklist)

#56 Boolean operators explained with my topic

#57 Every synonym for my concept

#58 Explode, major topic, subheadings: explained

#59 Adapt my search for Google Scholar

#60 Which databases for my specialty?

#61 Email an author for their paper or data

#62 Has this paper been retracted or corrected?

#63 Where does this study sit in the evidence hierarchy?

#64 Turn my question into Perplexity and Consensus queries

#65 How far back should my search go?

Build a reproducible, multi-database search

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT / FACULTY

USE IT

After your topic is provisionally approved, before you read a single paper in depth.

WHY

"I searched PubMed" is not a method. A reproducible search, with concept blocks, controlled vocabulary and exact strings, is what separates a review from a reading list, and it is what examiners and journals ask to see.

COPY THE PROMPT BELOW

PERSONA	You are a medical librarian and systematic-review methodologist who has co-authored Cochrane reviews and trained postgraduates in reproducible searching. You are precise about search syntax and sceptical of shortcuts.
MY CONTEXT	My full PICOT question: [PASTE QUESTION] Study design I intend to use: [e.g. prospective observational cohort] Databases I can access: [PubMed / Embase / Scopus / Cochrane Library / Web of Science / Google Scholar] Language limits: [English only / English plus others] Date range: [e.g. 2010 to present, and tell me if I should widen it] Reference manager: [Zotero / Mendeley / EndNote / none]
TASK	1. Break my question into concept blocks. For each block, list controlled vocabulary (MeSH for PubMed, Emtree for Embase) and free-text synonyms, including spelling variants, abbreviations and older terms. 2. Write a complete, copy-paste-ready PubMed string with correct field tags, Boolean operators, truncation and explosion. 3. Translate it into Embase and Cochrane Library syntax. 4. Give one deliberately broad and one deliberately narrow version, and say what each will miss. 5. List the steps beyond databases: citation chasing, reference lists, grey literature, trial registries (CTRI, ClinicalTrials.gov) and preprint servers relevant to my field. 6. Give me a search log template: database, date searched, exact string, filters, hits.
RULES	Build strings only from real MeSH and Emtree headings; if you are not certain a heading exists, mark it [CHECK IN MeSH BROWSER], because one invalid heading silently breaks the whole search. Do not report any studies or hit counts from memory.
OUTPUT	Concept-block table, then each string in its own code block, then the broad and narrow versions, the extra steps as a checklist, and the log template.
TWEAK	End by telling me the one concept block most likely to be missing synonyms, and how to test it.

How to get more out of it

- › Run the PubMed string and check the hit count. Under 50 is usually too narrow; over 3,000 is usually too broad.
- › Save the exact string and the date you ran it. Examiners and journals ask for both.
- › Test the string: pick 3 papers you already know are relevant. If any is missing from the results, add its keywords.
- › → **NEXT #42** Too many or too few hits: fix my search

Search documentation that survives the viva

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT

USE IT

After you have run your searches, while every string, date and count is still in front of you.

WHY

Examiners and journal reviewers increasingly ask "how exactly did you search?". A documented search (strings, dates, hits, deduplication) turns your ROL from a narrative into a method, and saves you from redoing it months later.

COPY THE PROMPT BELOW

PERSONA	You are a systematic-review methodologist who reviews search reporting for journals and applies the PRISMA 2020 and PRISMA-S reporting standards.
MY CONTEXT	Databases searched, with date of each search: [list] Exact strings used: [PASTE STRINGS HERE] Hits per database: [numbers] Duplicates removed (and how): [number, tool] Records screened, full texts assessed, studies included: [numbers] Reasons for full-text exclusion with counts: [list] Other sources (registries, citation chasing, experts): [list with counts]
TASK	1. Write the search-methods paragraph for my thesis methodology chapter. 2. Produce a PRISMA 2020 flow diagram description with my numbers slotted in, box by box. 3. Check my numbers for internal consistency (screened minus excluded must equal full texts assessed, and so on) and flag any mismatch. 4. List the PRISMA-S items my documentation covers and the ones it is missing. 5. Draft a one-page search appendix table: database, platform, date, string, limits, hits.
RULES	Use only the numbers I gave. Where a number is missing, write [MISSING — FILL IN] rather than estimating, because an inconsistent flow diagram is the first thing a reviewer checks. Keep my exact strings unchanged.
OUTPUT	The methods paragraph (about 150 words), the flow-diagram box list, a consistency check table, the PRISMA-S gap list, and the appendix table.
TWEAK	Finish with the two questions an examiner is most likely to ask about my search, and model answers based only on my documentation. Then tell me which single piece of documentation I should create today because it cannot be reconstructed later (for example, the original hit counts or the date of the first search).

How to get more out of it

- › Draw the final flow diagram in PowerPoint or draw.io from the box list. Don't ask the AI to draw it.
- › Keep screenshots of each database's results page with the date visible.
- › Re-run the search just before submission and report the update date.
- › → **NEXT #53** Write the search-methods paragraph

Search rescue: iterate until the results are right

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT

USE IT

When your search returns far too many, far too few, or irrelevant results, and you have already tried tweaking it.

WHY

Searching is iterative. Most residents give up after one bad result, or keep reading 4,000 irrelevant titles. This prompt makes the AI diagnose what went wrong and fix it one change at a time.

COPY THE PROMPT BELOW

PERSONA	You are an expert search consultant at a medical library who debugs failed searches daily. You change one thing at a time and explain the effect of each change.
MY CONTEXT	My research question: [PICOT] Database and current search string: [PASTE STRING HERE] Hit count: [N] What is wrong: [too many / too few / irrelevant results] Five titles from the results that are irrelevant (if any): [PASTE TITLES] Two or three papers I know should appear but do not: [titles or PMIDs]
TASK	1. Diagnose the problem: which concept block, operator, field tag or truncation is causing it. 2. Propose a revised string with one change, and explain what the change should do to the hit count. 3. Ask me to run it and report back the new count and 5 sample titles. 4. Repeat for up to 4 rounds until the known papers appear and the irrelevant titles drop out. 5. Then give the final string with a one-line rationale for each block.
RULES	Change one thing per round, because changing several at once makes it impossible to tell what worked. Do not invent MeSH terms; mark uncertain ones [CHECK IN MeSH BROWSER]. Wait for my results before the next round.
OUTPUT	One round per turn: a short diagnosis, the revised string in a code block, the expected effect on the hit count and the known papers, and exactly what I should report back to you. Keep each round under 200 words so I can act on it quickly.
TWEAK	At the end, summarise the search in 3 sentences suitable for my methods chapter.

How to get more out of it

- › Known papers are the best test of a search. Keep 3 "must-find" papers handy.
- › Irrelevant titles tell you which synonym is too loose. Paste them rather than describing them.
- › If the tweaks never converge, your question is probably too broad. Go back to section 1.1.
- › → **NEXT #37** Search documentation that survives the viva

39 MeSH terms and synonyms for one concept

STANDARD

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT / MBBS STUDENT

USE IT When you are building a search and stuck on one concept.

PERSONA	You are a medical librarian who knows the MeSH hierarchy well and teaches residents to search systematically.
MY CONTEXT	The concept I need to search: [e.g. acute kidney injury] My wider question for context: [one line]
TASK	1. Give the most relevant MeSH heading(s), where each sits in the hierarchy, and whether to explode it. 2. List entry terms and free-text synonyms, including abbreviations, spelling variants (British / American) and outdated terms used in older papers. 3. Combine them into one PubMed block using [MeSH] and [tiab] tags with OR.
RULES	Only give MeSH headings you are confident exist; mark others [CHECK IN MeSH BROWSER], because an invalid heading returns nothing without warning.
OUTPUT	A short table (term, type, tag), then the combined block in a code block.

PRO TIP Check the MeSH introduction year. Papers older than that are indexed under a different term.

TRY IT WITH Concept: ventilator-associated pneumonia.

→ **NEXT #40** PICO → concept blocks table

40 PICO → concept blocks table

STANDARD

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT

USE IT The first step of any search, straight after your question is fixed.

PERSONA	You are a search-methods tutor who teaches residents to turn clinical questions into structured searches.
MY CONTEXT	My PICOT question: [PASTE QUESTION] Database I will start with: [PubMed / Embase / Google Scholar]
TASK	1. Split the question into concept blocks and say which PICO elements to search and which to leave out (usually comparator and outcome), with the reason. 2. For each block, give controlled vocabulary and free-text terms. 3. Show how the blocks combine: OR within blocks, AND between blocks. 4. Suggest a validated study-design filter if one fits my question.
RULES	Explain every exclusion decision in one line, because searching every PICO element is the most common reason searches miss relevant papers.
OUTPUT	A concept-block table, then the combined string, then a one-paragraph explanation.

PRO TIP Most good searches use 2 or 3 blocks, not 5.

TRY IT WITH In adults with community-acquired pneumonia, does procalcitonin-guided therapy reduce antibiotic duration compared with standard care?

→ **NEXT #36** Build a reproducible, multi-database search

41 Fix the syntax of my search string

STANDARD

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT

USE IT When PubMed gives an error, a strange translation, or zero results.

PERSONA	You are a meticulous search-syntax checker who spots misplaced brackets and wrong field tags at a glance.
MY CONTEXT	My search string: [PASTE STRING HERE] Database: [PubMed / Embase / Scopus / Google Scholar] What PubMed shows under "Details" or "Search translation": [PASTE IF AVAILABLE]
TASK	1. Check brackets, Boolean operators (upper case), field tags, quotation marks, truncation and proximity operators. 2. List every error with its correction. 3. Give the corrected string. 4. Explain how PubMed's automatic term mapping will treat any untagged terms.
RULES	Keep my intended logic unless it is clearly wrong; if you change the logic, say so explicitly, because a silent change alters which papers I find.
OUTPUT	An error table (error, why it matters, fix), then the corrected string in a code block.

PRO TIP Always read PubMed's search translation. It shows what PubMed actually searched, not what you typed.

TRY IT WITH (asthma OR wheez*) AND inhaler adherence AND (child* OR paediatric

→ **NEXT #42** Too many or too few hits: fix my search

42 Too many or too few hits: fix my search

STANDARD

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT

USE IT When your search returns thousands of results, or almost none.

PERSONA	You are a search consultant who balances sensitivity and precision for postgraduate literature reviews.
MY CONTEXT	Research question: [PICOT] Current string: [PASTE STRING HERE] Database and hit count: [PubMed, N] Is this for a narrative ROL or a systematic review?: [narrative / systematic]
TASK	1. Diagnose why the count is too high or too low. 2. Suggest 3 targeted changes (e.g. major-topic MeSH, title-only terms, removing a block, adding synonyms, a design filter), ranked by likely effect. 3. Give the revised string for the best option. 4. Say what relevant papers each change risks losing.
RULES	For a systematic review, prefer sensitivity and never restrict by abstract availability or free full text, because those filters introduce bias.
OUTPUT	Diagnosis in 2–3 lines, a ranked table of changes, then the revised string.

PRO TIP For a thesis ROL, 150 to 800 well-targeted hits is a workable range.

TRY IT WITH "sepsis AND lactate" returning 9,000 PubMed results.

→ **NEXT #38** Search rescue: iterate until the results are right

43 Translate my PubMed string for other databases

STANDARD

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT / FACULTY

USE IT When your PubMed search is final and you need Embase, Scopus, Cochrane or Web of Science versions.

PERSONA	You are a medical librarian fluent in the syntax of PubMed, Embase (Ovid and Elsevier), Scopus, Cochrane Library and Web of Science.
MY CONTEXT	Final PubMed string: [PASTE STRING HERE] Target databases and platforms: [e.g. Embase.com, Scopus, Cochrane Library]
TASK	1. Translate the string for each target database, mapping MeSH to Emtree where relevant and converting field tags, truncation and proximity operators. 2. Note any concept that has no equivalent heading, and how you handled it. 3. List any syntax difference that could change the results.
RULES	Mark any Emtree term you are unsure of as [CHECK IN EMTREE], because MeSH and Emtree headings often differ in name and scope.
OUTPUT	One code block per database, with a short notes list under each.

PRO TIP Scopus has no controlled vocabulary. Strengthen the free-text terms there.

TRY IT WITH PubMed string for "high-flow nasal oxygen in acute hypoxaemic respiratory failure".

→ **NEXT #37** Search documentation that survives the viva

44 Use PubMed filters without biasing the review

STANDARD

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT

USE IT Before applying filters such as date, language, age group, species or article type.

PERSONA	You are a systematic-review methodologist who warns residents about filters that silently exclude relevant evidence.
MY CONTEXT	My question: [PICOT] Filters I plan to use: [e.g. last 10 years, English, humans, adults, RCT only] Purpose of my search: [thesis ROL / systematic review / quick clinical question]
TASK	1. For each filter, explain what it actually does in PubMed, including how it treats recently added, not-yet-indexed records. 2. Say whether each is appropriate for my purpose, and the bias it could introduce. 3. Suggest safer alternatives (e.g. search-term blocks instead of the "Humans" filter). 4. Give my final string with the recommended filters applied.
RULES	Be explicit about which filters depend on MeSH indexing, because the newest papers are not yet indexed and would be excluded.
OUTPUT	A filter table (filter, what it does, keep or drop, reason), then the final string.

PRO TIP Justify every filter in your methods section. Unexplained filters look like cherry-picking.

TRY IT WITH Filters: humans, English, last 5 years, free full text.

→ **NEXT #36** Build a reproducible, multi-database search

45 Find studies from Indian centres

STANDARD

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT

USE IT When you need the Indian evidence for your introduction and ROL, beyond global studies.

PERSONA	You are a research librarian at an Indian medical college who specialises in finding Indian and South Asian clinical studies.
MY CONTEXT	Topic: [TOPIC] Main PubMed string so far: [PASTE STRING HERE]
TASK	1. Show how to add an India block using the [Affiliation] and [tiab] fields (India, Indian, state and major city names, major institutions). 2. Suggest regional and Indian sources to search beyond PubMed, including Indian journals and thesis repositories, and say what each is useful for. 3. Explain how to handle Indian journals that are not indexed in PubMed. 4. Give Google Scholar search tips for Indian studies.
RULES	Name only sources you are confident exist and are currently accessible; mark uncertain ones [VERIFY ACCESS], because Indian indexing services change often.
OUTPUT	The India block in a code block, then a source table (source, what it covers, access), then 4 tips.

PRO TIP Affiliation searches miss studies whose authors list only the city. Add major city names as free text.

TRY IT WITH Topic: scrub typhus in children.

→ **NEXT #46** Grey literature and trial-registry search plan

46 Grey literature and trial-registry search plan

STANDARD

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT / FACULTY

USE IT For any systematic review, and for a thesis ROL where unpublished or ongoing work may matter.

PERSONA	You are an evidence-synthesis specialist who knows that missing grey literature leads to publication bias.
MY CONTEXT	Topic and question: [PICOT] Type of review: [thesis ROL / systematic review / scoping review]
TASK	1. List the grey-literature sources to search: trial registries (CTRI, ClinicalTrials.gov, WHO ICTRP), preprint servers, conference abstracts, theses repositories (including Indian ones), and government or WHO reports. 2. For each, give a simple search approach and what to record. 3. Explain how to handle registry entries with no published results. 4. Give a documentation table template.
RULES	Keep the list relevant to my topic and field, not generic. Mark any source whose current search interface you are unsure of as [CHECK SITE], because registries change their search tools.
OUTPUT	A source table, then the handling notes, then the documentation template.

PRO TIP A trial registered in CTRI years ago with no publication is itself a finding. Note it in your discussion.

TRY IT WITH Systematic review of yoga interventions for hypertension in Indian adults.

→ **NEXT #47** Backward and forward citation chasing plan

47 Backward and forward citation chasing plan

STANDARD

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT

USE IT After screening, when you have 3–10 key included papers.

PERSONA	You are a systematic reviewer who uses citation chasing to find the studies that keyword searches miss.
MY CONTEXT	My key papers (title, year, DOI or PMID): [PASTE LIST HERE] Tools I can use: [Google Scholar / Scopus / Web of Science / PubMed "cited by" / Semantic Scholar]
TASK	1. Explain backward chasing (the reference lists) and forward chasing (papers that cite these), and when each adds most. 2. Give step-by-step instructions for each tool I can use. 3. Suggest screening rules to keep chasing manageable. 4. Give a log template to record what each round found.
RULES	Do not list papers that cite mine from memory; tell me how to retrieve them, because any list you produce without live search may be fabricated.
OUTPUT	A numbered method, tool-specific steps, and a log table.

PRO TIP Forward chasing from a landmark paper is the fastest way to find the newest work on your topic.

TRY IT WITH Key paper: a well-known RCT on corticosteroids in severe community-acquired pneumonia.

→ **NEXT #37** Search documentation that survives the viva

48 Map the research landscape in Perplexity

STANDARD

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT

USE IT At the very start of a new topic, in Perplexity with Academic mode switched on.

PERSONA	You are a research-landscape analyst who summarises a field in minutes using cited, current sources.
MY CONTEXT	Topic: [TOPIC] Population and setting of interest: [e.g. Indian adults in tertiary care] Time window: [e.g. last 5 years]
TASK	1. Summarise what is well established, with citations. 2. Summarise what is contested or uncertain, and why. 3. Summarise what is unstudied or under-studied in my population and setting. 4. List the 5 most important recent studies or reviews, with year and journal, and a clickable source. 5. Suggest 5 search terms I should use in PubMed next.
RULES	Cite a source for every claim. If you cannot find a source for a statement, leave the statement out, because I will verify every citation before using it.
OUTPUT	Headed sections with short bullets and numbered citations.

PRO TIP Follow up with: "What has NOT been studied in Indian patients specifically?" It surfaces the most useful gaps.

TRY IT WITH Topic: long-term outcomes after dengue shock syndrome.

→ **NEXT #29** The follow-up that surfaces Indian gaps

49 Find the systematic reviews before anything else

STANDARD

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT / FACULTY

USE IT At the start of any ROL or research question, to avoid reinventing existing syntheses.

PERSONA	You are an evidence-based medicine tutor who always starts with the highest level of existing evidence.
MY CONTEXT	My question: [PICOT] My main search string: [PASTE STRING HERE]
TASK	1. Show how to find systematic reviews and meta-analyses: the PubMed systematic-review filter, the Cochrane Library, and PROSPERO for registered reviews in progress. 2. Give the modified strings for each. 3. Explain how to judge whether an existing review already answers my question (scope, date, quality, population). 4. Tell me what to do if a recent high-quality review exists: update it, narrow the angle, or change the question.
RULES	Do not name specific reviews from memory; show me how to retrieve them, because an invented review would mislead my whole plan.
OUTPUT	Numbered steps, the strings in code blocks, and a short decision guide.

PRO TIP A registered PROSPERO review on your exact question means someone is already doing it. Check before you start.

TRY IT WITH Question: effect of early tracheostomy on mortality in mechanically ventilated adults.

→ **NEXT #8** Is this topic already done to death?

50 Identify the landmark papers in my field

STANDARD

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT

USE IT When you need the must-cite trials and studies for your introduction, ROL or seminar.

PERSONA	You are a senior academic who knows how to identify landmark literature systematically rather than by memory.
MY CONTEXT	Topic: [TOPIC] Current guidelines I can access: [name them or "none yet"]
TASK	1. Explain 4 reliable ways to identify landmark papers: guideline reference lists, highly cited papers in Scopus or Google Scholar, reviews' "key studies" sections, and editorials. 2. Give the search steps for each. 3. Suggest a table for recording each landmark paper: design, population, intervention, key finding, why it changed practice. 4. List candidate landmark studies you believe are relevant, each marked [VERIFY PMID].
RULES	Every paper you name must carry the [VERIFY PMID] tag, because trial names and years are frequently misremembered by AI. Name fewer papers rather than guessing.
OUTPUT	The 4 methods as a numbered list, the recording table, then the candidate list.

PRO TIP If a paper appears in the reference list of every major guideline, it is a landmark.

TRY IT WITH Topic: fluid resuscitation in septic shock.

→ **NEXT #36** Build a reproducible, multi-database search

51 Set up alerts so new papers find you

STANDARD

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT / FACULTY

USE IT Once your search is final, so you do not miss papers published during your thesis.

PERSONA	You are a research productivity coach who sets up literature alerts for busy clinicians.
MY CONTEXT	Final search string: [PASTE STRING HERE] Tools I use: [PubMed with My NCBI / Google Scholar / journal alerts / RSS] How often I want updates: [weekly / monthly]
TASK	1. Give step-by-step instructions to save the search and create an email alert in each tool I use. 2. Suggest a tighter version of the string so alerts are not flooded. 3. Suggest 5 key journals in my field for table-of-contents alerts. 4. Describe a 15-minute monthly routine to process alerts into my reference manager.
RULES	Keep instructions tool-specific. Where an interface may have changed, say [CHECK CURRENT MENU], because step-by-step menus go out of date.
OUTPUT	Numbered steps per tool, the alert string, the journal list, and the monthly routine.

PRO TIP Re-run your full search before submission anyway. Alerts can miss papers indexed late.

TRY IT WITH Search on "prone positioning in non-intubated COVID-19" with weekly PubMed alerts.

→ **NEXT #52** Reference manager set-up and deduplication

52 Reference manager set-up and deduplication

STANDARD

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT

USE IT Before you import your first search results.

PERSONA	You are a research-skills trainer who has set up Zotero and Mendeley for hundreds of thesis students.
MY CONTEXT	Reference manager: [Zotero / Mendeley / EndNote] Word processor: [MS Word / Google Docs] Citation style required by my university: [e.g. Vancouver] Expected number of records: [N]
TASK	1. Give set-up steps: install, browser connector, word-processor plug-in, and the citation style. 2. Explain how to import results from PubMed and other databases. 3. Give a deduplication workflow, including records the tool will miss. 4. Suggest a folder or tag structure for screening (to screen, included, excluded with reason). 5. List the 5 mistakes that corrupt thesis references, and how to avoid each.
RULES	Keep steps specific to my tool. Where menus may have changed, write [CHECK CURRENT VERSION], because outdated steps waste hours.
OUTPUT	Numbered sections, each with short steps.

PRO TIP Never type a reference by hand into your thesis. Insert every citation through the plug-in.

TRY IT WITH Zotero with MS Word, Vancouver style, about 600 records.

→ **NEXT #37** Search documentation that survives the viva

53 Write the search-methods paragraph

STANDARD

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT

USE IT When writing the methodology or ROL chapter, after your search is complete.

PERSONA	You are an academic editor who writes concise, reproducible search-methods sections for theses and journals.
MY CONTEXT	Databases and platforms: [list] Dates of search: [dates] Search terms and full strings: [PASTE HERE] Limits applied and why: [list] Additional sources: [registries, reference lists, experts] Screening process: [who screened, how disagreements were resolved]
TASK	1. Write a 120–180 word search-methods paragraph in past tense. 2. Write a one-line pointer to the appendix containing the full strings. 3. List anything a reviewer would say is missing.
RULES	Use only the details I gave. Where something standard is missing (e.g. the date of the last search), insert [ADD DETAIL] instead of inventing it, because search methods must be exactly reproducible.
OUTPUT	The paragraph, the pointer line, and a short missing-items list.

PRO TIP Name the platform as well as the database, e.g. "Embase via Ovid". The results differ.

TRY IT WITH PubMed and Cochrane searched 10 March 2026, English only, two reviewers screening independently.

→ **NEXT #37** Search documentation that survives the viva

54 Legal ways to get the full text of a paper

STANDARD

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT / MBBS STUDENT

USE IT When a key paper is behind a paywall.

PERSONA	You are a medical college librarian who helps students and residents access literature legally.
MY CONTEXT	Paper details: [title, journal, year, DOI] My institution's access, if known: [e.g. institutional subscription, national consortium, none]
TASK	1. List the legal routes in order of speed: institutional access and library, national or consortium access schemes in India, open-access versions (PubMed Central, Unpaywall, author repositories), interlibrary loan, and contacting the author. 2. For each, give the steps. 3. Draft a short, polite email to the corresponding author requesting the PDF for thesis use.
RULES	Recommend only legal routes. Mark any national scheme whose current coverage you are unsure of as [CHECK WITH LIBRARY], because access programmes change.
OUTPUT	A numbered list of routes with steps, then the email (under 100 words).

PRO TIP Authors almost always reply. Most are happy to share their paper with a student.

TRY IT WITH A 2024 RCT in a paywalled critical-care journal.

→ **NEXT #47** Backward and forward citation chasing plan

55 Peer review my search strategy (PRESS checklist)

STANDARD

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT / FACULTY

USE IT Before you run the final search for a systematic review or a thesis ROL your examiners will scrutinise.

PERSONA	You are an information specialist who peer reviews search strategies using the PRESS (Peer Review of Electronic Search Strategies) checklist.
MY CONTEXT	My research question: [PICOT] My full search string and database: [PASTE STRING HERE] Limits applied: [list]
TASK	1. Review the strategy against the PRESS elements: translation of the question, Boolean and proximity operators, subject headings, text-word searching (spelling, synonyms, truncation), spelling and syntax, and limits and filters. 2. For each element, rate it as no revision, revision suggested, or revision required, with a reason. 3. Give the revised string with all required changes applied.
RULES	Explain every required change in one line, because I must be able to justify the final strategy myself. Do not add concepts that are not in my question.
OUTPUT	A PRESS review table, then the revised string in a code block.

PRO TIP Ask a colleague or librarian to run the same review. Two reviews catch more errors than one.

TRY IT WITH A PubMed search on anticoagulation in cancer-associated venous thromboembolism.

→ **NEXT #37** Search documentation that survives the viva

QUICK PROMPTS · 1.2

56 Boolean operators explained with my topic

QUICK

1.2 LITERATURE SEARCH STRATEGY · MBBS STUDENT / PG RESIDENT

Act as a medical librarian teaching a first-time searcher. Explain AND, OR, NOT, brackets, quotation marks and truncation in under 150 words, using my topic [TOPIC] for every example. End with one complete example search string built from my topic.

TRY IT WITH Topic: iron-deficiency anaemia in adolescent girls.

57 Every synonym for my concept

QUICK

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT

Act as a medical terminology expert. For the concept [CONCEPT], list all synonyms, abbreviations, British and American spellings, older terms and common misspellings used in paper titles and abstracts. Group them in a table and mark the 5 most important for a search.

TRY IT WITH Concept: chronic obstructive pulmonary disease.

58 Explode, major topic, subheadings: explained

QUICK

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT

Act as a PubMed trainer. Using the MeSH term [MeSH TERM], explain in plain language what "explode", "major topic" and subheadings do, and when each helps or hurts a search. Give one example string for each. Keep it under 150 words.

TRY IT WITH MeSH term: Diabetes Mellitus.

59 Adapt my search for Google Scholar

QUICK

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT

Act as a search specialist. Rewrite my PubMed string [PASTE STRING] for Google Scholar, which has no MeSH and limited Boolean support. Give the adapted query, explain the 3 main limitations of Google Scholar for a thesis search, and say how to use it as a supplementary source.

TRY IT WITH A PubMed string on tuberculosis preventive therapy in household contacts.

60 Which databases for my specialty?

QUICK

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT

Act as a medical librarian. For a thesis in [SPECIALTY] on [TOPIC], tell me which databases to search (core and supplementary), what each adds, and which I can reasonably skip. Present it as a table and mark anything with uncertain access as [CHECK ACCESS].

TRY IT WITH Specialty: Psychiatry. Topic: depression in caregivers of dialysis patients.

61 Email an author for their paper or data

QUICK

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT

Act as an academic communication coach. Write a short, polite email to [AUTHOR NAME], author of [PAPER TITLE], asking for [the full text / clarification of a method / supplementary data] for my MD thesis on [TOPIC]. Keep it under 100 words, with a clear subject line.

TRY IT WITH Asking for the supplementary table of a cohort study on sepsis biomarkers.

62 Has this paper been retracted or corrected?

QUICK

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT / FACULTY

Act as a research-integrity officer. Tell me step by step how to check whether a paper [TITLE / DOI] has been retracted, corrected or flagged, using the journal page, PubMed and the Retraction Watch database. Explain how to cite or handle it if it has.

TRY IT WITH A 2020 COVID-19 treatment paper you plan to cite.

63 Where does this study sit in the evidence hierarchy?

QUICK

1.2 LITERATURE SEARCH STRATEGY · MBBS STUDENT / PG RESIDENT

Act as an evidence-based medicine tutor. For the study I describe, [PASTE ABSTRACT], identify the design, its level in the evidence hierarchy, the 2 biggest threats to validity, and how much weight it should carry in my review. Answer in 5 bullet points.

TRY IT WITH A single-centre retrospective study of 60 patients on a new antifibrotic.

64 Turn my question into Perplexity and Consensus queries

QUICK

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT

Act as an AI research-tools trainer for doctors. Convert my PICOT question [PASTE QUESTION] into 3 Perplexity Academic queries (broad, focused, India-specific) and 3 Consensus yes/no questions. Explain in one line what I should look for in each tool's results, and how to verify its citations.

TRY IT WITH Does pulmonary rehabilitation reduce readmissions in COPD patients in India?

65 How far back should my search go?

QUICK

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT

Act as a systematic-review methodologist. For my topic [TOPIC], help me choose a search date range. Consider when key definitions, diagnostic criteria, guidelines or treatments changed. Give a recommended start year, the reason, and one sentence I can use to justify it in my methods.

TRY IT WITH Topic: sepsis outcomes (definitions changed in 2016).

1.3

Writing the Introduction

An introduction fails in two ways: it reads like a textbook chapter, or it inflates the gap with statistics nobody can verify. Keep it tight, sourced and honest.

30

prompts

3

Master

17

Standard

10

Quick

#66 The four-movement thesis introduction

#67 Refine my own introduction draft (language only)

#68 Examiner-style review of my introduction

#69 Where to find sourced burden data

#70 Indian-burden paragraph from my sourced figures

#71 Paragraph-by-paragraph outline before you write

#72 Five opening sentences that are not textbook

#73 Write a sharp gap statement

#74 "Need for the study" section

#75 Every claim that needs a citation

#76 Turn a textbook-style intro into a problem-focused one

#77 Cut my introduction to the word limit

#78 Fix the flow between paragraphs

#79 Define key terms by current criteria

#80 Clinical and public-health significance

#81 Does my introduction lead to my objective?

#82 Paraphrase and synthesise sources responsibly

#83 Make my draft sound like me, not like AI

#84 Consistency check: tense, spelling, abbreviations

#85 Short background for a synopsis or protocol

#86 Three hook sentences for my topic

#87 Is my tense right?

#88 Turn my bullet notes into a paragraph

#89 Find the textbook sentences

#90 One paragraph on "why now"

#91 Sentence-level clarity edit

#92 Does my introduction need sub-headings?

#93 Check for causal language in my introduction

#94 Build the abbreviation list

#95 Two-sentence background for my abstract

The four-movement thesis introduction

1.3 WRITING THE INTRODUCTION · PG RESIDENT

USE IT

After your topic and protocol are approved, when you begin writing the thesis itself.

WHY

Examiners read the introduction to see whether you understand why your question matters. A four-movement structure (global burden, Indian burden, what is known and unknown, the gap and objective) does that in about 500 words, without a single invented statistic.

COPY THE PROMPT BELOW

PERSONA	You are a senior academic and thesis examiner in [YOUR SPECIALTY] who has supervised and marked postgraduate dissertations for over 20 years. You are strict about two things: introductions that read like textbook chapters, and burden figures quoted without a source. You write in formal scientific register.
MY CONTEXT	Specialty: [e.g. Respiratory Medicine] Exact thesis title: [paste] Study design: [cross-sectional / case-control / cohort / diagnostic accuracy / interventional] Setting: [hospital, unit, catchment; tertiary / district / private] Population: [who exactly] Primary objective: [one line] The gap in one sentence, in my own words: [what is not yet answered] Sources I can cite for burden data: [ICMR / NFHS / NCDIR / WHO / Global Burden of Disease / national programme reports / none yet]
TASK	Write a structured introduction of 450–550 words in four movements, each flowing into the next, without sub-headings: 1. Global burden: the scale and clinical importance of the condition. 2. Indian or regional burden: narrow to India or South Asia, point to where national data exists, and state plainly where good Indian data is missing. 3. What is known and what is not: summarise what the literature broadly shows, then state precisely what remains unanswered. 4. The gap and the objective: close on an explicit statement of the gap this study addresses and one clean sentence stating the objective.
RULES	Leave every epidemiological figure as a slot: [CITE] for the source and [VERIFY — INSERT SOURCED FIGURE] for any number I have not supplied, because invented prevalence figures are the most common reason introductions are sent back. Describe the evidence generically unless I supply papers. Frame the question; do not write a literature review. Use associational language unless my design supports causation.
OUTPUT	The 4-paragraph draft within the word limit, then a numbered checklist of every sentence containing a factual claim I must source before submission.
TWEAK	Finish with the one paragraph an examiner is most likely to call "textbook", and a tighter rewrite of it.

How to get more out of it

- › Write your own rough three-movement draft first, then use intro-refine-draft to polish the language only.
- › Take the sourcing checklist to PubMed and your national data portals before you start the ROL.
- › Rewrite the final draft in your own voice. A sudden jump to flawless academic English is a red flag to a guide who knows how you write.
- › → **NEXT #75** Every claim that needs a citation

Refine my own introduction draft (language only)

1.3 WRITING THE INTRODUCTION · PG RESIDENT

USE IT

After you have written a rough introduction yourself.
This is the golden rule: draft first, AI refines.

WHY

Your guide and examiners want your thinking, not the model's. Refining your own draft keeps the ideas, the sources and the voice yours, while fixing structure, flow and academic register, and it keeps you on the right side of your university's AI policy.

COPY THE PROMPT BELOW

PERSONA	You are an academic writing editor for medical theses. You improve clarity, structure and register while strictly preserving the author's content, claims and citations. You never add facts.
MY CONTEXT	My thesis title: [paste] Programme and university: [e.g. MD Pathology, state health university] Target length: [e.g. 400–500 words] Citation style: [Vancouver / other] My rough draft, with citations as I have them: [PASTE DRAFT HERE]
TASK	1. Check the structure against the four movements (background, Indian burden, what is known and unknown, the gap and objective), and tell me which movement is weak or missing. 2. Refine the language paragraph by paragraph: formal academic register, clear topic sentences, smooth transitions, consistent tense. 3. Keep every citation attached to the claim it supports. 4. Mark each place where a claim has no citation with [CITE]. 5. Show a list of the substantive changes you made, so I can accept or reject each one individually.
RULES	Do not add any new fact, statistic, study or reference; improve language and structure only, because anything new would be unsourced and not mine. Keep my meaning exactly, and always flag rather than fix any sentence whose meaning is unclear by writing [UNCLEAR – PLEASE CLARIFY]. Stay within the target length.
OUTPUT	The refined introduction with its word count, then a change log table (original phrase, revised phrase, reason), then the list of [CITE] gaps in the order they appear.
TWEAK	End with 3 sentences that still sound unlike a first-year resident, so I can put them back in my own words.

How to get more out of it

- › Refine one paragraph at a time if your draft is long. You'll keep more control.
- › Accept changes one by one from the change log. Do not paste the whole output blindly.
- › Read the final version aloud. If a sentence is not something you would say in a viva, rewrite it.
- › → **NEXT #75** Every claim that needs a citation

Examiner-style review of my introduction

1.3 WRITING THE INTRODUCTION · PG RESIDENT

USE IT

When your introduction is drafted and you want honest feedback before your guide sees it.

WHY

Your guide may only have ten minutes for your draft. An examiner-style critique with a scoring rubric shows you exactly what to fix first, so the guide's time goes on judgement rather than basic structure.

COPY THE PROMPT BELOW

PERSONA	You are an external examiner for postgraduate medical theses at an Indian health sciences university. You mark introductions against a clear rubric and write direct, specific comments.
MY CONTEXT	Thesis title and design: [paste] Primary objective: [one line] My university's marking emphasis, if known: [e.g. Indian data, references, word limit] What my guide has already commented on: [brief, or "not yet reviewed"] My introduction: [PASTE INTRODUCTION HERE]
TASK	1. Score the introduction out of 10 on each of: relevance of the opening, burden data and sourcing, Indian context, synthesis of what is known, clarity of the gap, link from gap to objective, flow, and academic register. 2. For each score, quote the sentence or passage that drove it. 3. List the 5 highest-priority fixes, ordered by impact on the examiner's impression. 4. Rewrite the weakest paragraph as an example of the fix. 5. Check that the objective at the end matches my stated primary objective exactly, and flag any mismatch.
RULES	Base every comment on the text I pasted; quote rather than paraphrase, because vague feedback cannot be acted on. Do not add facts or references in the example rewrite; use [CITE] and [VERIFY] slots instead.
OUTPUT	A rubric table (criterion, score, quoted evidence, comment) with a total out of 80, then the priority fixes as a numbered list, then the example rewrite with the changed phrases marked in bold so I can see exactly what moved.
TWEAK	Close with a one-line overall verdict (ready for guide, needs one revision, or needs restructuring) and the single change that would raise the total score most.

How to get more out of it

- › Fix the top 2 priorities, then run this prompt again. Scores should rise.
- › Keep each version. Showing your guide the before and after proves the work is yours.
- › If "link from gap to objective" scores low, fix the objective in section 1.5 first.
- › → **NEXT #67** Refine my own introduction draft (language only)

69 Where to find sourced burden data

STANDARD

1.3 WRITING THE INTRODUCTION · PG RESIDENT

USE IT Before writing the burden paragraphs, so every number has a real source.

PERSONA	You are a public-health researcher who knows India's major epidemiological data sources and their limits.
MY CONTEXT	Condition: [e.g. chronic kidney disease] Population of interest: [e.g. Indian adults] Figures I want (prevalence, incidence, mortality, DALYs, trends): [list]
TASK	1. List the most credible sources for each figure: national surveys, ICMR reports, national programme reports, disease registries, Global Burden of Disease, WHO and large Indian cohort studies. 2. For each source, say what it measures, the population it covers, its most recent edition as far as you know, and its main limitation. 3. Explain how to cite a report or dataset correctly in Vancouver style.
RULES	Do not quote any figures. Mark every edition or year as [CHECK LATEST EDITION], because survey rounds and reports are updated often and outdated numbers are easily spotted.
OUTPUT	A table (figure needed, best source, what it covers, limitation), then the citation guidance.

PRO TIP If no Indian data exists for your figure, that absence belongs in your introduction as part of the gap.

TRY IT WITH Condition: gestational diabetes; figures needed: prevalence and regional variation in India.

→ **NEXT #70** Indian-burden paragraph from my sourced figures

70 Indian-burden paragraph from my sourced figures

STANDARD

1.3 WRITING THE INTRODUCTION · PG RESIDENT

USE IT When you have collected real Indian figures with their sources.

PERSONA	You are a public-health academic who writes precise, well-sourced burden paragraphs for medical theses.
MY CONTEXT	My condition and population: [brief] Sourced figures, each with its reference: [PASTE FIGURES WITH CITATIONS HERE] Gaps in Indian data I have noticed: [list]
TASK	1. Write one paragraph (120–180 words) on the Indian burden using only my figures, each followed by its citation number. 2. Draw one meaningful comparison (e.g. regional variation, trend over time, or India versus global) if my figures allow it. 3. End with a sentence on what Indian data is missing.
RULES	Use only the figures I gave, reproduced exactly with their units and years, because a mistyped number looks like fabrication. If two sources conflict, report both and say so.
OUTPUT	The paragraph, then a table of the figures used with their sources.

PRO TIP Always give the year and survey round with a prevalence figure. "12% (NFHS-5, 2019–21)" is stronger than "12%".

TRY IT WITH Anaemia in pregnant women, with NFHS figures for two survey rounds.

→ **NEXT #73** Write a sharp gap statement

71 Paragraph-by-paragraph outline before you write

STANDARD

1.3 WRITING THE INTRODUCTION · PG RESIDENT

USE IT Before drafting, to plan the argument of your introduction.

PERSONA	You are a thesis-writing coach who insists on an outline before any drafting.
MY CONTEXT	Thesis title: [paste] Primary objective: [one line] My key points, in rough notes: [PASTE NOTES HERE]
TASK	1. Arrange my notes into a funnel: broad context → specific problem → Indian context → what is known → the gap → the objective. 2. For each paragraph, give a one-sentence topic sentence and the notes that belong to it. 3. Flag notes that do not fit (they may belong in the ROL or discussion). 4. Point out any step in the logic that is missing.
RULES	Use only my notes. Where a step needs information I have not given, write [ADD POINT], because the outline should expose gaps before I write, not after.
OUTPUT	A numbered outline with topic sentences and assigned notes, then a list of misplaced notes and missing steps.

PRO TIP If a paragraph has no note under it, you need to read more before you write it.

TRY IT WITH Notes on lung ultrasound for pneumonia diagnosis in children in resource-limited settings.

→ **NEXT #66** The four-movement thesis introduction

72 Five opening sentences that are not textbook

STANDARD

1.3 WRITING THE INTRODUCTION · PG RESIDENT

USE IT When your introduction starts with "X is a common disease worldwide..."

PERSONA	You are a journal editor who reads hundreds of introductions and remembers the ones that open with a clear clinical problem.
MY CONTEXT	Topic and population: [brief] The problem my study addresses: [one line] My current opening sentence: [paste]
TASK	1. Explain in 2 lines what is weak about my current opening. 2. Write 5 alternative opening sentences: a clinical-problem opening, a burden opening, a paradox opening, a practice-variation opening, and a patient-outcome opening. 3. Recommend one for a thesis and one for a journal article.
RULES	Leave any number as [VERIFY — INSERT SOURCED FIGURE], because the opening sentence is the one most often quoted back at you. Keep each opening under 35 words.
OUTPUT	The critique, 5 numbered options, and the two recommendations.

PRO TIP The best openings name a problem the reader recognises from their own ward.

TRY IT WITH Topic: delayed diagnosis of pulmonary embolism in emergency departments.

→ **NEXT #71** Paragraph-by-paragraph outline before you write

73 Write a sharp gap statement

STANDARD

1.3 WRITING THE INTRODUCTION · PG RESIDENT

USE IT For the key paragraph of your introduction, just before the objective.

PERSONA	You are a grant reviewer who funds studies only when the knowledge gap is specific and credible.
MY CONTEXT	What is known (my summary): [brief] What is not known: [brief] Why it matters in my setting: [brief] My objective: [one line]
TASK	1. Write the gap statement in 3 sentences: what is known, what is missing, why it matters clinically in my setting. 2. Write the bridging sentence that leads into my objective. 3. Give a weaker and a stronger version, so I can see the difference. 4. Check whether my objective actually addresses the gap as stated.
RULES	Do not overstate the gap with phrases such as "no study has ever examined"; use "few studies" or "limited data" unless I have evidence, because absolute claims invite an examiner to prove you wrong.
OUTPUT	The recommended gap statement and bridge, then the weak versus strong comparison, then the alignment check.

PRO TIP Your gap statement, objective and conclusion should read like three parts of one argument.

TRY IT WITH Known: high-flow nasal oxygen reduces intubation in some adults. Missing: data from Indian district ICUs.

→ **NEXT #81** Does my introduction lead to my objective?

74 "Need for the study" section

STANDARD

1.3 WRITING THE INTRODUCTION · PG RESIDENT

USE IT When your university's thesis format has a separate "Need for the study" or "Rationale" heading.

PERSONA	You are an Indian medical university thesis evaluator who reads many "Need for the study" sections and knows what makes one convincing.
MY CONTEXT	Thesis title and design: [paste] The gap (from my introduction): [paste] Local factors that make this study relevant here: [e.g. high disease burden, unique population, practice variation, cost] Expected benefit of the results: [brief]
TASK	1. Write a "Need for the study" section of 150–200 words that answers: why this question, why now, why here, and who benefits. 2. End with a sentence linking to the aims. 3. List any statement that needs a citation.
RULES	Keep claims of benefit realistic for a single-centre thesis. Mark any statistic as [VERIFY], because overclaimed impact is a common examiner criticism.
OUTPUT	The section, then a citation checklist.

PRO TIP "Why here" is your strongest argument in an Indian setting. Make it specific to your hospital's patients.

TRY IT WITH Rationale for studying post-TB lung disease in a TB-endemic district.

→ **NEXT #81** Does my introduction lead to my objective?

75 Every claim that needs a citation

STANDARD

1.3 WRITING THE INTRODUCTION · PG RESIDENT

USE IT After drafting any section, before you submit it to your guide.

PERSONA	You are a meticulous research-integrity reviewer who checks that every factual claim in a thesis is supported.
MY CONTEXT	My text: [PASTE INTRODUCTION OR SECTION HERE]
TASK	1. List every sentence that contains a factual claim (numbers, associations, guideline statements, "it is well known that..."). 2. For each, say whether it currently has a citation, and whether the citation type fits (primary study, review, guideline, dataset). 3. Flag claims that are too strong for any single source, and suggest softer wording. 4. Group the unsupported claims into a sourcing to-do list.
RULES	Do not supply references yourself; tell me what kind of source each claim needs, because citations you generate may not exist.
OUTPUT	A numbered table (sentence, claim, cited yes or no, source type needed, wording note), then the to-do list.

PRO TIP Work through the to-do list on PubMed before your guide meeting. Every unsupported claim is a viva question waiting to happen.

TRY IT WITH Paste a draft paragraph on vitamin D deficiency prevalence and its link with TB.

→ **NEXT #69** Where to find sourced burden data

76 Turn a textbook-style intro into a problem-focused one

STANDARD

1.3 WRITING THE INTRODUCTION · PG RESIDENT

USE IT When your guide says your introduction reads like a textbook chapter.

PERSONA	You are a thesis supervisor who has rescued many textbook-style introductions by refocusing them on the research problem.
MY CONTEXT	My current introduction: [PASTE INTRODUCTION HERE] My research question and objective: [paste]
TASK	1. Mark every sentence that is textbook content (definitions, pathophysiology, classification, general treatment) not needed to understand the problem. 2. Suggest what to delete, what to move to the ROL, and what to keep. 3. Restructure the remaining content into a problem-focused introduction of no more than [WORD LIMIT] words. 4. Show the new paragraph order with one-line purposes.
RULES	Keep my existing citations with their claims. Add no new content; where the restructured text needs a link, write [ADD LINKING SENTENCE], because new content must come from my own reading.
OUTPUT	A marked-up sentence list (delete / move / keep), then the restructured introduction, then the paragraph map.

PRO TIP If a sentence would appear in any textbook chapter on the disease, it usually doesn't belong in your introduction.

TRY IT WITH An introduction that spends 3 paragraphs on the pathophysiology of preeclampsia before the research question.

→ **NEXT #68** Examiner-style review of my introduction

77 Cut my introduction to the word limit

STANDARD

1.3 WRITING THE INTRODUCTION · PG RESIDENT

USE IT When your section is over the word limit set by your university or journal.

PERSONA	You are a precise academic editor who cuts words without cutting substance.
MY CONTEXT	Current text: [PASTE TEXT HERE] Current word count and target: [e.g. 780 → 500] Parts that must stay: [e.g. the Indian data paragraph]
TASK	1. Identify redundancy, throat-clearing phrases, repeated ideas and content that belongs in another chapter. 2. Produce the trimmed version at or just under the target. 3. List what you removed or merged, so I can restore anything important.
RULES	Keep every citation with its claim and keep all numbers exactly as written, because trimming must never change the evidence. Do not add new content.
OUTPUT	The trimmed text with its word count, then a list of cuts.

PRO TIP Cut whole ideas before cutting words. Trimming every sentence slightly makes the text worse.

TRY IT WITH A 900-word introduction on stroke thrombolysis that needs to be 550 words.

→ **NEXT #78** Fix the flow between paragraphs

78 Fix the flow between paragraphs

STANDARD

1.3 WRITING THE INTRODUCTION · PG RESIDENT

USE IT When each paragraph is fine on its own but the introduction feels disjointed.

PERSONA	You are an academic writing coach who specialises in coherence and argument flow.
MY CONTEXT	My text: [PASTE TEXT HERE]
TASK	1. Summarise the main point of each paragraph in one line, to expose the argument. 2. Identify where the logic jumps, repeats or reverses. 3. Suggest a reordering if needed. 4. Write improved first and last sentences for each paragraph so they connect.
RULES	Change only topic sentences, closing sentences and order; leave the rest of each paragraph as I wrote it, because the content and voice must stay mine.
OUTPUT	The paragraph-point map, a list of problems, then the revised opening and closing sentences for each paragraph.

PRO TIP Read only the first sentence of each paragraph in order. If that doesn't tell the story, the flow is broken.

TRY IT WITH A 5-paragraph introduction that jumps from global burden to treatment options and back to Indian burden.

→ **NEXT #68** Examiner-style review of my introduction

79 Define key terms by current criteria

STANDARD

1.3 WRITING THE INTRODUCTION · PG RESIDENT

USE IT When your introduction or methods needs formal definitions (e.g. sepsis, AKI, ARDS, metabolic syndrome).

PERSONA	You are a clinical academic who insists that every key term in a thesis is defined by a named, current criterion.
MY CONTEXT	Terms I need to define: [list] Specialty and population: [brief]
TASK	1. For each term, name the current standard definition or criteria set (organisation and year) and summarise it. 2. Note any older definition still used in papers I may cite, and how the change affects comparisons. 3. Suggest one sentence for each term suitable for my introduction or methods.
RULES	Mark every criteria name and year as [VERIFY FROM ORIGINAL SOURCE], because definitions are updated and misremembered versions are common. Do not paraphrase numeric thresholds unless you are certain; otherwise write [CHECK THRESHOLD].
OUTPUT	A table (term, current definition, organisation and year, older versions), then the sentences.

PRO TIP Use the same definition throughout your thesis. Mixing criteria between chapters is a classic examiner catch.

TRY IT WITH Terms: sepsis, septic shock, acute kidney injury, delirium.

→ **NEXT #66** The four-movement thesis introduction

80 Clinical and public-health significance

STANDARD

1.3 WRITING THE INTRODUCTION · PG RESIDENT

USE IT For the paragraph that explains who will benefit from your findings.

PERSONA	You are a health-services researcher who writes realistic significance statements for single-centre studies.
MY CONTEXT	Research question: [PICOT] Possible findings and what each would mean: [brief] Stakeholders: [e.g. clinicians, hospital administrators, national programme]
TASK	1. Write a significance paragraph of 100–150 words covering clinical practice, patients, the health system and future research. 2. Give a version for a thesis and a more concise version for a journal article. 3. Point out any benefit that overreaches for a single-centre design.
RULES	Frame benefits conditionally ("if confirmed, these findings could..."), because single-centre results rarely change practice alone, and examiners know it.
OUTPUT	The two versions, then the overreach check.

PRO TIP Name the specific decision your results could change, e.g. "whether to repeat chest X-ray at 48 hours".

TRY IT WITH Study of a bedside frailty score predicting post-operative complications in elderly surgical patients.

→ **NEXT #74** "Need for the study" section

81 Does my introduction lead to my objective?

STANDARD

1.3 WRITING THE INTRODUCTION · PG RESIDENT

USE IT After drafting the introduction and objectives, before submission to your guide.

PERSONA	You are a methodologist who checks the logical chain from problem to gap to objective in research proposals.
MY CONTEXT	My introduction: [PASTE INTRODUCTION HERE] My aims and objectives: [PASTE AIMS AND OBJECTIVES HERE]
TASK	1. Extract the gap as the introduction states it. 2. Compare the gap with each objective: does each objective address it, go beyond it, or ignore it? 3. Check that the population, exposure and outcome wording is identical between the introduction and the objectives. 4. Suggest the minimum edits to align them.
RULES	Point to exact phrases; do not rewrite whole sections, because the fix is usually a few words.
OUTPUT	An alignment table (gap element, objective wording, match yes or no, suggested edit).

PRO TIP Examiners often read the last paragraph of your introduction and your objectives side by side.

TRY IT WITH The introduction says the gap is Indian data; the objective does not mention the setting.

→ **NEXT #138** Primary and secondary objectives in SMART form

82 Paraphrase and synthesise sources responsibly

STANDARD

1.3 WRITING THE INTRODUCTION · PG RESIDENT

USE IT When you have key passages from papers and need to use their ideas in your own words, with citations.

PERSONA	You are an academic integrity adviser who teaches synthesis: combining ideas from several sources in the writer's own words.
MY CONTEXT	Source passages, each with its citation: [PASTE PASSAGES WITH CITATIONS HERE] The point I want to make: [one line]
TASK	1. Identify the shared idea and any disagreement across the passages. 2. Write a synthesis of 2–4 sentences in academic language, combining the sources and citing each claim. 3. Show which part of the synthesis came from which source. 4. Point out any phrase that still sits too close to the original wording.
RULES	Synthesise ideas rather than rewording sentence by sentence, and cite every source used, because the goal is honest scholarship, not getting past a similarity checker. Add no claims beyond the passages.
OUTPUT	The synthesis, then a source-mapping table, then the list of phrases to reword.

PRO TIP Close the source before writing your own version. Then compare. That habit prevents accidental plagiarism.

TRY IT WITH Three passages on the association between air pollution and COPD exacerbations.

→ **NEXT #75** Every claim that needs a citation

83 Make my draft sound like me, not like AI

STANDARD

1.3 WRITING THE INTRODUCTION · PG RESIDENT

USE IT When your text, even though you wrote it, has picked up generic AI-style phrasing after editing.

PERSONA	You are an editor who helps academic writers sound precise and natural rather than generic.
MY CONTEXT	My text: [PASTE TEXT HERE] My natural writing style (a paragraph I wrote without help): [PASTE SAMPLE HERE]
TASK	1. List the phrases that sound generic or AI-like (e.g. "delve", "plays a crucial role", "in today's world", empty transitions). 2. Suggest precise alternatives for each, matching my sample's style. 3. Point out sentences where every sentence has the same length and rhythm, and vary them. 4. Return the revised text.
RULES	Keep all facts, numbers and citations exactly as written. This is style editing of my own work; do not add content, because academic integrity requires that the ideas and sources stay mine.
OUTPUT	A phrase table (original, why it's weak, alternative), then the revised text.

PRO TIP Specific beats impressive. "Mortality was 18%" is better than "mortality remains a significant concern".

TRY IT WITH An introduction paragraph full of "crucial", "pivotal" and "it is important to note".

→ **NEXT #68** Examiner-style review of my introduction

84 Consistency check: tense, spelling, abbreviations

STANDARD

1.3 WRITING THE INTRODUCTION · PG RESIDENT

USE IT Before submitting any chapter, as a final polish.

PERSONA	You are a copy-editor for medical theses who catches inconsistencies authors stop seeing.
MY CONTEXT	My text: [PASTE TEXT HERE] Required conventions: [e.g. British spelling, Vancouver citations, SI units]
TASK	1. Check tense (present for established knowledge, past for specific studies). 2. Check spelling consistency (British or American), abbreviation use (defined at first use, then used consistently), units, number formatting and citation placement. 3. List every inconsistency with its location and correction. 4. Produce a list of the abbreviations used and their expansions.
RULES	Report inconsistencies rather than silently fixing them, because I need to apply the same corrections across all my chapters.
OUTPUT	An issue table (location, issue, correction), then the abbreviation list.

PRO TIP Build one abbreviation list for the whole thesis and check every chapter against it.

TRY IT WITH A methods and introduction section mixing "haemoglobin" and "hemoglobin".

→ **NEXT #77** Cut my introduction to the word limit

85 Short background for a synopsis or protocol

STANDARD

1.3 WRITING THE INTRODUCTION · PG RESIDENT

USE IT When you need a 150–250 word background for a synopsis, IEC form or CTRI registration.

PERSONA	You are a research committee member who reads hundreds of synopses and values brevity.
MY CONTEXT	Thesis title: [paste] Gap and objective: [paste] Key sourced facts I want to include: [PASTE FACTS WITH CITATIONS] Word limit: [e.g. 200 words]
TASK	1. Write a background within the word limit: the problem, Indian relevance, the gap and the objective. 2. Write a 50-word version for form fields with tight limits. 3. List the citations used.
RULES	Use only the facts I gave with their citations; write [CITE] where a needed claim has none, because synopsis claims are checked by the committee.
OUTPUT	The full background, the 50-word version, then the citation list.

PRO TIP Keep the same wording for the gap and objective across synopsis, protocol, IEC form and CTRI. Committees compare them.

TRY IT WITH Synopsis background for a cross-sectional study of depression in post-stroke patients.

→ **NEXT #296** Complete study protocol, IEC-ready

QUICK PROMPTS · 1.3

86 Three hook sentences for my topic

QUICK

1.3 WRITING THE INTRODUCTION · PG RESIDENT

Act as a medical writing editor. Write 3 opening sentences for an introduction on [TOPIC] in [POPULATION]: one clinical-problem hook, one burden hook with a [VERIFY] slot for the number, and one practice-gap hook. Keep each under 30 words and formal.

TRY IT WITH Topic: sepsis-associated acute kidney injury in Indian ICUs.

87 Is my tense right?

QUICK

1.3 WRITING THE INTRODUCTION · PG RESIDENT

Act as an academic copy-editor. Check the tenses in this paragraph, [PASTE PARAGRAPH], using the convention of present tense for established knowledge and past tense for specific studies and my own methods. List each error with its correction, then give the corrected paragraph.

TRY IT WITH A paragraph mixing "Smith found" and "studies show" inconsistently.

88 Turn my bullet notes into a paragraph

QUICK

1.3 WRITING THE INTRODUCTION · PG RESIDENT

Act as an academic writing coach. Turn these notes, [PASTE NOTES WITH CITATIONS], into one coherent paragraph of about 120 words in formal thesis style. Keep every citation with its claim, add no new facts, and mark any missing link as [ADD LINK].

TRY IT WITH Five bullet notes on the burden of snakebite envenoming in rural India.

89 Find the textbook sentences

QUICK

1.3 WRITING THE INTRODUCTION · PG RESIDENT

Act as a strict thesis examiner. In this introduction, [PASTE TEXT], mark every sentence that is general textbook content rather than part of the argument for my research question [QUESTION]. List them with a one-line reason and say whether to delete each one or move it to the ROL.

TRY IT WITH An introduction on thyroid disorders in pregnancy that opens with thyroid physiology.

90 One paragraph on "why now"

QUICK

1.3 WRITING THE INTRODUCTION · PG RESIDENT / FACULTY

Act as a research strategist. In under 100 words, explain why my research question [QUESTION] is timely now: a recent guideline change, a new test or therapy, a policy change, or an emerging burden. Use [CITE] slots for every claim and do not invent dates.

TRY IT WITH Question on point-of-care HbA1c testing in primary health centres.

91 Sentence-level clarity edit

QUICK

1.3 WRITING THE INTRODUCTION · PG RESIDENT

Act as a medical copy-editor. Edit this paragraph, [PASTE PARAGRAPH], for clarity only: shorter sentences, active voice where natural, precise words and no jargon stacking. Keep all facts and citations unchanged, and show the edited version followed by the 3 most important changes.

TRY IT WITH A 90-word sentence describing the pathogenesis and burden of NAFLD.

92 Does my introduction need sub-headings?

QUICK

1.3 WRITING THE INTRODUCTION · PG RESIDENT

Act as an Indian medical university thesis evaluator. My university is [UNIVERSITY] and my introduction is [WORD COUNT] words. Tell me whether to use sub-headings, which headings are conventional in Indian MD/MS theses if so, and how to keep the argument continuous across them. Mark university-specific rules as [CHECK YOUR UNIVERSITY FORMAT].

TRY IT WITH A 1,200-word introduction for an MS Orthopaedics thesis.

93 Check for causal language in my introduction

QUICK

1.3 WRITING THE INTRODUCTION · PG RESIDENT

Act as an epidemiology reviewer. Check this text, [PASTE TEXT], for causal words (causes, leads to, results in, prevents, due to) attached to observational evidence. List each instance with a correct associational rewrite, and leave the causal wording where it is supported by RCT evidence I have cited.

TRY IT WITH A paragraph claiming obesity "causes" poor asthma control, citing cross-sectional studies.

94 Build the abbreviation list

QUICK

1.3 WRITING THE INTRODUCTION · PG RESIDENT

Act as a thesis copy-editor. From this text, [PASTE TEXT], extract every abbreviation, give its expansion, note where it is first used and whether it was expanded there, and flag any abbreviation used only once. Return an alphabetical table ready for my thesis front matter.

TRY IT WITH Introduction and methods of a thesis on COPD exacerbations.

95 Two-sentence background for my abstract

QUICK

1.3 WRITING THE INTRODUCTION · PG RESIDENT

Act as a journal abstract editor. From my introduction, [PASTE INTRODUCTION], write the Background of a structured abstract in exactly two sentences: the problem with its relevance, and the gap leading to the objective. Stay under 50 words, add no new facts, and keep any number exactly as written.

TRY IT WITH An introduction on post-COVID pulmonary fibrosis in Indian patients.

1.4

Review of Literature

A literature review is not a summary of ten papers you happened to find. It is a reproducible search, an evidence table and a defensible gap statement.

40

prompts

5

Master

23

Standard

12

Quick

#96 Screen, extract and synthesise: the ROL engine

#97 Thematic ROL draft from my evidence table

#98 Evidence table from uploaded PDFs, with quotes

#99 Global and Indian studies: the classic thesis ROL

#100 Examiner-style critique of my ROL chapter

#101 Inclusion and exclusion criteria for screening

#102 Triage a batch of abstracts

#103 Design my data-extraction sheet

#104 Extract one paper into my evidence table

#105 Which risk-of-bias tool for which study?

#106 Apply a risk-of-bias tool to one paper

#107 Find the themes in my evidence table

#108 Explain why studies disagree

#109 One study, one paragraph, done properly

#110 Convert a chronological ROL into a thematic one

#111 Build a literature synthesis matrix

#112 Heading structure for my ROL chapter

#113 The closing paragraph that leads to my study

#114 Summarise a paper, grounded in quotes

#115 Summary table of key studies

#116 Check the numbers in my ROL against the sources

#117 Format and check my reference list

#118 The questions to ask NotebookLM about my papers

#119 Quick critical appraisal of one paper

#120 Read and summarise a meta-analysis

#121 Historical review sub-section without errors

#122 Update my ROL before submission

#123 Register a systematic review on PROSPERO

#124 Pull the PICO out of an abstract

#125 One-line takeaway of a paper

#126 The limitation that matters most

#127 Compare two studies in three sentences

#128 What design is this study really?

#129 Interpret an effect size and its CI

#130 Results in plain language

#131 Does this reference support my sentence?

#132 Format one reference in Vancouver style

#133 How much should this study count?

#134 Customise a NotebookLM audio overview

#135 Is this ROL paragraph a list or a synthesis?

Screen, extract and synthesise: the ROL engine

1.4 REVIEW OF LITERATURE · PG RESIDENT / FACULTY

USE IT

After your search is complete (section 1.2), when you have a pile of records to turn into a review.

WHY

Most thesis ROLs are a string of study summaries. Examiners want to see explicit criteria, a structured evidence table, a quality assessment and a synthesis that ends on the gap. This prompt builds all four from your own papers.

COPY THE PROMPT BELOW

PERSONA	You are a systematic-review methodologist who has co-authored Cochrane reviews and supervised dozens of postgraduate literature reviews. You are sceptical of weak evidence and insist on explicit, reproducible decisions.
MY CONTEXT	My PICOT question: [PASTE QUESTION] Study designs I expect to find: [e.g. cohorts, case-control, a few RCTs] Number of records after deduplication: [N] Papers I have uploaded to this chat: [list, or "none yet"]
TASK	Part A: screen and extract 1. Draft explicit inclusion and exclusion criteria, phrased so a second reviewer could apply them without asking me questions. 2. Design a data-extraction table: author, year, country, setting, design, sample size, population, exposure or intervention, comparator, outcome definition, effect estimate with 95% CI, follow-up, funding, key limitation. 3. Name the right risk-of-bias tool for each design I will encounter, and list its domains. Part B: synthesise 4. Give a skeleton for the review narrative: how to open, how to group studies by theme rather than chronology, how to handle conflicting findings, how to close on the gap. 5. Show how to write a three-sentence gap statement: what is known, what is missing, why it matters in my setting. 6. Draft a PRISMA-style flow description with placeholder numbers.
RULES	Do not give any specific citation, author, journal, year, sample size or result unless I uploaded that paper or you retrieved it through live search with a PMID or DOI, because an invented study in a thesis is academic misconduct. Where you describe what the literature broadly shows, label it [UNVERIFIED BACKGROUND – CONFIRM INDEPENDENTLY]. Never estimate a statistic from memory.
OUTPUT	Criteria as two numbered lists, the extraction table as a column list, a tool table, the narrative skeleton, the gap-statement template, and the flow description.
TWEAK	End with the 3 decisions in this plan an examiner is most likely to question, and how to justify each.

How to get more out of it

- › Upload the actual PDFs before asking for any comparison across papers. That is the difference between synthesis and fabrication.
- › Pilot the extraction table on 3 papers before extracting all of them. Columns always need adjusting.
- › Keep a list of excluded full texts with reasons. It feeds your PRISMA flow diagram.
- › → **NEXT #102** Triage a batch of abstracts

Thematic ROL draft from my evidence table

1.4 REVIEW OF LITERATURE · PG RESIDENT

USE IT

When your extraction table is complete and you are ready to write the review chapter.

WHY

Writing from an evidence table, rather than from memory or AI knowledge, guarantees every sentence traces back to a real paper you read. Grouping by theme shows examiners you synthesised the literature rather than listed it.

COPY THE PROMPT BELOW

PERSONA	You are a senior academic who writes and examines literature reviews for medical theses. You synthesise across studies, compare findings, and always end each theme with an interpretive sentence.
MY CONTEXT	Thesis title and objective: [paste] Citation style: [Vancouver / Harvard] Target length of the ROL chapter: [e.g. 2,500–3,500 words] My completed extraction table (author, year, design, n, setting, key finding with numbers, limitation, reference number): [PASTE TABLE HERE] Themes I think exist (optional): [list] My guide's expectations for the chapter, if known: [e.g. separate Indian studies section, 25+ references]
TASK	- Propose 4–6 themes that organise the studies, and assign each study to a theme. - For each theme, write 2–4 paragraphs that compare studies: agreement, disagreement and the likely reasons (population, setting, definitions, design, sample size). - Include a short subsection on Indian studies, or say explicitly that Indian data is lacking in a theme. - End each theme with a sentence saying what it means for my study. - Close the chapter with a gap paragraph that leads to my objective. - After each theme, list the reference numbers used, so I can check coverage against my table.
RULES	Use only the studies and numbers in my table, reproduced exactly with their reference numbers, because every figure will be checked against the source. If a theme has a single study, say so rather than generalising. Add no new studies; where support is missing, write [CITATION NEEDED — FIND SOURCE].
OUTPUT	A theme map table first, then the chapter with theme headings, within the target length.
TWEAK	Finish with the 3 places where my evidence is thinnest, so I can search again before submission.

How to get more out of it

- Write one theme per session. Long outputs lose accuracy.
- Spot-check 5 numbers in the draft against your PDFs before you trust the rest.
- Put your own interpretation into the closing sentence of each theme. That is where your voice matters most.
- → **NEXT #116** Check the numbers in my ROL against the sources

Evidence table from uploaded PDFs, with quotes

1.4 REVIEW OF LITERATURE · PG RESIDENT / FACULTY

USE IT

In NotebookLM, Claude or ChatGPT with file upload, once you have 5–15 full-text PDFs.

WHY

Asking an AI to "summarise the literature" invites invented studies. Asking it to extract from PDFs you uploaded, and to quote the sentence each data point came from, lets you verify every cell in minutes.

COPY THE PROMPT BELOW

PERSONA	You are a meticulous research assistant trained in data extraction for systematic reviews. You extract only what is written, you quote your source, and you say "not reported" when information is absent.
MY CONTEXT	My research question: [PICOT] Number of PDFs uploaded: [N] Variables I need from each paper: [e.g. design, country, sample size, population, exposure definition, outcome definition, main effect estimate with 95% CI, follow-up, limitations] Tool I am using: [NotebookLM / Claude / ChatGPT / Gemini] How I will use the table: [thesis ROL / systematic review / journal club]
TASK	1. For each uploaded paper, extract every variable I listed into a table, one row per paper. 2. For each cell, give the supporting quotation from the paper with its section (Methods, Results, Table 2 and so on). 3. Write "NOT REPORTED" where the paper does not state the information. 4. After the table, list any inconsistency within a paper (e.g. abstract and results numbers differ). 5. Summarise in 5 bullets what the set of papers shows collectively, citing papers by first author and year. 6. Flag any paper whose design or population makes it a poor fit for my question.
RULES	Extract only from the uploaded documents; do not supplement from your general knowledge, because I will cite these values in my thesis. Reproduce numbers exactly, including decimals and units. If a PDF is unreadable or partial, say which one.
OUTPUT	The extraction table, then a quotation appendix grouped by paper, then the inconsistency list and the 5-bullet summary.
TWEAK	End by telling me which paper's data you are least confident about, and why.

How to get more out of it

- › In NotebookLM, click the citation numbers to jump to the exact passage. Check 3 per paper.
- › Upload papers in batches of 5 to 8. Very large batches reduce extraction accuracy.
- › Save the quotation appendix. It is your proof if an examiner asks where a number came from.
- › → **NEXT #97** Thematic ROL draft from my evidence table

Global and Indian studies: the classic thesis ROL

1.4 REVIEW OF LITERATURE · PG RESIDENT

USE IT

When your university expects the traditional Indian thesis ROL format, with global studies, Indian studies and gaps as sub-headings.

WHY

Many Indian universities and guides expect studies described one by one (author, year, sample, finding, limitation) under global and Indian headings. This prompt keeps that format and makes it analytical instead of a list, using only papers you supply.

COPY THE PROMPT BELOW

PERSONA	You are a senior Indian medical academic who has guided and examined theses for 20 years and knows the conventional ROL format. You insist that each study description ends with its relevance to the present study.
MY CONTEXT	Thesis title: [paste] Primary objective: [one line] Global studies, with the extracted details for each: [PASTE STUDY DETAILS HERE] Indian studies, with the extracted details for each: [PASTE STUDY DETAILS HERE] Citation style: [Vancouver] Approximate length expected for the chapter: [e.g. 2,000–3,000 words]
TASK	1. Write a sub-section "Global studies" covering my global studies in chronological order, one short paragraph each: author(s), year, setting, design, sample size, key finding with numbers, one limitation, and one sentence on relevance to my study. 2. Write a sub-section "Indian studies" in the same format. 3. After each sub-section, write a synthesis paragraph comparing the studies: agreement, disagreement and why. 4. Write a sub-section "Gaps leading to the present study" in 3–5 sentences. 5. Add a summary table of all studies.
RULES	Use only the study details I pasted, with numbers reproduced exactly. Where a detail is missing, write [NOT REPORTED] rather than filling it in, because examiners cross-check these paragraphs against the original papers. Write in paragraphs, not bullet points. Use associational language for observational studies, and keep the order of studies consistent between the text and the summary table.
OUTPUT	Three headed sub-sections in formal thesis prose, then the summary table (author, year, country, design, n, key finding, limitation).
TWEAK	Close with a note on any study whose limitation seriously weakens its relevance to my question.

How to get more out of it

- › Most universities expect 15 or more references in the ROL. Check your guide's expectation.
- › The synthesis paragraphs are what raise this from a list to a review. Don't skip them.
- › If your Indian studies are few, say so plainly. That scarcity supports your gap.
- › → **NEXT #113** The closing paragraph that leads to my study

Examiner-style critique of my ROL chapter

1.4 REVIEW OF LITERATURE · PG RESIDENT

USE IT

When your ROL chapter is drafted and before it goes to your guide.

WHY

An ROL is judged on coverage, synthesis, critical appraisal and how clearly it leads to the gap. An honest examiner-style critique tells you which of these is weak while there is still time to search and rewrite.

COPY THE PROMPT BELOW

PERSONA	You are an external examiner for MD/MS theses who has marked hundreds of Review of Literature chapters. You are specific and fair, and you quote the text you are criticising.
MY CONTEXT	Thesis title and objective: [paste] Number of references in the chapter: [N] Date range of the included studies: [e.g. 2008–2025] Citation style used: [Vancouver / other] What my guide has already commented on: [brief, or "not yet reviewed"] My ROL chapter: [PASTE CHAPTER HERE]
TASK	1. Score the chapter out of 10 on each of: coverage, recency, inclusion of Indian evidence, synthesis versus listing, critical appraisal of study quality, handling of conflicting findings, logical flow towards the gap, and accuracy of reporting (numbers and designs described clearly). 2. Quote the passage behind each score. 3. List the 6 most important fixes, in priority order. 4. Identify paragraphs that merely list studies, and rewrite one as a synthesis example. 5. Flag any claim that is not attributed to a reference. 6. Check that the studies emphasised at the end of the chapter are the ones that support my gap statement.
RULES	Base all comments on the text I pasted. In the example rewrite, add no new studies or numbers; use [CITE] slots, because the rewrite must stay within my evidence.
OUTPUT	A rubric table with a total out of 80, the priority fixes as a numbered list, the flagged claims, and the example rewrite.
TWEAK	End with a one-line verdict (ready for guide, needs one revision, or needs restructuring) and the single fix with the biggest impact.

How to get more out of it

- › Run this again after revising. Your synthesis score should rise the most.
- › If "recency" scores low, re-run your search for the last 2 years before rewriting.
- › Share the rubric with your guide along with the chapter. It speeds up their review.
- › → **NEXT #97** Thematic ROL draft from my evidence table

101 Inclusion and exclusion criteria for screening

STANDARD

1.4 REVIEW OF LITERATURE · PG RESIDENT

USE IT Before screening titles and abstracts.

PERSONA	You are a systematic reviewer who writes screening criteria clear enough for two independent reviewers to agree.
MY CONTEXT	My question: [PICOT] Type of review: [thesis ROL / systematic review / scoping review] Designs, languages and dates I intend to include: [brief]
TASK	1. Write inclusion criteria for population, exposure or intervention, comparator, outcomes, study design, setting, language and date. 2. Write exclusion criteria, each with a one-line reason. 3. Give 3 borderline examples and how the criteria would decide each. 4. Suggest a pilot: screen 50 records together and calculate agreement.
RULES	Make each criterion testable from a title and abstract where possible, because criteria that need full text slow screening down badly.
OUTPUT	Two numbered lists, a borderline-examples table, then the pilot plan.

PRO TIP Screen generously at title and abstract stage. Be strict at full text.

TRY IT WITH Question: effect of pulmonary rehabilitation on quality of life in post-TB lung disease.

→ **NEXT #102** Triage a batch of abstracts

102 Triage a batch of abstracts

STANDARD

1.4 REVIEW OF LITERATURE · PG RESIDENT

USE IT When screening titles and abstracts against your criteria.

PERSONA	You are a careful second reviewer on a systematic review team who applies criteria consistently and explains every decision.
MY CONTEXT	My inclusion and exclusion criteria: [PASTE CRITERIA HERE] A batch of 10–20 titles and abstracts, numbered: [PASTE ABSTRACTS HERE]
TASK	1. Classify each record as include, exclude or unsure. 2. For each exclusion, name the criterion it failed. 3. For each unsure record, state exactly what the full text needs to show. 4. Summarise the counts.
RULES	When in doubt, mark unsure rather than exclude, because wrongly excluded studies cannot be recovered later. Decide only from the text given.
OUTPUT	A table (record number, decision, reason or criterion), then the counts.

PRO TIP You remain the first reviewer. Use this as a second opinion and resolve disagreements yourself.

TRY IT WITH 15 abstracts retrieved on "procalcitonin antibiotic duration ICU".

→ **NEXT #98** Evidence table from uploaded PDFs, with quotes

103 Design my data-extraction sheet

STANDARD

1.4 REVIEW OF LITERATURE · PG RESIDENT

USE IT Before extracting data from included papers.

PERSONA	You are a data-extraction lead for systematic reviews who designs sheets that capture everything needed and nothing more.
MY CONTEXT	My question: [PICOT] Designs included: [list] Planned analysis: [narrative synthesis / meta-analysis] Tool: [Excel / Google Sheets]
TASK	1. List the extraction columns grouped as study details, population, exposure or intervention, outcomes, results and quality. 2. For each column, give a definition, format (text, number, drop-down) and example entry. 3. Add columns needed if I later want a meta-analysis (e.g. events and totals per group, or means and SDs). 4. Suggest how to pilot and version the sheet.
RULES	Keep outcome columns aligned with my primary and secondary outcomes, because unmatched columns lead to missing data at synthesis.
OUTPUT	A column table (group, column, definition, format, example), then the piloting steps.

PRO TIP Use drop-downs for design and country. They make later summary tables effortless.

TRY IT WITH Diagnostic accuracy studies of lung ultrasound for pneumonia in children.

→ **NEXT #98** Evidence table from uploaded PDFs, with quotes

104 Extract one paper into my evidence table

STANDARD

1.4 REVIEW OF LITERATURE · PG RESIDENT

USE IT While reading each included paper, to fill your table accurately.

PERSONA	You are a precise research assistant who extracts data only from the text provided and marks everything else as not reported.
MY CONTEXT	My extraction columns: [PASTE COLUMN LIST HERE] The paper (full text or relevant sections): [PASTE OR UPLOAD PAPER HERE]
TASK	1. Fill each column from the paper. 2. Give the section or table each value came from. 3. Mark missing items as NOT REPORTED. 4. Note any discrepancy between the abstract and the results. 5. Write a one-sentence limitation relevant to my question.
RULES	Reproduce numbers exactly, including confidence intervals and units, because a transcription error in the evidence table propagates through the whole ROL.
OUTPUT	A two-column table (column, value with source location), then the discrepancy notes and the limitation sentence.

PRO TIP Extract the primary outcome first. If the paper doesn't report it clearly, reconsider inclusion.

TRY IT WITH A cohort study of NT-proBNP and mortality in acute heart failure.

→ **NEXT #106** Apply a risk-of-bias tool to one paper

105 Which risk-of-bias tool for which study?

STANDARD

1.4 REVIEW OF LITERATURE · PG RESIDENT

USE IT Before appraising the quality of your included studies.

PERSONA	You are an evidence-synthesis methodologist who knows the major quality-appraisal tools and their correct use.
MY CONTEXT	Designs in my review, with counts: [e.g. 3 RCTs, 8 cohorts, 2 diagnostic accuracy studies] Type of review: [thesis ROL / systematic review]
TASK	1. Match each design to the recommended tool (for example RoB 2 for RCTs, ROBINS-I or Newcastle-Ottawa for non-randomised studies, QUADAS-2 for diagnostic accuracy, AMSTAR 2 for systematic reviews, JBI checklists for others). 2. List each tool's domains and the output (overall judgement). 3. Explain how to present the results in a thesis (table or traffic-light plot). 4. Mention simpler options if a full tool is too heavy for a narrative thesis ROL.
RULES	Name only established tools and describe their domains accurately; if unsure of a detail, write [CHECK TOOL MANUAL], because misapplying a tool is worse than not using one.
OUTPUT	A design-to-tool table with domains, then the presentation guidance.

PRO TIP robvis is a free tool that produces traffic-light plots from your judgements.

TRY IT WITH 4 RCTs and 6 prospective cohorts on early mobilisation in the ICU.

→ **NEXT #106** Apply a risk-of-bias tool to one paper

106 Apply a risk-of-bias tool to one paper

STANDARD

1.4 REVIEW OF LITERATURE · PG RESIDENT

USE IT When appraising each included study.

PERSONA	You are a trained risk-of-bias assessor who justifies every judgement with a quote from the paper.
MY CONTEXT	Tool: [RoB 2 / ROBINS-I / Newcastle-Ottawa / QUADAS-2 / JBI] The paper (methods and results): [PASTE OR UPLOAD PAPER HERE] My outcome of interest: [outcome]
TASK	1. Assess each domain of the tool for my outcome. 2. Give the judgement for each domain (e.g. low, some concerns, high) with a supporting quote. 3. Give the overall judgement using the tool's own algorithm. 4. Write one sentence I can use in my ROL describing the study's quality.
RULES	Base judgements only on what the paper reports; where information is missing, say so and apply the tool's rule for missing information, because judgements without quotes cannot be defended.
OUTPUT	A domain table (domain, judgement, quote, reason), then the overall judgement and the ROL sentence.

PRO TIP Have a second person assess 2 or 3 papers independently. Report your agreement in the methods.

TRY IT WITH RoB 2 for an RCT of dexamethasone in severe pneumonia, outcome 28-day mortality.

→ **NEXT #107** Find the themes in my evidence table

107 Find the themes in my evidence table

STANDARD

1.4 REVIEW OF LITERATURE · PG RESIDENT

USE IT After extraction, before writing, to decide the structure of your ROL.

PERSONA	You are a qualitative synthesis expert who groups research findings into meaningful themes.
MY CONTEXT	My research question: [PICOT] My extraction table: [PASTE TABLE HERE]
TASK	1. Propose 4–6 themes, each with a one-line definition. 2. Assign every study to one or more themes. 3. For each theme, summarise in 2 lines what the evidence shows and how consistent it is. 4. Identify studies that don't fit any theme, and what that means. 5. Suggest the order of themes that best leads to my research gap.
RULES	Build themes from the studies in the table only; if a theme rests on a single study, flag it as thin evidence, because thin themes are easily challenged.
OUTPUT	A theme table (theme, definition, studies, consistency), then the suggested order.

PRO TIP Good themes are usually questions: "Does X predict Y?", "In whom?", "How is Y measured?"

TRY IT WITH An extraction table of 18 studies on neutrophil-lymphocyte ratio in sepsis.

→ **NEXT #97** Thematic ROL draft from my evidence table

108 Explain why studies disagree

STANDARD

1.4 REVIEW OF LITERATURE · PG RESIDENT / FACULTY

USE IT When included studies report opposite or very different results.

PERSONA	You are a clinical epidemiologist skilled at explaining heterogeneity between studies.
MY CONTEXT	Studies that disagree, each with design, population, setting, definitions, sample size and result: [PASTE DETAILS HERE]
TASK	1. Compare the studies side by side on population, setting, exposure or intervention definition, outcome definition, measurement method, timing, design and sample size. 2. Identify the most likely explanations for the disagreement, ranked. 3. Say whether chance (small samples, wide confidence intervals) could explain it. 4. Write a paragraph for my ROL that presents the disagreement and the likely reasons. 5. Say what my study could add to resolve it.
RULES	Use only the details I gave; where a comparison needs missing information, write [NOT REPORTED], because speculation presented as fact weakens the ROL.
OUTPUT	A comparison table, the ranked explanations, then the ROL paragraph.

PRO TIP Differences in outcome definitions explain more disagreement than most residents expect.

TRY IT WITH Two cohorts disagree on whether obesity increases mortality in ARDS (the "obesity paradox").

→ **NEXT #97** Thematic ROL draft from my evidence table

109 One study, one paragraph, done properly

STANDARD

1.4 REVIEW OF LITERATURE · PG RESIDENT

USE IT When writing individual study descriptions in the traditional ROL format.

PERSONA	You are a thesis-writing mentor who writes compact, informative study descriptions.
MY CONTEXT	Study details (author, year, country, design, sample size, population, key finding with numbers, limitation): [PASTE DETAILS HERE] My research question: [one line]
TASK	1. Write a 70–100 word paragraph: who, where, design, sample, key finding with its numbers, one limitation, and one sentence on relevance to my study. 2. Write a 40-word version for a summary table or a shorter ROL.
RULES	Reproduce numbers and terms exactly as given, and use the design's correct language (associated with, not caused, for observational designs), because study paragraphs are cross-checked against the paper.
OUTPUT	The two versions, labelled.

PRO TIP The relevance sentence is what separates a review from a bibliography. Never skip it.

TRY IT WITH A 2021 Indian cross-sectional study of 312 patients on vitamin D and COPD exacerbations.

→ **NEXT #99** Global and Indian studies: the classic thesis ROL

110 Convert a chronological ROL into a thematic one

STANDARD

1.4 REVIEW OF LITERATURE · PG RESIDENT

USE IT When your guide says your ROL is "just a list of studies by year".

PERSONA	You are an academic editor who restructures list-style literature reviews into synthesised, thematic ones.
MY CONTEXT	My current ROL: [PASTE ROL HERE] My research question: [one line]
TASK	1. Identify the themes present across the studies I have described. 2. Reassign each study paragraph to a theme. 3. Rewrite the ROL thematically: for each theme, an opening sentence, comparison across studies, and an interpretive closing sentence. 4. Keep a short chronological note only where historical development matters.
RULES	Use only the studies and numbers already in my text, with their citations, because restructuring must not introduce new evidence.
OUTPUT	A theme map, then the restructured ROL.

PRO TIP Keep your old version. Some universities still want a brief chronological overview before the thematic review.

TRY IT WITH A 15-paragraph ROL on hypothyroidism in pregnancy ordered from 1995 to 2024.

→ **NEXT #100** Examiner-style critique of my ROL chapter

111 Build a literature synthesis matrix

STANDARD

1.4 REVIEW OF LITERATURE · PG RESIDENT

USE IT When planning the ROL, to see which studies speak to which sub-questions.

PERSONA	You are a research-methods teacher who uses synthesis matrices to show residents how studies connect.
MY CONTEXT	Sub-questions or themes of my review: [list] Included studies with key findings: [PASTE LIST HERE]
TASK	1. Build a matrix with studies as rows and themes as columns, and a short finding in each relevant cell. 2. Mark empty cells: these are evidence gaps. 3. Summarise each column in one sentence. 4. Identify the column with the least evidence, and relate it to my research gap.
RULES	Fill cells only from the findings I gave; leave the rest blank rather than inferring, because empty cells are the insight.
OUTPUT	The matrix table, column summaries, then the gap note.

PRO TIP Put the matrix in your thesis appendix. Examiners like to see the synthesis visually.

TRY IT WITH 12 studies on AKI biomarkers across themes: diagnosis, prognosis, cost, Indian data.

→ **NEXT #107** Find the themes in my evidence table

112 Heading structure for my ROL chapter

STANDARD

1.4 REVIEW OF LITERATURE · PG RESIDENT

USE IT Before you write, so the chapter has a logical architecture.

PERSONA	You are a senior thesis guide who has seen what structures examiners find clear.
MY CONTEXT	Thesis title and objectives: [paste] Themes or areas my reading covers: [list] University format requirements, if any: [e.g. must include a historical review]
TASK	1. Propose a heading and sub-heading structure for the chapter (for example: historical perspective, definitions, epidemiology, pathophysiology relevant to the question, themed evidence, Indian studies, gaps). 2. Give a suggested length and number of references for each section. 3. Mark which sections are essential and which are optional for my question. 4. Explain the logic that links the sections.
RULES	Keep pathophysiology and textbook content only where it is needed to understand the research question, because examiners penalise padding.
OUTPUT	A numbered outline with lengths and reference counts, then the linking logic.

PRO TIP If a section cannot be linked to your objective in one sentence, drop it.

TRY IT WITH Thesis: serum magnesium and outcomes in critically ill patients.

→ **NEXT #97** Thematic ROL draft from my evidence table

113 The closing paragraph that leads to my study

STANDARD

1.4 REVIEW OF LITERATURE · PG RESIDENT

USE IT For the final paragraph of your ROL.

PERSONA	You are a thesis examiner who reads the end of the ROL to judge whether the student understands their own study's purpose.
MY CONTEXT	Summary of what my ROL showed: [brief] Gaps identified: [list] My objectives: [paste]
TASK	1. Write a closing paragraph of 120–180 words that summarises the state of the evidence, names the specific gaps, and states how the present study addresses them. 2. Check that the gaps named here match the gap in my introduction. 3. Suggest one sentence that connects to my methodology chapter.
RULES	Keep the gap claims modest ("limited data", "few Indian studies") unless my ROL demonstrates absence, because absolute claims are easy to disprove.
OUTPUT	The paragraph, then the consistency check, then the connecting sentence.

PRO TIP This paragraph and your introduction's gap paragraph should say the same thing in different words.

TRY IT WITH ROL on steroid use in tuberculous meningitis; gap: outcome data from Indian district hospitals.

→ **NEXT #81** Does my introduction lead to my objective?

114 Summarise a paper, grounded in quotes

STANDARD

1.4 REVIEW OF LITERATURE · PG RESIDENT / MBBS STUDENT

USE IT When you need a reliable summary of a paper you will cite.

PERSONA	You are a critical reader who summarises papers faithfully and never overstates their findings.
MY CONTEXT	The paper: [PASTE OR UPLOAD PAPER HERE] Why I am reading it: [e.g. for my ROL on X]
TASK	1. First, extract 5–8 direct quotes that capture the aim, design, population, main results and the authors' stated limitations. 2. Then write a 120-word summary based only on those quotes. 3. Note what the authors concluded, and whether the results fully support that conclusion.
RULES	Base the summary only on the quotes you extracted, because quote-first summarising prevents misrepresentation of the paper.
OUTPUT	The quotes with section labels, then the summary, then the conclusion check.

PRO TIP If the conclusion claims more than the results show, note it. It's useful for your discussion chapter.

TRY IT WITH A cohort paper on the association of sleep apnoea and resistant hypertension.

→ **NEXT #104** Extract one paper into my evidence table

115 Summary table of key studies

STANDARD

1.4 REVIEW OF LITERATURE · PG RESIDENT

USE IT For the table that closes or accompanies your ROL.

PERSONA	You are an academic editor who produces clear, journal-quality summary tables.
MY CONTEXT	Key studies with extracted details: [PASTE DETAILS HERE] Columns my guide wants (if specified): [list] Page orientation limit: [portrait / landscape]
TASK	1. Build a summary table: author and year, country, design, sample size, population, key finding with numbers, limitation. 2. Order rows logically (by theme, then year). 3. Write a table title and footnotes explaining abbreviations. 4. Suggest how to split the table if it is too wide.
RULES	Reproduce numbers exactly and write NR where a value is not reported, because table errors are highly visible to examiners.
OUTPUT	The table in Markdown, then the title and footnotes.

PRO TIP Keep each "key finding" cell to one line with a number. Long text cells make tables unreadable.

TRY IT WITH 10 studies on the diagnostic accuracy of GeneXpert for extrapulmonary TB.

→ **NEXT #97** Thematic ROL draft from my evidence table

116 Check the numbers in my ROL against the sources

STANDARD

1.4 REVIEW OF LITERATURE · PG RESIDENT

USE IT Before submitting your ROL or discussion, to catch transcription errors.

PERSONA	You are a fact-checker for a medical journal who compares every number in a manuscript with its source.
MY CONTEXT	My ROL text: [PASTE TEXT HERE] The source passages or tables for each cited study: [PASTE SOURCE EXTRACTS HERE]
TASK	1. List every number in my text with its citation. 2. Match each against the source passage. 3. Mark each as correct, wrong (give the correct value) or not verifiable from the extracts provided. 4. Flag any rounding, unit or percentage-versus-proportion error.
RULES	Compare only against the extracts I gave; do not "correct" numbers from memory, because the source document is the only authority.
OUTPUT	A table (number, citation, source value, status), then a count of errors.

PRO TIP Check the numbers you copied from abstracts. Abstracts sometimes differ from the results tables.

TRY IT WITH A ROL paragraph quoting prevalence, odds ratios and sample sizes from 6 studies.

→ **NEXT #117** Format and check my reference list

117 Format and check my reference list

STANDARD

1.4 REVIEW OF LITERATURE · PG RESIDENT

USE IT Before submission, to get the reference list right.

PERSONA	You are a reference-list editor who applies Vancouver (ICMJE / NLM) style precisely.
MY CONTEXT	My reference list as it currently is: [PASTE REFERENCES HERE] Style required: [Vancouver / other]
TASK	1. Reformat every reference to the required style. 2. Flag references missing essential elements (authors, title, journal abbreviation, year, volume, pages, DOI). 3. Flag possible duplicates and numbering problems. 4. List references that should be checked for existence and accuracy on PubMed.
RULES	Do not invent missing details such as volume, pages or DOI; write [MISSING — CHECK PUBMED], because made-up bibliographic details make a real reference look fake.
OUTPUT	The reformatted list, then a problem table (reference number, issue).

PRO TIP Export references from Zotero or Mendeley instead of formatting them by hand wherever possible.

TRY IT WITH 25 references in mixed formats copied from Google Scholar.

→ **NEXT #52** Reference manager set-up and deduplication

118 The questions to ask NotebookLM about my papers

STANDARD

1.4 REVIEW OF LITERATURE · PG RESIDENT

USE IT After uploading your included papers into a NotebookLM notebook.

PERSONA	You are a research-tools trainer who teaches residents to interrogate their own document collections.
MY CONTEXT	My research question: [PICOT] Number and type of papers uploaded: [e.g. 12 cohorts, 3 RCTs] Sections of my ROL: [list]
TASK	1. Write 15 questions to ask NotebookLM, grouped by: study characteristics, findings, disagreements, methods quality, gaps. 2. For each group, say what a useful answer should contain. 3. Suggest a prompt for NotebookLM's audio overview focused on my question. 4. Explain how to verify answers using NotebookLM's inline citations.
RULES	Phrase every question so the answer must come from the uploaded sources, because NotebookLM is only reliable when it is grounded in them.
OUTPUT	A grouped question list, then the audio-overview prompt, then the verification steps.

PRO TIP Ask "Which papers disagree about X, and how?". It is the fastest way to find material for your synthesis.

TRY IT WITH 15 papers on post-operative delirium in elderly hip fracture patients.

→ **NEXT #98** Evidence table from uploaded PDFs, with quotes

119 Quick critical appraisal of one paper

STANDARD

1.4 REVIEW OF LITERATURE · PG RESIDENT / MBBS STUDENT

USE IT When deciding how much weight a single paper deserves in your review.

PERSONA	You are a journal-club moderator who appraises papers quickly and fairly.
MY CONTEXT	The paper: [PASTE OR UPLOAD PAPER HERE] My research question: [one line]
TASK	1. Identify the design and its natural strengths and weaknesses. 2. Assess selection, measurement, confounding and attrition. 3. Judge whether the results are precise (confidence intervals) and clinically meaningful. 4. Judge applicability to Indian patients and to my question. 5. Give an overall weight for my review: high, moderate or low, with one clear reason.
RULES	Quote the paper when judging a weakness, because unsupported criticism is easy to dismiss.
OUTPUT	Five short headed sections, then a one-line verdict.

PRO TIP Use the verdict line directly in your ROL table's "limitation" column.

TRY IT WITH An observational study claiming statins reduce mortality in sepsis.

→ **NEXT #106** Apply a risk-of-bias tool to one paper

120 Read and summarise a meta-analysis

STANDARD

1.4 REVIEW OF LITERATURE · PG RESIDENT

USE IT When a key source in your ROL is a systematic review or meta-analysis.

PERSONA	You are a biostatistician who explains forest plots and heterogeneity clearly to clinicians.
MY CONTEXT	The meta-analysis (abstract, results and forest plots, or upload): [PASTE OR UPLOAD HERE] My question: [one line]
TASK	1. State the pooled estimate, its CI and what it means clinically. 2. Explain the heterogeneity (I^2 , τ^2 , prediction interval if reported) and what it implies. 3. Summarise the certainty of evidence (GRADE) if reported. 4. Point out the main limitations (small-study effects, included study quality). 5. Write a 100-word ROL paragraph summarising the meta-analysis accurately.
RULES	Use only the reported numbers; if a statistic is not reported, say so rather than computing it, because an invented pooled estimate misrepresents the review.
OUTPUT	Five short answers, then the paragraph.

PRO TIP A pooled estimate with high heterogeneity is an average of different things. Say so in your ROL.

TRY IT WITH A meta-analysis of prone positioning in ARDS with I^2 of 62%.

→ **NEXT #97** Thematic ROL draft from my evidence table

121 Historical review sub-section without errors

STANDARD

1.4 REVIEW OF LITERATURE · PG RESIDENT

USE IT When your university's thesis format expects a historical review.

PERSONA	You are a medical historian and academic who insists that every historical claim is dated and sourced.
MY CONTEXT	Topic: [TOPIC] Milestones I have already sourced: [PASTE MILESTONES WITH REFERENCES] Word limit: [e.g. 300 words]
TASK	1. Arrange my milestones into a timeline table: date, milestone, significance. 2. Suggest other milestones I may be missing, each marked [VERIFY DATE AND SOURCE]. 3. Write a historical review of the stated length from my sourced milestones only. 4. End with one sentence linking history to the present research question.
RULES	The prose must use only milestones I sourced; suggested ones stay in the table until I verify them, because dates and "firsts" are frequently misattributed.
OUTPUT	The timeline table, the suggestion list, then the prose.

PRO TIP Keep history short. One well-sourced page beats five pages of unverified dates.

TRY IT WITH Topic: evolution of mechanical ventilation.

→ **NEXT #112** Heading structure for my ROL chapter

122 Update my ROL before submission

STANDARD

1.4 REVIEW OF LITERATURE · PG RESIDENT

USE IT 2–3 months before thesis submission, to add papers published since your first search.

PERSONA	You are a systematic reviewer who updates reviews efficiently and transparently.
MY CONTEXT	Original search date and strings: [PASTE HERE] New records retrieved since then (titles and abstracts): [PASTE HERE] My current ROL themes: [list]
TASK	1. Screen the new records against my criteria. 2. For each included record, say which theme it belongs to and whether it confirms, contradicts or extends existing findings. 3. Suggest exactly where each fits in the text, with a one-sentence insertion. 4. Draft the updated search-methods sentence with the new date.
RULES	Use only the new records I pasted; insertions must use [CITE] slots until I verify the full text, because abstracts alone can mislead.
OUTPUT	A screening table, then an insertion plan (theme, position, sentence), then the updated methods sentence.

PRO TIP If a major new study contradicts your findings, address it in your discussion too.

TRY IT WITH 12 new PubMed records on tele-rehabilitation after stroke since your 2025 search.

→ **NEXT #116** Check the numbers in my ROL against the sources

123 Register a systematic review on PROSPERO

STANDARD

1.4 REVIEW OF LITERATURE · PG RESIDENT / FACULTY

USE IT If your thesis or first paper is a systematic review, before you start screening.

PERSONA	You are a systematic-review methodologist who has registered and peer reviewed many PROSPERO records.
MY CONTEXT	Review question: [PICOT] Planned databases, dates and designs: [brief] Planned synthesis: [narrative / meta-analysis] Review team and roles: [brief]
TASK	1. Draft the main PROSPERO fields: review title, question, searches, condition, participants, intervention or exposure, comparator, outcomes (main and additional), data extraction, risk of bias, synthesis strategy and subgroup analyses. 2. Flag fields where my plan is too vague to register. 3. Explain what must be registered before screening starts and what counts as a deviation.
RULES	Mark field names or rules you are unsure of as [CHECK CURRENT PROSPERO FORM], because the registration form is revised periodically. Do not claim the review is novel; tell me to search PROSPERO first.
OUTPUT	A field-by-field draft, then the vagueness flags, then the rules on timing and deviations.

PRO TIP Registration before screening is what reviewers check. Register early, even if the protocol is short.

TRY IT WITH Systematic review of vitamin D supplementation for TB treatment outcomes.

→ **NEXT #101** Inclusion and exclusion criteria for screening

QUICK PROMPTS · 1.4

124 Pull the PICO out of an abstract

QUICK

1.4 REVIEW OF LITERATURE · MBBS STUDENT / PG RESIDENT

Act as an evidence-based medicine tutor. From this abstract, [PASTE ABSTRACT], extract the population, intervention or exposure, comparator, outcome, design and sample size into a 6-row table. Write NOT REPORTED where missing, and add one line on the main result with its numbers.

TRY IT WITH Abstract of an RCT comparing two regimens for H. pylori eradication.

125 One-line takeaway of a paper

QUICK

1.4 REVIEW OF LITERATURE · PG RESIDENT

Act as a senior academic. Read this abstract or paper, [PASTE TEXT], and give its takeaway in one sentence of no more than 30 words that states the design, population and main finding with its key number, without overstating causation.

TRY IT WITH A cohort study on handgrip strength and mortality in cirrhosis.

126 The limitation that matters most

QUICK

1.4 REVIEW OF LITERATURE · PG RESIDENT

Act as a critical peer reviewer. For this study, [PASTE ABSTRACT OR METHODS], identify the single limitation that most threatens its conclusion, explain in 2 sentences which direction it likely biases the result, and write one sentence I can use in my ROL.

TRY IT WITH A single-centre retrospective study of 80 patients on a new sepsis score.

127 Compare two studies in three sentences

QUICK

1.4 REVIEW OF LITERATURE · PG RESIDENT

Act as a thesis mentor. Compare these two studies, [PASTE DETAILS OF STUDY A AND STUDY B], in exactly three sentences: what each found with numbers, whether they agree, and the most likely reason for any difference. Use associational language for observational designs.

TRY IT WITH Two studies on HbA1c and surgical site infection with opposite results.

128 What design is this study really?

QUICK

1.4 REVIEW OF LITERATURE · MBBS STUDENT / PG RESIDENT

Act as an epidemiology examiner testing a postgraduate. From this methods section, [PASTE METHODS], identify the true study design (authors sometimes mislabel it), justify your answer in 3 short bullets citing the methods, and state what the design can and cannot conclude.

TRY IT WITH A "prospective study" that actually analysed existing hospital records.

129 Interpret an effect size and its CI

QUICK

1.4 REVIEW OF LITERATURE · MBBS STUDENT / PG RESIDENT

Act as a biostatistics tutor. Interpret this result for a clinician: [PASTE RESULT, e.g. OR 1.8, 95% CI 1.1–2.9]. Explain what it means, whether it is statistically and clinically meaningful, and how to phrase it correctly in my ROL in one sentence.

TRY IT WITH Adjusted hazard ratio 0.72 (95% CI 0.55–0.94) for mortality with early antibiotics.

130 Results in plain language

QUICK

1.4 REVIEW OF LITERATURE · MBBS STUDENT / PG RESIDENT

Act as a medical educator teaching interns on the ward. Translate this results passage, [PASTE PASSAGE], into plain language for a junior colleague in under 80 words. Keep every number accurate and explain each statistical term briefly in plain brackets.

TRY IT WITH A results paragraph with sensitivity, specificity, AUC and likelihood ratios.

131 Does this reference support my sentence?

QUICK

1.4 REVIEW OF LITERATURE · PG RESIDENT

Act as a citation auditor. Here is my sentence, [SENTENCE], and the abstract or passage of the reference I cited, [PASTE SOURCE TEXT]. Tell me if the source fully supports, partly supports or does not support my sentence, and rewrite the sentence to match the source exactly.

TRY IT WITH A sentence claiming a drug "reduces mortality", citing a study that reported length of stay only.

132 Format one reference in Vancouver style

QUICK

1.4 REVIEW OF LITERATURE · PG RESIDENT / MBBS STUDENT

Act as a reference editor for medical theses. Format this reference in Vancouver (ICMJE / NLM) style: [PASTE REFERENCE DETAILS]. List any element that is missing rather than inventing it, and show the in-text citation format my thesis should use.

TRY IT WITH Details of a 2023 journal article with 8 authors, copied from Google Scholar.

133 How much should this study count?

QUICK

1.4 REVIEW OF LITERATURE · PG RESIDENT

Act as a systematic reviewer. Given this study's design, size and quality, [PASTE BRIEF DETAILS], tell me how much weight it should carry in my ROL compared with a large RCT or a meta-analysis, and give me one hedged sentence to describe it.

TRY IT WITH A 40-patient pilot study with a striking positive result.

134 Customise a NotebookLM audio overview

QUICK

1.4 REVIEW OF LITERATURE · PG RESIDENT / MBBS STUDENT

Act as a NotebookLM power user. Write a customisation instruction for an Audio Overview of my uploaded papers on [TOPIC] for a [PG resident / exam aspirant]. Focus it on findings, disagreements and gaps, skip history, and keep it under 80 words.

TRY IT WITH 10 uploaded papers on non-invasive ventilation in COPD exacerbations.

135 Is this ROL paragraph a list or a synthesis?

QUICK

1.4 REVIEW OF LITERATURE · PG RESIDENT

Act as an ROL examiner. Read this paragraph, [PASTE PARAGRAPH], and tell me whether it synthesises or merely lists studies, in one line. Then give 3 specific edits that would make it compare and interpret the studies, without adding new evidence.

TRY IT WITH A paragraph describing four studies one after another in date order.

SECTION 1.5

1.5

Aims, objectives & hypotheses

Your objectives are the contract for the whole thesis: every method, every table and every conclusion must trace back to them. Get them precise now.

20

prompts

2

Master

12

Standard

6

Quick

#136 Aims, objectives and hypotheses, fully aligned

#137 Committee review of my objectives

#138 Primary and secondary objectives in SMART form

#139 Null and alternative hypotheses, correctly stated

#140 Aim versus objective: fix the confusion

#141 Map each objective to outcome, measurement and analysis

#142 Objectives for a descriptive study

#143 Objectives for a diagnostic accuracy study

#144 Too many objectives: trim to what is feasible

#145 Secondary objectives that add value at no extra cost

#146 Fix vague verbs in my objectives

#147 From PICOT question to a full set of objectives

#148 Objectives for an interventional study or RCT

#149 Objectives for qualitative or mixed-methods studies

#150 Write my aim in one sentence

#151 Rewrite one vague objective

#152 Is my hypothesis testable?

#153 One-sided or two-sided test?

#154 Which of my objectives should be primary?

#155 Objectives squeezed into a form field

Aims, objectives and hypotheses, fully aligned

1.5 AIMS, OBJECTIVES & HYPOTHESES · PG RESIDENT

USE IT

After your topic is approved and before you write the protocol or synopsis.

WHY

Vague objectives ("to study", "to see") produce vague methods and unanswerable viva questions. Precise objectives, each mapped to an outcome, a measurement and an analysis, make the rest of the thesis almost write itself.

COPY THE PROMPT BELOW

PERSONA	You are a research methodologist and thesis committee member at an Indian medical university who has reviewed hundreds of synopses. You insist that every objective is measurable, feasible and linked to a specific analysis.
MY CONTEXT	Programme and specialty: [e.g. MD Community Medicine] Thesis title: [paste] Research question (PICOT): [paste] Study design: [cross-sectional / case-control / cohort / diagnostic accuracy / interventional] Sample size planned and time available: [N, months] Variables I plan to collect: [list]
TASK	1. Write one overall aim in a single sentence. 2. Write one primary objective and up to 3 secondary objectives in SMART form, each starting with a precise verb (to estimate, to compare, to determine the association between...). 3. Write the null and alternative hypotheses for the primary objective (and for secondary objectives where a comparison is made), and say whether the test is one-sided or two-sided, with the reason. 4. Build an alignment table: objective → outcome variable → how and when it is measured → statistical analysis. 5. Flag any objective my design or sample size cannot support.
RULES	Use my design's language: descriptive designs estimate, analytical designs compare or associate, and only randomised designs test effects, because objectives that claim more than the design can deliver are the most common synopsis rejection. Keep each objective to one sentence and one outcome. Where a timing or measurement detail is missing, write [SPECIFY] instead of assuming it.
OUTPUT	Aim, numbered objectives, hypotheses, then the alignment table, then the flags.
TWEAK	Finish with the 3 questions a synopsis committee is most likely to ask about these objectives, and short answers.

How to get more out of it

- Your conclusion will answer the primary objective in one sentence. Write that sentence now as a test.
- Keep secondary objectives few. Every extra objective adds variables, time and multiple-testing problems.
- Paste the alignment table into your protocol. It shows the committee your thinking in one glance.
- → **NEXT #141** Map each objective to outcome, measurement and analysis

Committee review of my objectives

1.5 AIMS, OBJECTIVES & HYPOTHESES · PG RESIDENT

USE IT

When your objectives are drafted, before synopsis submission.

WHY

Synopsis committees return objectives for the same reasons every term: vague verbs, unmeasurable outcomes, too many objectives and a mismatch with the design. This prompt runs that review before the committee does.

COPY THE PROMPT BELOW

PERSONA	You are the member-secretary of a postgraduate research committee at an Indian medical college. You review objectives strictly against the design and feasibility, and you write clear, actionable comments.
MY CONTEXT	Thesis title and design: [paste] Sample size and duration: [brief] My aims, objectives and hypotheses as drafted: [PASTE HERE] My introduction's gap statement: [paste] Comments already received from my guide: [brief, or "none yet"] My university's synopsis format, if it prescribes one: [brief]
TASK	1. Review each objective for: a precise verb, a single measurable outcome, a defined population and time frame, fit with the design, and feasibility with my sample size. 2. Mark each as approved, minor revision or major revision, with a reason. 3. Check that the primary objective directly answers the gap statement. 4. Check the hypotheses: correctly stated, matched to the primary objective, testable with my design. 5. Rewrite every objective that needs revision, keeping as much of my original wording as possible. 6. Recommend whether to cut, merge or demote any objective to "exploratory", and explain the effect on my workload and analysis.
RULES	Comment only on what I wrote, and quote the words you object to, because vague committee feedback is useless to a resident. Do not change my research question; if the objectives cannot be fixed without changing it, say so plainly.
OUTPUT	A review table (objective, verdict, issue, rewrite), then the hypothesis check, then the cut, merge or demote advice.
TWEAK	End with the revised full set of aims, objectives and hypotheses, ready to paste into my synopsis.

How to get more out of it

- › Take the revised set to your guide with the review table. It shows why each change was made.
- › If a secondary objective needs an extra test or visit, weigh it against your timeline before keeping it.
- › Keep the objective wording identical across synopsis, protocol, IEC form and thesis.
- › → **NEXT #138** Primary and secondary objectives in SMART form

138 Primary and secondary objectives in SMART form

STANDARD

1.5 AIMS, OBJECTIVES & HYPOTHESES · PG RESIDENT

USE IT When drafting objectives for your synopsis or protocol.

PERSONA	You are a thesis guide who writes objectives that are specific, measurable, achievable, relevant and time-bound.
MY CONTEXT	I am doing [MD / MS / DNB] in [SPECIALTY] in India. My thesis topic is [TOPIC] and my design is [cross-sectional / case-control / cohort / interventional]. Setting: [hospital]. Duration: [months].
TASK	1. Write one primary aim. 2. Write 3 specific secondary objectives. 3. Write one null and one alternative hypothesis for the primary objective. 4. After each objective, name the outcome variable it measures in brackets.
RULES	One sentence each, formal academic language, and a verb that matches my design (estimate, compare, determine the association), because the verb tells the committee what analysis to expect.
OUTPUT	A numbered list: aim, objectives, hypotheses.

PRO TIP If you can't name the outcome variable for an objective, the objective is not ready.

TRY IT WITH MD Paediatrics; topic: vitamin D levels and recurrent wheeze in children aged 1–5; cross-sectional.

→ **NEXT #139** Null and alternative hypotheses, correctly stated

139 Null and alternative hypotheses, correctly stated

STANDARD

1.5 AIMS, OBJECTIVES & HYPOTHESES · PG RESIDENT

USE IT When your protocol or synopsis needs formal hypotheses.

PERSONA	You are a biostatistician who teaches residents to state hypotheses that match their analysis.
MY CONTEXT	Primary objective: [paste] Design and comparison groups: [brief] Primary outcome and its type (continuous / binary / time-to-event): [brief]
TASK	1. Write the null (H0) and alternative (H1) hypotheses in words and in parameter form (e.g. $\mu_1 = \mu_2$). 2. Recommend a one-sided or two-sided test, with the reason. 3. Name the test that will evaluate the hypothesis. 4. Say whether a hypothesis is needed at all (purely descriptive objectives may not need one).
RULES	State hypotheses about population parameters, not sample results, because that is the most common error examiners catch. Default to two-sided unless there is a strong, pre-specified reason.
OUTPUT	The hypotheses, the sidedness decision, the test, then the "is it needed" note.

PRO TIP A non-inferiority or equivalence study needs a different hypothesis structure and a margin. Ask separately if yours is one.

TRY IT WITH Comparing mean HbA1c reduction between two oral regimens over 12 weeks.

→ **NEXT #141** Map each objective to outcome, measurement and analysis

140 Aim versus objective: fix the confusion

STANDARD

1.5 AIMS, OBJECTIVES & HYPOTHESES · MBBS STUDENT / PG RESIDENT

USE IT When your aims and objectives say the same thing twice, or mix levels.

PERSONA	You are a research-methods teacher who explains the hierarchy from aim to objectives to hypotheses clearly.
MY CONTEXT	My current aims and objectives: [PASTE HERE] My research question: [paste] My programme: [MD / MS / DNB / MBBS project]
TASK	1. Explain the difference between an aim, an objective and a hypothesis in 3 lines. 2. Classify each of my statements as aim, objective, hypothesis or none. 3. Rewrite them into a clean hierarchy: one aim, then objectives, then hypotheses.
RULES	Keep my research intent unchanged; only restructure and reword, because the content must still match my approved topic.
OUTPUT	The 3-line explanation, a classification table, then the rewritten set.

PRO TIP An aim is the "why"; objectives are the "what exactly"; hypotheses are the "what I expect".

TRY IT WITH Aims: "to study thyroid function in CKD" and "to see if thyroid is affected in CKD patients".

→ **NEXT #138** Primary and secondary objectives in SMART form

141 Map each objective to outcome, measurement and analysis

STANDARD

1.5 AIMS, OBJECTIVES & HYPOTHESES · PG RESIDENT

USE IT After finalising objectives, before designing the proforma and the analysis plan.

PERSONA	You are a methodologist who builds the traceability table linking objectives to data and analysis.
MY CONTEXT	My objectives: [PASTE OBJECTIVES HERE] Variables I plan to collect: [list] Software: [SPSS / R / Stata / Jamovi]
TASK	1. For each objective, give the outcome variable, its type, the measurement tool, the time point, and the statistical test or effect measure. 2. Flag any objective with no matching variable, and any variable with no matching objective. 3. Suggest what to add or drop.
RULES	Choose tests that fit the variable type and design, and mark any uncertain choice as [CONFIRM WITH STATISTICIAN], because this table will become part of your analysis plan.
OUTPUT	A mapping table, then the orphan objectives and variables, then the recommendations.

PRO TIP Variables with no objective should usually leave your proforma. They cost time and add nothing.

TRY IT WITH Objectives on prevalence of depression in dialysis patients and its association with serum albumin.

→ **NEXT #196** Pre-specified statistical analysis plan

142 Objectives for a descriptive study

STANDARD

1.5 AIMS, OBJECTIVES & HYPOTHESES · PG RESIDENT / MBBS STUDENT

USE IT When your study describes prevalence, patterns or characteristics rather than testing a comparison.

PERSONA	You are an epidemiologist who writes objectives for cross-sectional and descriptive studies without forcing hypotheses on them.
MY CONTEXT	Topic and population: [brief] What I want to describe: [prevalence / pattern / distribution / characteristics] Any associations I also want to explore: [list]
TASK	1. Write objectives that estimate (with precision) rather than test. 2. Write any exploratory association objectives, labelled as such. 3. Say whether formal hypotheses are needed for each. 4. Suggest how sample size should be based on precision (margin of error) rather than power.
RULES	Use "to estimate" and "to describe" for descriptive aims, because "to prove" or "to test" signals a design error to the committee.
OUTPUT	A numbered list of objectives with their type, then the hypothesis and sample-size notes.

PRO TIP A descriptive study with a clear prevalence estimate and confidence interval is a perfectly good thesis.

TRY IT WITH Prevalence and pattern of dermatological adverse drug reactions in a tertiary hospital.

→ **NEXT #202** Sample size to estimate a proportion (prevalence)

143 Objectives for a diagnostic accuracy study

STANDARD

1.5 AIMS, OBJECTIVES & HYPOTHESES · PG RESIDENT

USE IT When your thesis evaluates a test against a reference standard.

PERSONA	You are a diagnostic-research methodologist familiar with the STARD reporting guideline.
MY CONTEXT	Index test: [e.g. bedside lung ultrasound] Reference standard: [e.g. CT chest] Target condition and population: [brief] Clinical role of the test: [replacement / triage / add-on]
TASK	1. Write the primary objective in terms of sensitivity and specificity (with 95% CI) of the index test against the reference standard. 2. Write secondary objectives (predictive values, likelihood ratios, AUC for continuous tests, inter-observer agreement, time or cost). 3. Explain which measures depend on prevalence and how to phrase them. 4. Note what "role of the test" means for the objectives.
RULES	Make the reference standard explicit in every accuracy objective, because accuracy without a named reference standard is meaningless.
OUTPUT	A numbered objective list, then the prevalence note, then the role note.

PRO TIP If your reference standard is imperfect, say so in your objectives' limitation note, not only in the discussion.

TRY IT WITH Point-of-care ultrasound versus chest X-ray for pneumothorax in trauma patients.

→ **NEXT #205** Sample size for a diagnostic accuracy study

144 Too many objectives: trim to what is feasible

STANDARD

1.5 AIMS, OBJECTIVES & HYPOTHESES · PG RESIDENT

USE IT When your list of objectives is long and your guide says "focus".

PERSONA	You are a pragmatic thesis supervisor who protects residents from over-ambitious protocols.
MY CONTEXT	My objectives: [PASTE OBJECTIVES HERE] Sample size and duration: [brief] Tests or visits each objective requires: [brief] Who pays for each test: [hospital / patient / project]
TASK	1. Estimate the workload, cost and statistical burden of each objective. 2. Rank the objectives by importance to the research question. 3. Recommend which to keep, merge, demote to exploratory, or drop. 4. Present the trimmed set.
RULES	Keep the primary objective unchanged; explain every cut in one line, because I'll need to justify it to my guide.
OUTPUT	A ranking table (objective, burden, importance, recommendation), then the trimmed set.

PRO TIP Three objectives done well beat eight done badly.

TRY IT WITH Eight objectives for a 2-year study of 100 patients with liver cirrhosis.

→ **NEXT #136** Aims, objectives and hypotheses, fully aligned

145 Secondary objectives that add value at no extra cost

STANDARD

1.5 AIMS, OBJECTIVES & HYPOTHESES · PG RESIDENT

USE IT When your primary objective is set and you want useful secondary questions from data you already collect.

PERSONA	You are an efficient clinical researcher who makes every data point count.
MY CONTEXT	Primary objective: [paste] Data I will already collect: [list] Design and sample size: [brief] Specialty and setting: [brief]
TASK	1. Suggest 5 secondary objectives answerable with the data I already collect. 2. For each, give the outcome, the analysis and what it adds clinically. 3. Warn which would be underpowered, and how to label them in my protocol.
RULES	Suggest no objective that needs new tests, visits or consent changes, because the point is zero extra cost. Label underpowered analyses exploratory.
OUTPUT	A table (objective, data used, analysis, added value, power warning).

PRO TIP Subgroup objectives are almost always underpowered. Call them exploratory from the start.

TRY IT WITH Primary: 30-day mortality in sepsis; data: vitals, CBC, lactate, SOFA, comorbidities.

→ **NEXT #141** Map each objective to outcome, measurement and analysis

146 Fix vague verbs in my objectives

STANDARD

1.5 AIMS, OBJECTIVES & HYPOTHESES · PG RESIDENT / MBBS STUDENT

USE IT When your objectives use verbs such as "to study", "to see", "to know" or "to evaluate".

PERSONA	You are an academic editor for research proposals who replaces vague verbs with precise ones.
MY CONTEXT	My objectives: [PASTE OBJECTIVES HERE] Design: [design] Primary outcome: [outcome]
TASK	1. Mark each vague verb. 2. Replace it with a precise verb that matches the design and the intended analysis (estimate, compare, determine the association, assess the accuracy, evaluate the effect). 3. Add any missing population, outcome, setting or time frame. 4. Give the final wording for each objective.
RULES	Match the verb to the design, because "to evaluate the effect" in an observational study promises causation you cannot deliver.
OUTPUT	A before-and-after table, then the final list.

PRO TIP Read each objective and ask, "What single number will answer this?". If you can't say, it's still vague.

TRY IT WITH "To study the role of CRP in appendicitis."

→ **NEXT #138** Primary and secondary objectives in SMART form

147 From PICOT question to a full set of objectives

STANDARD

1.5 AIMS, OBJECTIVES & HYPOTHESES · PG RESIDENT

USE IT When your question is fixed and you need everything that follows from it.

PERSONA	You are a research mentor who derives objectives systematically from the research question.
MY CONTEXT	My PICOT question: [PASTE QUESTION] Design: [design] Sample size and follow-up I can manage: [brief]
TASK	1. Derive the primary objective directly from the PICOT elements. 2. Suggest 2–3 secondary objectives that follow naturally (other outcomes, time points or subgroups). 3. Write the hypotheses. 4. Check that the population, comparison and outcome wording is identical across question, objectives and hypotheses.
RULES	Do not introduce new populations or outcomes absent from my PICOT without flagging them as additions, because the objectives must stay within my approved question.
OUTPUT	The derived set, then the consistency check.

PRO TIP Copy exact phrases from your PICOT. Paraphrasing the population in the objectives creates confusion later.

TRY IT WITH In adults with COPD exacerbation (P), does NIV started in the ward (I) versus in the ICU (C) change 30-day mortality (O) over 30 days (T)?

→ **NEXT #139** Null and alternative hypotheses, correctly stated

148 Objectives for an interventional study or RCT

STANDARD

1.5 AIMS, OBJECTIVES & HYPOTHESES · PG RESIDENT

USE IT When your thesis compares an intervention with a control, randomised or not.

PERSONA	You are a clinical trials methodologist who writes trial objectives that follow CONSORT and SPIRIT conventions.
MY CONTEXT	Intervention and comparator: [brief] Population and setting: [brief] Primary endpoint and time point: [e.g. treatment failure at day 7] Randomised?: [yes / no] Is the aim superiority, non-inferiority or equivalence?: [state or "unsure"]
TASK	1. Write the primary objective with the endpoint, time point and comparison. 2. Write secondary objectives, including safety and adverse events. 3. Write hypotheses matching the aim (superiority, non-inferiority with margin, or equivalence). 4. Explain what changes if the study is not randomised.
RULES	For a non-inferiority or equivalence aim, include a margin slot marked [JUSTIFY MARGIN FROM LITERATURE], because an unjustified margin is the first thing reviewers attack.
OUTPUT	Objectives, hypotheses, then the notes on randomisation and aim type.

PRO TIP Always include a safety objective in an interventional thesis. The IEC will ask for it.

TRY IT WITH Nebulised hypertonic saline versus normal saline in infants with bronchiolitis, length of stay.

→ **NEXT #139** Null and alternative hypotheses, correctly stated

149 Objectives for qualitative or mixed-methods studies

STANDARD

1.5 AIMS, OBJECTIVES & HYPOTHESES · PG RESIDENT / FACULTY

USE IT When part or all of your study uses interviews, focus groups or open-ended data.

PERSONA	You are a mixed-methods researcher in health services who writes objectives fit for qualitative inquiry.
MY CONTEXT	Topic: [TOPIC] Quantitative component (if any): [brief] Qualitative component: [e.g. in-depth interviews with 20 caregivers] Sequence of the components: [parallel / quantitative first / qualitative first]
TASK	1. Write the overall aim. 2. Write quantitative objectives in measurable form. 3. Write qualitative objectives using exploratory verbs (to explore, to understand, to describe experiences of). 4. Explain how the two components integrate (explanatory, exploratory or convergent design).
RULES	Don't give qualitative objectives numeric targets or hypotheses, because that misunderstands qualitative research and reviewers notice.
OUTPUT	Aim, quantitative objectives, qualitative objectives, then the integration note.

PRO TIP Name your mixed-methods design explicitly. Reviewers look for it.

TRY IT WITH Adherence to antiretroviral therapy: a survey of 200 patients plus interviews with 15 non-adherent patients.

→ **NEXT #136** Aims, objectives and hypotheses, fully aligned

150 Write my aim in one sentence

QUICK

1.5 AIMS, OBJECTIVES & HYPOTHESES · PG RESIDENT / MBBS STUDENT

Act as a research-methods tutor for postgraduates. From my topic **[TOPIC]** and design **[DESIGN]**, write a single-sentence study aim that names the population, the main variable and the setting. Give 3 versions and mark the most precise one, with a one-line reason.

TRY IT WITH Topic: sleep quality in resident doctors; design: cross-sectional survey.

151 Rewrite one vague objective

QUICK

1.5 AIMS, OBJECTIVES & HYPOTHESES · PG RESIDENT

Act as a strict thesis committee reviewer. Rewrite this objective, **[OBJECTIVE]**, so it is specific, measurable and matched to my design **[DESIGN]**. Give the rewrite, name the outcome variable and its unit, and explain the change in one short line.

TRY IT WITH "To assess the outcome of patients with dengue."

152 Is my hypothesis testable?

QUICK

1.5 AIMS, OBJECTIVES & HYPOTHESES · PG RESIDENT

Act as a biostatistician who reviews thesis synopses. Tell me whether this hypothesis, **[HYPOTHESIS]**, is testable with my design **[DESIGN]** and data **[PRIMARY OUTCOME TYPE]**. If not, explain why in 2 lines and give a testable version with its null hypothesis.

TRY IT WITH "Yoga improves the quality of life of asthmatic patients."

153 One-sided or two-sided test?

QUICK

1.5 AIMS, OBJECTIVES & HYPOTHESES · PG RESIDENT

Act as a biostatistics examiner. For my hypothesis, **[HYPOTHESIS]**, tell me whether a one-sided or two-sided test is appropriate, justify it in 3 lines, and explain how the choice affects my sample size and how examiners usually view one-sided tests.

TRY IT WITH A new dressing is expected to reduce wound infection rates compared with standard dressing.

154 Which of my objectives should be primary?

QUICK

1.5 AIMS, OBJECTIVES & HYPOTHESES · PG RESIDENT

Act as a thesis guide. From these objectives, **[PASTE OBJECTIVES]**, tell me which should be the primary objective and why (clinical importance, measurability, feasibility, sample size basis). Rank the rest as secondary objectives in order of importance, with one line each.

TRY IT WITH Three objectives on mortality, ICU stay and ventilator days in ARDS.

155 Objectives squeezed into a form field

QUICK

1.5 AIMS, OBJECTIVES & HYPOTHESES · PG RESIDENT

Act as a research-forms expert. Condense my objectives, **[PASTE OBJECTIVES]**, to fit a **[WORD OR CHARACTER LIMIT]** field on the IEC or CTRI form, keeping the primary objective complete and the secondary objectives recognisable. Show the character count of the final text.

TRY IT WITH Four objectives condensed to 500 characters for a CTRI registration.

1.6

Study design & methodology

The methods chapter is the document you will be held to for three years. Every ambiguity here becomes an unanswerable viva question later.

40

prompts

5

Master

23

Standard

12

Quick

- #156 Choose and defend the right study design
- #157 Write the full methodology chapter
- #158 Operational definitions for every variable
- #159 Case record form (proforma) with a data-entry codebook
- #160 Bias and confounding audit of my design
- #161 PICO framework and inclusion/exclusion criteria
- #162 Make eligibility criteria bedside-proof
- #163 Choose and describe my sampling method
- #164 Randomisation, allocation concealment and blinding
- #165 Choosing and matching controls in a case-control study
- #166 Follow-up plan that keeps loss to follow-up low
- #167 Tighten a cross-sectional study design
- #168 Design a diagnostic accuracy study (STARD)
- #169 Record-review (retrospective) methodology
- #170 Identify confounders and draw my causal diagram
- #171 Inter-observer agreement and quality-control plan
- #172 Translate and validate a questionnaire
- #173 Study procedure as a patient journey
- #174 Blinding when full blinding is impossible
- #175 Before-after and quasi-experimental studies done right

- #176 Plan a pilot study for my thesis
- #177 Data management plan for my thesis
- #178 Laboratory sample handling and pre-analytical control
- #179 Standardised imaging acquisition and reading protocol
- #180 Which reporting guideline applies to my study?
- #181 Describe the intervention so others can replicate it (TIDieR)
- #182 Designing studies in patients who cannot consent
- #183 Convert my protocol into thesis methods
- #184 Name my study design correctly
- #185 Prospective or retrospective: what's the difference for me?
- #186 Strengths and weaknesses of my design
- #187 Operationally define one variable
- #188 Draw a simple random sample in Excel
- #189 Allocation concealment in my trial
- #190 Can I apply this criterion at the bedside?
- #191 List the likely confounders
- #192 Cohort or case-control? Decide now
- #193 Could observer bias affect my results?
- #194 Write the "study setting" paragraph
- #195 My design in one line for the title page

Choose and defend the right study design

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

USE IT

After your research question is fixed and before you write the protocol.

WHY

The design decides what you can conclude. Choosing it by habit ("everyone does cross-sectional") leads to objectives your data cannot answer. This prompt compares designs against your actual question and resources, and gives you the defence for the viva.

COPY THE PROMPT BELOW

PERSONA	You are a clinical epidemiologist who teaches research methodology to postgraduates and sits on thesis committees. You are practical about what can be done in a 2-year MD programme at an Indian government medical college, and precise about what each design can and cannot conclude.
MY CONTEXT	Programme and specialty: [e.g. MD General Medicine] Thesis topic: [TOPIC] Research question: [what I want to find out, in PICOT form if possible] Eligible patients per month and months available: [N, X] Follow-up I can realistically do: [none / in-hospital / 30 days / 6 months] Resources: [routine tests only / some special tests / funded] Can I assign the intervention myself?: [yes / no / not applicable]
TASK	1. List the designs that could answer my question (cross-sectional, case-control, prospective or retrospective cohort, diagnostic accuracy, before-after, randomised trial). 2. For each, state what it could conclude, its main bias, its sample-size and time implications, and its feasibility in my setting. 3. Recommend the single most appropriate design and explain why it fits my question better than the obvious alternative. 4. State its key limitation and how to reduce it in the protocol. 5. Say whether a simpler design could answer a slightly narrower version of my question, and what I would lose. 6. Write a 100-word design justification paragraph for my methods chapter.
RULES	Match conclusions to the design: association for observational designs and effect for randomised ones, because overclaiming the design is a classic viva trap. Where feasibility depends on local facts I have not given, write [CHECK LOCALLY] rather than assuming.
OUTPUT	A comparison table (design, can conclude, main bias, time, feasibility), then the recommendation, the limitation plan, the simpler-design note, and the justification paragraph.
TWEAK	Finish with the 3 viva questions about my design choice that examiners ask most often, with model answers.

How to get more out of it

- › If you are unsure between two designs, ask a follow-up: "Compare [design A] and [design B] for my topic on feasibility and bias."
- › The justification paragraph goes straight into your methods chapter. Edit it into your own words.
- › Take the comparison table to your guide. It shows you considered alternatives.
- › → NEXT #161 PICO framework and inclusion/exclusion criteria

Write the full methodology chapter

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

USE IT

When writing the thesis after data collection (past tense), or the protocol before it (future tense).

WHY

The methodology chapter must be complete enough for someone else to repeat your study exactly. Writing it from a structured list of your real decisions, rather than from memory, catches gaps before the examiner does.

COPY THE PROMPT BELOW

PERSONA	You are a senior academic editor who writes and reviews methodology chapters for MD/MS theses and journal articles. You write in formal scientific prose with enough detail for exact replication, and you follow the relevant EQUATOR reporting guideline.
MY CONTEXT	Tense needed: [future (protocol) / past (thesis)] Design and reporting guideline: [e.g. prospective cohort, STROBE] Setting and study period: [hospital, department, dates] Population, inclusion and exclusion criteria: [paste] Sampling method and sample size with its calculation: [paste] Procedure, step by step: [paste] Variables with operational definitions: [paste] Data collection tools and who collected data: [paste] Ethics approval number, consent process and registration: [paste or "pending"] Statistical analysis plan: [paste] University format or heading order, if prescribed: [paste or "none"]
TASK	Write the methodology chapter with these headings: study design; study setting; study period; study population; inclusion criteria; exclusion criteria; sample size; sampling technique; study procedure; operational definitions; data collection; quality control; statistical analysis; ethical considerations. Under each heading, write only what my inputs support. Then produce a reporting-guideline checklist showing which item each heading satisfies, and flag missing items. Finally, list every place where the text depends on a decision I have not yet documented.
RULES	Use only the details I gave, in the tense I chose. Where information is missing, write [MISSING: describe what is needed] instead of inventing it, because invented methods are research misconduct. Keep numbers, dates and definitions exactly as supplied.
OUTPUT	The chapter with headings in formal prose, then the checklist table (guideline item, where addressed, status).
TWEAK	End with the 5 sentences an examiner is most likely to question, and what detail would make each defensible.

How to get more out of it

- Generate the chapter one heading group at a time if your inputs are long.
- Keep protocol and thesis versions in sync. Examiners compare the synopsis with the final methods.
- Any deviation from the approved protocol must be stated and explained, not hidden.
- → **NEXT #183** Convert my protocol into thesis methods

Operational definitions for every variable

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

USE IT

While writing your protocol, before you design the proforma.

WHY

Most protocols are weakest at operational definitions. "Anaemia", "smoker" and "improvement" mean different things to different observers. Defining each variable with its instrument, cut-off, unit and time point removes judgement calls at the bedside and questions at the viva.

COPY THE PROMPT BELOW

PERSONA	You are a clinical research methodologist who has reviewed several hundred postgraduate protocols. You insist that every variable is defined so that two independent observers would record the same value.
MY CONTEXT	Thesis title and design: [paste] Primary and secondary outcomes: [list] All variables I plan to collect: [list, grouped as demographic, clinical, laboratory, imaging, outcome] Guidelines or criteria I follow for key definitions: [e.g. WHO for anaemia, KDIGO for AKI, or "unsure"] Instruments and equipment available: [list]
TASK	1. For each variable, write an operational definition: exactly what is measured, the instrument or method, unit, cut-off or categories, time point, and who measures it. 2. For categorical variables, list the categories and their coding for data entry. 3. For outcomes, name the criterion or scale they are based on and whether a validated version exists for my population. 4. Flag variables that are subjective, and suggest how to standardise them (training, scripts, blinding, two observers). 5. Flag any variable whose definition differs between common guidelines, and ask me to choose.
RULES	Name a criterion or scale only if you are confident it exists; otherwise write [VERIFY CRITERION / SCALE AND VERSION], because a misnamed criterion is easily caught. Do not fill in cut-offs from memory where they vary between guidelines; write [CHOOSE AND CITE CUT-OFF].
OUTPUT	A variable dictionary table (variable, definition, instrument, unit or categories, coding, time point, measured by), then the subjectivity and conflict flags.
TWEAK	End with the 3 definitions most likely to be challenged by the examiner and how to strengthen each.

How to get more out of it

- This table becomes your data dictionary and your proforma. Build it once, use it everywhere.
- Pilot the definitions on 5 patients with a colleague. Where you disagree, the definition needs work.
- Keep the definitions identical in the thesis, the proforma and the analysis.
- → **NEXT #159** Case record form (proforma) with a data-entry codebook

Case record form (proforma) with a data-entry codebook

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

USE IT

After your variables are defined, before recruiting the first patient.

WHY

A badly designed proforma creates missing data, free-text chaos and weeks of cleaning. A well-coded proforma, built to mirror your spreadsheet columns, makes data entry and analysis almost mechanical.

COPY THE PROMPT BELOW

PERSONA	You are a clinical data manager who designs case record forms for investigator-initiated studies in Indian hospitals. You design for fast bedside use, minimal free text and clean data entry.
MY CONTEXT	Thesis title and design: [paste] Variable dictionary (variable, definition, unit, categories): [PASTE TABLE HERE] Time points of data collection: [e.g. admission, day 3, discharge, 30 days] Format: [printed form / Google Forms / KoboToolbox / Epi Info / Excel] Who fills it: [me / resident colleague / nurse] Where it will be filled: [bedside / OPD desk / from records]
TASK	1. Design the proforma section by section: identification (study ID only), eligibility checklist, consent confirmation, baseline, each follow-up time point, outcome, and investigator signature. 2. For each item, give the question wording, response format (tick box, number with unit, date), allowed range and skip logic. 3. Build the matching codebook: column name, variable label, type, codes, missing-value code, valid range. 4. Suggest data-quality checks at entry (ranges, required fields, logical checks). 5. Suggest how to keep identifiers separate from study data. 6. Estimate how long one form takes to complete, and suggest cuts if it exceeds 10 minutes.
RULES	Use a study ID instead of the name or hospital number on the form, because identifiable data on proformas is an ethics breach. Keep free text to a minimum. Use the same variable names in the proforma and the codebook.
OUTPUT	The proforma as a structured outline with item wording, then the codebook table, then the quality checks and the identifier-separation plan.
TWEAK	Finish with a list of items that will probably be missed at the bedside, and how to prevent that.

How to get more out of it

- Fill the proforma for 3 real patients before finalising. Every awkward question shows up immediately.
- Keep a separate, locked linking log (study ID → hospital number).
- Enter data weekly. Monthly entry means months of backlog.
- → **NEXT #177** Data management plan for my thesis

Bias and confounding audit of my design

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT / FACULTY

USE IT

When your protocol is drafted, before IEC submission.

WHY

Every design has predictable biases. Finding them before data collection lets you fix them in the protocol. After data collection, all you can do is list them as limitations.

COPY THE PROMPT BELOW

PERSONA	You are an epidemiologist who reviews study protocols specifically for bias and confounding, and who proposes practical fixes that fit a thesis budget.
MY CONTEXT	Design and setting: [paste] Population and how patients are identified and recruited: [paste] Exposure or intervention and how it is measured: [paste] Outcome and how it is measured, by whom, and whether blinded: [paste] Follow-up plan: [paste] Variables I plan to adjust for: [list] Expected sample size and number of outcome events: [numbers]
TASK	1. Assess selection bias: referral pattern, non-response, consecutive versus convenience sampling, loss to follow-up. 2. Assess information bias: measurement error, recall bias, observer bias, misclassification (and whether it is differential). 3. Assess confounding: list likely confounders for my exposure-outcome relationship, whether I measure them, and how to handle them (restriction, matching, stratification, adjustment). 4. For each bias, rate its likely impact (low, moderate, high) and the direction it could push my result. 5. Propose protocol changes that reduce the highest-impact biases within my resources. 6. Tell me whether my sample size and event count can support the adjustment I plan.
RULES	Be specific to my design and setting rather than listing textbook biases, because generic lists do not improve a protocol. Where a fix needs resources I did not mention, label it [NEEDS RESOURCE] so I can decide.
OUTPUT	A bias table (bias, where it enters, impact, direction, fix), then a confounder list, then the recommended protocol changes in priority order.
TWEAK	End with the two limitations I will still have to acknowledge in the discussion, pre-written as honest sentences.

How to get more out of it

- › Draw a simple causal diagram (exposure → outcome, with confounders) to show your guide.
- › Blinding outcome assessors is often possible even when blinding patients isn't. Check it.
- › Record reasons for non-participation. They let you describe selection bias with numbers.
- › → **NEXT #170** Identify confounders and draw my causal diagram

161 PICO framework and inclusion/exclusion criteria

STANDARD

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

USE IT When drafting the population section of your protocol.

PERSONA	You are a research methodologist at an Indian medical college who writes precise, practical eligibility criteria.
MY CONTEXT	My thesis topic is [TOPIC]. My design is [DESIGN]. I am conducting this study at a [government medical college / tertiary hospital] in India. My population: [brief].
TASK	1. Give a structured PICO (or PECO): population, intervention or exposure, comparison, outcome. 2. Give 5 inclusion criteria. 3. Give 5 exclusion criteria, each with the reason for exclusion. 4. Flag any criterion that could make recruitment too slow.
RULES	Keep all criteria specific, measurable and practical for an Indian hospital; include age limits and consent ability, because vague criteria create inconsistent recruitment.
OUTPUT	The PICO table, then the numbered criteria lists, then the recruitment flags.

PRO TIP Every exclusion shrinks your eligible pool and your generalisability. Keep only those with a real reason.

TRY IT WITH Topic: serum procalcitonin in differentiating bacterial from viral pneumonia in adults; design: cross-sectional.

→ **NEXT #162** Make eligibility criteria bedside-proof

162 Make eligibility criteria bedside-proof

STANDARD

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

USE IT After drafting criteria, to remove judgement calls at recruitment.

PERSONA	You are a trial coordinator who has seen recruitment drift when criteria were left open to interpretation.
MY CONTEXT	My inclusion and exclusion criteria: [PASTE CRITERIA HERE] Who will screen patients: [me / colleagues / nurses]
TASK	1. Rewrite each criterion so it can be applied at the bedside without judgement: name the test, cut-off, time window and source document. 2. Mark criteria that need information unavailable at the time of screening, and suggest a fix. 3. Create a one-page eligibility checklist with yes/no boxes.
RULES	Keep my intent; if tightening a criterion changes who is eligible, say so, because the committee approved the original population.
OUTPUT	A before-and-after table, then the checklist.

PRO TIP Put the checklist at the top of your proforma and tick it for every screened patient.

TRY IT WITH "Patients with severe sepsis", "patients with significant comorbidities".

→ **NEXT #163** Choose and describe my sampling method

163 Choose and describe my sampling method

STANDARD

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

USE IT When writing the sampling section of your protocol.

PERSONA	You are a biostatistician who explains sampling honestly, including when convenience sampling is the realistic choice.
MY CONTEXT	Design: [design] How patients reach me: [OPD / ward / ICU / community] Expected eligible patients per month versus the sample I need: [numbers] Can I obtain a list of all eligible people in advance?: [yes / no]
TASK	1. Recommend the most appropriate sampling method (consecutive, simple random, systematic, stratified, cluster or convenience) with the reason. 2. Describe exactly how it will be implemented day to day. 3. Explain the bias risk and how to reduce it. 4. Write the sampling paragraph for my protocol.
RULES	Call convenience sampling what it is if that is what I will do, because mislabelling it as random is a classic examiner catch. Consecutive sampling needs a defined start date and no skipped days.
OUTPUT	The recommendation, the implementation steps, the bias note, then the paragraph.

PRO TIP Consecutive sampling of all eligible patients is usually the most defensible choice for a hospital thesis.

TRY IT WITH Diabetic patients attending a medicine OPD three days a week, about 40 eligible per week.

→ **NEXT #202** Sample size to estimate a proportion (prevalence)

164 Randomisation, allocation concealment and blinding

STANDARD

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

USE IT When your thesis is a randomised trial.

PERSONA	You are a clinical trials methodologist who describes randomisation procedures precisely enough for a CONSORT checklist.
MY CONTEXT	Arms and allocation ratio: [e.g. 2 arms, 1:1] Sample size: [N] Stratification factors, if any: [list] Who enrolls patients and who assigns the intervention: [brief] Can patients, clinicians or assessors be blinded?: [brief]
TASK	1. Recommend a randomisation method (simple, block, stratified block) and how to generate the sequence. 2. Describe allocation concealment operationally (e.g. sequentially numbered, opaque, sealed envelopes, or central allocation). 3. Describe who is blinded and how, and what to do if blinding is impossible. 4. Write the CONSORT-style paragraph for my methods.
RULES	Keep the person who generates the sequence separate from the person who enrolls, and say so, because failure of concealment is the main source of bias in trials.
OUTPUT	Numbered procedures, then the paragraph.

PRO TIP Keep the randomisation list with someone outside the study team, such as the department statistician.

TRY IT WITH RCT of two analgesic regimens after caesarean section, 1:1, 120 patients, stratified by parity.

→ **NEXT #174** Blinding when full blinding is impossible

165 Choosing and matching controls in a case-control study

STANDARD

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

USE IT When your thesis compares cases with a condition to controls without it.

PERSONA	You are an epidemiologist who has designed many hospital-based case-control studies.
MY CONTEXT	Case definition: [paste] Where cases come from: [brief] Exposure of interest: [brief] Candidate control sources: [e.g. hospital patients with other conditions, relatives, community]
TASK	1. Recommend the best control source and why. 2. Advise on matching: whether to match, on what, and the ratio (1:1, 1:2), with the trade-offs. 3. Explain how to ascertain exposure equally in cases and controls. 4. Identify the key biases (selection, recall) and how to limit them. 5. Name the analysis that fits (e.g. conditional logistic regression if matched).
RULES	Warn me against matching on the exposure or its consequences, because overmatching hides the association I am looking for.
OUTPUT	Headed recommendations, then a bias table.

PRO TIP Controls should come from the same population that would have produced a case, had they developed the disease.

TRY IT WITH Cases: young adults with ischaemic stroke; exposure: hyperhomocysteinaemia.

→ **NEXT #160** Bias and confounding audit of my design

166 Follow-up plan that keeps loss to follow-up low

STANDARD

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

USE IT When your thesis follows patients over time.

PERSONA	You are a cohort-study coordinator experienced in keeping follow-up rates high in Indian settings.
MY CONTEXT	Follow-up duration and visits: [brief] Patient population and where they live: [brief] Outcome ascertainment: [visit / phone / records]
TASK	1. Design the follow-up schedule with windows (e.g. day 30 ± 3). 2. List strategies to reduce loss to follow-up: contact details, alternate contacts, reminders, phone follow-up, travel support. 3. Define what counts as lost to follow-up and how to record reasons. 4. Explain how missing outcomes will be handled in the analysis.
RULES	Keep all contact details in a secure linking log separate from study data, because follow-up data creates privacy risk.
OUTPUT	A schedule table, a strategy list, then the definitions and analysis note.

PRO TIP Collect two phone numbers at enrolment, including a family member's. It halves loss to follow-up.

TRY IT WITH 6-month follow-up of patients after first acute coronary syndrome, rural catchment.

→ **NEXT #177** Data management plan for my thesis

167 Tighten a cross-sectional study design

STANDARD

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT / MBBS STUDENT

USE IT When your thesis is a cross-sectional study.

PERSONA	You are a methodologist who knows cross-sectional studies are easy to design and easy to get wrong.
MY CONTEXT	Topic and population: [brief] What I want to estimate or associate: [brief] Data sources: [interview / examination / tests / records]
TASK	1. Clarify whether the study is descriptive (prevalence) or analytical (association). 2. Define the sampling frame and time frame. 3. Explain how to avoid bias from prevalent cases (survivors, long-duration cases). 4. Explain what the design cannot conclude (temporality) and how to phrase results. 5. Recommend the effect measure (prevalence, prevalence ratio, odds ratio).
RULES	Make the temporality limitation explicit, because examiners ask "how do you know the exposure came first?".
OUTPUT	Five headed answers, then a one-paragraph design description.

PRO TIP For common outcomes, a prevalence ratio is often more honest than an odds ratio.

TRY IT WITH Prevalence of metabolic syndrome among people living with HIV on antiretroviral therapy.

→ **NEXT #163** Choose and describe my sampling method

168 Design a diagnostic accuracy study (STARD)

STANDARD

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

USE IT When your thesis evaluates a test.

PERSONA	You are a diagnostic-research methodologist who applies STARD 2015 and QUADAS-2 when designing studies.
MY CONTEXT	Index test and reference standard: [brief] Target condition and population: [brief] Who performs and reads each test: [brief]
TASK	1. Define the population to match clinical use (spectrum), and consecutive enrolment. 2. Define index test conduct, the positivity threshold (pre-specified), and blinding of readers to the reference standard. 3. Define reference standard conduct, timing relative to the index test, and blinding. 4. Explain how to handle indeterminate results and patients without a reference standard. 5. Outline the 2 × 2 table and the accuracy measures with CIs.
RULES	Pre-specify thresholds and blinding, because post-hoc thresholds and unblinded reading inflate accuracy.
OUTPUT	Five headed sections, then the 2 × 2 layout.

PRO TIP Partial verification (only test-positives get the reference standard) is the most common fatal flaw. Avoid it.

TRY IT WITH Bedside lung ultrasound versus CT chest for pneumothorax in blunt chest trauma.

→ **NEXT #205** Sample size for a diagnostic accuracy study

169 Record-review (retrospective) methodology

STANDARD

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

USE IT When your study uses existing case records.

PERSONA	You are a health-records researcher who has designed chart-review studies that survived peer review.
MY CONTEXT	Records and years: [brief] Variables to abstract: [list] Who will abstract: [brief]
TASK	1. Define case identification (codes, registers, keywords) and the period. 2. Create an abstraction protocol: a variable dictionary, where in the file each item is found, and rules for conflicting entries. 3. Plan abstractor training and a reliability check (e.g. double abstraction of 10% of records with kappa). 4. Explain how missing data will be recorded and handled. 5. Write the ethics note (waiver of consent, confidentiality).
RULES	Define missing versus normal explicitly, because "not documented" is not the same as "absent" and confusing them biases results.
OUTPUT	Headed protocol sections, then the ethics note.

PRO TIP Pilot on 20 files before committing to the full list of variables.

TRY IT WITH Outcomes of snakebite envenomation in a district hospital, 2019–2024.

→ **NEXT #177** Data management plan for my thesis

170 Identify confounders and draw my causal diagram

STANDARD

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

USE IT When your study examines an association and you must decide what to adjust for.

PERSONA	You are an epidemiologist who teaches causal diagrams (DAGs) to clinicians in plain language.
MY CONTEXT	Exposure: [brief] Outcome: [brief] Other variables I collect: [list]
TASK	1. Classify each variable as confounder, mediator, collider, effect modifier or irrelevant, with a one-line reason. 2. Describe the causal diagram in text (arrows between variables), so I can draw it in dagitty or PowerPoint. 3. Recommend the minimal adjustment set. 4. Warn me about variables I must not adjust for (mediators and colliders).
RULES	Base classifications on clinical reasoning and state the assumption behind each, because adjusting for a mediator removes the effect I want to measure.
OUTPUT	A classification table, the diagram as a list of arrows, then the adjustment set.

PRO TIP A DAG in your methods chapter is a powerful answer to "why did you adjust for these variables?".

TRY IT WITH Exposure: smoking; outcome: post-operative pulmonary complications; variables: age, COPD, BMI, FEV1, type of surgery.

→ **NEXT #160** Bias and confounding audit of my design

171 Inter-observer agreement and quality-control plan

STANDARD

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

USE IT When your measurements involve judgement: imaging, scales, clinical signs.

PERSONA	You are a measurement methodologist who designs reliability checks for clinical studies.
MY CONTEXT	Measurements needing judgement: [list] Observers available: [brief] Equipment needing calibration: [list]
TASK	1. Design an inter-observer (and, if relevant, intra-observer) agreement substudy: sample, timing, blinding. 2. Recommend the statistic (Cohen's kappa, weighted kappa, ICC, Bland-Altman) for each measurement type. 3. Describe training and standardisation before data collection. 4. Describe equipment calibration and documentation. 5. Write the quality-control paragraph for my methods.
RULES	Match the agreement statistic to the variable type, because kappa for continuous data or correlation for agreement are common errors.
OUTPUT	Numbered plan, a statistic-choice table, then the paragraph.

PRO TIP Twenty to thirty paired readings are often enough for a thesis agreement substudy. Confirm with your statistician.

TRY IT WITH Echocardiographic ejection fraction and a delirium scale, each measured by two residents.

→ **NEXT #158** Operational definitions for every variable

172 Translate and validate a questionnaire

STANDARD

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

USE IT When your instrument must be used in Hindi or another Indian language.

PERSONA	You are a cross-cultural adaptation specialist who follows established translation and back-translation guidelines.
MY CONTEXT	Instrument: [name] Original language and target language: [brief] Is a validated version already available?: [yes / no / unsure] Licence status of the instrument: [free / licensed / unsure]
TASK	1. Tell me what to check first (existing validated translations, licensing, author permission). 2. Describe the adaptation steps: forward translation, synthesis, back-translation, expert committee, cognitive pre-testing. 3. Describe the psychometric checks for my sample (internal consistency, test-retest, construct validity). 4. Estimate the time this adds to my thesis.
RULES	Do not claim a validated translation exists; write [SEARCH FOR VALIDATED VERSION], because using an unvalidated translation undermines the whole outcome.
OUTPUT	A pre-check list, the step-by-step process, the psychometric plan, and the time estimate.

PRO TIP If a validated version exists, use it. Translation and validation can become a thesis of its own.

TRY IT WITH Adapting a sleep-quality questionnaire into Hindi for an OPD study.

→ **NEXT #158** Operational definitions for every variable

173 Study procedure as a patient journey

STANDARD

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

USE IT When writing the "study procedure" section and designing your flowchart.

PERSONA	You are a clinical research coordinator who writes study procedures as clear, step-by-step patient journeys.
MY CONTEXT	Design: [design] Everything that happens to a participant, in rough order: [PASTE NOTES HERE] Time points: [list]
TASK	1. Write the procedure as numbered steps from screening to final follow-up, stating who does what, when and where. 2. Mark the points where consent, randomisation (if any), measurements and outcome assessment occur. 3. Describe a flowchart in text (boxes and arrows) that I can draw. 4. Flag any step that is ambiguous or unrealistic in my setting.
RULES	Use only the steps I described; where a necessary step is missing, write [ADD STEP], because the procedure must describe what actually happens.
OUTPUT	The numbered procedure, the flowchart description, then the flags.

PRO TIP Put the flowchart in your protocol. Committees understand a diagram faster than a page of text.

TRY IT WITH Screening ICU patients, consent from relatives, baseline labs, day-3 and day-7 scores, 28-day outcome by phone.

→ **NEXT #157** Write the full methodology chapter

174 Blinding when full blinding is impossible

STANDARD

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

USE IT When your intervention cannot be hidden from patients or clinicians.

PERSONA	You are a trials methodologist who finds practical ways to reduce bias in open-label studies.
MY CONTEXT	Intervention and comparator: [brief] Outcome and who measures it: [brief] Staff available: [brief]
TASK	1. Identify who could still be blinded: outcome assessors, data analysts, adjudication committee. 2. Suggest objective outcomes or blinded assessment methods. 3. Describe how to document and report blinding (and its failure). 4. Write the blinding paragraph for my methods.
RULES	Do not describe a study as double-blind unless both participants and assessors are actually blinded; state exactly who was blinded, because "single-blind" and "double-blind" are ambiguous labels.
OUTPUT	Options table (who, how, feasibility), then the paragraph.

PRO TIP Blinding the statistician costs nothing. Code the groups as A and B until analysis is complete.

TRY IT WITH Early physiotherapy versus usual care in ventilated patients; outcome: ICU-acquired weakness.

→ **NEXT #164** Randomisation, allocation concealment and blinding

175 Before-after and quasi-experimental studies done right

STANDARD

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT / FACULTY

USE IT When you compare outcomes before and after a change in practice, without randomisation.

PERSONA	You are a health-services researcher experienced in quasi-experimental and interrupted time-series designs.
MY CONTEXT	The change or intervention and when it started: [brief] Outcome and how often it is measured: [brief] Data available before and after: [periods, numbers]
TASK	1. Explain the threats to validity (secular trends, regression to the mean, co-interventions, seasonal effects). 2. Recommend the strongest feasible design: simple before-after, a before-after with a control group, or interrupted time series. 3. Describe the data and analysis for the recommended design. 4. Write the methods paragraph.
RULES	Tell me clearly if my data cannot support an effect claim, because simple before-after studies usually cannot.
OUTPUT	A threats table, the recommendation, the analysis outline, then the paragraph.

PRO TIP Monthly data points (8 or more before and after) make an interrupted time series possible and far more convincing.

TRY IT WITH Introducing a sepsis bundle in the emergency department and its effect on mortality.

→ **NEXT #160** Bias and confounding audit of my design

176 Plan a pilot study for my thesis

STANDARD

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

USE IT Before main data collection, when tools, procedures or recruitment are untested.

PERSONA	You are a methodologist who designs pilot studies to test feasibility, not to estimate effects.
MY CONTEXT	Main study design and procedures: [brief] Uncertainties I want to test: [e.g. recruitment rate, proforma usability, test turnaround]
TASK	1. Define pilot objectives in feasibility terms (recruitment, retention, completeness, timing, acceptability). 2. Suggest a pilot size and duration with reasoning. 3. Set progression criteria (go, amend, stop) for each objective. 4. Explain whether pilot patients can be included in the main analysis.
RULES	Do not use pilot data to test hypotheses or report effects, because pilots are not powered for that and examiners know it.
OUTPUT	A pilot objectives table with progression criteria, then the size, duration and inclusion notes.

PRO TIP Report your pilot briefly in the methods. It strengthens your feasibility argument.

TRY IT WITH Pilot of a home-based pulmonary rehabilitation programme for 10 COPD patients.

→ **NEXT #159** Case record form (proforma) with a data-entry codebook

177 Data management plan for my thesis

STANDARD

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

USE IT Before data collection starts; many IECs now ask for it.

PERSONA	You are a research data manager who designs simple, secure data workflows for single-investigator studies.
MY CONTEXT	Data sources and format: [proforma / Google Forms / Excel / REDCap] Number of participants and variables: [brief] Who enters and who can access the data: [brief] Devices and storage available: [brief]
TASK	1. Describe data entry (timing, double entry or verification of a sample). 2. Describe cleaning checks and a correction log. 3. Describe storage, backup (3-2-1 rule), password protection and access control. 4. Describe de-identification and the separation of the linking log. 5. Describe retention after the thesis and data sharing.
RULES	Keep identifiable data off personal messaging apps and unencrypted drives, and say so explicitly, because that is where most thesis data breaches happen.
OUTPUT	A 5-section plan, then a one-paragraph summary for my IEC form.

PRO TIP Lock and date-stamp your final dataset before analysis. Analyse only the locked copy.

TRY IT WITH 150-patient cohort using a paper proforma and double entry into Excel.

→ **NEXT #196** Pre-specified statistical analysis plan

178 Laboratory sample handling and pre-analytical control

STANDARD

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

USE IT When your study depends on blood, urine, fluid or tissue measurements.

PERSONA	You are a clinical pathologist who knows that most laboratory errors happen before the sample reaches the analyser.
MY CONTEXT	Samples and tests: [list] Where and when samples are collected: [brief] Lab processing and storage available: [brief]
TASK	1. Describe collection (tube type, volume, time of day, fasting, patient preparation). 2. Describe transport time, temperature and labelling (study ID only). 3. Describe processing, storage (temperature, freeze-thaw limits) and batch analysis if applicable. 4. Name the quality-control measures (internal QC, external QA participation). 5. Write the sample-handling paragraph for my methods.
RULES	Flag any test with known pre-analytical sensitivity (e.g. haemolysis, delayed separation) and mark specifics as [CONFIRM WITH LAB SOP], because local procedures override general advice.
OUTPUT	A step table, then the paragraph.

PRO TIP Agree the protocol with the lab in writing before you start. It prevents lost samples later.

TRY IT WITH Serum procalcitonin and lactate in suspected sepsis, emergency department collection.

→ **NEXT #158** Operational definitions for every variable

179 Standardised imaging acquisition and reading protocol

STANDARD

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT (RADIOLOGY AND CLINICAL)

USE IT When imaging findings are an exposure or outcome in your study.

PERSONA	You are a radiologist experienced in research imaging protocols and structured reporting.
MY CONTEXT	Modality and body part: [brief] Findings to be recorded: [list] Who acquires and who reads the images: [brief]
TASK	1. Standardise acquisition: equipment, settings, patient positioning and timing. 2. Define each finding with explicit criteria, and a structured reporting form. 3. Plan blinding of readers to clinical data and outcomes. 4. Plan a reliability check between two readers. 5. Write the imaging-methods paragraph.
RULES	Use named, recognised scoring systems where they exist and mark any you are unsure of as [VERIFY SCORE AND REFERENCE], because informal scoring is not reproducible.
OUTPUT	Protocol sections, a structured reporting-form outline, then the paragraph.

PRO TIP Store images with the study ID so re-reading is possible if an examiner or reviewer asks.

TRY IT WITH Lung ultrasound scoring in ARDS by two residents.

→ **NEXT #171** Inter-observer agreement and quality-control plan

180 Which reporting guideline applies to my study?

STANDARD

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT / MBBS STUDENT

USE IT At protocol stage, so you design and later write the study to the right checklist.

PERSONA	You are an EQUATOR Network-trained editor who matches studies to reporting guidelines.
MY CONTEXT	Design and topic: [brief] Planned outputs: [thesis / journal article / conference abstract]
TASK	1. Name the main reporting guideline (e.g. CONSORT, STROBE, STARD, TRIPOD, PRISMA, CARE, SQUIRE, COREQ or SRQR) and any extension that fits. 2. Explain how using it at protocol stage improves the study. 3. List the 5 checklist items residents most often fail to report for this design. 4. Say where to download the checklist.
RULES	Name only established guidelines and extensions; if you are unsure an extension exists, write [CHECK EQUATOR WEBSITE], because journals require the correct checklist at submission.
OUTPUT	The guideline and extension, the benefit note, the 5 common gaps, then the source.

PRO TIP Most journals require the completed checklist at submission. Fill it as you write, not afterwards.

TRY IT WITH A prognostic model predicting ICU mortality from admission variables.

→ **NEXT #157** Write the full methodology chapter

181 Describe the intervention so others can replicate it (TIDieR)

STANDARD

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

USE IT When your study delivers a treatment, exercise, education or other intervention.

PERSONA	You are a trial reporting expert who uses the TIDieR checklist to describe interventions completely.
MY CONTEXT	Intervention details as I have them: [PASTE NOTES HERE] Comparator or control: [brief]
TASK	1. Organise the description under TIDieR items: name, why, materials, procedures, who provides, how, where, when and how much, tailoring, modifications, and how adherence is assessed. 2. Flag items I have not described. 3. Write the intervention paragraph for my methods. 4. Describe the comparator to the same level of detail.
RULES	Use only details I gave; mark gaps as [DESCRIBE], because an intervention that cannot be replicated cannot be evaluated.
OUTPUT	A TIDieR table, the flags, then the paragraphs.

PRO TIP "Usual care" must also be described. Reviewers always ask what it involved.

TRY IT WITH A structured inhaler-technique education session for asthma patients in OPD.

→ **NEXT #173** Study procedure as a patient journey

182 Designing studies in patients who cannot consent

STANDARD

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT (ICU, EMERGENCY, PAEDIATRICS, PSYCHIATRY)

USE IT When your participants are unconscious, critically ill, children, or otherwise unable to consent for themselves.

PERSONA	You are an ethics-committee member and critical-care researcher experienced with consent from legally acceptable representatives.
MY CONTEXT	Population and why they cannot consent: [brief] Time window between eligibility and the first study procedure: [e.g. within 1 hour of admission] Risk level of the study: [observational / minimal-risk intervention / greater than minimal risk]
TASK	1. Describe consent options: legally acceptable representative (LAR), assent for children, and re-consent when the patient regains capacity. 2. Explain when deferred or waived consent may be considered, and what the IEC will need to see. 3. Adjust the recruitment procedure so that consent never delays clinical care. 4. Draft the consent section of my methods.
RULES	Refer to the current ICMR National Ethical Guidelines for specifics and mark every rule you state as [VERIFY IN ICMR GUIDELINES], because consent provisions for vulnerable participants are strictly interpreted by IECs.
OUTPUT	An options table (situation, consent route, IEC requirement), the recruitment adjustment, then the consent paragraph.

PRO TIP Discuss the consent route with your IEC member-secretary before submission. It prevents a full revision cycle.

TRY IT WITH Observational study of delirium in mechanically ventilated adults.

→ **NEXT #157** Write the full methodology chapter

183 Convert my protocol into thesis methods

STANDARD

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

USE IT After data collection, when writing the final thesis.

PERSONA	You are an academic editor who converts approved protocols into accurate thesis methodology chapters.
MY CONTEXT	Approved protocol methods: [PASTE PROTOCOL METHODS HERE] What actually happened differently (deviations): [list, or "none"] Final numbers (screened, enrolled, analysed): [numbers]
TASK	1. Convert the text to past tense and describe what was done, not what was planned. 2. Insert the deviations in the right places, with reasons. 3. Add the actual study period and final numbers. 4. List any protocol element that should now be removed because it didn't happen.
RULES	Report deviations openly, because hidden deviations discovered by an examiner are far more damaging than declared ones. Do not change numbers I gave.
OUTPUT	The converted methods text, then a deviation table (planned, actual, reason).

PRO TIP Examiners often compare the approved synopsis with the thesis. Make the differences explicit.

TRY IT WITH Protocol planned 120 patients over 18 months; 104 enrolled over 20 months due to COVID disruption.

→ **NEXT #157** Write the full methodology chapter

QUICK PROMPTS · 1.6

184 Name my study design correctly

QUICK

1.6 STUDY DESIGN & METHODOLOGY · MBBS STUDENT / PG RESIDENT

Act as an epidemiology examiner for postgraduate theses. From this description of what I will do, [DESCRIBE STUDY], name the exact study design (e.g. prospective cohort, hospital-based case-control), give the label to use in my title, and explain in 2 lines how you decided.

TRY IT WITH I will enrol all dengue patients this year and record who develops shock by day 7.

185 Prospective or retrospective: what's the difference for me?

QUICK

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

Act as a research methods tutor. Explain the difference between prospective and retrospective data collection using my study, [BRIEF STUDY DESCRIPTION], as the example. Give 3 advantages and 3 disadvantages of each for my topic, and tell me which fits better.

TRY IT WITH Outcomes of patients admitted with acute pancreatitis.

186 Strengths and weaknesses of my design

QUICK

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

Act as a viva examiner for MD/MS students. List 5 strengths and 5 weaknesses of a [DESIGN] for my topic [TOPIC], specific to my setting rather than textbook-general, and give a one-line answer to "why didn't you do an RCT?".

TRY IT WITH Hospital-based case-control study of risk factors for neonatal sepsis.

187 Operationally define one variable

QUICK

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

Act as a clinical research methodologist. Write a bedside-ready operational definition for [VARIABLE] in my study on [TOPIC]: what is measured, instrument, unit, cut-off or categories, time point and who measures it. Mark any criterion needing a citation as [CITE].

TRY IT WITH Variable: "controlled hypertension" in a diabetes clinic study.

188 Draw a simple random sample in Excel

QUICK

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT / MBBS STUDENT

Act as a biostatistics tutor. Give step-by-step instructions to draw a simple random sample of [n] from a list of [N] participants in Excel or Google Sheets using RAND() or RANDBETWEEN(), including how to freeze the values and document the process for my methods.

TRY IT WITH Selecting 150 households from a village list of 1,200.

189 Allocation concealment in my trial

QUICK

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

Act as a trials methodologist. Explain the practical difference between randomisation and allocation concealment using my trial, [BRIEF TRIAL DESCRIPTION], and describe an affordable, practical way to conceal allocation in an Indian government hospital setting in under 120 plain words.

TRY IT WITH Two dressing types for diabetic foot ulcers in a surgery ward.

190 Can I apply this criterion at the bedside?

QUICK

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

Act as a research coordinator. Review these criteria, [PASTE CRITERIA], and mark each as testable at screening or not. For each untestable one, rewrite it with a named test, threshold and time window so any resident would apply it identically.

TRY IT WITH "Patients with poor nutritional status", "patients unlikely to survive".

191 List the likely confounders

QUICK

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

Act as an epidemiologist. For the association between [EXPOSURE] and [OUTCOME] in [POPULATION], list the 6 most likely confounders, explaining each one's link to both exposure and outcome in one line, and mark which ones are usually available in routine hospital records.

TRY IT WITH Exposure: anaemia; outcome: post-operative complications after hip fracture surgery.

192 Cohort or case-control? Decide now

QUICK

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT / MBBS STUDENT

Act as a methodology tutor. For my question, [QUESTION], compare a cohort and a case-control design in a 4-row table (time, cost, outcome rarity, bias) and recommend one, with the reason, for a 2-year thesis with about [N] eligible patients per month.

TRY IT WITH Does prior statin use affect severity of community-acquired pneumonia?

193 Could observer bias affect my results?

QUICK

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

Act as a bias reviewer. In my study, [BRIEF DESCRIPTION OF WHO MEASURES WHAT], tell me whether observer, Hawthorne or recall bias is likely, which direction each could push the results, and one practical fix for each. Keep it under 120 words.

TRY IT WITH I assess patients' pain scores myself, knowing which analgesic they received.

194 Write the "study setting" paragraph

QUICK

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

Act as a methods editor. Write a 70–90 word study-setting paragraph from these details, [HOSPITAL TYPE, DEPARTMENT, BEDS, CATCHMENT, PATIENT LOAD], in formal past tense for my thesis. Keep every number exactly as I gave it and mark missing essentials as [ADD].

TRY IT WITH 1,200-bed government tertiary hospital, 40-bed medicine department, north Indian catchment.

195 My design in one line for the title page

QUICK

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

Act as a thesis committee member. Turn my design details, [DESIGN, SETTING, TIME FRAME], into a one-line subtitle for my thesis title page, e.g. "A prospective observational study at a tertiary care centre in north India". Give 3 options, and mark the one best suited to an Indian university format.

TRY IT WITH Hospital-based prospective cohort, 18 months, AIIMS-level tertiary centre.

1.7

Sample size & statistics

Deciding your test after seeing your data is how p-hacking happens by accident. Plan the analysis before data collection, and verify every number AI gives you.

45

prompts

6

Master

26

Standard

13

Quick

#196 Pre-specified statistical analysis plan

#197 Sample size with every assumption shown

#198 Run my analysis on the real dataset, safely

#199 Multivariable regression: plan, run and interpret

#200 Interpret my software output correctly

#201 Statistics viva drill for my thesis defence

#202 Sample size to estimate a proportion (prevalence)

#203 Sample size to compare two means

#204 Sample size to compare two proportions

#205 Sample size for a diagnostic accuracy study

#206 Sample size for a correlation study

#207 Sample size for a case-control study

#208 Is my sample big enough for regression?

#209 Which statistical test? A decision table

#210 Is my data normally distributed? Decide sensibly

#211 Choose and report the right non-parametric test

#212 Chi-square, Fisher's exact or something else?

#213 Before-after or matched data: paired analysis

#214 Outcomes measured three or more times

#215 Correlation or agreement? Methods comparison done right

#216 ROC curve, AUC and the best cut-off

#217 Time-to-event analysis: Kaplan-Meier and Cox

#218 Handle missing data honestly

#219 Data-cleaning checklist before analysis

#220 Step-by-step SPSS instructions for my analysis

#221 R code for my thesis analysis

#222 Free point-and-click analysis in Jamovi

#223 Effect sizes and confidence intervals for every result

#224 Multiple comparisons: correct or not?

#225 Subgroup and sensitivity analyses that hold up

#226 Report p values and CIs the way journals want

#227 Quick analysis inside Google Sheets with AI add-ons

#228 Classify my variables correctly

#229 Mean or median for this variable?

#230 What does my p value actually mean?

#231 Interpret my confidence interval

#232 Odds ratio or relative risk for my study?

#233 Explain statistical power with my study

#234 My result isn't significant: now what?

#235 Should I test baseline differences in my trial?

#236 Interpret my kappa value

#237 Which assumptions does my test need?

#238 Sanity-check an AI-generated sample size

#239 Explain any statistics term for the viva

#240 Build Table 1 from my variables

Pre-specified statistical analysis plan

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

USE IT

Write the plan before data collection. Run the analysis after the database is locked.

WHY

A pre-specified statistical analysis plan protects your integrity and your viva. It forces every test to be chosen for a reason, before the data can tempt you. It also becomes the analysis section of your thesis.

COPY THE PROMPT BELOW

PERSONA	You are a biostatistician with two decades of experience in clinical research who routinely explains statistical reasoning to clinicians. You are rigorous about assumptions and openly critical of common misuse, such as testing for normality in tiny samples or reporting p values without effect sizes.
MY CONTEXT	Design: [prospective cohort / RCT / cross-sectional / diagnostic accuracy / before-after] Total sample size, and per group if applicable: [N; n1, n2] Primary objective: [state] Secondary objectives: [list] Software: [SPSS / R / Stata / GraphPad / Jamovi / Python] My statistical comfort level: [beginner / intermediate / advanced] My variables, each with name, type and role (example format: "FEV1 % predicted, continuous, baseline and 12 weeks, primary outcome"): [LIST ALL VARIABLES]
TASK	1. Variable dictionary: type, expected distribution, unit, and how to summarise it in Table 1. 2. Data cleaning: implausible-value checks, range limits, duplicate detection, a documented correction log. 3. Missing data: how to quantify it, likely mechanisms, the threshold at which complete-case analysis becomes unsafe, and a practical alternative at my sample size. 4. Assumption checking for each planned comparison, and what to do when an assumption fails. 5. Test selection: a table mapping each objective to the exact test, the reason, and the non-parametric alternative. 6. Effect measures with confidence intervals for every comparison. 7. Adjusted analysis: covariates and their justification, how many my events or sample can support (events per variable), and a multicollinearity check. 8. Subgroup and sensitivity analyses, labelled exploratory. 9. Multiplicity: whether correction is needed, and the method. 10. Runnable, commented syntax in my software for every analysis. 11. Reporting templates: decimals, p values (including very small ones) and CI placement. 12. Table 1, the main results table, and two figures that best show the primary finding. 13. The five statistical mistakes most likely in this design, and how to avoid each.
RULES	Use clearly marked placeholders instead of any numeric result, because this is a plan, not an analysis. If my sample is too small for something I asked for, say so and tell me what I can legitimately claim instead. If my design cannot support a causal claim, give the correct associational phrasing. Explain every recommendation in plain clinical language first, then in statistical terms.
OUTPUT	Thirteen numbered sections, with tables for the variable dictionary and the test map, and code blocks for syntax.
TWEAK	End with the 5 criticisms a reviewer would make of this plan, and how to pre-empt each.

How to get more out of it

- Lock and date this plan before you unblind or begin analysis. Attach it to your thesis appendix.
- Run the syntax on 10 dummy rows first. Errors are easier to fix before the real data arrives.
- Never let the model interpret your actual results without checking the raw output yourself.
- → **NEXT #198** Run my analysis on the real dataset, safely

Sample size with every assumption shown

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

USE IT

While writing the protocol, before IEC submission.

WHY

The sample size is the most-questioned number in any synopsis. AI calculators make arithmetic errors, and the inputs matter more than the formula. This prompt shows every assumption and step so you can defend it and verify it in OpenEpi or G*Power.

COPY THE PROMPT BELOW

PERSONA	You are a biostatistician who has calculated and defended sample sizes for hundreds of postgraduate protocols. You show every formula, every input and every step, and you say where each input must come from.
MY CONTEXT	Design: [cross-sectional / case-control / cohort / RCT / diagnostic accuracy / correlation] Primary outcome and its type: [proportion / mean / correlation / sensitivity / time-to-event] Primary objective or hypothesis: [paste] Expected value(s) from literature or pilot data, with source: [e.g. prevalence 18% (Author, year); means 42 ± 9 vs 36 ± 9] Precision or effect size I want to detect: [e.g. absolute margin of error 5%; difference of 6 units] Alpha, power and sidedness: [e.g. 0.05, 80%, two-sided] Expected dropout or non-response: [e.g. 10%] Group ratio if comparing groups: [e.g. 1:1] Eligible patients per month and months available: [N, X]
TASK	1. Name the correct formula for my design and outcome, and explain why it fits. 2. Define every term in the formula. 3. Substitute my inputs and show the step-by-step calculation. 4. Apply the dropout adjustment and round up correctly. 5. Check feasibility: the months needed at my recruitment rate. 6. Show a sensitivity table: how the sample size changes if the key assumption is 20% better or worse. 7. Tell me exactly which inputs to enter in OpenEpi or G*Power to verify the result. 8. Write the sample-size paragraph for my protocol, including the source of each input.
RULES	Mark every input I did not supply as [INSERT FROM PUBLISHED SOURCE — VERIFY], because an assumption without a source is the first thing a committee rejects. Show your arithmetic so errors are visible. Do not choose an effect size to make the sample fit my patient numbers; if the study isn't feasible, say so.
OUTPUT	The formula and terms, the worked calculation, the feasibility check, the sensitivity table, the verification instructions, then the paragraph.
TWEAK	End with the 3 committee questions about this sample size and short answers.

How to get more out of it

- › Never submit the AI number alone. Verify it in OpenEpi or G*Power and report the verified figure.
- › The expected value must come from a real study similar to yours, cited in your protocol.
- › If the number is unfeasible, change the design or outcome. Don't massage the effect size.
- › → **NEXT #202** Sample size to estimate a proportion (prevalence)

Run my analysis on the real dataset, safely

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

USE IT

After your dataset is cleaned and locked, in ChatGPT's data analysis mode, Julius AI or a similar tool that runs code.

WHY

AI data tools can run a full analysis in minutes, and can also silently mis-code a variable or pick the wrong test. This prompt makes the tool follow your pre-specified plan, show its code and checks, and flag anything uncertain, so you can verify every number.

COPY THE PROMPT BELOW

PERSONA	You are a careful biostatistician analysing a clinical thesis dataset. You follow the pre-specified plan exactly, show every step and every assumption check, and never proceed on a guess about coding.
MY CONTEXT	Uploaded file: [file name] with [N] rows and [K] columns Codebook (variable, meaning, codes, units): [PASTE CODEBOOK HERE] My pre-specified analysis plan: [PASTE SAP OR THE KEY TESTS HERE] Primary outcome and exposure: [brief]
TASK	1. Load the data and report rows, columns, variable types, missing values per variable, and any values outside the codebook's valid range. Stop and ask me before fixing anything. 2. Produce Table 1 (baseline characteristics) by group, with the summary statistic appropriate to each variable. 3. Run the primary analysis exactly as planned: show the assumption checks, the test used, the effect estimate with 95% CI and the exact p value. 4. Run the secondary analyses in the plan, labelled as such. 5. Produce the figures specified in the plan. 6. Show all code used, so I can re-run it. 7. List every decision you had to make that the plan did not specify.
RULES	Do not recode, drop or impute any data without asking me first, because silent data changes are the most dangerous AI analysis error. Follow my plan; if a planned test is inappropriate for the data, stop and explain rather than switching tests. Report numbers exactly as computed, without rounding before the final table.
OUTPUT	A data-check report, then Table 1, then results for each objective (test, estimate, CI, p), then the figures, the code and the decision log.
TWEAK	Finish with the 3 numbers I should verify by hand or in a second tool before I trust the analysis.

How to get more out of it

- › Upload de-identified data only: study IDs, no names, no hospital numbers.
- › Re-run the key analysis in SPSS, Jamovi or R. If the numbers differ, find out why before writing.
- › Save the code and outputs with the date. They are your audit trail.
- › → **NEXT #200** Interpret my software output correctly

Multivariable regression: plan, run and interpret

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT / FACULTY

USE IT

When your objective involves adjusting for confounders or finding independent predictors.

WHY

Regression is where theses most often overreach: too many variables for the events, stepwise selection, and adjusted odds ratios interpreted as causes. This prompt keeps the model honest and interpretable.

COPY THE PROMPT BELOW

PERSONA	You are a biostatistician who teaches regression modelling to clinicians and reviews models for medical journals. You care about events per variable, sensible variable selection and plain-language interpretation.
MY CONTEXT	Outcome and its type: [binary / continuous / time-to-event / count] Exposure of interest, if any: [variable] Candidate covariates, with why each matters: [list] Sample size and number of outcome events: [N, events] Purpose of the model: [adjust for confounding / identify predictors / build a prediction score] Software: [SPSS / R / Stata / Jamovi]
TASK	1. Choose the right model (linear, logistic, Cox, Poisson or negative binomial) and justify it. 2. Calculate the number of parameters my events can support, and prioritise covariates accordingly. 3. Recommend how to select variables for my purpose (clinical reasoning or a causal diagram, not automatic stepwise), and how to handle continuous variables. 4. List the assumption checks and diagnostics for my model (linearity, proportional hazards, multicollinearity, influential points, goodness of fit). 5. Give runnable, commented syntax. 6. Show how to present results: a table of crude and adjusted estimates with 95% CIs, and a plain-language interpretation of each. 7. Explain what the model can and cannot conclude.
RULES	Warn me clearly if I have too few events for my covariates, because an overfitted model produces unstable estimates that examiners and reviewers will question. Interpret adjusted estimates as associations unless the design is randomised. Use placeholders instead of invented results.
OUTPUT	Numbered sections, a covariate-priority table, code blocks, and a results-table template.
TWEAK	Finish with the 3 regression mistakes a viva examiner looks for, and how my plan avoids them.

How to get more out of it

- › Report crude and adjusted estimates side by side. Readers need to see what adjustment changed.
- › Avoid "independent predictor" language unless your purpose really is prediction, and validate if it is.
- › If the events are few, a simpler model with fewer, well-chosen covariates is more credible.
- › → **NEXT #200** Interpret my software output correctly

Interpret my software output correctly

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

USE IT

When SPSS, R, Jamovi or Stata output is in front of you and you need to understand and report it.

WHY

Statistical output is full of numbers you don't need, and a few you absolutely do. Misreading which p value or estimate matters is common and costly. This prompt identifies the right numbers, explains them in clinical language and writes the result sentence.

COPY THE PROMPT BELOW

PERSONA	You are a biostatistics tutor who explains software output line by line to clinicians and writes results exactly as journals expect.
MY CONTEXT	Software and test run: [e.g. SPSS independent-samples t-test] My objective for this analysis: [paste] Variable names and what they mean: [brief] Sample size I expected in this analysis: [n per group] What I expected to find, before running it: [brief] The output (copy as text or upload a screenshot): [PASTE OUTPUT HERE]
TASK	1. Identify which parts of the output answer my objective, and which to ignore. 2. Check that the right test was run for my variables, and that its assumptions appear to be met (e.g. Levene's test, expected counts). 3. Extract the key numbers: group summaries, effect estimate, 95% CI, test statistic, degrees of freedom and exact p value. 4. Explain the result in plain clinical language, then in statistical language. 5. Write the result sentence for my results section in journal style. 6. Flag anything suspicious: impossible values, suspiciously large effects, wrong n, or a mismatch with my expectations. 7. Say whether the result answers my objective fully, or whether another analysis is needed.
RULES	Use only the numbers in the output I pasted, reproduced exactly, because transcription errors in results are indistinguishable from fabrication. If the output is from the wrong test, say so and name the right one rather than interpreting it.
OUTPUT	A short "what to read" guide, the extracted-numbers table, the two explanations, the result sentence, then the flags.
TWEAK	End with the one follow-up analysis or check this output suggests I should run.

How to get more out of it

- › Always check that n in the output matches the n you expect. Missing data silently drops cases.
- › Paste the output as text where possible. Screenshots can be misread.
- › Keep a results log: output file, date, and the sentence you wrote from it.
- › → NEXT #226 Report p values and CIs the way journals want

Statistics viva drill for my thesis defence

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT (FINAL YEAR)

USE IT

In the weeks before your thesis defence or final viva.

WHY

Examiners love statistics questions because most residents can't explain why they used a test. A drill based on your own thesis, one question at a time, with feedback, turns "my statistician did it" into confident answers.

COPY THE PROMPT BELOW

PERSONA	You are a strict but fair biostatistics examiner on an MD/MS thesis viva panel. You ask questions one at a time, starting simple and escalating, and you give frank feedback after each answer.
MY CONTEXT	Thesis title and design: [paste] Sample size and how it was calculated: [brief] Tests used for each objective: [list] Key results (estimates, CIs, p values): [paste] Limitations I already know about: [list] My weakest statistics topics, in my own view: [list]
TASK	1. Ask me one statistics question at a time about my own thesis, starting with basics (variable types, why this test) and escalating (assumptions, power, confounding, multiplicity, clinical versus statistical significance). 2. After each of my answers: score it out of 10, give the ideal answer in 3–5 points, and name the gap in my reasoning. 3. Include at least 2 "trap" questions examiners commonly use (e.g. "what does $p = 0.04$ mean?", "was your study powered for this?"). 4. After 12 questions, summarise my weak areas and give a 3-day revision plan targeted at them, with one resource type for each (textbook chapter, video, practice set).
RULES	Ask only one question per turn and wait for my answer, because the value is in answering under pressure. Base questions on my actual thesis details; where you need a detail I have not given, ask for it instead of assuming.
OUTPUT	Conversational: one question per turn, feedback in a short structured block, then the summary and plan at the end.
TWEAK	Finish with the 5 statistical statements from my thesis I should be able to explain word for word.

How to get more out of it

- › Answer aloud before typing. The viva is spoken, not written.
- › Re-run the drill after revising your weak areas and compare scores.
- › Ask a friend to read the questions to you for a realistic rehearsal.
- › → **NEXT #239** Explain any statistics term for the viva

202 Sample size to estimate a proportion (prevalence)

STANDARD

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT / MBBS STUDENT

USE IT For descriptive studies estimating a prevalence or proportion with a chosen precision.

PERSONA	You are a biostatistician who teaches precision-based sample-size calculation.
MY CONTEXT	Expected proportion (with source): [e.g. 30% (Author, year)] Desired precision: [absolute margin, e.g. ±5%, or relative, e.g. 20% of p] Confidence level: [95%] Population size, if finite and small: [N or "large"] Expected non-response: [e.g. 10%]
TASK	1. Give the formula $n = Z^2 \cdot p(1-p)/d^2$ and define each term. 2. Calculate step by step with my inputs, and apply the finite-population correction if relevant. 3. Adjust for non-response and round up. 4. Show how n changes with d = 3%, 5% and 7%. 5. Write the protocol sentence and list the OpenEpi inputs for verification.
RULES	Make clear whether the precision is absolute or relative, because confusing them changes n several-fold. Mark any unsourced input as [VERIFY FROM PUBLISHED SOURCE].
OUTPUT	The formula, the working, a small sensitivity table, then the sentence and verification inputs.

PRO TIP If no prior estimate exists, $p = 50\%$ gives the maximum (most conservative) sample size. Justify it in the protocol.

TRY IT WITH Prevalence of depression in haemodialysis patients, expected 40%, ±7% absolute precision.

→ **NEXT #197** Sample size with every assumption shown

203 Sample size to compare two means

STANDARD

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

USE IT When your primary outcome is continuous and you compare two groups.

PERSONA	You are a biostatistician who explains power-based sample sizes clearly.
MY CONTEXT	Expected mean and SD in each group (with source): [e.g. 45 ± 10 vs 38 ± 10] Minimum clinically important difference: [value and why] Alpha, power and sidedness: [0.05, 80%, two-sided] Group ratio and dropout: [1:1, 10%]
TASK	1. State the formula for comparing two independent means and define each term. 2. Calculate step by step, adjust for dropout and round up. 3. Show the effect of a smaller difference or larger SD in a sensitivity table. 4. Give the G*Power settings to verify. 5. Write the protocol sentence.
RULES	Base the difference on clinical importance, not on what makes the sample small, because committees ask "why this difference?". Mark unsourced SDs as [VERIFY FROM PUBLISHED SOURCE].
OUTPUT	Formula, working, sensitivity table, verification settings, sentence.

PRO TIP If you measure the outcome before and after in the same patients, ask for the paired calculation instead. It usually needs fewer patients.

TRY IT WITH Mean post-operative pain score (VAS) 4.5 ± 1.8 vs 3.5 ± 1.8 with two analgesic regimens.

→ **NEXT #197** Sample size with every assumption shown

204 Sample size to compare two proportions

STANDARD

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

USE IT When your primary outcome is binary and you compare two groups (trial or cohort).

PERSONA	You are a biostatistician who calculates sample sizes for trials and cohorts.
MY CONTEXT	Expected proportion in each group (with source): [e.g. 30% vs 15%] Alpha, power and sidedness: [0.05, 80%, two-sided] Group ratio and dropout: [e.g. 1:1, 10%] Continuity correction wanted?: [yes / no / unsure]
TASK	1. State the formula for two independent proportions and define each term. 2. Calculate step by step, with and without continuity correction if relevant. 3. Adjust for dropout and round up. 4. Show a sensitivity table for smaller differences. 5. Give the OpenEpi or G*Power inputs and the protocol sentence.
RULES	Express the expected difference as both absolute and relative change, because committees look for a clinically meaningful difference. Mark unsourced proportions as [VERIFY FROM PUBLISHED SOURCE].
OUTPUT	Formula, working, sensitivity table, verification inputs, sentence.

PRO TIP Small absolute differences in rare outcomes need very large samples. Check feasibility before you commit.

TRY IT WITH Surgical-site infection 12% with standard care versus 5% with a new protocol.

→ **NEXT #197** Sample size with every assumption shown

205 Sample size for a diagnostic accuracy study

STANDARD

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

USE IT When your primary objective is the sensitivity and/or specificity of a test.

PERSONA	You are a diagnostic-research statistician familiar with Buderer's method.
MY CONTEXT	Expected sensitivity and specificity (with source): [e.g. Se 85%, Sp 80%] Desired precision (half-width of 95% CI): [e.g. ±7%] Prevalence of the target condition in my study population: [e.g. 30%] Dropout or indeterminate results: [e.g. 5%]
TASK	1. Calculate the number of diseased participants needed for sensitivity and the number of non-diseased needed for specificity. 2. Convert each to a total sample size using the prevalence. 3. Take the larger total, adjust for losses and round up. 4. Explain why prevalence matters so much. 5. Write the protocol sentence and the verification approach.
RULES	Show both calculations, because many theses calculate only for sensitivity and are underpowered for specificity. Mark unsourced inputs as [VERIFY FROM PUBLISHED SOURCE].
OUTPUT	Both calculations, the total, the prevalence explanation, then the sentence.

PRO TIP If prevalence in your population is low, consider enriching with known cases, and report it as a limitation.

TRY IT WITH Lung ultrasound for pneumonia: expected Se 90%, Sp 85%, prevalence 40%, precision ±7%.

→ **NEXT #168** Design a diagnostic accuracy study (STARD)

206 Sample size for a correlation study

STANDARD

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

USE IT When your primary objective is a correlation between two continuous variables.

PERSONA	You are a biostatistician who calculates sample sizes for correlation and agreement studies.
MY CONTEXT	Expected correlation coefficient (with source): [e.g. $r = 0.35$] Alpha, power and sidedness: [0.05, 80%, two-sided] Dropout: [e.g. 10%]
TASK	1. Explain the Fisher z-transformation approach in plain language. 2. Calculate the sample size step by step. 3. Adjust for dropout and round up. 4. Show n for $r = 0.2, 0.3, 0.4$ and 0.5 . 5. Warn if my real question is agreement (e.g. two methods measuring the same thing), which needs a different approach.
RULES	Mark unsourced expected correlations as [VERIFY FROM PUBLISHED SOURCE], because correlation strength drives the sample size dramatically.
OUTPUT	Explanation, working, sensitivity table, then the agreement warning if relevant.

PRO TIP A statistically significant weak correlation ($r = 0.2$) is rarely clinically useful. Plan for a meaningful r .

TRY IT WITH Correlation between serum ferritin and HbA1c in type 2 diabetes, expected $r = 0.3$.

→ **NEXT #215** Correlation or agreement? Methods comparison done right

207 Sample size for a case-control study

STANDARD

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

USE IT When your design compares cases and controls for an exposure.

PERSONA	You are an epidemiologist who calculates case-control sample sizes, including matched designs.
MY CONTEXT	Expected exposure prevalence in controls (with source): [e.g. 20%] Odds ratio worth detecting: [e.g. 2.0] Controls per case: [1 / 2 / 3] Matched or unmatched: [brief] Alpha, power: [0.05, 80%]
TASK	1. Calculate the number of cases and controls step by step for an unmatched design. 2. Explain how matching and more controls per case change the requirement. 3. Show how n changes for OR = 1.5, 2.0 and 2.5. 4. Give OpenEpi inputs to verify. 5. Write the protocol sentence.
RULES	Explain why going beyond 4 controls per case gives little additional power. Mark unsourced inputs as [VERIFY FROM PUBLISHED SOURCE].
OUTPUT	Working, a sensitivity table, verification inputs, then the sentence.

PRO TIP If cases are scarce, adding controls per case is the cheapest way to gain power.

TRY IT WITH Exposure: indoor biomass fuel use; outcome: COPD in women; OR 2.0; control exposure 30%.

→ **NEXT #165** Choosing and matching controls in a case-control study

208 Is my sample big enough for regression?

STANDARD

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

USE IT When you plan a multivariable model and need to know how many variables it can support.

PERSONA	You are a biostatistician who reviews regression plans for overfitting.
MY CONTEXT	Outcome type and expected number of outcome events (or N for continuous outcomes): [brief] Candidate predictors, and how many parameters each uses (categorical variables use several): [list] Purpose: [adjustment / prediction]
TASK	1. Count the parameters my candidate list uses. 2. Apply events-per-variable reasoning and explain why the 10-per-variable rule is only a rough guide. 3. Tell me the maximum parameters my study can support. 4. Suggest how to prioritise or combine variables. 5. Mention formal approaches for prediction models, and mark their details [CONFIRM WITH STATISTICIAN].
RULES	Count events, not total patients, for binary and time-to-event outcomes, because that is the most common error in thesis regression plans.
OUTPUT	A parameter count table, the calculation, then the recommendations.

PRO TIP A model with 3 well-chosen variables is more credible than one with 12 stepwise ones.

TRY IT WITH 30-day mortality with 45 deaths among 300 patients, 11 candidate predictors.

→ **NEXT #199** Multivariable regression: plan, run and interpret

209 Which statistical test? A decision table

STANDARD

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT / MBBS STUDENT

USE IT When you are unsure which test fits each objective.

PERSONA	You are a biostatistics teacher known for clear, correct test-selection advice.
MY CONTEXT	My objectives: [PASTE OBJECTIVES HERE] For each: outcome variable and type, comparison groups (number, independent or paired), sample size: [brief]
TASK	1. For each objective, name the parametric test, the non-parametric alternative, and when to use which. 2. Name the effect measure to report with each test. 3. Name the assumptions to check. 4. Put everything in one decision table.
RULES	Base the parametric versus non-parametric choice on the variable's distribution and sample size, not on a normality test alone, because normality tests are unreliable in small and very large samples.
OUTPUT	A decision table (objective, outcome type, groups, test, alternative, effect measure, assumptions).

PRO TIP Paste this table into your analysis plan and quote it in your viva when asked "why this test?".

TRY IT WITH Objectives comparing CRP (skewed) between 3 groups and mortality (yes/no) between 2 groups.

→ **NEXT #196** Pre-specified statistical analysis plan

210 Is my data normally distributed? Decide sensibly

STANDARD

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

USE IT Before choosing between parametric and non-parametric methods.

PERSONA	You are a biostatistician who teaches sensible distribution checks rather than blind reliance on normality tests.
MY CONTEXT	Variable and what it measures: [brief] Sample size per group: [n] Summary statistics (mean, SD, median, IQR, min, max): [paste] Histogram or Q-Q plot description, or the Shapiro-Wilk result: [paste]
TASK	1. Assess the distribution from the summaries (e.g. SD large relative to the mean for a positive-only variable suggests skew) and the plot. 2. Explain what the normality test does and does not tell me at my sample size. 3. Recommend how to summarise the variable (mean $\pm$ SD or median with IQR) and which test family to use. 4. Mention transformation (e.g. log) as an option, with its trade-offs.
RULES	Combine clinical knowledge of the variable with the data, because a normality test alone is underpowered in small samples and oversensitive in large ones.
OUTPUT	The assessment, the explanation, the recommendation, then the transformation note.

PRO TIP Hospital-stay length, CRP and most lab counts are usually skewed. Expect medians.

TRY IT WITH ICU length of stay: mean 6.2, SD 5.8, median 4, IQR 3–8, n = 90.

→ **NEXT #211** Choose and report the right non-parametric test

211 Choose and report the right non-parametric test

STANDARD

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

USE IT When your data is skewed or ordinal and parametric tests don't fit.

PERSONA	You are a biostatistician who explains non-parametric methods in plain language.
MY CONTEXT	Outcome and its type: [skewed continuous / ordinal] Groups and whether independent or paired: [brief] Sample sizes: [n per group]
TASK	1. Name the right test (Mann-Whitney U, Wilcoxon signed-rank, Kruskal-Wallis, Friedman) and why. 2. Explain what it actually tests in plain language. 3. Tell me which summary and effect measure to report (median, IQR, Hodges-Lehmann estimate or rank-biserial correlation). 4. Give the post-hoc approach if there are more than 2 groups. 5. Show the reporting sentence template.
RULES	Avoid saying the test compares medians unless the distributions have similar shapes, because that is a common reporting error.
OUTPUT	The test choice, explanation, reporting guidance, then the template.

PRO TIP Report the median and IQR per group alongside the p value. A p value alone tells readers nothing about the size of the difference.

TRY IT WITH Pain scores (0–10) compared among three anaesthesia techniques.

→ **NEXT #223** Effect sizes and confidence intervals for every result

212 Chi-square, Fisher's exact or something else?

STANDARD

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT / MBBS STUDENT

USE IT When comparing proportions or categorical variables between groups.

PERSONA	You are a biostatistics tutor who explains categorical-data tests precisely.
MY CONTEXT	Variables and categories: [brief] Table layout and cell counts (paste the cross-table if you have it): [PASTE TABLE] Are the groups independent or paired?: [brief]
TASK	1. Decide between chi-square, chi-square with continuity correction, Fisher's exact, chi-square for trend and McNemar, based on my table and design. 2. Explain the expected-count rule and check it on my table. 3. Name the effect measure to report (risk difference, relative risk or odds ratio with 95% CI). 4. Show the reporting sentence template.
RULES	Base the expected-count check on expected, not observed, counts, because that distinction is commonly confused.
OUTPUT	The decision, the expected-count check, the effect measure, then the template.

PRO TIP For ordered categories (mild, moderate, severe), a trend test uses the order and has more power.

TRY IT WITH A 2 × 2 table of smoking status by post-operative pneumonia with one cell containing 3.

→ **NEXT #223** Effect sizes and confidence intervals for every result

213 Before-after or matched data: paired analysis

STANDARD

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

USE IT When the same patients are measured twice, or cases are matched to controls.

PERSONA	You are a biostatistician who catches residents analysing paired data as independent groups.
MY CONTEXT	What is measured and when: [brief] Outcome type: [continuous / binary / ordinal] Sample size: [n pairs]
TASK	1. Confirm whether the data is truly paired, and why that matters. 2. Name the test: paired t-test, Wilcoxon signed-rank or McNemar. 3. Name the effect measure: mean difference with 95% CI, median difference, or discordant-pair odds ratio. 4. Explain the assumption checks (e.g. distribution of the differences). 5. Show the reporting sentence template.
RULES	Check the distribution of the differences, not of each time point separately, because that is the assumption the paired t-test actually makes.
OUTPUT	Numbered answers, then the template.

PRO TIP Paired designs need fewer patients. Say so when justifying your sample size.

TRY IT WITH HbA1c before and after 3 months of a lifestyle intervention in 60 patients.

→ **NEXT #214** Outcomes measured three or more times

214 Outcomes measured three or more times

STANDARD

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

USE IT When the same patients have an outcome measured at several time points.

PERSONA	You are a biostatistician who teaches repeated-measures analysis without unnecessary complexity.
MY CONTEXT	Outcome and time points: [brief] Groups (if comparing): [brief] Sample size and missing follow-ups: [brief] Software: [SPSS / R / Jamovi]
TASK	1. Explain why running separate t-tests at each time point is wrong. 2. Compare repeated-measures ANOVA, a linear mixed model and the Friedman test for my data. 3. Recommend one, with its assumptions and how it handles missing time points. 4. Give the software steps or syntax. 5. Show how to present the results (a line graph with error bars, and a table).
RULES	Recommend a mixed model if follow-ups are missing, and explain why, because repeated-measures ANOVA drops every patient with any missing time point.
OUTPUT	The explanation, the comparison, the recommendation, the steps, then the presentation guide.

PRO TIP A clear line graph of means over time with 95% CIs often explains your result better than any p value.

TRY IT WITH SOFA score at day 1, 3 and 7 in two sepsis treatment groups.

→ **NEXT #196** Pre-specified statistical analysis plan

215 Correlation or agreement? Methods comparison done right

STANDARD

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

USE IT When comparing two methods of measuring the same thing, or relating two variables.

PERSONA	You are a measurement statistician who explains why correlation is not agreement.
MY CONTEXT	The two measurements and what they measure: [brief] Purpose: [do they agree? / are they related?] Sample size: [n]
TASK	1. Tell me whether my question is about association (correlation) or agreement (interchangeability). 2. For association: choose Pearson or Spearman and explain why. 3. For agreement: explain the Bland-Altman analysis (bias, limits of agreement) and ICC, and when to use each. 4. Define in advance what limits of agreement would be clinically acceptable. 5. Show how to report the result.
RULES	Do not use correlation to claim agreement, because two methods can correlate perfectly and still disagree systematically.
OUTPUT	The diagnosis of my question, the method, then the reporting template.

PRO TIP Decide the clinically acceptable limits of agreement before looking at your data.

TRY IT WITH Comparing a pulse oximeter with arterial blood gas SaO₂ in ICU patients.

→ **NEXT #171** Inter-observer agreement and quality-control plan

216 ROC curve, AUC and the best cut-off

STANDARD

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

USE IT When your study evaluates a continuous marker or score for diagnosis or prognosis.

PERSONA	You are a diagnostic statistician who explains ROC analysis honestly, including its limits.
MY CONTEXT	Marker or score: [brief] Outcome (reference standard): [brief] Sample size and number with the outcome: [brief] Software: [SPSS / MedCalc / R / Jamovi]
TASK	1. Explain what the ROC curve and AUC represent in plain language. 2. Give the steps to run the analysis in my software. 3. Explain how to choose a cut-off (Youden index versus clinically chosen sensitivity), and report sensitivity, specificity, predictive values and likelihood ratios at it. 4. Warn about optimism when the cut-off is derived and tested in the same data. 5. Show the reporting template.
RULES	Report the AUC with its 95% CI, and call a data-derived cut-off exploratory unless it is validated, because optimistic cut-offs are a common reviewer criticism.
OUTPUT	Explanation, steps, cut-off guidance, then the template.

PRO TIP Compare your marker's AUC with an existing score's AUC in the same patients. It makes the result meaningful.

TRY IT WITH Serum lactate for predicting 28-day mortality in sepsis.

→ **NEXT #205** Sample size for a diagnostic accuracy study

217 Time-to-event analysis: Kaplan-Meier and Cox

STANDARD

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

USE IT When your outcome is time until an event (death, relapse, readmission) and follow-up varies.

PERSONA	You are a biostatistician who teaches survival analysis to clinicians clearly.
MY CONTEXT	Event and time origin: [brief] Follow-up duration and how censoring occurs: [brief] Groups and covariates: [brief] Number of events: [n] Software: [SPSS / R / Stata]
TASK	1. Define time zero, the event and censoring for my study. 2. Explain Kaplan-Meier curves and the log-rank test for group comparison. 3. Explain the Cox model, the hazard ratio, and the proportional-hazards check. 4. Say how many covariates my events can support. 5. Give software steps and a reporting template (median survival, HR with 95% CI, number at risk under the curve).
RULES	Warn me if informative censoring is likely (e.g. sicker patients lost to follow-up), because it biases survival estimates.
OUTPUT	Definitions, methods, the covariate limit, steps, then the template.

PRO TIP Always show the number at risk beneath your Kaplan-Meier plot. Reviewers expect it.

TRY IT WITH Time to first exacerbation after discharge in COPD, comparing two inhaler regimens.

→ **NEXT #199** Multivariable regression: plan, run and interpret

218 Handle missing data honestly

STANDARD

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

USE IT At the analysis-planning stage, and again when your dataset is complete.

PERSONA	You are a biostatistician who treats missing data as a bias problem, not a nuisance.
MY CONTEXT	Variables with missing values and the percentage missing: [PASTE TABLE] Why values are missing, as far as I know: [brief] Sample size: [N]
TASK	1. Classify the likely mechanism for each variable (MCAR, MAR or MNAR) in plain language. 2. Recommend an approach for each: complete-case analysis, multiple imputation or sensitivity analysis. 3. Explain what to report in the thesis (amount missing, pattern, handling). 4. Warn against single mean imputation and last observation carried forward. 5. Give software steps if imputation is needed.
RULES	Recommend reporting missingness transparently even when it is small, because silent exclusion is an examiner favourite.
OUTPUT	A variable table (variable, % missing, likely mechanism, approach), then the reporting guidance and steps.

PRO TIP Compare the characteristics of patients with and without missing outcomes. It is a simple, powerful sensitivity check.

TRY IT WITH 12% missing 30-day outcomes, 8% missing baseline albumin in a 200-patient cohort.

→ **NEXT #196** Pre-specified statistical analysis plan

219 Data-cleaning checklist before analysis

STANDARD

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

USE IT After data entry is complete, before any analysis.

PERSONA	You are a clinical data manager who has cleaned hundreds of thesis datasets.
MY CONTEXT	Dataset description (variables, types, N): [brief or PASTE CODEBOOK] Software: [Excel / SPSS / R]
TASK	1. Give a step-by-step checklist: duplicates, IDs, ranges, impossible values, date logic (discharge after admission), unit consistency, coding consistency and missing-value codes. 2. For each step, give the formula or software method. 3. Describe how to log each correction with its source document. 4. Describe how to lock the final dataset.
RULES	Tell me never to overwrite the raw data file, and to correct only against the source document, because undocumented corrections are indistinguishable from data manipulation.
OUTPUT	A numbered checklist with methods, then a correction-log template.

PRO TIP Frequency tables of every categorical variable catch half of all coding errors in five minutes.

TRY IT WITH 180-row Excel sheet of ICU patients with vitals, labs and outcomes.

→ **NEXT #198** Run my analysis on the real dataset, safely

220 Step-by-step SPSS instructions for my analysis

STANDARD

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

USE IT When you run the analysis yourself in SPSS.

PERSONA	You are an SPSS trainer who gives exact menu paths and explains which output to read.
MY CONTEXT	SPSS version: [e.g. 26 / 29] Analysis needed: [e.g. independent t-test of BMI by group, chi-square of mortality by group] Variable names in my file: [list]
TASK	1. Give the exact menu path for each analysis (Analyze → ...). 2. Say which boxes to tick and options to select (including effect sizes and CIs). 3. Explain which output tables to read and which rows or columns matter. 4. Give the equivalent syntax so I can save and re-run it.
RULES	Mark any menu item that may differ by version as [MAY DIFFER BY VERSION], because SPSS menus change between releases.
OUTPUT	Numbered steps per analysis, then the syntax block, then an output-reading guide.

PRO TIP Always paste the syntax into a syntax file. It is your record of exactly what you ran.

TRY IT WITH Mann-Whitney U test of hospital stay by diabetes status, SPSS 27.

→ **NEXT #200** Interpret my software output correctly

221 R code for my thesis analysis

STANDARD

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT / FACULTY

USE IT When you want free, reproducible analysis in R or RStudio.

PERSONA	You are an R programmer and biostatistician who writes clean, commented code for clinical researchers.
MY CONTEXT	Data file and format: [e.g. thesis.csv] Variable names and types: [list] Analyses needed: [list] My R experience: [none / basic / comfortable]
TASK	1. Write a complete, commented script: load packages, import data, check structure, Table 1, primary analysis, secondary analyses, and figures. 2. Use widely used packages (e.g. tidyverse, gtsummary) and explain why. 3. Include assumption checks. 4. Explain how to run it step by step for my experience level.
RULES	Use my exact variable names and don't recode anything without a comment explaining it, because hidden recoding changes results.
OUTPUT	One code block with section comments, then running instructions.

PRO TIP Run the script on a copy of the data first. Keep the raw file untouched.

TRY IT WITH CSV with age, sex, group, CRP, and a binary outcome "death28".

→ **NEXT #200** Interpret my software output correctly

222 Free point-and-click analysis in Jamovi

STANDARD

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT / MBBS STUDENT

USE IT When you don't have an SPSS licence and want a free, simple tool.

PERSONA	You are a statistics trainer who teaches Jamovi to clinicians.
MY CONTEXT	Analyses needed: [list] Variable names and types: [list]
TASK	1. Explain how to import my data and set the measure type (nominal, ordinal, continuous) for each variable. 2. Give the exact steps for each analysis, including effect sizes and assumption checks. 3. Explain how to export tables and plots for my thesis. 4. Tell me how to cite Jamovi and its R packages in my methods.
RULES	Mark any menu wording that may have changed as [CHECK CURRENT VERSION] , because the interface is updated regularly.
OUTPUT	Numbered steps per analysis, then the export and citation notes.

PRO TIP Setting the measure type correctly is the step most beginners skip. It decides which tests Jamovi offers.

TRY IT WITH Independent t-test, chi-square, and a Pearson correlation.

→ **NEXT #200** Interpret my software output correctly

223 Effect sizes and confidence intervals for every result

STANDARD

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

USE IT When planning or writing results, so you report more than p values.

PERSONA	You are a biostatistician who insists that every comparison is reported with an effect size and CI.
MY CONTEXT	My comparisons, each with outcome type and design: [list]
TASK	1. For each comparison, name the right effect measure (mean difference, risk difference, relative risk, odds ratio, hazard ratio, correlation, AUC). 2. Explain what each means clinically in one line. 3. Explain how to obtain its 95% CI in my software. 4. Show a reporting sentence for each.
RULES	Choose relative risk over odds ratio where the design allows and the outcome is common, and explain why, because odds ratios exaggerate effects when outcomes are common.
OUTPUT	A table (comparison, effect measure, clinical meaning, CI method), then the sentences.

PRO TIP A CI crossing the line of no effect but including large effects means "inconclusive", not "no effect".

TRY IT WITH Comparisons of mortality (binary), ICU stay (skewed) and SOFA change (continuous) between two groups.

→ **NEXT #226** Report p values and CIs the way journals want

224 Multiple comparisons: correct or not?

STANDARD

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

USE IT When you test many outcomes, subgroups or time points.

PERSONA	You are a biostatistician who explains multiplicity pragmatically.
MY CONTEXT	Number and type of tests planned: [brief] Primary versus secondary outcomes: [brief] Sample size: [N]
TASK	1. Explain the multiplicity problem using my number of tests. 2. Recommend whether to correct (Bonferroni, Holm, false discovery rate) or to pre-specify and label outcomes as secondary or exploratory. 3. Explain the trade-off (power versus false positives). 4. Write a methods sentence describing my approach.
RULES	Keep the primary outcome uncorrected if there is only one, and say so, because over-correcting a single primary outcome is a common error.
OUTPUT	The explanation, the recommendation, the trade-off, then the sentence.

PRO TIP Pre-specifying one primary outcome is the simplest multiplicity strategy there is.

TRY IT WITH 12 lab parameters compared between 3 groups.

→ **NEXT #225** Subgroup and sensitivity analyses that hold up

225 Subgroup and sensitivity analyses that hold up

STANDARD

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

USE IT When planning extra analyses beyond the primary one.

PERSONA	You are a trial statistician who is sceptical of subgroup claims.
MY CONTEXT	Primary analysis: [brief] Subgroups I am interested in: [list] Assumptions I am unsure about (missing data, outliers, definitions): [list]
TASK	1. Recommend which subgroups to pre-specify, with a biological rationale. 2. Explain why subgroup analyses should use an interaction test rather than separate p values. 3. Design 2–3 sensitivity analyses that test my key assumptions. 4. Explain how to label and report them.
RULES	Label all subgroup results exploratory unless the study was powered for them, because spurious subgroup findings are among the commonest misleading results in medicine.
OUTPUT	A subgroup table (subgroup, rationale, method), a sensitivity-analysis list, then reporting guidance.

PRO TIP A forest plot of subgroup estimates with CIs shows readers how consistent your result is.

TRY IT WITH Primary: effect of early antibiotics on mortality; subgroups: age, diabetes, source of infection.

→ **NEXT #196** Pre-specified statistical analysis plan

226 Report p values and CIs the way journals want

STANDARD

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT / MBBS STUDENT

USE IT When writing results, tables and abstracts.

PERSONA	You are a statistical editor for a medical journal who corrects p-value reporting daily.
MY CONTEXT	My results as I have written them: [PASTE RESULTS TEXT OR TABLE] Target: [thesis / journal name]
TASK	1. Check each reported result for: exact p values (not "NS" or " $p < 0.05$ " when an exact value is available), the correct number of decimals, CIs alongside estimates, and correct presentation of very small p values (e.g. $p < 0.001$). 2. Check wording: "significant" qualified as statistical, no "trend towards significance", and no " $p = 0.000$ ". 3. Return the corrected text.
RULES	Change formatting and wording only; never change a number, because the numbers must match the software output exactly.
OUTPUT	An issue table (original, problem, corrected), then the corrected text.

PRO TIP Report the effect estimate first and the p value last: "RR 0.72 (95% CI 0.55–0.94; $p = 0.02$)".

TRY IT WITH "Mortality was significantly lower ($p = 0.000$) and there was a trend towards shorter stay ($p = 0.07$)."

→ **NEXT #241** Write the Results chapter from my analysis output

227 Quick analysis inside Google Sheets with AI add-ons

STANDARD

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT / MBBS STUDENT

USE IT For quick descriptive statistics and simple tests while your data is still in Google Sheets or Excel.

PERSONA	You are a practical data coach who uses Gemini in Sheets, ChatGPT for Sheets and similar add-ons, and who knows their limits.
MY CONTEXT	Sheet structure (columns and types): [brief] Analyses I need: [e.g. descriptives by group, a chi-square test, a correlation] Tool: [Gemini in Sheets / ChatGPT for Sheets / Julius AI]
TASK	1. Write the exact prompts to type into the tool's side panel for each analysis, e.g. "Compare mean systolic BP between the intervention and control columns. Check normality first, tell me which test is appropriate, then run it and report the exact p value." 2. Explain how to check each result by hand or with a spreadsheet formula. 3. List the analyses I should move to proper statistical software instead.
RULES	Tell me to verify every key number in a second tool before it goes into my thesis, because AI add-ons can produce realistic but wrong statistics.
OUTPUT	A prompt list per analysis, verification methods, then the "move to software" list.

PRO TIP Use add-ons for exploration and data cleaning. Use proper software for the final thesis analysis.

TRY IT WITH A sheet of 120 diabetic patients with HbA1c, BMI, treatment group and complications.

→ **NEXT #198** Run my analysis on the real dataset, safely

228 Classify my variables correctly

QUICK

1.7 SAMPLE SIZE & STATISTICS · MBBS STUDENT / PG RESIDENT

Act as a biostatistics tutor. Classify each of my variables, [PASTE VARIABLE LIST], as continuous, discrete, binary, nominal or ordinal, name the right summary statistic for each (mean $\pm$ SD, median with IQR, n with %), and flag any variable I should not convert into categories.

TRY IT WITH Age, sex, BMI, NYHA class, smoking (yes/no), number of admissions, CRP.

229 Mean or median for this variable?

QUICK

1.7 SAMPLE SIZE & STATISTICS · MBBS STUDENT / PG RESIDENT

Act as a biostatistician reviewing a thesis Table 1. For my variable [VARIABLE] with these summaries, [MEAN, SD, MEDIAN, IQR, MIN, MAX, n], tell me whether to report the mean $\pm$ SD or the median with IQR, and explain your reasoning in 3 lines using the numbers.

TRY IT WITH Hospital stay: mean 7.1, SD 6.4, median 5, IQR 3–9, range 1–42, n = 110.

230 What does my p value actually mean?

QUICK

1.7 SAMPLE SIZE & STATISTICS · MBBS STUDENT / PG RESIDENT

Act as a biostatistics examiner. Explain what my result [e.g. $p = 0.03$ for the difference in X] does and does not mean, in plain language, in under 100 words. Include the 2 most common misinterpretations and why they are wrong.

TRY IT WITH $p = 0.04$ for a difference in mortality between two antibiotic regimens.

231 Interpret my confidence interval

QUICK

1.7 SAMPLE SIZE & STATISTICS · MBBS STUDENT / PG RESIDENT

Act as a clinical epidemiologist teaching residents. Interpret this confidence interval, [ESTIMATE AND 95% CI], for a clinician in under 100 words: what range of true effects is plausible, whether it excludes no effect, and whether it excludes a clinically important effect.

TRY IT WITH Mean difference in systolic BP -6 mmHg (95% CI -11 to -1).

232 Odds ratio or relative risk for my study?

QUICK

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

Act as an epidemiology tutor. For my design [DESIGN] and outcome frequency [e.g. 25%], tell me whether to report an odds ratio or a relative risk, why, and how much an odds ratio would overstate the relative risk here. Answer in under 120 words.

TRY IT WITH Prospective cohort, outcome occurring in 30% of patients.

233 Explain statistical power with my study

QUICK

1.7 SAMPLE SIZE & STATISTICS · MBBS STUDENT / PG RESIDENT

Act as a biostatistics teacher. Explain statistical power, alpha and beta using my study, [BRIEF STUDY AND SAMPLE SIZE], as the example. Include what happens if my study is underpowered, and how to phrase a non-significant result honestly. Keep it under 150 words.

TRY IT WITH Two-arm trial of 60 patients comparing two antiemetics.

234 My result isn't significant: now what?

QUICK

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

Act as a biostatistician and thesis guide. My primary result was [ESTIMATE, 95% CI, p]. Explain in plain words what this means, whether the study was likely underpowered, how to write it honestly in results and discussion, and which phrases to avoid (e.g. "trend towards").

TRY IT WITH RR 0.80 (95% CI 0.55–1.16), $p = 0.24$, in a 150-patient cohort.

235 Should I test baseline differences in my trial?

QUICK

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

Act as a trial statistician. Explain whether I should report p values for baseline characteristics in Table 1 of my randomised trial [BRIEF DESCRIPTION], what CONSORT recommends, and what to do instead if the groups look imbalanced. Keep it under 120 words.

TRY IT WITH A 2-arm RCT of 80 patients where mean age differs by 4 years between groups.

236 Interpret my kappa value

QUICK

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

Act as a measurement statistician. Interpret my inter-observer agreement result, [KAPPA OR WEIGHTED KAPPA WITH 95% CI], for my measurement [MEASUREMENT], explain what the value means in practice, mention the prevalence paradox, and give the sentence for my results section.

TRY IT WITH Kappa 0.62 (95% CI 0.48–0.76) for two readers grading chest X-ray consolidation.

237 Which assumptions does my test need?

QUICK

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

Act as a biostatistics tutor. For the test [TEST] on my variables [VARIABLES], list each assumption, exactly how to check it in [SOFTWARE], what violation looks like, and the fallback test if it fails. Present it as a short table.

TRY IT WITH One-way ANOVA of FEV1 across three smoking categories in SPSS.

238 Sanity-check an AI-generated sample size

QUICK

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

Act as a biostatistician. An AI tool gave me a sample size of [n] for [DESIGN AND OUTCOME] with these inputs, [INPUTS]. Recalculate it step by step, tell me whether the number is right, and give the exact OpenEpi or G*Power inputs to confirm it.

TRY IT WITH $n = 96$ for estimating a prevalence of 50% with $\pm 10\%$ precision.

239 Explain any statistics term for the viva

QUICK

1.7 SAMPLE SIZE & STATISTICS · MBBS STUDENT / PG RESIDENT

Act as a viva examiner who wants clear answers. Explain [STATISTICAL TERM] clearly in 3 levels for my thesis viva: one sentence, a 60-word explanation with an example from my thesis on [TOPIC], and the examiner's likely follow-up question with its answer.

TRY IT WITH Term: confounding. Thesis: vitamin D and TB treatment outcomes.

240 Build Table 1 from my variables

QUICK

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

Act as a clinical biostatistician. From my variable list with types, [PASTE LIST], build a Table 1 template by group: rows, the summary format for each (n (%), mean $\pm$ SD, median [IQR]), units in the row labels, and footnotes. Leave the values as placeholders.

TRY IT WITH Age, sex, BMI, diabetes, hypertension, SOFA, lactate, in survivors versus non-survivors.

1.8

Results, tables & figures

Results report what you found, in past tense, with no interpretation. Every number must match the output, and every table must earn its place.

25

prompts

3

Master

14

Standard

8

Quick

#241 Write the Results chapter from my analysis output

#242 Plan every table and figure in my results

#243 Consistency audit: text, tables, figures and abstract

#244 Describe Table 1 in two short paragraphs

#245 Report the primary outcome properly

#246 Report secondary outcomes without clutter

#247 Participant flow diagram and paragraph

#248 The best figure for this result

#249 Build the chart step by step in my software

#250 Format my table to journal standard

#251 Titles, legends and footnotes for my tables and figures

#252 Remove interpretation from my results

#253 Stop repeating my tables in the text

#254 Report a non-significant result honestly

#255 Report diagnostic accuracy results (STARD)

#256 Present regression results clearly

#257 Condense results for an abstract or poster

#258 One result, one sentence

#259 Check my percentages and denominators

#260 How many decimal places?

#261 Table or figure for this result?

#262 Check my chart for common errors

#263 Describe a figure in the text

#264 Label an exploratory analysis correctly

#265 Fix the tense in my results

Write the Results chapter from my analysis output

1.8 RESULTS, TABLES & FIGURES · PG RESIDENT

USE IT

After your analysis is complete and verified, before writing the discussion.

WHY

The results chapter is where transcription errors, interpretation creep and text that repeats the tables creep in. Writing it from your verified numbers in a fixed order (flow, baseline, primary, secondary) keeps it accurate, readable and defensible.

COPY THE PROMPT BELOW

PERSONA	You are a medical journal editor who writes and edits results sections. You report findings in past tense, in a logical order, with effect estimates and confidence intervals, and you never interpret in the results.
MY CONTEXT	Design and reporting guideline: [e.g. prospective cohort, STROBE] Participant flow numbers (screened, eligible, enrolled, lost, analysed): [numbers] Table 1 baseline characteristics (paste the table): [PASTE TABLE] Primary outcome result with effect estimate, 95% CI and p: [paste exactly] Secondary outcome results: [paste exactly] Tables and figures I have, with numbers and titles: [list] Target length: [e.g. 800–1,200 words for a thesis; 500–700 for a journal]
TASK	1. Open with participant flow, referring to the flow diagram. 2. Describe baseline characteristics briefly, highlighting only clinically important features and referring to Table 1. 3. Report the primary outcome with its effect estimate, 95% CI and exact p value. 4. Report secondary outcomes in the order of the objectives. 5. Report any pre-specified subgroup or sensitivity analyses, labelled as such. 6. Refer to each table and figure at the right point, and highlight rather than repeat their contents. 7. After the draft, list every number used with its source (table or output), for cross-checking.
RULES	Use only the numbers I pasted, reproduced exactly, because a single mistyped number undermines the credibility of the whole chapter. No interpretation, no comparison with literature and no causal words. Report non-significant results with their estimates and CIs, never as "no effect". Qualify "significant" as statistically significant.
OUTPUT	The results text with sub-headings by objective, then the number-source table.
TWEAK	End with the 3 results an examiner will look at first, and whether my text presents each one clearly.

How to get more out of it

- Write results before the discussion. It stops you writing results to fit your story.
- Check every number in the text against its table. Mismatches are the commonest results error.
- If the text simply repeats a table, cut the text to the key finding and refer to the table.
- → **NEXT #243** Consistency audit: text, tables, figures and abstract

Plan every table and figure in my results

1.8 RESULTS, TABLES & FIGURES · PG RESIDENT

USE IT

After analysis and before writing, to decide how your results will be displayed.

WHY

Examiners and reviewers read tables and figures first. A planned set, where each item answers an objective and nothing is duplicated, makes results easy to follow. A random pile of SPSS outputs and pie charts makes them hard work.

COPY THE PROMPT BELOW

PERSONA	You are a scientific data-visualisation editor for a medical journal. You match each result to the display that shows it most honestly and clearly, and you follow journal conventions for tables and figures.
MY CONTEXT	Objectives: [PASTE OBJECTIVES] Results for each objective (brief, with the variable types): [PASTE SUMMARY] Number of tables and figures allowed (thesis or journal): [e.g. no limit / 4 tables + 2 figures] Software I'll use for figures: [Excel / GraphPad / R / SPSS] My university's or journal's figure requirements, if any: [e.g. 300 dpi, grayscale, maximum width]
TASK	1. Propose the complete set: which results go in tables, which in figures, and which only in text. 2. For each table: title, rows, columns, summary format, footnotes and abbreviations. 3. For each figure: chart type and why (bar with error bars, box plot, forest plot, Kaplan-Meier, ROC, scatter with fit line, flow diagram), axes, units, labels and legend. 4. Flag displays to avoid for my data (pie charts, 3D charts, dynamite plots for skewed data, dual axes). 5. Give step-by-step instructions to create each figure in my software. 6. Number everything in order of mention in the text. 7. Say which displays could go in an appendix instead of the main chapter.
RULES	Choose displays that show the distribution or uncertainty (CIs, IQRs) rather than hiding it, because misleading displays are a frequent examiner criticism. Use placeholders instead of values.
OUTPUT	A display plan table (number, type, objective served, content), then table templates, figure specifications, the "avoid" list and the software steps.
TWEAK	End with the single figure that best tells the story of my thesis, and why.

How to get more out of it

- › One figure that shows your primary result clearly is worth more than ten decorative charts.
- › Keep fonts, colours and decimal places consistent across all tables and figures.
- › Make every table understandable on its own, with a full title and footnotes.
- › → **NEXT #249** Build the chart step by step in my software

Consistency audit: text, tables, figures and abstract

1.8 RESULTS, TABLES & FIGURES · PG RESIDENT / FACULTY

USE IT

When the thesis or manuscript is drafted, before submission.

WHY

Inconsistent numbers between the abstract, text, tables and figures are among the most common reasons reviewers lose trust in a paper, and examiners spot them fast. A systematic audit catches them all at once.

COPY THE PROMPT BELOW

PERSONA	You are a meticulous statistical reviewer for a medical journal who cross-checks every number in a manuscript for consistency and plausibility.
MY CONTEXT	Abstract: [PASTE ABSTRACT] Results text: [PASTE RESULTS TEXT] Tables (as text): [PASTE TABLES] Figure legends and any numbers shown in figures: [PASTE LEGENDS] Total N and group sizes: [numbers] Missing data per key variable: [brief] Version of the document and date: [e.g. draft 3, 12 May]
TASK	1. Extract every number from each section and match it across sections. 2. Flag any number that differs between sections. 3. Check internal arithmetic: percentages match their numerators and denominators, subgroup counts sum to totals, means lie within their ranges, and CIs contain their point estimates. 4. Check that p values are consistent with the CIs (a 95% CI excluding the null should go with $p < 0.05$). 5. Check that the denominators change sensibly where there is missing data, and that the missing data is reported somewhere. 6. Check that every table and figure is cited in the text, in order. 7. Check that the abstract's conclusion matches the primary result as reported in the results text.
RULES	Report discrepancies without deciding which value is right, because only the original source output can settle that. Do not alter any numbers yourself.
OUTPUT	A discrepancy table (number, location A, location B, issue), then the arithmetic and consistency check results, then the citation-order check and the abstract check.
TWEAK	Finish with the 3 discrepancies most likely to be noticed by an examiner or reviewer, to fix first.

How to get more out of it

- › Fix discrepancies from the original output, never by adjusting the text to match a table.
- › Re-run this audit after every major revision. New inconsistencies appear with each edit.
- › Ask a colleague to check 10 random numbers against the raw output as an independent check.
- › → NEXT #241 Write the Results chapter from my analysis output

244 Describe Table 1 in two short paragraphs

STANDARD

1.8 RESULTS, TABLES & FIGURES · PG RESIDENT

USE IT When writing the baseline characteristics paragraph of your results.

PERSONA	You are a journal editor who keeps baseline descriptions short and informative.
MY CONTEXT	Table 1 (paste as text): [PASTE TABLE] Design (randomised or observational): [brief] Groups: [brief] Target: [thesis / journal]
TASK	1. Write 2 short paragraphs describing the study population: key demographics, key clinical features, and notable differences between groups. 2. Highlight only clinically important features, not every row. 3. For an RCT, describe balance without p values; for observational studies, note the differences that may confound results.
RULES	Reproduce numbers exactly and refer to Table 1 rather than repeating it, because text that repeats a table wastes the reader's time.
OUTPUT	The two paragraphs.

PRO TIP Readers use Table 1 to judge whether your results apply to their patients. Make the setting and severity clear.

TRY IT WITH Table 1 of a 200-patient sepsis cohort comparing survivors and non-survivors.

→ **NEXT #245** Report the primary outcome properly

245 Report the primary outcome properly

STANDARD

1.8 RESULTS, TABLES & FIGURES · PG RESIDENT

USE IT For the most important paragraph of your results.

PERSONA	You are a medical statistician who writes primary-outcome paragraphs that are accurate and complete.
MY CONTEXT	Primary objective: [paste] Result: event counts or group summaries, the effect estimate with 95% CI, and exact p value: [paste exactly] Adjusted result, if any: [paste]
TASK	1. Write the primary-outcome paragraph: summaries in each group, the effect estimate with 95% CI and p value, and the adjusted estimate if applicable. 2. Refer to the relevant table or figure. 3. Write a one-sentence version for the abstract.
RULES	Report the estimate and CI before the p value, and use past tense without interpretation, because readers need the size and precision of the effect, not just significance.
OUTPUT	The paragraph, then the abstract sentence.

PRO TIP Give absolute numbers as well as relative measures, e.g. "18/100 vs 30/100 (RR 0.60, 95% CI ...)".

TRY IT WITH 28-day mortality 22% vs 31%, RR 0.71 (95% CI 0.50–0.99), p = 0.04.

→ **NEXT #246** Report secondary outcomes without clutter

246 Report secondary outcomes without clutter

STANDARD

1.8 RESULTS, TABLES & FIGURES · PG RESIDENT

USE IT When you have several secondary outcomes to report.

PERSONA	You are an editor who reports multiple outcomes clearly and briefly.
MY CONTEXT	Secondary objectives: [list] Results for each (estimates, CIs, p values): [PASTE RESULTS] Table where they are shown: [e.g. Table 3] Which analyses were pre-specified and which were added later: [brief]
TASK	1. Report each secondary outcome in the order of the objectives, in 1–2 sentences each. 2. Group related outcomes where sensible. 3. Refer to the table rather than repeating every number. 4. Label exploratory analyses clearly, and keep them after the pre-specified ones.
RULES	Report non-significant secondary outcomes as fully as significant ones, because selective reporting is a form of bias that examiners recognise.
OUTPUT	The paragraphs.

PRO TIP If a secondary outcome surprised you, report it plainly here. The interpretation belongs in the discussion.

TRY IT WITH Secondary outcomes: ICU stay, ventilator days, AKI incidence, and 90-day mortality.

→ **NEXT #241** Write the Results chapter from my analysis output

247 Participant flow diagram and paragraph

STANDARD

1.8 RESULTS, TABLES & FIGURES · PG RESIDENT

USE IT For the first figure and paragraph of your results.

PERSONA	You are a reporting-guideline expert who builds CONSORT and STROBE flow diagrams.
MY CONTEXT	Numbers: assessed for eligibility, excluded (with reasons and counts), enrolled, allocated (if applicable), lost to follow-up (with reasons), analysed: [numbers] Design: [RCT / cohort / cross-sectional / diagnostic]
TASK	1. Describe the flow diagram box by box with my numbers, following CONSORT or STROBE conventions. 2. Check that the numbers add up at every step. 3. Write the participant-flow paragraph for the results.
RULES	Flag any step where the numbers don't add up instead of adjusting them, because flow-diagram arithmetic errors are easy to spot.
OUTPUT	The box list with arrows, the arithmetic check, then the paragraph.

PRO TIP Draw the diagram in PowerPoint or draw.io from the box list. It's quicker than fixing an AI-drawn one.

TRY IT WITH 312 screened, 94 excluded (not meeting criteria 61, declined 33), 218 enrolled, 12 lost, 206 analysed.

→ **NEXT #242** Plan every table and figure in my results

248 The best figure for this result

STANDARD

1.8 RESULTS, TABLES & FIGURES · PG RESIDENT / MBBS STUDENT

USE IT When you are unsure how to display a particular result.

PERSONA	You are a data-visualisation specialist for medical research.
MY CONTEXT	The result to display: [brief] Variable types and groups: [brief] Where it will appear: [thesis / journal / poster / slide] Sample size per group: [n]
TASK	1. Recommend the best chart type and explain why. 2. Name 2 alternatives and when they would be better. 3. Name the chart types to avoid for this result. 4. Specify the axes, labels, error bars or intervals, and legend.
RULES	Prefer displays that show individual data or uncertainty for small samples, because bar charts of means hide the distribution.
OUTPUT	The recommendation, alternatives, avoid list, then the specification.

PRO TIP For skewed data in small groups, a box plot with the individual points overlaid is usually best.

TRY IT WITH CRP levels (skewed) in three groups of 25 patients each.

→ **NEXT #249** Build the chart step by step in my software

249 Build the chart step by step in my software

STANDARD

1.8 RESULTS, TABLES & FIGURES · PG RESIDENT

USE IT When you know which figure you want and need to make it.

PERSONA	You are a practical graphics trainer who creates publication-quality charts in common software.
MY CONTEXT	Chart type and data: [brief] Software: [Excel / Google Sheets / GraphPad / R ggplot2 / SPSS] Output target: [thesis print / journal / slide]
TASK	1. Give step-by-step instructions (or code) to create the chart. 2. Specify formatting: font sizes, axis titles with units, removing chart junk, colour-blind-safe colours, and error bars or CIs. 3. Explain how to export at the right resolution (e.g. 300 dpi for print). 4. Write the figure title and legend.
RULES	Keep the axes honest: start bar charts at zero, and use the same scale when figures are compared, because distorted axes mislead readers.
OUTPUT	Numbered steps or a code block, formatting rules, export settings, then the legend.

PRO TIP Make the chart in grayscale first. If it still works, colour will only improve it.

TRY IT WITH Kaplan-Meier curves for two groups with the number at risk, in R.

→ **NEXT #251** Titles, legends and footnotes for my tables and figures

250 Format my table to journal standard

STANDARD

1.8 RESULTS, TABLES & FIGURES · PG RESIDENT

USE IT When your tables are copied from software output and look messy.

PERSONA	You are a journal production editor who formats tables to house style.
MY CONTEXT	My table (as text): [PASTE TABLE] Target: [thesis / journal name] What the table is meant to show: [one line]
TASK	1. Restructure the table: a clear title, logical row order, grouped variables, units in the row labels, and consistent decimals. 2. Remove software clutter (test statistics columns readers don't need, redundant rows). 3. Add footnotes: abbreviations, the tests used, and symbols. 4. Return the formatted table.
RULES	Keep every value exactly as given; only restructure and format, because tables must still match the software output.
OUTPUT	The formatted table, then a list of the changes made.

PRO TIP Use horizontal lines only (top, below the header, bottom). Journals rarely accept vertical lines.

TRY IT WITH An SPSS crosstab output pasted with all its default columns.

→ **NEXT #251** Titles, legends and footnotes for my tables and figures

251 Titles, legends and footnotes for my tables and figures

STANDARD

1.8 RESULTS, TABLES & FIGURES · PG RESIDENT

USE IT When finalising tables and figures.

PERSONA	You are a scientific editor who writes self-explanatory table titles and figure legends.
MY CONTEXT	Table or figure content and purpose: [brief or PASTE] Abbreviations used: [list] Tests used: [list] Target: [thesis / journal / poster]
TASK	1. Write a title that states what, who, where and when. 2. Write a legend or footnotes: abbreviations, statistical tests, what the error bars represent, and sample sizes. 3. Check that the item can be understood without reading the text.
RULES	State exactly what error bars or ranges represent (SD, SE or 95% CI), because unlabelled error bars are uninterpretable.
OUTPUT	The title, then the legend or footnotes.

PRO TIP A good title tells the reader the finding's context. A great one lets them skip the text.

TRY IT WITH A bar chart of mean ventilator days by group, with error bars.

→ **NEXT #242** Plan every table and figure in my results

252 Remove interpretation from my results

STANDARD

1.8 RESULTS, TABLES & FIGURES · PG RESIDENT

USE IT When your guide says your results read like a discussion.

PERSONA	You are a strict methods editor who keeps results and discussion separate.
MY CONTEXT	My results text: [PASTE TEXT] My objectives, so you can check the order: [PASTE OBJECTIVES] Target: [thesis / journal]
TASK	1. Mark every sentence that interprets, explains, compares with literature or speculates. 2. Suggest moving each to the discussion or deleting it. 3. Return the cleaned results text in neutral past tense, ordered by objective. 4. Collect the removed interpretive sentences in a separate list for my discussion.
RULES	Keep all numbers and factual findings unchanged, because the cleaning must remove only interpretation, not results.
OUTPUT	A table of flagged sentences (sentence, problem, move or delete), then the cleaned text.

PRO TIP Words such as "interestingly", "surprisingly", "suggesting", "due to" and "because" usually signal interpretation.

TRY IT WITH A results paragraph saying "mortality was lower, probably due to earlier antibiotics".

→ **NEXT #241** Write the Results chapter from my analysis output

253 Stop repeating my tables in the text

STANDARD

1.8 RESULTS, TABLES & FIGURES · PG RESIDENT

USE IT When your results text restates every row of the tables.

PERSONA	You are a journal editor who cuts results sections to their essential points.
MY CONTEXT	Results text: [PASTE TEXT] Tables it refers to: [PASTE TABLES] Target length: [words]
TASK	1. Identify the text that duplicates table contents. 2. Rewrite the text to highlight only the key findings (the primary result, important differences, surprises) with a pointer to the table. 3. Give the word count before and after. 4. List any table row that should be moved to an appendix.
RULES	Keep the primary outcome's numbers in the text even if they are in a table, because readers expect the key result stated in full.
OUTPUT	The rewritten text, then the word counts.

PRO TIP A useful rule: the text tells the story, the tables hold the evidence.

TRY IT WITH 600 words describing every row of a 20-row Table 2.

→ **NEXT #241** Write the Results chapter from my analysis output

254 Report a non-significant result honestly

STANDARD

1.8 RESULTS, TABLES & FIGURES · PG RESIDENT

USE IT When your primary or key result is not statistically significant.

PERSONA	You are a research-integrity-minded statistician who reports null results clearly and without spin.
MY CONTEXT	The result (estimate, 95% CI, p): [paste] Sample size and whether the study was powered for this outcome: [brief]
TASK	1. Write the result sentence neutrally with the estimate and CI. 2. Explain what the CI does and does not rule out. 3. List spin phrases to avoid (e.g. "trend towards", "marginally significant", "no effect"). 4. Suggest how the discussion should handle it (and point to that section).
RULES	Do not suggest any re-analysis to reach significance, because that is p-hacking.
OUTPUT	The result sentence, the CI explanation, the avoid list, then the discussion pointer.

PRO TIP A well-reported null result from a well-conducted study is publishable. Spin is what gets it rejected.

TRY IT WITH OR 1.35 (95% CI 0.88–2.07), p = 0.17 for diabetes and wound dehiscence.

→ **NEXT #266** Write the discussion: ten paragraphs, no overclaiming

255 Report diagnostic accuracy results (STARD)

STANDARD

1.8 RESULTS, TABLES & FIGURES · PG RESIDENT

USE IT When your thesis evaluates a test.

PERSONA	You are a diagnostic-research editor who applies STARD reporting.
MY CONTEXT	2 × 2 table (true positives, false positives, false negatives, true negatives): [numbers] Indeterminate results and how they were handled: [brief] Pre-specified threshold: [value]
TASK	1. Calculate sensitivity, specificity, PPV, NPV, LR+ and LR- with 95% CIs, showing the formulas. 2. Write the results paragraph in STARD style. 3. Present the 2 × 2 table cleanly. 4. Remind me which measures depend on prevalence.
RULES	Show the formulas and arithmetic so I can check them, and recommend confirming the CIs in MedCalc or OpenEpi, because hand-calculated CIs are error-prone.
OUTPUT	The calculations, the table, then the paragraph.

PRO TIP Report the prevalence in your sample. PPV and NPV mean nothing without it.

TRY IT WITH TP 72, FP 14, FN 8, TN 106 for lung ultrasound versus CT.

→ **NEXT #216** ROC curve, AUC and the best cut-off

256 Present regression results clearly

STANDARD

1.8 RESULTS, TABLES & FIGURES · PG RESIDENT

USE IT When reporting a multivariable model in your results.

PERSONA	You are a biostatistical editor who formats regression tables for clinical journals.
MY CONTEXT	Model type and outcome: [brief] Output with crude and adjusted estimates and CIs for each variable: [PASTE OUTPUT] Number of observations and events: [numbers]
TASK	1. Build a table: variable, category or reference, crude estimate (95% CI), adjusted estimate (95% CI), p value. 2. Add footnotes: model type, variables adjusted for, n and events, and fit statistics if relevant. 3. Write a results paragraph describing the key adjusted associations.
RULES	Mark reference categories clearly and report estimates exactly as in the output, because reversed or missing reference categories flip the meaning.
OUTPUT	The table, the footnotes, then the paragraph.

PRO TIP Put the exposure of interest in the first row. Readers look for it first.

TRY IT WITH Logistic regression for 30-day mortality with age, lactate, SOFA and diabetes.

→ **NEXT #200** Interpret my software output correctly

257 Condense results for an abstract or poster

STANDARD

1.8 RESULTS, TABLES & FIGURES · PG RESIDENT

USE IT When you need the key results in 80–150 words.

PERSONA	You are an abstract editor for a medical conference who picks the numbers that matter.
MY CONTEXT	Full results text or key numbers: [PASTE HERE] Word limit: [e.g. 100 words] Format: [structured abstract results / poster panel / slide]
TASK	1. Select the key results: sample, primary outcome with its estimate and CI, and one or two important secondary results. 2. Write them within the limit. 3. List what was left out.
RULES	Keep numbers exactly as in the source, and include the primary outcome even if it is non-significant, because selective abstract reporting is a form of spin.
OUTPUT	The condensed results with word count, then the left-out list.

PRO TIP One primary result stated clearly beats five results crammed in.

TRY IT WITH A 900-word results chapter condensed to 120 words for a conference abstract.

→ **NEXT #320** Structured abstract within the word limit

258 One result, one sentence

QUICK

1.8 RESULTS, TABLES & FIGURES · MBBS STUDENT / PG RESIDENT

Act as a medical journal editor. Turn this result, [PASTE RESULT WITH NUMBERS], into one correctly reported sentence in past tense for my thesis results: group values, effect estimate with 95% CI and exact p value, no interpretation and no causal words.

TRY IT WITH Mean LOS 6.2 vs 8.1 days, difference -1.9 (95% CI -3.4 to -0.4), p = 0.01.

259 Check my percentages and denominators

QUICK

1.8 RESULTS, TABLES & FIGURES · PG RESIDENT

Act as a numbers checker for a thesis. Check these counts and percentages, [PASTE TABLE OR TEXT], for arithmetic errors, wrong denominators (especially with missing data) and inconsistent rounding. List each error with its correct value and the denominator used.

TRY IT WITH "Diabetes 42 (35%), hypertension 51 (41%)" with a total of 120 patients.

260 How many decimal places?

QUICK

1.8 RESULTS, TABLES & FIGURES · PG RESIDENT / MBBS STUDENT

Act as a statistical editor. Tell me how many decimal places to use for means, SDs, percentages, odds ratios, CIs and p values in my [thesis / journal] results, with the reasoning, and correct each of these examples: [PASTE EXAMPLES].

TRY IT WITH "Mean age 45.2367 years, 33.33333% male, OR 2.5, p = 0.0001."

261 Table or figure for this result?

QUICK

1.8 RESULTS, TABLES & FIGURES · PG RESIDENT

Act as a data-presentation editor. For this result, [DESCRIBE RESULT], tell me whether a table, a figure or a single sentence of text shows it best, why, and exactly what the recommended display should contain. Answer in under 100 words.

TRY IT WITH Prevalence of 6 comorbidities in 150 patients.

262 Check my chart for common errors

QUICK

1.8 RESULTS, TABLES & FIGURES · PG RESIDENT

Act as a figure reviewer for a medical journal. From this description of my chart, [DESCRIBE CHART OR PASTE SETTINGS], check axis labels and units, the axis starting point, error-bar definition, legend, colour use and chart junk. List each problem with its fix.

TRY IT WITH A 3D pie chart of the distribution of blood groups.

263 Describe a figure in the text

QUICK

1.8 RESULTS, TABLES & FIGURES · PG RESIDENT

Act as a results editor. Write 1–2 sentences for my results text that point the reader to [FIGURE NUMBER] and state the main pattern it shows, using these numbers: [KEY NUMBERS]. No interpretation, past tense, under 50 words, and refer to the figure by number.

TRY IT WITH Figure 2 shows Kaplan–Meier curves; median time to event 14 vs 22 days.

264 Label an exploratory analysis correctly

QUICK

1.8 RESULTS, TABLES & FIGURES · PG RESIDENT

Act as a trial statistician. Write the sentence that introduces my exploratory or post-hoc analysis, [DESCRIBE ANALYSIS], in the results, making clear that it was not pre-specified and is only hypothesis-generating, and give the matching caution sentence for the discussion.

TRY IT WITH A subgroup analysis by diabetes status that was added after seeing the data.

265 Fix the tense in my results

QUICK

1.8 RESULTS, TABLES & FIGURES · PG RESIDENT

Act as a copy-editor for medical theses. Correct the tense in this results text, [PASTE TEXT], using past tense for findings and present tense only when referring to tables and figures ("Table 2 shows"). Return the corrected text and list the changes.

TRY IT WITH "The mortality is lower in group A and Figure 3 showed the trend."

1.9

Discussion, conclusion & limitations

The discussion is where good data gets ruined by overclaiming, and it is the only section where your scientific judgement is visible. Structure it deliberately.

30

prompts

4

Master

17

Standard

9

Quick

#266 Write the discussion: ten paragraphs, no overclaiming

#267 Limitations that are honest, ranked and specific

#268 Hostile-reviewer attack on my discussion

#269 Conclusion, recommendations and summary chapter

#270 The first paragraph of the discussion

#271 Compare one finding with the literature

#272 Why my result differs from previous studies

#273 The mechanism paragraph, clearly labelled as speculation

#274 Clinical implications without overreach

#275 Strengths without inflation

#276 Future research that someone could actually do

#277 A conclusion that doesn't exceed the data

#278 Scan my discussion for overclaiming

#279 Who do my results apply to?

#280 Discuss a surprising result

#281 Discuss a non-significant primary result

#282 Check the structure of my discussion

#283 The "Summary" chapter of an Indian thesis

#284 What my findings mean for Indian practice

#285 Cut my discussion to the word limit

#286 Practical recommendations from my thesis

#287 Rewrite "further studies are needed"

#288 Calibrate my hedging

#289 One take-home message

#290 Qualify every "significant"

#291 Which way does this bias push my result?

#292 Swap causal words for correct ones

#293 One comparison sentence with a study

#294 Is this really a strength?

#295 Do my conclusions match my objectives?

Write the discussion: ten paragraphs, no overclaiming

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT

USE IT

After results are final. This is the section examiners and reviewers read most closely.

WHY

The discussion is where good data gets ruined by overclaiming, and it is the only section where your scientific judgement is visible. A fixed structure (answer, compare, explain, implications, strengths, limitations, future, conclusion) keeps it honest and complete.

COPY THE PROMPT BELOW

PERSONA	You are a journal editor and senior academic who has written and revised hundreds of discussion sections and rejected many more. You are ruthless about overinterpretation, unsupported causal language, and limitation sections that hide the real weaknesses.
MY CONTEXT	Design: [state] · Sample size: [state] Primary objective: [state] Primary result with effect estimate and CI: [paste exactly as computed] Secondary results: [paste, with numbers] Findings that surprised me or contradicted my hypothesis: [state] Findings that were not significant: [state] Known limitations: [list] Setting: [Indian tertiary / rural / private, etc.] · Target: [thesis / journal name] I have uploaded the [N] most relevant published papers. Use only these, plus anything you retrieve through live search with a verifiable PMID or DOI.
TASK	Draft a discussion of about [800–1,200] words: 1. Paragraph 1: answer the research question in 1–3 sentences with the principal finding and its numbers. No literature. 2. Paragraphs 2–4: for each key comparison, state what the prior study found, whether mine agrees or differs, and the likely reason (population, setting, definitions, measurement, sample size, follow-up). End every comparison with an interpretive sentence. 3. Paragraph 5: the most plausible mechanism, marked clearly where speculative. 4. Paragraph 6: clinical implications: who could act on this and how; say plainly if it is not yet practice-changing. 5. Paragraph 7: strengths, without inflation. 6. Paragraph 8: limitations in order of threat to the conclusion, each with the likely direction of bias. 7. Paragraph 9: specific, doable future research, not "further studies are needed". 8. Paragraph 10: a 2–3 sentence conclusion that does not exceed the data.
RULES	Use associational language unless my design is randomised; "caused", "led to", "resulted in" and "prevented" are banned otherwise, because causal overreach is the most common reviewer criticism. Qualify every "significant" as statistical or clinical. Never call a non-significant result a "trend". Where a comparison needs a source I have not supplied, write [CITATION NEEDED — FIND SOURCE]. Introduce no result that is not in my results section.
OUTPUT	The discussion in paragraphs with no sub-headings unless my target requires them, within the word range.
TWEAK	Then list separately the three sentences a hostile reviewer would attack first, and how to rewrite each one defensively.

How to get more out of it

- › Give the model your actual numbers. A discussion written around vague findings reads exactly as vague as it was written.
- › The hostile-reviewer list is worth more than the draft. Act on it before submission.
- › If the draft sounds more confident than you feel about your own data, the draft is wrong, not your instinct.
- › → **NEXT #267** Limitations that are honest, ranked and specific

Limitations that are honest, ranked and specific

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT

USE IT

When writing the limitations paragraph, and before your viva.

WHY

Examiners judge your scientific maturity by your limitations. A generic list ("small sample, single centre") signals you didn't think. Limitations ranked by threat, each with the direction of bias and what you did to reduce it, signal that you did.

COPY THE PROMPT BELOW

PERSONA	You are a clinical epidemiologist and journal reviewer who evaluates limitation sections for honesty and insight. You expect each limitation to state how it could have changed the result.
MY CONTEXT	Design and setting: [paste] How participants were selected and how many declined or were lost: [brief with numbers] How the exposure and outcome were measured, and by whom: [brief] Confounders measured and not measured: [lists] Sample size, and whether the study was powered for each reported outcome: [brief] Primary result: [estimate with CI] Limitations I have thought of: [list]
TASK	1. Identify all important limitations under selection, information (measurement), confounding, sample size or precision, and generalisability. 2. Rank them by how much they threaten my main conclusion. 3. For each, state the likely direction of bias (towards or away from the null, or unpredictable) and whether it could explain my result entirely. 4. For each, state what I did in the design or analysis to reduce it, if anything. 5. Write the limitations paragraph (180–250 words) in ranked order, honest but not self-defeating. 6. Separately, list 2 limitations I should acknowledge in the viva even if they are not in the thesis text.
RULES	Be specific to my study, because generic limitations suggest you haven't understood your own design. Don't invent limitations my design doesn't have, and don't hide real ones behind softer wording. Keep numbers exactly as I gave them.
OUTPUT	A ranked limitation table (limitation, type, direction of bias, mitigation, impact), then the paragraph, then the viva list.
TWEAK	End with a one-line statement of how confident a reader should be in my primary conclusion, given these limitations.

How to get more out of it

- Start with the limitation that matters most, not the most comfortable one.
- "Single-centre" is only a limitation if you explain why your centre might differ from others.
- Pair each limitation with what the next study should do differently. It leads naturally into future research.
- **NEXT #268** Hostile-reviewer attack on my discussion

Hostile-reviewer attack on my discussion

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT / FACULTY

USE IT

When your discussion is drafted, before your guide or a journal sees it.

WHY

The kindest thing you can do for your discussion is let someone attack it first. A structured hostile review finds overclaims, missing comparisons and weak logic, and it hands you defensive rewrites you can use on paper and at the viva.

COPY THE PROMPT BELOW

PERSONA	You are Reviewer 2: a senior, sceptical peer reviewer for a high-impact clinical journal. You are fair but relentless, and you target overinterpretation, selective comparisons and conclusions that outrun the data.
MY CONTEXT	Design, sample size and primary result: [brief with numbers] Target: [thesis / journal name and its readership] My results section: [PASTE RESULTS] My discussion: [PASTE DISCUSSION]
TASK	1. Write a reviewer report on the discussion with major comments (threats to validity or interpretation) and minor comments (wording, clarity, missing references), each quoting the text it refers to. 2. Identify every sentence that claims more than the results show, and every causal claim unsupported by the design. 3. Identify comparisons with literature that seem selective, where only agreeing studies are cited, and say what kind of study is missing. 4. Identify results reported in the discussion that do not appear in the results. 5. Identify important limitations the discussion leaves out or understates. 6. For each major comment, write a defensive rewrite of the relevant text and a one-line response I could use in a reply to reviewers or at the viva.
RULES	Quote the exact text for every criticism, because unquoted criticism cannot be acted on. Rewrites must not add new data or references; use [CITE] slots where a reference is needed. Keep my numbers unchanged.
OUTPUT	The reviewer report (major, then minor), then a table (quoted sentence, problem, rewrite, one-line response).
TWEAK	End with an overall verdict (accept, minor revision, major revision) with the single change that would most improve the discussion.

How to get more out of it

- Run this before your guide reads the draft. Their time is better spent on judgement than on catching overclaims.
- Keep the responses. They are ready-made answers for viva questions about your interpretation.
- Run it again after revising. The major comments should turn into minor ones.
- → **NEXT #278** Scan my discussion for overclaiming

Conclusion, recommendations and summary chapter

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT

USE IT

For the final chapters of your thesis: summary, conclusion and recommendations.

WHY

Many Indian theses end with separate summary, conclusion and recommendations chapters, and examiners read them first when time is short. They must match the objectives exactly and claim no more than the data shows.

COPY THE PROMPT BELOW

PERSONA	You are a senior examiner for MD/MS theses who reads the summary and conclusion first to judge whether the student understood their study. You insist that every conclusion maps to an objective and every recommendation is proportionate to the evidence.
MY CONTEXT	Thesis title: [paste] Objectives (numbered): [PASTE OBJECTIVES] Key results for each objective (with numbers): [PASTE RESULTS] Main limitations: [list] My university's required end chapters and word limits: [e.g. Summary 500 words, Conclusion, Recommendations] Design and setting: [brief]
TASK	1. Write the summary chapter: background (2 lines), aims, methods (design, setting, sample, key measurements), key results by objective with numbers, and the conclusion, within the word limit. 2. Write the conclusion: one numbered statement per objective, each answering it directly and in associational terms where appropriate. 3. Write recommendations in 3 groups: for clinical practice (only if justified), for the department or health system, and for future research, each specific and feasible. 4. Build a check table showing each objective, its conclusion statement and the result that supports it. 5. Flag any result that is important but not captured by any conclusion.
RULES	Every conclusion must answer an objective using a result I gave, because conclusions without matching results are a classic examiner criticism. Recommendations for practice must be proportionate: a single-centre observational study can suggest or support, not establish. Keep numbers exactly as supplied.
OUTPUT	The summary, the numbered conclusions, the grouped recommendations, then the check table.
TWEAK	End with the one conclusion statement most likely to be challenged at the viva, and a safer wording for it.

How to get more out of it

- Read your conclusion and your objectives side by side. They should mirror each other.
- Future-research recommendations should name the design, population and outcome.
- Keep the summary free of any number that doesn't appear in your results chapter.
- → **NEXT #283** The "Summary" chapter of an Indian thesis

270 The first paragraph of the discussion

STANDARD

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT

USE IT To open your discussion.

PERSONA	You are a journal editor who wants the discussion to open with the answer, not with background.
MY CONTEXT	Primary objective: [paste] Primary result with numbers: [paste] Most important secondary result: [paste] Design: [brief]
TASK	1. Write the opening paragraph (60–100 words) stating the answer to the research question with the key numbers. 2. Add one sentence on the most important secondary finding. 3. Give an alternative version for a journal article of 50 words or fewer.
RULES	No literature, no background and no interpretation in this paragraph, and use associational language for observational designs, because the first sentence is read as your main claim.
OUTPUT	The two versions.

PRO TIP If you can't state your main finding in one sentence, your results section needs work first.

TRY IT WITH Primary: in-hospital mortality 18% vs 27% between early and late mobilisation groups (RR 0.67, 95% CI 0.45–0.99).

→ **NEXT #271** Compare one finding with the literature

271 Compare one finding with the literature

STANDARD

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT

USE IT For each key comparison paragraph in your discussion.

PERSONA	You are a clinical academic who writes comparison paragraphs that interpret rather than list.
MY CONTEXT	My finding with numbers: [paste] Comparable studies I have read, with design, setting, population and result: [PASTE DETAILS OR UPLOAD PAPERS]
TASK	1. Compare my finding with each study: agree, partly agree or differ. 2. For differences, give the most likely explanations (population, setting, definitions, measurement, timing, sample size). 3. Write a paragraph of 120–180 words that compares and interprets, ending with what the comparison means.
RULES	Use only the studies I supplied, citing each with its reference number, because comparisons with unseen or invented studies are academic misconduct. Present disagreeing studies too, not only supporting ones.
OUTPUT	A comparison table, then the paragraph.

PRO TIP One paragraph per key finding is a good discipline. Don't mix three findings in one paragraph.

TRY IT WITH My finding: prevalence of sarcopenia 38% in COPD; three studies reported 15%, 25% and 40%.

→ **NEXT #272** Why my result differs from previous studies

272 Why my result differs from previous studies

STANDARD

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT

USE IT When your result disagrees with published findings and you must explain it.

PERSONA	You are an epidemiologist skilled at explaining discrepancies between studies without special pleading.
MY CONTEXT	My study: design, population, setting, definitions, sample size and result: [brief] The study or studies I differ from, with the same details: [brief]
TASK	1. Compare the studies systematically on each methodological dimension. 2. Rank the plausible explanations for the difference. 3. Consider chance (sample size, CI overlap) explicitly. 4. Write a discussion paragraph presenting the difference and the most likely explanations, without dismissing either result.
RULES	Don't claim my study is "better" without a specific methodological reason, and consider that my own study might be the outlier, because examiners respect balanced reasoning.
OUTPUT	The comparison table, the ranked explanations, then the paragraph.

PRO TIP Overlapping confidence intervals often mean the "difference" isn't really a difference. Check that first.

TRY IT WITH My study found no association between vitamin D and TB outcome; two earlier studies found one.

→ **NEXT #271** Compare one finding with the literature

273 The mechanism paragraph, clearly labelled as speculation

STANDARD

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT

USE IT When your discussion needs a biological or clinical explanation for your finding.

PERSONA	You are a clinician-scientist who explains mechanisms carefully and labels speculation clearly.
MY CONTEXT	My finding: [paste] Relevant physiology or mechanisms I know of, with sources: [PASTE NOTES WITH CITATIONS]
TASK	1. Suggest the 2–3 most plausible mechanisms for my finding. 2. For each, say how strong the supporting evidence is (established, plausible, speculative). 3. Write a paragraph of 100–150 words presenting them with hedged language. 4. Suggest what study could test the leading mechanism.
RULES	Mark every mechanistic claim needing support as [CITE], and phrase speculation as speculation ("may", "could", "one possible explanation"), because mechanisms presented as fact are a common overreach.
OUTPUT	The mechanism list with evidence strength, then the paragraph, then the testing suggestion.

PRO TIP Keep this paragraph short. Examiners value an honest "the mechanism is unclear" over a confident guess.

TRY IT WITH Finding: low serum magnesium associated with higher ICU mortality.

→ **NEXT #274** Clinical implications without overreach

274 Clinical implications without overreach

STANDARD

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT

USE IT For the implications paragraph of your discussion.

PERSONA	You are a guideline-panel member who judges how far a single study can change practice.
MY CONTEXT	Main findings: [paste] Design, sample size and setting: [brief] Who could use the findings (clinicians, administrators, programmes): [list]
TASK	1. State what the findings could mean for practice, and for whom. 2. Distinguish what can be acted on now, what needs confirmation, and what is hypothesis-generating. 3. Write the implications paragraph (100–150 words). 4. Flag any implication that overreaches the design.
RULES	Use proportionate verbs (suggests, supports, may inform) for single-centre observational results, because "should be adopted" claims are rarely justified by one study.
OUTPUT	The three-level classification, the paragraph, then the overreach flags.

PRO TIP Name the specific clinical decision your findings touch, e.g. "whether to repeat lactate at 6 hours".

TRY IT WITH Finding: a bedside frailty score predicted post-operative delirium in elderly surgical patients.

→ **NEXT #275** Strengths without inflation

275 Strengths without inflation

STANDARD

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT

USE IT For the strengths paragraph of your discussion.

PERSONA	You are a sceptical reviewer who dislikes inflated strength statements.
MY CONTEXT	Design features: [e.g. prospective, consecutive recruitment, blinded outcome assessment, validated tools, low loss to follow-up] Anything unique about my setting or population: [brief]
TASK	1. Identify genuine methodological strengths and explain why each matters for validity. 2. Remove claims that are not real strengths (e.g. "first study in India" unless verified, or "large sample" when it isn't). 3. Write the strengths paragraph (80–120 words).
RULES	Tie each strength to a bias it reduces, because a strength without a reason is just a claim. Do not claim novelty unless I confirm it has been checked; write [VERIFY NOVELTY] instead.
OUTPUT	A strengths table (feature, bias reduced), the removed claims, then the paragraph.

PRO TIP "Prospective, consecutive recruitment with complete follow-up" is a real strength. Say it plainly.

TRY IT WITH A prospective cohort of 180 patients with 97% follow-up and blinded outcome assessment.

→ **NEXT #267** Limitations that are honest, ranked and specific

276 Future research that someone could actually do

STANDARD

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT

USE IT Instead of writing "further studies are needed".

PERSONA	You are a research funder who only backs specific, feasible next studies.
MY CONTEXT	My findings and limitations: [brief] Open questions my study raised: [list] My setting and the resources a follow-up study could realistically use: [brief]
TASK	1. Propose 3–4 specific future studies, each with design, population, setting, outcome and why it follows from my findings. 2. Rank them by importance and feasibility, and say which one could be a thesis for a junior resident in my department. 3. Write the future-research paragraph (80–120 words).
RULES	Make each proposal specific enough to become someone's thesis topic, because vague calls for research add nothing.
OUTPUT	A proposal table, then the paragraph.

PRO TIP Link each future study to one of your limitations. It shows you understand what your study couldn't answer.

TRY IT WITH My cross-sectional study found an association between sleep apnoea and poor asthma control.

→ **NEXT #277** A conclusion that doesn't exceed the data

277 A conclusion that doesn't exceed the data

STANDARD

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT

USE IT For the final paragraph of your discussion or abstract.

PERSONA	You are a journal editor who rewrites overreaching conclusions every day.
MY CONTEXT	Primary objective: [paste] Primary result: [paste] Design and setting: [brief] My current conclusion: [paste] Target: [thesis / journal abstract / poster]
TASK	1. Check my conclusion against the result and design, and list every overreach. 2. Rewrite it in 2–3 sentences: what was found, in whom, and what it suggests. 3. Write a one-sentence version for the abstract.
RULES	Match the strength of language to the design and precision of the result, because the conclusion is the most-quoted sentence of any study.
OUTPUT	The overreach list, the rewritten conclusion, then the abstract sentence.

PRO TIP Your conclusion should answer your primary objective in the same words the objective used.

TRY IT WITH Current: "Early antibiotics save lives in sepsis." Design: retrospective cohort.

→ **NEXT #269** Conclusion, recommendations and summary chapter

278 Scan my discussion for overclaiming

STANDARD

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT

USE IT As a final check on any discussion or conclusion.

PERSONA	You are a research-integrity editor who scans manuscripts for spin.
MY CONTEXT	My design: [design] My primary result: [estimate, CI, p] My discussion text: [PASTE DISCUSSION]
TASK	1. Flag causal language used with an observational design. 2. Flag spin: "trend towards", "marginally significant", emphasis on secondary results over a null primary result, and extrapolation beyond the population. 3. Flag unqualified "significant". 4. Give a corrected version of each flagged sentence.
RULES	Keep the meaning where it is justified and change only the overreaching wording, because the goal is accuracy, not timidity.
OUTPUT	A table (sentence, type of spin, corrected sentence), then a count by type.

PRO TIP Search your text for "caus", "effect", "impact", "prevent", "led to", "due to". Most overclaims hide there.

TRY IT WITH A discussion claiming a supplement "improved outcomes" based on a before-after study.

→ **NEXT #268** Hostile-reviewer attack on my discussion

279 Who do my results apply to?

STANDARD

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT

USE IT For the generalisability part of your limitations or implications.

PERSONA	You are an epidemiologist who thinks carefully about external validity.
MY CONTEXT	Setting and how patients reached my hospital: [brief] Population characteristics (severity, age, comorbidity, socio-economic profile): [brief] Exclusions: [list] Proportion of screened patients who were enrolled: [numbers]
TASK	1. Describe the population my results apply to most directly. 2. Identify populations to whom they may not apply, and why. 3. Discuss referral bias for a tertiary centre, if relevant, and the likely direction of its effect on my estimates. 4. Write a generalisability paragraph (80–120 words).
RULES	Base the assessment on my actual setting and exclusions, because generic statements about "single-centre studies" add nothing.
OUTPUT	Headed notes, then the paragraph.

PRO TIP Tertiary-centre patients are usually sicker. Say whether that makes your estimates higher or lower than in district practice.

TRY IT WITH ICU cohort at a referral hospital receiving transfers from 6 districts.

→ **NEXT #267** Limitations that are honest, ranked and specific

280 Discuss a surprising result

STANDARD

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT

USE IT When a result contradicts your hypothesis or expectations.

PERSONA	You are a thoughtful senior researcher who treats surprising results as information, not embarrassment.
MY CONTEXT	What I expected, and why: [brief] What I found, with numbers: [paste] Possible issues with my data or methods: [brief]
TASK	1. List the explanations in order: chance, bias, confounding, measurement error, and a genuine effect. 2. Evaluate each against my study details. 3. Suggest a check I can still run (e.g. a sensitivity analysis) if appropriate. 4. Write a paragraph (100–150 words) presenting the finding honestly.
RULES	Don't explain away the finding, and don't overstate it, because examiners respect a balanced treatment of surprises.
OUTPUT	The explanations table, the check, then the paragraph.

PRO TIP Surprising findings often make the best future-research questions. Say so.

TRY IT WITH Expected obesity to worsen ARDS outcomes; found lower mortality in obese patients.

→ **NEXT #266** Write the discussion: ten paragraphs, no overclaiming

281 Discuss a non-significant primary result

STANDARD

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT

USE IT When your primary outcome was not statistically significant.

PERSONA	You are a statistician-editor who helps authors write honest, publishable null-result discussions.
MY CONTEXT	Primary result (estimate, 95% CI, p): [paste] Sample size and whether it met the calculated target: [brief] Expected effect used for the sample size: [brief]
TASK	1. Interpret the CI: what effect sizes are ruled out, and what remain plausible. 2. Discuss power: whether the study could have detected a clinically important effect. 3. Compare with the literature honestly. 4. Write the opening paragraph of the discussion for this result.
RULES	Don't call the result a trend or imply a hidden effect, and don't call it proof of no effect, because both misrepresent a null result.
OUTPUT	The CI interpretation, the power note, then the paragraph.

PRO TIP "Our study was unable to demonstrate..." is better than "there was no difference".

TRY IT WITH RR 0.85 (95% CI 0.62–1.17), p = 0.32, sample of 180 against a target of 200.

→ **NEXT #266** Write the discussion: ten paragraphs, no overclaiming

282 Check the structure of my discussion

STANDARD

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT

USE IT When your discussion is drafted and feels disorganised.

PERSONA	You are a thesis-writing coach who maps discussions against a standard structure.
MY CONTEXT	My discussion: [PASTE DISCUSSION] My primary result: [paste] Target structure, if my guide or journal prescribes one: [brief]
TASK	1. Map each paragraph to a structural element: principal findings, comparison, explanation, mechanism, implications, strengths, limitations, future research or conclusion. 2. Identify missing, repeated or misplaced elements. 3. Suggest a reordering. 4. Check that the first paragraph states the principal finding and the last states a proportionate conclusion.
RULES	Suggest structural changes only, keeping my text as it is, because the content and voice must remain mine.
OUTPUT	A paragraph map table, the problems, then the suggested order.

PRO TIP If your discussion opens with background, move it. The first paragraph must answer the question.

TRY IT WITH A discussion that opens with 2 paragraphs of epidemiology before mentioning the results.

→ **NEXT #266** Write the discussion: ten paragraphs, no overclaiming

283 The "Summary" chapter of an Indian thesis

STANDARD

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT

USE IT When your university requires a separate summary chapter.

PERSONA	You are an MD/MS thesis examiner who reads the summary first.
MY CONTEXT	Title and objectives: [paste] Methods in brief: [paste] Key results by objective, with numbers: [paste] Conclusion: [paste] Word or page limit: [e.g. 500 words]
TASK	1. Write the summary within the limit, with short paragraphs for background, aims, methods, results and conclusion. 2. Keep results in the order of the objectives, with the key numbers. 3. Check that every number appears in my results chapter.
RULES	Add no information that isn't elsewhere in the thesis, because the summary must be a faithful miniature of it.
OUTPUT	The summary with its word count, then the number check.

PRO TIP Write the summary last and read it aloud. If it doesn't flow, your thesis argument doesn't either.

TRY IT WITH Summary for a thesis on thyroid dysfunction in chronic kidney disease, 500 words.

→ **NEXT #269** Conclusion, recommendations and summary chapter

284 What my findings mean for Indian practice

STANDARD

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT / FACULTY

USE IT For a paragraph relating your results to Indian clinical practice, programmes or policy.

PERSONA	You are a public-health physician familiar with Indian health programmes and resource settings.
MY CONTEXT	My findings: [brief] Relevant national programmes, guidelines or practices I know of: [list with sources] Resource context of typical Indian hospitals for this condition: [brief]
TASK	1. Relate the findings to current Indian practice and resource realities. 2. Identify which national programme or guideline, if any, the findings are relevant to. 3. Discuss feasibility of acting on the findings in district and primary settings. 4. Write the paragraph (100–150 words).
RULES	Mention a programme or guideline only if I listed it or you mark it [VERIFY PROGRAMME / GUIDELINE], because misnamed programmes undermine credibility.
OUTPUT	Headed notes, then the paragraph.

PRO TIP Examiners in India value this paragraph. It shows you thought beyond your own hospital.

TRY IT WITH Finding: low-cost bedside ultrasound detected pneumonia as accurately as X-ray in children.

→ **NEXT #274** Clinical implications without overreach

285 Cut my discussion to the word limit

STANDARD

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT

USE IT When your discussion is over the limit for your thesis or target journal.

PERSONA	You are a journal editor who cuts discussions without losing substance.
MY CONTEXT	My discussion: [PASTE DISCUSSION] Current and target word count: [e.g. 1,900 → 1,200] Parts I must keep: [list]
TASK	1. Identify repetition of results, background that belongs in the introduction, over-long literature summaries and redundant limitations. 2. Produce the trimmed discussion at or under the target. 3. List what was cut, so I can restore anything essential.
RULES	Keep all citations with their claims and all numbers unchanged, because trimming must not alter the evidence. Keep the standard structure intact.
OUTPUT	The trimmed discussion with its word count, then the cut list.

PRO TIP Repeated results are the easiest cut. The discussion should interpret results, not restate them.

TRY IT WITH A 2,000-word thesis discussion needing to become a 1,100-word journal discussion.

→ **NEXT #268** Hostile-reviewer attack on my discussion

286 Practical recommendations from my thesis

STANDARD

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT

USE IT When your thesis format requires a recommendations section.

PERSONA	You are a health-policy adviser who writes recommendations that are proportionate to the evidence.
MY CONTEXT	Key findings: [brief] Design strength and limitations: [brief] Stakeholders: [clinicians, hospital, programme, researchers] My university's format for this section, if any: [brief]
TASK	1. Write 4–6 recommendations grouped by stakeholder. 2. Grade each by the strength of support from my study (strong, moderate or preliminary). 3. Make each specific and feasible, with who should act, how, and what it would cost in effort.
RULES	Base practice recommendations only on results my design can support, and phrase the rest as research recommendations, because examiners question recommendations that outrun the data.
OUTPUT	The grouped, graded recommendations.

PRO TIP One specific, feasible recommendation is worth more than five generic ones.

TRY IT WITH Thesis finding: 40% of diabetic patients had undiagnosed peripheral neuropathy in OPD screening.

→ **NEXT #269** Conclusion, recommendations and summary chapter

QUICK PROMPTS · 1.9

287 Rewrite "further studies are needed"

QUICK

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT

Act as a journal editor. Replace the phrase "further studies are needed" in my discussion with one specific sentence naming the design, population, outcome and reason, based on my study's main limitation, [LIMITATION], and main finding, [FINDING]. Give 2 alternatives.

TRY IT WITH Limitation: single-centre, 6-week follow-up.
Finding: early mobilisation reduced delirium.

288 Calibrate my hedging

QUICK

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT

Act as an academic editor. For each of these sentences, [PASTE SENTENCES], tell me whether the hedging is too strong, too weak or right for my design [DESIGN], and rewrite it at the right level. Keep claims supported and avoid both overclaiming and excessive qualifiers.

TRY IT WITH "It is possible that this may perhaps suggest..." and "This proves that..."

289 One take-home message

QUICK

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT / MBBS STUDENT

Act as a senior clinician giving a grand-rounds summary. From my findings, [PASTE KEY FINDINGS], write the single take-home message a busy clinician on the ward should remember, in one sentence of under 25 words, accurate to my design and not overstated.

TRY IT WITH Findings on the association between delayed antibiotic administration and mortality in sepsis.

290 Qualify every "significant"

QUICK

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT

Act as a statistical copy-editor. In this text, [PASTE TEXT], find every use of "significant" or "significantly" and rewrite each to say whether it means statistically significant (with the p value or CI) or clinically important. Return the corrected text with changes highlighted.

TRY IT WITH A discussion with "significantly higher", "a significant finding" and "significant patients".

291 Which way does this bias push my result?

QUICK

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT

Act as an epidemiologist examining my thesis. For this limitation, [LIMITATION], in my study of [EXPOSURE] and [OUTCOME], tell me whether it likely biases my estimate towards or away from the null, why, and give one sentence I can use in my limitations paragraph.

TRY IT WITH Limitation: non-differential misclassification of smoking status by self-report.

292 Swap causal words for correct ones

QUICK

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT

Act as an epidemiology editor. Rewrite these sentences, [PASTE SENTENCES], replacing causal words (causes, leads to, results in, prevents, due to) with accurate associational language suited to my [observational] design, while keeping the meaning as strong as the evidence allows.

TRY IT WITH "Anaemia leads to poor wound healing" from a cross-sectional study.

293 One comparison sentence with a study

QUICK

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT

Act as a thesis writing coach. Write one discussion sentence comparing my result, [MY RESULT WITH NUMBERS], with this study's result, [THEIR RESULT, DESIGN, SETTING, CITATION NUMBER], noting agreement or difference and one likely methodological reason. Keep it under 45 words.

TRY IT WITH My prevalence 34% vs their 22% in a community sample.

294 Is this really a strength?

QUICK

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT

Act as a sceptical reviewer. For each strength I plan to claim, [PASTE LIST], tell me whether it is a genuine methodological strength, what bias it reduces, or why it isn't really a strength. Suggest an accurate wording for each one worth keeping.

TRY IT WITH "First study of its kind", "large sample of 90 patients", "prospective design".

295 Do my conclusions match my objectives?

QUICK

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT

Act as a thesis examiner. Compare my objectives, [PASTE OBJECTIVES], with my conclusions, [PASTE CONCLUSIONS]. For each objective, say whether a conclusion answers it directly, partly or not at all, and suggest the missing or corrected conclusion statement for each gap.

TRY IT WITH Three objectives but only two conclusions, one of which introduces a new claim.

1.10

Protocol, ethics & synopsis

The protocol is the document you will be held to for three years, and the IEC is the gate you must pass first. Decide everything in advance.

20

prompts

3

Master

11

Standard

6

Quick

#296 Complete study protocol, IEC-ready

#297 IEC submission pack and pre-review

#298 Participant information sheet and consent form

#299 The ethics section of my protocol or thesis

#300 Respond to IEC queries

#301 CTRI registration, field by field

#302 University synopsis in the required format

#303 Month-by-month thesis timeline

#304 Realistic thesis budget

#305 Risk-benefit assessment and adverse-event plan

#306 Confidentiality and data-protection plan

#307 Request a waiver of consent

#308 Protocol amendment request to the IEC

#309 Funding and conflict-of-interest declarations

#310 Rewrite a consent paragraph in plain language

#311 What documents does my IEC need?

#312 Does my study involve vulnerable participants?

#313 Compensation and insurance: what do I write?

#314 Assent form for children

#315 Version control for my study documents

Complete study protocol, IEC-ready

1.10 PROTOCOL, ETHICS & SYNOPSIS · PG RESIDENT

USE IT

Before ethics committee submission and before university synopsis deadlines.

WHY

The protocol is the document you will be held to for the next three years. Ambiguity here becomes an unanswerable question at your viva. This prompt makes you decide everything in advance, in the format Indian IECs expect.

COPY THE PROMPT BELOW

PERSONA	You are a clinical trials methodologist and ethics committee member who has reviewed several hundred postgraduate protocols. You know exactly which sections get sent back for revision and why. You write in formal scientific register suitable for institutional submission.
MY CONTEXT	Title: [working title] Research question (PICOT): [paste] Design: [prospective observational cohort / single-arm interventional / randomised controlled / diagnostic accuracy / cross-sectional] Setting: [hospital, unit, catchment] Duration approved for the study: [X months] Expected eligible patients per month: [N] Primary outcome and how it is measured: [state] Secondary outcomes: [list] Intervention or exposure, fully described (dose, timing, device, operator): [describe] Comparator: [describe]
TASK	Draft a complete protocol in this order, to a standard that would pass institutional ethics review at an Indian medical college: 1. Title-page fields for institutional submission. 2. Structured synopsis of 300 words. 3. Introduction and rationale ending in an explicit gap statement, with [CITE] slots. 4. Aims and objectives: one primary, numbered secondaries, each measurable and time-bound. 5. Null and alternative hypotheses. 6. Methodology: design and justification; setting and period; source population; operationally defined inclusion criteria; exclusion criteria with reasons; sampling and how it will be implemented; sample size (formula, every term, inputs marked [INSERT FROM PUBLISHED SOURCE — VERIFY], attrition); randomisation, concealment and blinding if applicable; the study procedure as a patient journey; operational definitions of every variable; data collection tools and who collects what; quality control and inter-observer agreement. 7. Statistical analysis plan in outline, with tests named against objectives. 8. Ethical considerations: consent (and assent), confidentiality and data storage, risks and mitigation, benefits, adverse-event provisions, compensation, conflicts of interest, and withdrawal. 9. Informed consent document in plain language (about eighth-grade reading level), suitable for translation. 10. A month-by-month Gantt-style timeline from ethics approval to thesis submission. 11. A budget with realistic line items. 12. CTRI registration requirements: what, when, and which fields will be difficult. 13. Limitations to acknowledge up front. 14. A numbered placeholder reference list.
RULES	Use [CITE] and [VERIFY] tags instead of any invented reference, prevalence figure, effect size or guideline statement, because the IEC checks them. Flag every section where an ethics committee is likely to raise a query, and say how to phrase it to avoid one. State methodological weaknesses plainly rather than writing around them. Formal scientific language throughout.
OUTPUT	The full protocol with numbered headings, then a separate list of likely IEC queries with pre-emptive fixes.
TWEAK	Finish with a follow-up I should run: "Act as the ethics committee reviewer and list every objection you would raise to this protocol."

How to get more out of it

- The operational-definitions section is where most protocols are weak. Don't shorten it.
- Get the sample size confirmed by a statistician. AI sample-size output is a starting draft, never the final number.
- Keep one master version with a version number and date. Every committee copy must match it.
- → NEXT #297 IEC submission pack and pre-review

IEC submission pack and pre-review

1.10 PROTOCOL, ETHICS & SYNOPSIS · PG RESIDENT / FACULTY

USE IT

When your protocol is final and you are preparing the ethics committee submission.

WHY

IEC rejections and revision cycles cost months. Most queries are predictable: consent wording, risk-benefit, vulnerable participants, data protection, compensation and inconsistency between documents. This prompt reviews your pack the way the committee will.

COPY THE PROMPT BELOW

PERSONA	You are the member-secretary of an institutional ethics committee at an Indian medical college, experienced with the ICMR National Ethical Guidelines and the committee's usual queries. You review submissions strictly but constructively.
MY CONTEXT	Study type and risk level (in my view): [observational / minimal risk / greater than minimal risk] Participants, including any vulnerable groups: [brief] Protocol summary: [PASTE SUMMARY] Consent documents (English version): [PASTE PIS AND CONSENT FORM] Data handling and storage plan: [brief] Funding and conflicts of interest: [brief] Documents I plan to submit: [list]
TASK	1. Classify the likely review category (exempt, expedited or full committee) with reasoning, marked [CONFIRM WITH YOUR IEC]. 2. Build a submission checklist of documents Indian IECs commonly require (application form, protocol, PIS and consent in English and local language, CRF, investigator CVs and GCP certificates, undertakings, budget, CTRI plan) and mark what I am missing. 3. Review the consent documents for completeness (purpose, procedures, duration, risks, benefits, alternatives, confidentiality, voluntariness, withdrawal, compensation, contacts) and readability. 4. Review the risk-benefit assessment and the adverse-event plan. 5. Check consistency of title, objectives, sample size and procedures across all documents. 6. Write the 10 queries this committee is most likely to raise, each with a model response and the document change that resolves it.
RULES	Refer to the current ICMR National Ethical Guidelines and mark every specific rule you cite as [VERIFY IN ICMR GUIDELINES], because committees check citations against the source. Don't guarantee approval or classification; the committee decides.
OUTPUT	The category note, the checklist table (document, required, status), the consent review, the risk-benefit note, the consistency table, then the 10 queries with responses.
TWEAK	End with the 3 changes that would most reduce the chance of a revision cycle.

How to get more out of it

- › Mismatched titles and sample sizes between documents are the commonest avoidable query.
- › Get the local-language consent form translated and back-translated. Committees check this.
- › Keep your GCP training certificate current. Many IECs require it from all investigators.
- › → **NEXT #300** Respond to IEC queries

Participant information sheet and consent form

1.10 PROTOCOL, ETHICS & SYNOPSIS · PG RESIDENT

USE IT

While preparing your IEC submission.

WHY

A consent form written in dense medical English is not informed consent. Committees reject forms that are too technical, incomplete or inconsistent with the protocol. A plain-language, complete, translation-ready document gets approved and actually informs patients.

COPY THE PROMPT BELOW

PERSONA	You are a research-ethics specialist who writes participant information sheets and consent forms for Indian hospital studies. You write at about an eighth-grade reading level, in short sentences, with no jargon.
MY CONTEXT	Study title (formal and a lay version): [paste] What participants will experience, step by step: [paste] Time required and number of visits: [brief] Samples, tests or procedures, including any extra blood volume: [brief] Risks and discomforts, and benefits (direct and indirect): [brief] Alternatives to participation: [brief] Data confidentiality arrangements: [brief] Compensation for participation or injury: [brief] Investigator and IEC contact details: [placeholders] Participants who need a legally acceptable representative, or assent (children): [brief] Local language(s) needed: [e.g. Hindi, Marathi]
TASK	1. Write the participant information sheet under simple headings: why this study, what will happen, how long, risks, benefits, alternatives, privacy, your rights, compensation, and who to contact. 2. Write the consent form with the standard statements to initial, plus signature, thumb-impression and witness sections, and an impartial-witness provision for participants who cannot read. 3. If relevant, write an LAR consent version and a short assent form for children. 4. Check readability: flag any sentence over 20 words and any medical term, with a simpler version. 5. Give translation and back-translation instructions for the local language versions.
RULES	Use only the procedures, risks and arrangements I described, because anything added or omitted makes the form inconsistent with the protocol. Mark every regulatory requirement you mention as [VERIFY IN ICMR GUIDELINES]. Never promise benefits or confidentiality beyond what the protocol provides.
OUTPUT	The information sheet, the consent form, the LAR and assent versions if needed, the readability flags table, then the translation instructions.
TWEAK	End with the 3 questions a patient is most likely to ask when you take consent, with simple answers.

How to get more out of it

- › Read the form aloud to a non-medical relative. Every place they frown needs rewriting.
- › Say "you may stop at any time without affecting your treatment" in every language version.
- › Keep a signed copy for the participant. Many IECs require this.
- › → NEXT #310 Rewrite a consent paragraph in plain language

299 The ethics section of my protocol or thesis

STANDARD

1.10 PROTOCOL, ETHICS & SYNOPSIS · PG RESIDENT

USE IT When writing the ethical considerations section.

PERSONA	You are an ethics committee member at an Indian government medical college who writes clear, compliant ethics sections.
MY CONTEXT	I am conducting a [STUDY DESIGN] study on [TOPIC] at a government medical college in India. My participants are [e.g. adult COPD patients]. Data collection involves [e.g. blood tests, interviews, chart review, imaging].
TASK	Write the ethics section covering: 1. The type of informed consent required and how it will be obtained. 2. The data confidentiality and anonymisation plan. 3. The risk-benefit analysis for participants. 4. Vulnerable populations, if applicable. 5. Compliance with the ICMR National Ethical Guidelines. 6. A statement on conflicts of interest.
RULES	Write in formal academic language suitable for IEC submission, and mark the guideline edition and any specific provision as [VERIFY CURRENT ICMR EDITION], because guideline editions are updated and committees check them.
OUTPUT	Six headed paragraphs, about 300 words in total.

PRO TIP Mirror the committee's own application-form headings where you can. It makes the reviewer's job easier.

TRY IT WITH Cross-sectional study using blood tests and questionnaires in adults with type 2 diabetes.

→ **NEXT #297** IEC submission pack and pre-review

300 Respond to IEC queries

STANDARD

1.10 PROTOCOL, ETHICS & SYNOPSIS · PG RESIDENT

USE IT When the ethics committee returns your submission with queries.

PERSONA	You are an experienced investigator who writes clear, respectful, point-by-point responses to ethics committees.
MY CONTEXT	The committee's queries, verbatim: [PASTE QUERIES] Relevant sections of my protocol and consent form: [PASTE SECTIONS]
TASK	1. Draft a point-by-point response letter: each query quoted, then my response, then the exact change made with page and section. 2. Suggest the revised protocol or consent text for each change. 3. Flag any query I should discuss with my guide before responding. 4. Draft the covering letter.
RULES	Address every query fully, and never argue with a query without a clear, respectful reason, because unresolved queries delay approval. Propose only changes consistent with my study.
OUTPUT	The response table (query, response, change, location), the revised text, the flags, then the covering letter.

PRO TIP Highlight all changes in the revised documents and submit a clean copy as well.

TRY IT WITH Queries on consent for illiterate patients, blood volume per visit, and data storage.

→ **NEXT #308** Protocol amendment request to the IEC

301 CTRI registration, field by field

STANDARD

1.10 PROTOCOL, ETHICS & SYNOPSIS · PG RESIDENT / FACULTY

USE IT When your study must be registered with the Clinical Trials Registry – India.

PERSONA	You are a research coordinator who has registered many studies on CTRI.
MY CONTEXT	Study type: [interventional / observational] Protocol summary: [PASTE SUMMARY] IEC approval status: [approved on date / pending] Sites and investigators: [brief]
TASK	1. Tell me whether my study needs CTRI registration and when (before enrolling the first participant), marked [VERIFY CURRENT CTRI AND ICMR REQUIREMENTS]. 2. Walk through the main registration fields and draft my entries: titles, sponsors, sites, conditions, intervention and comparator, inclusion and exclusion criteria, method of allocation and concealment, blinding, primary and secondary outcomes with timepoints, target sample size, phase, dates and brief summary. 3. Flag fields residents commonly get wrong. 4. Explain what happens after submission and how to update the record.
RULES	Keep every entry identical to my approved protocol, because discrepancies between the registry and the thesis are flagged by journals. Mark any field name you are unsure of as [CHECK CURRENT FORM].
OUTPUT	The requirement note, a field table with draft entries, the common-errors list, then the post-submission steps.

PRO TIP Journals increasingly require prospective registration for trials. Register before the first participant, not after.

TRY IT WITH Single-centre RCT of two physiotherapy protocols after knee replacement.

→ **NEXT #296** Complete study protocol, IEC-ready

302 University synopsis in the required format

STANDARD

1.10 PROTOCOL, ETHICS & SYNOPSIS · PG RESIDENT

USE IT When your university or college requires a thesis synopsis for approval.

PERSONA	You are a university thesis-cell member who reviews postgraduate synopses and knows the common format.
MY CONTEXT	My university and its synopsis format or headings, if known: [paste or "standard format"] Protocol details: [PASTE PROTOCOL OR KEY SECTIONS] Word or page limit: [e.g. 8 pages]
TASK	1. Arrange the content under the required headings (typically title, introduction, need for the study, aims and objectives, review of literature, materials and methods, and references). 2. Condense each section to fit the limit. 3. Check consistency with the protocol and IEC documents. 4. List the attachments the synopsis usually needs (guide's certificate, IEC approval, proforma).
RULES	Keep objectives, sample size and title identical to the IEC submission, because universities cross-check them. Mark headings I haven't confirmed with my university as [CHECK YOUR UNIVERSITY FORMAT].
OUTPUT	The formatted synopsis, then the consistency check and attachment list.

PRO TIP Ask a senior for an approved synopsis from your department. Matching the local format saves a revision.

TRY IT WITH MD Anaesthesiology synopsis for a prospective RCT on post-operative nausea.

→ **NEXT #85** Short background for a synopsis or protocol

303 Month-by-month thesis timeline

STANDARD

1.10 PROTOCOL, ETHICS & SYNOPSIS · PG RESIDENT

USE IT For your protocol, synopsis and personal planning.

PERSONA	You are a thesis project manager who builds realistic timelines with buffers.
MY CONTEXT	Course start and thesis submission deadline: [dates] Expected IEC approval date: [date] Recruitment rate and target: [N per month, total] Follow-up duration: [months] Known busy periods (exams, rotations, postings): [list]
TASK	1. Build a month-by-month timeline: approvals, registration, pilot, recruitment, follow-up, data entry, analysis, writing each chapter, guide review, corrections, binding and submission. 2. Add buffers where delays are likely. 3. Calculate whether recruitment fits, and flag it if it doesn't. 4. Present it as a Gantt-style table (tasks against months).
RULES	Build in at least one month of buffer before submission, and say if my deadline is unrealistic, because optimistic timelines are the commonest reason for late theses.
OUTPUT	The Gantt-style table, then the feasibility check and buffer notes.

PRO TIP Start writing the introduction and ROL during recruitment. Waiting until data collection ends is the classic mistake.

TRY IT WITH MD course starting July 2025, submission due January 2028, recruitment 8 per month for 120 patients.

→ **NEXT #18** What could kill this topic? A risk register

304 Realistic thesis budget

STANDARD

1.10 PROTOCOL, ETHICS & SYNOPSIS · PG RESIDENT / FACULTY

USE IT For your protocol, IEC form or a small grant application.

PERSONA	You are a research finance officer who builds realistic budgets for investigator-initiated studies.
MY CONTEXT	Tests, consumables and procedures per participant, beyond routine care: [list] Sample size: [N] Other costs: [printing, stationery, software, travel, translation, statistician, publication] Funding source: [self / department / grant]
TASK	1. Build a line-item budget: per-participant costs times N, plus fixed costs. 2. Mark who pays each item (hospital, investigator, grant). 3. Add a contingency line. 4. Flag any cost that would fall on patients, and how to avoid it.
RULES	Use placeholder unit costs marked [INSERT LOCAL COST] rather than inventing prices, because costs vary widely between institutions. Participants must not bear research-only costs, and the IEC will check this.
OUTPUT	The budget table, then the patient-cost flags.

PRO TIP Many state and national bodies offer small grants for postgraduate theses. Check your university's research cell.

TRY IT WITH 100 patients, each needing one extra serum biomarker test and a 6-month follow-up phone call.

→ **NEXT #296** Complete study protocol, IEC-ready

305 Risk-benefit assessment and adverse-event plan

STANDARD

1.10 PROTOCOL, ETHICS & SYNOPSIS · PG RESIDENT

USE IT When your study involves any procedure or intervention beyond routine care.

PERSONA	You are a clinical research safety officer who writes proportionate risk-benefit and adverse-event sections.
MY CONTEXT	Study procedures beyond routine care: [list] Participant population: [brief] Intervention, if any, and its known side effects: [brief]
TASK	1. List the risks of each study-specific procedure, with likelihood and severity. 2. Describe mitigation for each. 3. Describe direct and indirect benefits realistically. 4. Write the adverse-event plan: definitions, recording, grading, reporting timelines to the IEC, and management. 5. Summarise the overall risk category with justification.
RULES	Mark reporting timelines and definitions as [VERIFY IN ICMR / IEC SOP], because reporting requirements differ by study type and institution. Don't list routine-care risks as study risks.
OUTPUT	A risk table, the benefits paragraph, the AE plan, then the summary.

PRO TIP If a procedure is part of routine care anyway, say so clearly. It lowers the perceived risk accurately.

TRY IT WITH Study adding one extra 5 mL blood draw and a 6-minute walk test for COPD patients.

→ **NEXT #297** IEC submission pack and pre-review

306 Confidentiality and data-protection plan

STANDARD

1.10 PROTOCOL, ETHICS & SYNOPSIS · PG RESIDENT

USE IT For the confidentiality section of your IEC submission and protocol.

PERSONA	You are a health-data governance adviser familiar with Indian research-ethics requirements and data-protection principles.
MY CONTEXT	Data collected (identifiable and study data): [list] Where and how it will be stored: [brief] Who has access: [brief] Any data leaving the institution (cloud, AI tools, collaborators): [brief]
TASK	1. Describe de-identification: study IDs, a separate linking log, and restricted access. 2. Describe secure storage, backup, encryption and retention. 3. Describe the rules for using AI tools and cloud services with study data. 4. Describe what happens at study end (retention period, destruction). 5. Write the confidentiality paragraph for my protocol.
RULES	Recommend that no identifiable data ever be entered into consumer AI tools, and mark any legal requirement you mention (e.g. the Digital Personal Data Protection Act) as [VERIFY CURRENT LAW AND RULES], because legal obligations are specific and evolving.
OUTPUT	A 4-part plan, then the paragraph.

PRO TIP Keep the linking log (study ID → patient identity) on a different device from your study data.

TRY IT WITH Cohort of 150 patients with phone follow-up, data in Excel on a department laptop.

→ **NEXT #177** Data management plan for my thesis

307 Request a waiver of consent

STANDARD

1.10 PROTOCOL, ETHICS & SYNOPSIS · PG RESIDENT

USE IT For retrospective record-based studies or other minimal-risk research where consent is impracticable.

PERSONA	You are an IEC member who reviews waiver requests and knows what justifications committees accept.
MY CONTEXT	Study design and data source: [brief] Why obtaining consent is impracticable: [brief] Risk to participants: [brief] Confidentiality measures: [brief]
TASK	1. Assess whether my study plausibly qualifies for a waiver, with reasoning. 2. Write the waiver request: study description, justification of impracticability, minimal-risk argument, confidentiality safeguards, and how rights and welfare are protected. 3. List conditions the committee may impose.
RULES	Present the request honestly and mark the criteria you cite as [VERIFY IN ICMR GUIDELINES], because the committee alone decides eligibility for a waiver.
OUTPUT	The assessment, the request text, then the likely conditions.

PRO TIP Committees are far more comfortable granting waivers when data are anonymised at extraction.

TRY IT WITH Retrospective review of 5 years of ICU records on ventilator-associated pneumonia.

→ **NEXT #169** Record-review (retrospective) methodology

308 Protocol amendment request to the IEC

STANDARD

1.10 PROTOCOL, ETHICS & SYNOPSIS · PG RESIDENT

USE IT When you need to change the approved protocol (sample size, criteria, outcomes, procedures).

PERSONA	You are an experienced investigator who writes clear, well-justified amendment requests.
MY CONTEXT	Current approved protocol version and date: [version] Proposed changes and reasons: [list] Effect on participants, risks and consent: [brief]
TASK	1. Classify each change as minor or major, with reasoning, marked [CONFIRM WITH IEC SOP]. 2. Write the amendment request: a table of old text, new text and justification. 3. Identify documents that must be updated (protocol, consent form, CRF, CTRI record). 4. Note whether already-enrolled participants need re-consent.
RULES	Don't implement any change before approval unless it is needed to eliminate an immediate hazard, and say so, because unapproved changes are protocol violations.
OUTPUT	The classification, the amendment table, the documents list, then the re-consent note.

PRO TIP Update your CTRI record after approval. Journals compare the registry with the paper.

TRY IT WITH Increasing the sample size from 100 to 130 and adding a secondary outcome.

→ **NEXT #300** Respond to IEC queries

309 Funding and conflict-of-interest declarations

STANDARD

1.10 PROTOCOL, ETHICS & SYNOPSIS · PG RESIDENT / FACULTY

USE IT When your study receives any funding, free materials or support, or when you have any industry relationship.

PERSONA	You are a research-integrity officer who drafts transparent declarations.
MY CONTEXT	Funding sources and amounts: [brief] Free equipment, kits or drugs received: [brief] Any personal or institutional relationships with relevant companies: [brief] Role of funders in design, analysis or publication: [brief]
TASK	1. Identify everything that must be declared, including non-financial interests. 2. Draft the declarations for the protocol, IEC form, thesis and future manuscript. 3. Recommend safeguards if a sponsor provides materials (e.g. no sponsor role in analysis or publication decisions).
RULES	Err on the side of disclosure, because an undeclared interest discovered later damages credibility far more than a declared one.
OUTPUT	The list of interests, the declarations for each document, then the safeguards.

PRO TIP Free diagnostic kits from a manufacturer count as support. Declare them.

TRY IT WITH A company provided free point-of-care test cartridges for my diagnostic accuracy study.

→ **NEXT #297** IEC submission pack and pre-review

310 Rewrite a consent paragraph in plain language

QUICK

1.10 PROTOCOL, ETHICS & SYNOPSIS · PG RESIDENT

Act as a patient-communication specialist. Rewrite this consent-form paragraph, [PASTE PARAGRAPH], at about an eighth-grade reading level for Indian patients, with sentences under 20 words, no jargon and no added promises. Keep every fact exactly, then list the medical terms you replaced.

TRY IT WITH A paragraph describing venepuncture risks and the storage of serum samples.

311 What documents does my IEC need?

QUICK

1.10 PROTOCOL, ETHICS & SYNOPSIS · PG RESIDENT

Act as an IEC office coordinator at an Indian medical college. List the documents typically required for a [STUDY TYPE] postgraduate thesis submission, what each must contain, and which are most often missing. Mark anything institution-specific as [CHECK YOUR IEC SOP].

TRY IT WITH Prospective observational thesis with blood sampling.

312 Does my study involve vulnerable participants?

QUICK

1.10 PROTOCOL, ETHICS & SYNOPSIS · PG RESIDENT

Act as an ethics committee member. For my study population, [DESCRIBE POPULATION], identify any vulnerable groups under research-ethics principles, the extra protections each needs, and how to describe them in my protocol. Mark specific requirements as [VERIFY IN ICMR GUIDELINES].

TRY IT WITH ICU patients, some unconscious, including pregnant women and adolescents.

313 Compensation and insurance: what do I write?

QUICK

1.10 PROTOCOL, ETHICS & SYNOPSIS · PG RESIDENT

Act as a research-ethics adviser. For my [observational / interventional] thesis on [TOPIC], explain what I should state about compensation for research-related injury and about payment for participation, and draft the wording for my protocol. Mark regulatory specifics as [VERIFY WITH IEC].

TRY IT WITH Interventional thesis testing a physiotherapy protocol after abdominal surgery.

314 Assent form for children

QUICK

1.10 PROTOCOL, ETHICS & SYNOPSIS · PG RESIDENT (PAEDIATRICS)

Act as a paediatric research-ethics specialist. Write a short assent form for children aged [AGE RANGE] in my study, [BRIEF DESCRIPTION], using simple words, short sentences and a clear statement that they can say no. Mark age-band requirements as [VERIFY IN ICMR GUIDELINES].

TRY IT WITH Children aged 7–12 in a study requiring one finger-prick blood test.

315 Version control for my study documents

QUICK

1.10 PROTOCOL, ETHICS & SYNOPSIS · PG RESIDENT

Act as a research-documentation coordinator. Give me a simple version-control system for my protocol, consent forms, CRF and approvals: file naming, version numbers, dates, a change log and footer text, so every document the IEC sees matches my master copy.

TRY IT WITH Protocol v1.0, consent form v1.1, CRF v2.0 after an amendment.

1.11

Thesis to manuscript & publication

A thesis and a paper are different documents. Convert by cutting, comply with the journal's rules, and prepare for peer review before you submit.

35

prompts

4

Master

20

Standard

11

Quick

#316 Write the full manuscript for journal submission

#317 Thesis → paper cut plan (and the second paper inside it)

#318 Point-by-point response to peer reviewers

#319 Pre-submission peer review of my manuscript

#320 Structured abstract within the word limit

#321 Five manuscript titles for my paper

#322 Choose the right journal (and avoid predatory ones)

#323 Check my manuscript against the author guidelines

#324 Cover letter to the editor

#325 Keywords that help my paper get found

#326 Authorship order and CRediT contributions

#327 Disclose AI use in my manuscript

#328 Ethics, consent, funding, COI and data statements

#329 Compress my thesis methods for a journal

#330 Three-paragraph journal introduction

#331 Condense my thesis discussion for a journal

#332 Rejected: what now?

#333 Suggest reviewers properly

#334 Key points, highlights and "what this study adds"

#335 Brief for a graphical abstract

#336 Should I post a preprint?

#337 Write a letter to the editor about a published paper

#338 Check the galley proofs properly

#339 Read and fix a similarity (plagiarism) report honestly

#340 Running title under the limit

#341 Word budget for each section

#342 Cut my abstract to the limit

#343 Summarise my paper for social media

#344 Cut my references to the journal's limit

#345 Write the acknowledgements

#346 Polite status enquiry to the editor

#347 Data-availability statement

#348 Plain-language summary of my paper

#349 Reply to one tough reviewer comment

#350 Is splitting my thesis into two papers acceptable?

Write the full manuscript for journal submission

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT / FACULTY

USE IT

When the thesis is done and you are converting it into a paper you can submit.

WHY

A thesis and a paper are different documents. This prompt handles the conversion, reporting-guideline compliance and the submission package in one structured pass, using only your real numbers.

COPY THE PROMPT BELOW

PERSONA	You are a senior academic author and manuscript editor who has published extensively in clinical journals and has served as a peer reviewer. You know reporting guidelines thoroughly and you write in clear, economical scientific English for an international readership.
MY CONTEXT	Design: [state] · Current full title: [paste] Setting and period: [state] Sample size and participant flow: [screened, eligible, enrolled, completed, analysed] Methods summary: [paste from protocol] All results with exact numbers, effect estimates and CIs: [paste] Tables and figures I already have: [list with titles] Target journal: [name] · Word, abstract, reference, table and figure limits: [paste from author guidelines] Reporting guideline that applies: [CONSORT / STROBE / STARD / PRISMA / CARE / SQUIRE / other]
TASK	1. Journal fit: assess scope, design and novelty; if it is a poor fit, say so and suggest what type of journal fits better. Do not quote impact factors or acceptance rates from memory. 2. Title: five options (descriptive, finding-stating, PICO-style, short, conservative), with the best one for this journal. 3. Structured abstract within the limit, with my actual numbers. 4. Six keywords, preferring MeSH terms. 5. Introduction: three paragraphs (known, unknown and why it matters, objective), with [CITE] slots. 6. Methods: past tense, replicable, following my reporting guideline, noting which checklist item each subsection satisfies, with ethics, consent and registration placeholders. 7. Results: past tense, no interpretation; flow, baseline (Table 1), primary, then secondary outcomes, each with its effect estimate and CI. 8. Discussion: principal finding, comparison, explanation, implications, strengths, ranked limitations, future research, conclusion. 9. Conclusion, funding, conflicts of interest, CRediT author contributions, acknowledgements and data-availability statement. 10. A reporting-guideline checklist table, flagging unaddressed items. 11. A cover letter under 300 words. 12. A pre-submission critical review: major and minor comments in journal format, and which to fix before submitting.
RULES	Every citation slot stays [CITE] unless I uploaded the paper or you verified it with a PMID or DOI, because fabricated references end in retraction. Don't alter, round differently or recompute any number I give you. Add no findings, comparisons or claims beyond my data. Flag any claim my design cannot support. Note that AI cannot be an author and remind me to check the journal's AI-disclosure policy.
OUTPUT	Twelve numbered parts, with the manuscript sections in submission order.
TWEAK	If time is short, do step 12 first and tell me the 3 fixes that most improve my chance of acceptance.

How to get more out of it

- › Do step 12 first if you are short on time. A pre-submission critical review saves months of revision.
- › Paste the actual author guidelines. Generic manuscript advice produces desk rejections.
- › Convert thesis to paper by cutting, not rewriting. Most theses contain two papers hiding inside one document.
- › → **NEXT #317** Thesis → paper cut plan (and the second paper inside it)

Thesis → paper cut plan (and the second paper inside it)

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT

USE IT

Right after thesis submission, while the details are fresh.

WHY

Theses are long, comprehensive and repetitive; papers are short and focused. Most residents never publish because they try to shrink the whole thesis. This prompt finds the one or two papers inside it and tells you exactly what to cut.

COPY THE PROMPT BELOW

PERSONA	You are a senior clinical researcher who has turned dozens of MD/MS theses into published papers. You think in terms of "one paper, one message" and you cut ruthlessly.
MY CONTEXT	Thesis title and design: [paste] Objectives and key results for each: [PASTE OBJECTIVES AND RESULTS] Chapter word counts: [introduction, ROL, methods, results, discussion] Number of tables and figures in the thesis: [numbers] Target journal type and its word limit: [e.g. national indexed journal, 3,000 words] Co-authors and who will lead each paper: [brief]
TASK	1. Identify the main paper (primary objective) and any second paper (a distinct secondary objective, a substudy or a methodological finding), each with a one-line message. 2. For the main paper, map what to keep, condense or drop from each thesis chapter, with target word counts. 3. Select the 3–5 tables or figures that carry the main message. 4. Explain what the ROL becomes (usually two sentences in the introduction and comparisons in the discussion). 5. Draft the paper's single-sentence message and a working title. 6. Flag duplicate-publication risks if two papers share data, and how to avoid salami slicing. 7. List any additional analysis the paper needs that the thesis didn't include, marked [NEW ANALYSIS — CONFIRM WITH GUIDE].
RULES	Each proposed paper must stand on its own question, because splitting one result into several papers is salami publication. Use only results I gave; don't invent new analyses.
OUTPUT	A papers table (paper, question, message, key tables), then a chapter-by-chapter cut plan with word targets, then the salami-slicing check.
TWEAK	End with a 4-week writing schedule to get the main paper to submission.

How to get more out of it

- › Write the paper's abstract first. If it doesn't hold together, the paper won't either.
- › Keep the thesis methods chapter open. You'll compress it, not rewrite it.
- › Discuss authorship with your guide before you start writing, not after.
- › → **NEXT #316** Write the full manuscript for journal submission

Point-by-point response to peer reviewers

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT / FACULTY

USE IT

When you receive a "major" or "minor revision" decision.

WHY

A revision is an invitation, not a rejection. Most revisions are accepted when every comment is answered respectfully and specifically, with changes clearly located. Defensive or vague responses are how revisions become rejections.

COPY THE PROMPT BELOW

PERSONA	You are a senior author who has responded to hundreds of peer reviews and sits on editorial boards. You are gracious, precise and firm where the science warrants it.
MY CONTEXT	Editor's decision letter: [PASTE LETTER] Reviewer comments, verbatim: [PASTE ALL COMMENTS] My manuscript (relevant sections): [PASTE OR UPLOAD] Comments I disagree with, and why: [brief] Revision deadline: [date] Analyses or data I can realistically add in that time: [brief]
TASK	1. Number every comment and classify it: requires new analysis, requires text change, requires clarification, or respectful disagreement. 2. For each, draft a response: thank the reviewer where appropriate, answer the point directly, and state exactly what changed and where (page and line). 3. Draft the revised manuscript text for each change. 4. For disagreements, write a courteous, evidence-based rebuttal, and suggest a partial concession where reasonable, such as adding a limitation sentence. 5. Flag comments that need new data or analysis I may not have, and how to handle each honestly (e.g. acknowledging it as a limitation). 6. Draft the cover letter to the editor summarising the main changes.
RULES	Answer every comment, including the minor ones, because an unanswered comment is the commonest reason for a second revision round. Don't promise analyses I cannot do, and don't invent new results or references; use [CITE] or [NEW ANALYSIS NEEDED] slots.
OUTPUT	A response table (comment number, verbatim comment, response, change and location), the revised text blocks, then the cover letter.
TWEAK	End with the 2 responses most likely to decide acceptance, and whether they are strong enough.

How to get more out of it

- › Reply in a table. Editors find it far easier to check.
- › Highlight changes in a tracked or coloured version of the manuscript, and submit a clean copy too.
- › If a reviewer misunderstood something, the text was probably unclear. Fix the text, not just the reply.
- › → NEXT #324 Cover letter to the editor

Pre-submission peer review of my manuscript

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT / FACULTY

USE IT

One week before submission, when you think the manuscript is finished.

WHY

The fastest way to get published is to be your own harshest reviewer before the journal is. A structured pre-review, in journal format, catches the fatal flaws, the reporting gaps and the desk-rejection triggers while you can still fix them.

COPY THE PROMPT BELOW

PERSONA	You are an experienced peer reviewer and associate editor for clinical journals. You write structured reviews with major and minor comments, and you think about desk-rejection risk as an editor would.
MY CONTEXT	Target journal and its scope: [name, brief scope] Article type and word limit: [e.g. original article, 3,000 words] Reporting guideline: [e.g. STROBE] What I believe is the paper's main contribution: [one line] My biggest worry about the paper: [one line] My manuscript: [PASTE OR UPLOAD MANUSCRIPT]
TASK	1. Give an editor's first-read verdict: desk reject, send for review, or likely to be sent back for formatting, with reasons based on the title, abstract and cover-letter logic. 2. Write major comments on the question, design, analysis, interpretation and novelty, quoting the text. 3. Write minor comments on clarity, tables and figures, references and language. 4. Run a reporting-guideline check and list the items that are missing, with where each should go. 5. Check compliance with the author guidelines (word counts, abstract structure, reference style, declarations, figure formats). 6. Prioritise all issues: fix before submission, or can wait. 7. Say whether my stated main contribution comes across clearly in the title, abstract and conclusion.
RULES	Quote the manuscript for every comment, because specific comments can be acted on and vague ones can't. Don't rewrite the paper; identify and prioritise. Don't invent references or data.
OUTPUT	The verdict, major comments, minor comments, the guideline check, the compliance check, then a priority list.
TWEAK	End with the single weakness most likely to cause rejection, and the most efficient fix.

How to get more out of it

- › Ask a colleague outside your department to read the abstract and tell you the main finding. If they can't, rewrite it.
- › Fix every "before submission" item. Editors notice sloppy formatting within seconds.
- › Keep this review. Compare it with the real reviews when they arrive.
- › → **NEXT #322** Choose the right journal (and avoid predatory ones)

320 Structured abstract within the word limit

STANDARD

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT

USE IT For journal submission, conference abstracts and the thesis abstract.

PERSONA	You are a journal abstract editor who writes precise, complete structured abstracts.
MY CONTEXT	Target journal or conference and its abstract format (headings, word limit): [paste] Design, setting and period: [brief] Participants and sample size: [brief] Primary outcome result with estimate, CI and p: [paste] Key secondary results: [paste] Conclusion: [paste]
TASK	1. Write the abstract with the required headings, within the limit. 2. Include the design in the Methods, the primary result with numbers in the Results, and a proportionate Conclusion. 3. Give the word count. 4. Check that every number matches the full text.
RULES	Use only the numbers I gave, and make sure the conclusion follows from the primary result, because abstracts are read far more than papers and spin there misleads the most readers.
OUTPUT	The abstract, its word count, then the number check.

PRO TIP Many readers only read the abstract. Make sure it states your primary result even if it was null.

TRY IT WITH 250-word structured abstract for a prospective cohort of 210 patients on AKI after cardiac surgery.

→ **NEXT #321** Five manuscript titles for my paper

321 Five manuscript titles for my paper

STANDARD

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT

USE IT Before submission, or when your title feels flat.

PERSONA	You are a journal editor who knows which titles get read and cited.
MY CONTEXT	Main finding: [one line] Design and population: [brief] Target journal and any title limit: [name, characters] My current working title: [paste]
TASK	1. Write 5 titles: descriptive, finding-stating, question-form, short and punchy, and design-explicit (e.g. "...: a prospective cohort study"). 2. Check each for overclaiming, jargon, abbreviations and length. 3. Recommend one for my journal and explain why. 4. Suggest a running title.
RULES	Don't state causation in a title for an observational design, because editors flag it immediately.
OUTPUT	The 5 titles with character counts, the checks, the recommendation, then the running title.

PRO TIP Put the most important words at the start. Many databases truncate long titles.

TRY IT WITH Finding: frailty predicts 90-day mortality after emergency laparotomy in older Indian adults.

→ **NEXT #320** Structured abstract within the word limit

322 Choose the right journal (and avoid predatory ones)

STANDARD

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT / FACULTY

USE IT Before writing the final version, so you write for a specific journal.

PERSONA	You are a publication adviser who matches manuscripts to journals and protects authors from predatory publishers.
MY CONTEXT	Main finding and design: [brief] Specialty: [specialty] Priorities: [indexing needed (e.g. PubMed, Scopus), speed, cost, open access, national or international readership] Budget for article processing charges: [amount or "none"]
TASK	1. Describe the types of journals that fit my paper (specialty or general, national or international). 2. Explain how to find candidate journals: the reference list of my own paper, journal-finder tools, and indexing databases. 3. Give a checklist for verifying legitimacy: indexing claims checked at the source, a real editorial board, transparent peer review and charges, DOAJ listing for open-access journals. 4. List red flags of predatory journals. 5. Help me build a shortlist table from the journals I find.
RULES	Don't recommend specific journals, impact factors or acceptance rates from memory; mark any journal named as [VERIFY INDEXING AND SCOPE], because journal metrics and indexing change and predatory journals mimic real ones.
OUTPUT	Journal types, finding methods, the legitimacy checklist, red flags, then a shortlist template.

PRO TIP If a journal emails you inviting a submission with rapid publication, treat it as a red flag until proven otherwise.

TRY IT WITH Prospective observational study in pulmonology from an Indian medical college, no APC budget.

→ **NEXT #323** Check my manuscript against the author guidelines

323 Check my manuscript against the author guidelines

STANDARD

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT

USE IT Just before submission.

PERSONA	You are a journal's editorial assistant who checks submissions against the instructions for authors.
MY CONTEXT	The journal's author guidelines (relevant parts): [PASTE GUIDELINES] My manuscript: [PASTE OR UPLOAD]
TASK	1. Check article type, word counts (abstract and main text), heading structure, reference style and number, and table and figure limits and formats. 2. Check the required declarations: ethics, consent, registration, funding, conflicts of interest, data availability, AI disclosure and author contributions. 3. Check file requirements (title page, blinded manuscript, figures, checklist). 4. List every non-compliance with its fix.
RULES	Base every check on the guidelines I pasted, not on general assumptions, because journals differ in small details that cause technical returns.
OUTPUT	A compliance table (requirement, status, fix).

PRO TIP Technical returns cost weeks. Five minutes with this checklist saves them.

TRY IT WITH Original article for a national pulmonology journal with a 250-word abstract limit.

→ **NEXT #324** Cover letter to the editor

324 Cover letter to the editor

STANDARD

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT / FACULTY

USE IT For every journal submission.

PERSONA	You are a senior author whose cover letters get manuscripts sent out for review.
MY CONTEXT	Journal and editor's name: [name] Title and article type: [paste] Main question and finding: [brief] Why it matters to this journal's readers: [brief] Required declarations (original work, not under consideration elsewhere, ethics, conflicts): [brief]
TASK	1. Write a cover letter of 300 words or fewer: the question, the finding, why it matters to this journal's readership, and the standard declarations. 2. Add a line on any related prior publication or preprint, if applicable. 3. Keep the tone professional and specific.
RULES	Don't exaggerate novelty or impact, because editors discount inflated letters. Don't name suggested reviewers without checking their independence.
OUTPUT	The letter.

PRO TIP Name the editor and the journal correctly. Generic "Dear Sir/Madam" letters signal mass submission.

TRY IT WITH Cover letter for a diagnostic accuracy study of lung ultrasound in paediatric pneumonia.

→ **NEXT #319** Pre-submission peer review of my manuscript

325 Keywords that help my paper get found

STANDARD

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT

USE IT At submission, when the journal asks for keywords.

PERSONA	You are a medical librarian who chooses keywords for discoverability.
MY CONTEXT	Title and abstract: [PASTE] Number of keywords allowed: [e.g. 6] Target journal: [name]
TASK	1. Suggest keywords, preferring MeSH terms where they exist, plus 1–2 free-text terms readers would search. 2. Avoid words already in the title where the journal recommends that, and avoid overly broad terms. 3. Explain each choice in one line. 4. Suggest 2 alternative keywords for indexing services and search engines that don't use MeSH.
RULES	Mark any MeSH term you are unsure of as [CHECK IN MeSH BROWSER], because invalid terms reduce discoverability.
OUTPUT	A keyword list with the reason for each.

PRO TIP Think about how a clinician would search for your paper, not how you would describe it.

TRY IT WITH Abstract of a study on sarcopenia and outcomes in elderly COPD patients.

→ **NEXT #320** Structured abstract within the word limit

326 Authorship order and CRediT contributions

STANDARD

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT / FACULTY

USE IT Before writing the paper, and again at submission.

PERSONA	You are a research-integrity adviser who applies ICMJE authorship criteria fairly.
MY CONTEXT	People involved and what each did (conception, data collection, analysis, writing, supervision, funding): [list] Department customs about authorship order: [brief]
TASK	1. Apply the ICMJE criteria to each person: author, or acknowledgement. 2. Suggest an order based on contribution, with reasoning. 3. Write the CRediT contribution statement. 4. Draft the acknowledgements for contributors who don't qualify as authors. 5. Suggest how to discuss disagreements respectfully.
RULES	Apply the criteria as written, and flag gift or ghost authorship if my list suggests it, because authorship disputes and misconduct findings can follow a paper for years.
OUTPUT	An authorship table, the order, the CRediT statement, then the acknowledgements.

PRO TIP Agree authorship in writing before you start writing. It prevents the most painful disputes.

TRY IT WITH Resident, guide, co-guide, statistician, a colleague who helped recruit, and the head of department.

→ **NEXT #328** Ethics, consent, funding, COI and data statements

327 Disclose AI use in my manuscript

STANDARD

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT / FACULTY

USE IT When you used any AI tool while preparing the thesis or manuscript.

PERSONA	You are a publication-ethics editor familiar with ICMJE, COPE and journal policies on generative AI.
MY CONTEXT	AI tools used and for what (language editing, literature search, code, drafting): [list] Target journal's AI policy (paste if available): [PASTE]
TASK	1. Classify my uses by whether journals typically require disclosure. 2. Draft the disclosure statement in the journal's preferred location (methods, acknowledgements or a declaration). 3. Confirm the responsibilities that remain mine (accuracy, originality, citations). 4. Flag any use that may breach the journal's policy (e.g. AI-generated images or data).
RULES	Recommend full, honest disclosure of substantive AI use, and state that AI tools cannot be authors, because undisclosed AI use is increasingly treated as misconduct.
OUTPUT	The classification, the statement, the responsibilities, then the flags.

PRO TIP Keep a simple log of which AI tool you used for what and when. It makes disclosure easy.

TRY IT WITH Used Claude for language editing and ChatGPT for R code during the analysis.

→ **NEXT #328** Ethics, consent, funding, COI and data statements

328 Ethics, consent, funding, COI and data statements

STANDARD

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT

USE IT For the declarations section at submission.

PERSONA	You are an editorial assistant who drafts complete, journal-ready declarations.
MY CONTEXT	IEC name, approval number and date: [details] Consent type: [written informed / waiver] Registration: [CTRI number or not applicable] Funding: [details or none] Conflicts of interest: [details or none] Data availability plan: [on request / repository / not shareable, with reason]
TASK	1. Draft each declaration: ethics approval, consent to participate, consent for publication (if applicable), registration, funding, competing interests, data availability and author contributions. 2. Flag anything missing that the journal is likely to require.
RULES	Use only the details I gave; mark missing essentials as [ADD DETAIL], because incomplete declarations cause technical returns.
OUTPUT	Headed declarations, then the missing-items list.

PRO TIP Keep a copy of your IEC approval letter handy. Many journals ask for it at submission.

TRY IT WITH IEC approval dated March 2025, CTRI registration, no funding, no conflicts.

→ **NEXT #323** Check my manuscript against the author guidelines

329 Compress my thesis methods for a journal

STANDARD

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT

USE IT When turning the thesis methodology chapter into a 700–900 word methods section.

PERSONA	You are a manuscript editor who condenses methods without losing reproducibility.
MY CONTEXT	Thesis methodology chapter: [PASTE CHAPTER] Target word count: [e.g. 800] Reporting guideline: [e.g. STROBE]
TASK	1. Keep everything needed for replication and the reporting checklist. 2. Cut textbook explanations, repeated definitions and procedural detail that can go into a supplement. 3. Suggest what goes into supplementary material, and how to refer to it in the main text. 4. Return the compressed methods with the word count.
RULES	Don't change any definition, number or procedure, because the paper must describe exactly what the thesis did.
OUTPUT	The compressed methods, then the supplement list.

PRO TIP Cite your protocol or CTRI record for full details. It saves words legitimately.

TRY IT WITH A 4,000–word methods chapter from an MD thesis on AKI after cardiac surgery.

→ **NEXT #316** Write the full manuscript for journal submission

330 Three-paragraph journal introduction

STANDARD

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT

USE IT When converting your thesis introduction and ROL into a journal introduction.

PERSONA	You are a journal editor who expects introductions of about 400 words that lead straight to the question.
MY CONTEXT	Thesis introduction and gap statement: [PASTE] Key references I will cite (with numbers): [list] Objective: [one line]
TASK	1. Write paragraph 1: the problem and why it matters, with burden figures from my sources. 2. Write paragraph 2: what is known and what is not, with the specific gap. 3. Write paragraph 3: the objective, stated precisely. 4. Keep it to about 350–450 words.
RULES	Use only my references and figures, keeping [CITE] slots where they are missing, because journal introductions are checked against the reference list.
OUTPUT	The three paragraphs with the word count.

PRO TIP The last sentence of the introduction should read like the question your abstract answers.

TRY IT WITH Converting a 2,500-word thesis introduction and ROL on dengue shock predictors.

→ **NEXT #316** Write the full manuscript for journal submission

331 Condense my thesis discussion for a journal

STANDARD

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT

USE IT When your thesis discussion is far longer than the journal allows.

PERSONA	You are a journal editor who condenses discussions to the essential argument.
MY CONTEXT	Thesis discussion: [PASTE DISCUSSION] Target word count: [e.g. 1,000] The single main message of the paper: [one line]
TASK	1. Keep the structure: principal findings, key comparisons, explanation, implications, strengths and limitations, and conclusion. 2. Keep only the comparisons that matter for the main message. 3. Remove repetition of results and background. 4. Return the condensed discussion with the word count.
RULES	Keep citations with their claims and all numbers unchanged, and don't add new comparisons, because the paper must stay faithful to the thesis findings.
OUTPUT	The condensed discussion, then a list of the removed material.

PRO TIP A journal discussion usually cites 10–20 references, not the 60 in your thesis ROL.

TRY IT WITH A 3,500-word thesis discussion condensed to 1,100 words.

→ **NEXT #319** Pre-submission peer review of my manuscript

332 Rejected: what now?

STANDARD

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT / FACULTY

USE IT When a journal rejects your manuscript.

PERSONA	You are a mentor who has turned many rejections into publications.
MY CONTEXT	Rejection type: [desk rejection / after review] The editor's and reviewers' comments: [PASTE] My next-choice journals: [list]
TASK	1. Interpret the rejection: scope, quality or priority? 2. Identify the comments worth addressing before resubmitting elsewhere. 3. Make a revision plan in order of priority. 4. List the reformatting needed for the next journal. 5. Advise whether an appeal is reasonable (rarely), with reasons.
RULES	Be honest about fundamental flaws if the reviews identify them, because resubmitting an unchanged flawed manuscript wastes months.
OUTPUT	The interpretation, a prioritised revision plan, the reformatting list, then the appeal advice.

PRO TIP Take reviewers' comments seriously even after rejection. The next journal may send it to the same reviewer.

TRY IT WITH Rejected after review for "limited novelty and single-centre design".

→ **NEXT #322** Choose the right journal (and avoid predatory ones)

333 Suggest reviewers properly

STANDARD

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT / FACULTY

USE IT When the journal asks you to suggest reviewers.

PERSONA	You are an editor who knows how journals use and check suggested reviewers.
MY CONTEXT	Topic and methods of my paper: [brief] Authors of key papers I cited (from my reference list): [list] My co-authors' institutions and recent collaborators: [list]
TASK	1. Explain the criteria for a good suggested reviewer: expertise, independence and no recent collaboration. 2. Explain how to identify candidates from my reference list and PubMed. 3. Explain how to verify contact details from institutional pages. 4. Build a table of candidates from the authors I listed, with conflict checks.
RULES	Don't invent reviewer names or email addresses; work only from the authors I listed and tell me how to verify them, because fake or conflicted reviewer suggestions are treated as misconduct.
OUTPUT	The criteria, the method, then the candidate table.

PRO TIP Suggest reviewers from different countries and institutions. Editors trust diverse suggestions more.

TRY IT WITH Paper on high-flow nasal oxygen in COPD exacerbations.

→ **NEXT #324** Cover letter to the editor

334 Key points, highlights and "what this study adds"

STANDARD

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT

USE IT When the journal asks for highlights, key messages or a "what is known / what this adds" box.

PERSONA	You are a journal editor who writes crisp summary boxes.
MY CONTEXT	Journal's required format (number of points, character limits): [paste] Main findings with numbers: [paste] Gap addressed: [brief] Design and setting: [brief]
TASK	1. Write the box in the journal's required format: what is already known, what this study adds, and how it might affect practice or research. 2. Keep each point within the character limit and in plain language. 3. Check that nothing overclaims beyond the results, and that every number matches the abstract.
RULES	Keep claims proportionate to the design, because highlights are often quoted out of context.
OUTPUT	The box, then the character counts.

PRO TIP "What this adds" should be something a reader couldn't have said before your paper.

TRY IT WITH Findings: 1 in 3 elderly surgical patients developed delirium, and frailty predicted it.

→ **NEXT #320** Structured abstract within the word limit

335 Brief for a graphical abstract

STANDARD

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT

USE IT When the journal requests or encourages a graphical abstract.

PERSONA	You are a scientific illustrator who designs clear graphical abstracts.
MY CONTEXT	Question, design, population, main finding and conclusion: [brief] Journal's graphical-abstract specifications: [size, format] Tool I will use: [PowerPoint / Canva / BioRender]
TASK	1. Propose a left-to-right layout: population → method → key result → take-home. 2. Specify the icons, the text for each panel (under 15 words each) and the key numbers. 3. Suggest colours and fonts for readability. 4. List common mistakes to avoid.
RULES	Show the main result with its numbers accurately, and avoid visual exaggeration, because graphical abstracts are shared widely out of context.
OUTPUT	The panel-by-panel brief, then design notes and mistakes.

PRO TIP If it can't be understood in 10 seconds on a phone screen, simplify it.

TRY IT WITH RCT: early mobilisation vs usual care in ICU; outcome: delirium days.

→ **NEXT #334** Key points, highlights and "what this study adds"

336 Should I post a preprint?

STANDARD

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT / FACULTY

USE IT Before submission, if you are considering medRxiv or a similar server.

PERSONA	You are a publication strategist who knows preprint servers and journal policies.
MY CONTEXT	Type of study and any clinical implications: [brief] Target journals: [list] My reasons for considering a preprint: [brief]
TASK	1. Explain the benefits and risks of a preprint for my study. 2. Explain how to check whether my target journals accept preprinted manuscripts. 3. Explain the screening that medical preprint servers apply and what is required (e.g. ethics and registration statements). 4. Explain how to link the preprint to the final paper.
RULES	Mark journal-specific policies as [CHECK JOURNAL POLICY], because preprint policies vary and change.
OUTPUT	Benefits and risks, the policy check, the requirements, then the linking step.

PRO TIP Clinical findings in preprints can be misreported by the media. Word the conclusion carefully.

TRY IT WITH A diagnostic accuracy study of a new point-of-care test for TB.

→ **NEXT #322** Choose the right journal (and avoid predatory ones)

337 Write a letter to the editor about a published paper

STANDARD

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT / FACULTY

USE IT When you want to comment on a recently published article; it's often a first publication.

PERSONA	You are an experienced correspondent who writes concise, constructive letters that journals publish.
MY CONTEXT	The article I'm responding to (citation and key points): [PASTE] My comment: a methodological concern, an alternative interpretation, or relevant local data: [brief] Journal's letter limits (words, references, time window): [paste]
TASK	1. Structure the letter: reference to the article, the main point, supporting reasoning or data, and a constructive conclusion. 2. Write it within the limits. 3. Suggest a title.
RULES	Keep the tone collegial and evidence-based, and cite only sources I supply or [CITE], because letters that attack authors are rarely published.
OUTPUT	The title and the letter with its word count.

PRO TIP Submit quickly. Many journals only accept letters within a few weeks of publication.

TRY IT WITH Commenting on a trial that excluded patients with CKD, the majority in Indian practice.

→ **NEXT #324** Cover letter to the editor

338 Check the galley proofs properly

STANDARD

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT / FACULTY

USE IT When the journal sends proofs, usually with 48–72 hours to respond.

PERSONA	You are a meticulous production editor who knows what goes wrong at the proof stage.
MY CONTEXT	Proof PDF text or sections to check: [PASTE OR UPLOAD] My accepted manuscript for comparison: [PASTE OR UPLOAD]
TASK	1. Give a checklist: author names, affiliations and order; corresponding author details; title; abstract numbers; tables and figures (correct versions, legends, symbols); units and special characters; references; and declarations. 2. Compare the proof text with my accepted manuscript and list the differences. 3. Answer each typesetter's query clearly.
RULES	Request only corrections of errors, not new content, because journals refuse substantive changes at proof stage.
OUTPUT	The checklist, a difference list, then responses to the queries.

PRO TIP Check symbols ($\pm$, $\geq$, μ) and minus signs in tables. Typesetting often breaks them.

TRY IT WITH Proofs for an original article with 4 tables and a Kaplan–Meier figure.

→ **NEXT #319** Pre-submission peer review of my manuscript

339 Read and fix a similarity (plagiarism) report honestly

STANDARD

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT

USE IT When your university or journal returns a Turnitin, iThenticate or similar report.

PERSONA	You are a research-integrity officer who helps authors interpret similarity reports and fix genuine problems.
MY CONTEXT	Overall similarity percentage and the top matching sources: [PASTE SUMMARY] Highlighted passages with their matches: [PASTE PASSAGES] My university's or journal's threshold, if known: [e.g. 10%]
TASK	1. Separate acceptable matches (references, standard methods phrases, quotations with citation, my own registered protocol) from problematic ones (copied sentences without citation, close paraphrase). 2. For each problematic passage, explain the issue and how to rewrite it in my own words with the correct citation. 3. Advise how to report acceptable overlap, e.g. with my own thesis or preprint, and how to request exclusions properly.
RULES	Fix plagiarism by genuine rewriting and correct citation, never by tricks that hide text from the checker, because those are themselves misconduct. Mark institutional thresholds and rules as [CHECK YOUR UNIVERSITY / JOURNAL POLICY].
OUTPUT	A passage table (passage, match type, acceptable or not, fix), then the reporting advice.

PRO TIP Write methods and introduction in your own words from the start. Fixing overlap later takes far longer.

TRY IT WITH A 24% similarity report where most matches are from the methods and introduction.

→ **NEXT #82** Paraphrase and synthesise sources responsibly

340 Running title under the limit

QUICK

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT

Act as a journal editor. Write 3 running titles for my paper, [FULL TITLE], each under [e.g. 50] characters including spaces, keeping the core topic and population recognisable and avoiding abbreviations the journal hasn't defined. Show the character count for each.

TRY IT WITH Frailty and 90-day mortality after emergency laparotomy in older Indian adults.

341 Word budget for each section

QUICK

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT

Act as a manuscript editor. For a [ARTICLE TYPE] with a [WORD LIMIT] word limit, give a word budget for the introduction, methods, results and discussion, adjusted to my design, [DESIGN], and tell me which section usually runs over and how to control it.

TRY IT WITH Original article, 3,000 words, prospective cohort.

342 Cut my abstract to the limit

QUICK

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT

Act as an abstract editor. Cut this abstract, [PASTE ABSTRACT], from [CURRENT] to [TARGET] words without removing the design, setting, sample size, primary result with its numbers, or conclusion. Show the new word count and list the phrases you removed.

TRY IT WITH A 310-word abstract that must fit a 250-word limit.

343 Summarise my paper for social media

QUICK

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT / FACULTY

Act as a science communicator for doctors. Write a 3-post thread (each under 280 characters) summarising my published paper, [TITLE, KEY FINDING, DOI], accurately and without overclaiming, ending with the DOI link. Add one plain-language version for patients and families.

TRY IT WITH Paper on lung ultrasound for diagnosing pneumonia in children.

344 Cut my references to the journal's limit

QUICK

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT

Act as a manuscript editor. My paper has [N] references and the limit is [LIMIT]. From this list with the claims each supports, [PASTE LIST], recommend which to remove or combine, keeping primary sources for key claims and the most recent evidence.

TRY IT WITH 62 references for a journal with a 40-reference limit.

345 Write the acknowledgements

QUICK

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT

Act as an academic editor. Write a concise, professional acknowledgements paragraph thanking [LIST PEOPLE AND ROLES] who don't meet authorship criteria, plus all participants and any funding body, in under 80 words, warmly but formally, without claiming their endorsement of the conclusions.

TRY IT WITH Nursing staff, the lab technician, the statistician and all participants.

346 Polite status enquiry to the editor

QUICK

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT / FACULTY

Act as a senior author. Write a short, polite email to the editorial office of [JOURNAL] asking about the status of my manuscript [ID, TITLE], submitted on [DATE], which has been "under review" for [DURATION]. Keep it under 80 words.

TRY IT WITH Manuscript under review for 4 months with no update.

347 Data-availability statement

QUICK

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT

Act as a research-data editor. Write a data-availability statement for my study, [BRIEF DESCRIPTION], that reflects my plan, [SHARING PLAN], respects participant confidentiality and IEC conditions, and meets typical journal requirements. Give one short and one detailed version, and say which the journal likely prefers.

TRY IT WITH De-identified data available on reasonable request, subject to IEC approval.

348 Plain-language summary of my paper

QUICK

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT / FACULTY

Act as a patient-education writer. Write a 150-word plain-language summary of my paper, [PASTE ABSTRACT], for patients and the public: why the study was done, what was done, what was found and what it means, with no jargon and no overstatement.

TRY IT WITH Abstract of a study on inhaler technique errors in asthma patients.

349 Reply to one tough reviewer comment

QUICK

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT / FACULTY

Act as a senior author. Draft a courteous, specific reply to this reviewer comment, [PASTE COMMENT], given my position, [AGREE / PARTLY AGREE / DISAGREE AND WHY], stating exactly what I changed and where, or why I respectfully did not. Keep it under 150 words.

TRY IT WITH "The single-centre design severely limits generalisability; the conclusions should be tempered."

350 Is splitting my thesis into two papers acceptable?

QUICK

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT / FACULTY

Act as a publication-ethics adviser. I plan two papers from one dataset: [PAPER 1 QUESTION] and [PAPER 2 QUESTION]. Tell me whether this is legitimate or salami slicing, why, and how to cross-reference the papers and disclose the shared dataset.

TRY IT WITH Paper 1 on mortality predictors; paper 2 on the same predictors for ICU length of stay.

1.12

Case reports & case series

Most case reports are rejected because the case isn't novel or the timeline is unclear, not because the writing is weak. Check publishability first, then write to CARE.

30

prompts

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Master

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Standard

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Quick

#351 From interesting case to CARE-compliant case report

#352 Design and write a case series

#353 Case-report discussion grounded in the literature

#354 Is my case publishable?

#355 Build the case timeline table

#356 Consent for publication and de-identification

#357 Select and prepare clinical images

#358 Case report abstract

#359 Case report titles that follow the CARE convention

#360 Write the case presentation from my notes

#361 Show how differentials were excluded

#362 Learning points that readers remember

#363 "Images in clinical medicine" submission

#364 Journals for case reports: find and verify them

#365 Patient-level table for a case series

#366 Find previously reported similar cases

#367 Patient-perspective section

#368 Case report → conference poster

#369 Critique my case report draft

#370 Adverse drug reaction case report with causality assessment

#371 Clinicopathological conference (CPC) write-up

#372 One line on why my case is worth reading

#373 Keywords to search for similar cases

#374 Is my case report properly anonymised?

#375 Cut my case report to the word limit

#376 Legend for a clinical image

#377 Turn my case into a teaching caption

#378 Explain publication consent to my patient

#379 Case report or case series?

#380 The key sentence of my case-report cover letter

From interesting case to CARE-compliant case report

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT / FACULTY

USE IT

As soon as you suspect a case is unusual, while notes, images and consent are still easy to get.

WHY

Most case reports are rejected because the case isn't novel or the timeline is unclear, not because the writing is weak. This prompt checks publishability first, then drafts to the CARE guideline.

COPY THE PROMPT BELOW

PERSONA	You are the case-reports editor of an indexed [SPECIALTY] journal. You have handled hundreds of submissions and apply the CARE guidelines strictly. You are blunt about whether a case is actually publishable.
MY CONTEXT	Specialty and my level: [e.g. MD Respiratory Medicine, PG year 2] Setting: [tertiary / district / private] De-identified case summary (age band, sex, presentation, key investigations with dates as Day 0, Day 3..., treatment, outcome, follow-up): [PASTE] Why I think it is unusual: [state] Target journal and its case-report word limit: [name / limit] Patient consent for publication: [obtained / pending]
TASK	1. Publishability verdict: rare disease, rare presentation, new association, diagnostic lesson or management lesson? Score novelty out of 5 and justify it against my case. 2. Five title options that each contain the words "case report". 3. A CARE-structured draft: abstract (150 words or fewer), introduction, patient information, clinical findings, timeline table, diagnostic assessment (including the differentials you excluded and how), therapeutic intervention, follow-up and outcome, discussion, patient-perspective placeholder, and learning points. 4. A CARE checklist table showing where each item appears, flagging anything missing. 5. The consent and de-identification checklist I need before submission.
RULES	Use only the clinical details I gave; where something is missing, write [ASK ME: ...] instead of filling it in. Put [CITE] wherever literature is needed, and cite a paper only if I uploaded it or you can give a verified PMID or DOI, because fabricated references end in retraction. Keep every date relative (Day 0...) and remove anything identifying.
OUTPUT	Formal case-report register within my journal's word limit, then the two checklist tables.
TWEAK	Finish with the 3 comments a reviewer is most likely to make, and how to pre-empt each.

How to get more out of it

- If the verdict says "not novel", reframe it as a teaching image or a letter rather than forcing a full report.
- Build the timeline table first. Reviewers read it before the text.
- Upload 3–5 similar published cases so the discussion compares against real papers.
- → **NEXT #368** Case report → conference poster

Design and write a case series

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT / FACULTY

USE IT

When you have 3 or more patients with a shared unusual feature, diagnosis or treatment experience.

WHY

A case series is more than several case reports stapled together. Its value lies in a clear case definition, consistent data for every patient, and patterns across cases, presented honestly as descriptive evidence without a comparison group.

COPY THE PROMPT BELOW

PERSONA	You are a clinical researcher and journal editor who has published and reviewed many case series. You insist on a clear case definition, consecutive or complete inclusion where possible, and descriptive (not inferential) claims.
MY CONTEXT	Specialty and setting: [brief] The shared feature defining the series: [e.g. post-COVID mucormycosis with orbital involvement] Number of patients and the period: [N, dates] How the patients were identified (all consecutive cases, or selected): [brief] De-identified data for each patient (age band, sex, presentation, investigations, treatment, outcome, follow-up): [PASTE TABLE] Target journal and word limit: [name / limit] Ethics and consent status: [brief] Any patients excluded, and why: [brief]
TASK	1. Write a precise case definition and describe how cases were identified, including any selection. 2. Build a patient-level table (one row per patient) with consistent variables. 3. Summarise the patterns descriptively: counts, proportions, ranges and medians, and common and divergent features. 4. Draft the manuscript: abstract, introduction, methods (definition, identification, data sources, ethics), case descriptions or a summary table, results, discussion comparing with the literature, limitations, and conclusion. 5. State clearly what a case series can and cannot conclude.
RULES	Use only the data I gave. Avoid p values and causal claims, because a case series has no comparison group. Put [CITE] for literature unless I uploaded the paper or you give a verified PMID or DOI. Keep all details de-identified.
OUTPUT	The case definition, the patient table, the pattern summary, then the manuscript draft with headings.
TWEAK	End with the single most important pattern and the study design that would test it properly.

How to get more out of it

- › Include all eligible cases from the period, not only the interesting ones. Say how many were excluded, and why.
- › A clear, complete patient-level table is often the most-cited part of a case series.
- › Check whether your IEC requires approval for a retrospective case series. Many do.
- › → **NEXT #365** Patient-level table for a case series

Case-report discussion grounded in the literature

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT

USE IT

When the case description is written and you need the discussion and learning points.

WHY

The discussion is where a case report earns its place; it shows what the case adds to what is already published. Comparing with real, similar cases rather than a textbook summary is what reviewers look for.

COPY THE PROMPT BELOW

PERSONA	You are a senior clinician-academic who writes case-report discussions that teach. You compare the case with published cases systematically and draw practical lessons without overgeneralising.
MY CONTEXT	Case summary (de-identified): [PASTE] What makes it unusual: [brief] Similar published cases I have found (uploaded PDFs or pasted abstracts with citations): [PASTE OR UPLOAD] Target word count for the discussion: [e.g. 500 words] My specialty and the journal's readership: [brief] The diagnostic or management decision that was hardest in my case: [brief]
TASK	1. Summarise the published cases in a comparison table: age, sex, presentation, diagnosis, treatment and outcome, next to my case. 2. Identify what is similar, what differs, and what my case adds. 3. Discuss the diagnostic reasoning: why it was difficult, which clues mattered, which differentials were excluded and how. 4. Discuss management choices and the outcome in the light of the literature, including what might be done differently. 5. Write 3–5 learning points, each practical, specific and supported by the case. 6. Write the discussion within the word count, starting with the key reason the case is notable. 7. Suggest 2 recent reviews or guidelines I should search for and cite, described by topic rather than by title.
RULES	Compare only with the cases I supplied, citing each, because inventing similar cases is fabrication. Don't generalise from one case to practice recommendations; phrase lessons as "clinicians should consider..." rather than "should treat..."
OUTPUT	The comparison table, the discussion text, then the learning points.
TWEAK	End with the one sentence that best captures why this case deserves publication.

How to get more out of it

- › Search PubMed with the "Case Reports" article-type filter to find similar cases. Read them in full.
- › If very similar cases have been published recently, emphasise what differs about yours.
- › Learning points are often what readers remember. Make them specific enough to act on.
- › → **NEXT #362** Learning points that readers remember

354 Is my case publishable?

STANDARD

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT / FACULTY

USE IT Before you invest time writing, the moment a case seems interesting.

PERSONA	You are a case-reports editor who rejects most submissions for lack of novelty and knows what gets accepted.
MY CONTEXT	De-identified case summary: [PASTE] What I think is unusual: [brief] Similar cases I have found so far: [brief or "not searched yet"]
TASK	1. Classify the case's potential contribution: rare disease, rare presentation of a common disease, new association, diagnostic lesson, therapeutic lesson, adverse drug reaction, or educational value only. 2. Score novelty and educational value out of 5 each. 3. Recommend the best format: full case report, clinical image, letter, or teaching case only. 4. List the information and images I must collect now, before they are lost.
RULES	Base novelty on the case itself; where it depends on the literature, write [SEARCH PUBMED CASE REPORTS FIRST], because "first reported case" claims are often wrong.
OUTPUT	The classification, the scores, the format recommendation, then the collection checklist.

PRO TIP Collect the images, reports and consent now. Most unpublished case reports die because the data went missing.

TRY IT WITH Young woman with recurrent pneumothorax found to have thoracic endometriosis.

→ **NEXT #351** From interesting case to CARE-compliant case report

355 Build the case timeline table

STANDARD

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT

USE IT For every case report, before writing the narrative.

PERSONA	You are a meticulous clinical writer who builds CARE-style timelines.
MY CONTEXT	De-identified clinical events with their dates or intervals: [PASTE NOTES] Reference point (Day 0): [e.g. admission / symptom onset]
TASK	1. Convert all dates to relative days, weeks or months from Day 0. 2. Build a timeline table: time point, clinical event, investigations and key results, intervention, and response. 3. Flag gaps or inconsistencies in the sequence. 4. Suggest which time points need an image or a result shown.
RULES	Remove all actual calendar dates and identifying details, because timelines are a common source of accidental identification.
OUTPUT	The timeline table, then the flags and the image suggestions.

PRO TIP A clear timeline often lets you cut 200 words from the case narrative.

TRY IT WITH Notes on a patient with fever, jaundice and acute kidney injury later diagnosed with leptospirosis.

→ **NEXT #360** Write the case presentation from my notes

356 Consent for publication and de-identification

STANDARD

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT

USE IT Before submitting any case report or clinical image.

PERSONA	You are a publication-ethics editor who ensures patient privacy in case reports.
MY CONTEXT	Case details and images I plan to publish: [brief] Patient status: [able to consent / deceased / minor / lacks capacity] Journal's consent requirements (paste if available): [PASTE]
TASK	1. Explain what consent for publication must cover and who can give it in my patient's situation. 2. List everything that must be removed or masked: names, initials, dates, hospital numbers, locations, rare occupations, faces, tattoos, and metadata in image files. 3. Explain what to do if consent cannot be obtained, marked [CHECK JOURNAL POLICY]. 4. Draft a plain-language explanation to give the patient.
RULES	Treat consent as required unless the journal explicitly states otherwise, because publishing without it breaches patient rights and journal policy. Note that eye bars do not adequately de-identify faces.
OUTPUT	The consent explanation, the de-identification checklist, the no-consent guidance, then the patient-facing text.

PRO TIP Use the journal's own consent form if it has one. Many journals insist on it.

TRY IT WITH A case report with a facial photograph of a rare dermatological condition.

→ **NEXT #357** Select and prepare clinical images

357 Select and prepare clinical images

STANDARD

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT

USE IT When preparing images, radiographs or histology slides for a case report or poster.

PERSONA	You are a medical illustrator and journal image editor.
MY CONTEXT	Available images and what each shows: [list] Key finding each image must demonstrate: [brief] Journal requirements (resolution, format, size): [paste]
TASK	1. Recommend which images to include and in what order to tell the diagnostic story. 2. Explain how to crop, annotate (arrows, labels) and de-identify each image. 3. Explain file requirements (e.g. 300 dpi, TIFF or JPEG) and how to remove metadata. 4. Write a legend for each image.
RULES	Tell me to make only adjustments applied to the whole image (brightness, contrast) and never to alter findings, because image manipulation is misconduct.
OUTPUT	The image selection plan, the preparation steps, then the legends.

PRO TIP Remove patient details burned into radiographs and scan images before anything else.

TRY IT WITH Chest X-ray, CT and histology images for a case of pulmonary alveolar proteinosis.

→ **NEXT #356** Consent for publication and de-identification

358 Case report abstract

STANDARD

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT

USE IT When your case report needs an abstract.

PERSONA	You are a case-reports editor who writes clear abstracts.
MY CONTEXT	Case report draft or summary: [PASTE] Journal's abstract format (structured or unstructured) and word limit: [paste] The single lesson the case teaches: [one line] Target readership: [e.g. general physicians]
TASK	1. Write the abstract in the journal's required format: background (why the case matters), case presentation (key features), conclusion (the lesson). 2. Keep it within the limit. 3. Suggest 3–5 keywords, preferring MeSH terms.
RULES	Use only details in the draft, and make the conclusion a lesson rather than a generalisation, because one case cannot establish a rule.
OUTPUT	The abstract with its word count, then the keywords.

PRO TIP Put the unusual feature in the first sentence. That is what makes an editor keep reading.

TRY IT WITH Case of Kikuchi disease presenting as fever of unknown origin in a young man.

→ **NEXT #359** Case report titles that follow the CARE convention

359 Case report titles that follow the CARE convention

STANDARD

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT

USE IT When choosing your case report's title.

PERSONA	You are a journal editor who knows what titles attract readers and meet CARE conventions.
MY CONTEXT	Diagnosis and the unusual feature: [brief] Journal's title limits: [characters or words] Target readership: [e.g. general physicians / specialists]
TASK	1. Write 5 titles that include "case report" (or "a case report"). 2. Put the key diagnosis or phenomenon first. 3. Avoid overclaiming words such as "first", "unique" or "novel" unless verified. 4. Recommend one, with the reason.
RULES	Mark any "first report" claim as [VERIFY WITH LITERATURE SEARCH], because unverified priority claims are a common reason for reviewer criticism.
OUTPUT	Five titles with character counts, then the recommendation.

PRO TIP Titles that describe the lesson ("...mimicking...", "...presenting as...") are read more often.

TRY IT WITH Tuberculous pericarditis presenting as cardiac tamponade in a patient on dialysis.

→ **NEXT #358** Case report abstract

360 Write the case presentation from my notes

STANDARD

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT

USE IT For the main narrative of a case report.

PERSONA	You are a clinical writer who produces concise, chronological case presentations.
MY CONTEXT	De-identified notes: history, examination, investigations with results, management, and course: [PASTE NOTES] Word limit for this section: [e.g. 400 words]
TASK	1. Write the case presentation chronologically: presenting complaint, relevant history, examination, initial investigations, diagnostic reasoning and further tests, treatment, and outcome. 2. Include only details relevant to the diagnosis or lesson. 3. Report laboratory values with units and reference ranges where helpful. 4. Point to the timeline table and images at the right places.
RULES	Use only facts in my notes; where a relevant detail is missing, write [ASK ME: ...] , because invented clinical details in a case report are fabrication.
OUTPUT	The case presentation with its word count, then any [ASK ME] items listed.

PRO TIP Negative findings that ruled out important differentials are relevant. Include them.

TRY IT WITH Notes on a 30-year-old with stroke found to have antiphospholipid syndrome.

→ **NEXT #361** Show how differentials were excluded

361 Show how differentials were excluded

STANDARD

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT

USE IT For the diagnostic-assessment part of a case report.

PERSONA	You are a diagnostician who documents diagnostic reasoning transparently.
MY CONTEXT	Final diagnosis: [diagnosis] Differentials considered and the tests or findings used to exclude each: [PASTE NOTES] The investigation that finally confirmed the diagnosis: [brief]
TASK	1. Build a table: differential, why it was considered, the finding that excluded it, and the timing. 2. Identify any important differential that is not documented as excluded. 3. Write the diagnostic-assessment paragraph for the case report.
RULES	Use only the tests and findings I documented; flag any differential without a documented exclusion as [NOT EXCLUDED IN NOTES] , because reviewers will ask about it.
OUTPUT	The differential table, the missing-exclusion flags, then the paragraph.

PRO TIP Reviewers almost always ask "was X excluded?". This table answers it before they do.

TRY IT WITH Final diagnosis: adult-onset Still's disease; differentials: sepsis, lymphoma, SLE, TB.

→ **NEXT #353** Case-report discussion grounded in the literature

362 Learning points that readers remember

STANDARD

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT

USE IT For the "learning points" or "key messages" box of a case report.

PERSONA	You are a medical educator who writes practical, memorable learning points.
MY CONTEXT	Case summary and the lesson I think it teaches: [brief] Journal's format (number of points, word limit): [paste] Intended readers: [e.g. residents, general practitioners]
TASK	1. Write 3–5 learning points, each one sentence. 2. Make each specific and actionable (e.g. "Consider X in patients with Y who do not respond to Z"). 3. Avoid generic points ("a high index of suspicion is needed"). 4. Check that each point follows from the case.
RULES	Phrase points as considerations, not rules, because a single case can't establish practice.
OUTPUT	The learning points, then a one-line check of each.

PRO TIP If a point could apply to any case report, it's too generic. Rewrite it.

TRY IT WITH Case of methaemoglobinaemia after dapsone use, initially misdiagnosed as cyanotic heart disease.

→ **NEXT #353** Case-report discussion grounded in the literature

363 "Images in clinical medicine" submission

STANDARD

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT / FACULTY

USE IT When a case has one striking image but may not justify a full case report.

PERSONA	You are an editor of a clinical-images section who selects images that teach at a glance.
MY CONTEXT	The image(s) and what they show: [describe] Brief clinical context: [2–3 lines] Journal's clinical-image format (word limit, number of images, references): [paste]
TASK	1. Judge whether the image is teaching quality: classic, striking and clearly showing the finding. 2. Write the text within the limit: a brief clinical context, what the image shows, the diagnosis, and the outcome. 3. Write the image legend with annotations described. 4. Suggest a title.
RULES	Use only the details I gave; de-identify fully and mark consent as required, because image sections have the same consent requirements as case reports.
OUTPUT	The judgement, the title, the text, then the legend.

PRO TIP Image sections are a quick first publication for residents. The image must be excellent.

TRY IT WITH Chest X-ray showing a classic "air crescent" sign in invasive aspergillosis.

→ **NEXT #356** Consent for publication and de-identification

364 Journals for case reports: find and verify them

STANDARD

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT

USE IT When deciding where to submit a case report.

PERSONA	You are a publication adviser who knows the case-report publishing landscape and its predatory corners.
MY CONTEXT	Specialty and the type of case: [brief] Budget for publication charges: [amount or none] Indexing needed: [PubMed / Scopus / none required]
TASK	1. Describe the types of venues that publish case reports: specialty journals with case sections, dedicated case-report journals, and clinical-image sections. 2. Explain how to find candidates and verify their indexing and legitimacy. 3. List predatory red flags specific to case-report publishing. 4. Build a shortlist table from the journals I find.
RULES	Don't name journals, indexing status or charges from memory; mark anything named as [VERIFY], because case-report publishing has many predatory outlets.
OUTPUT	Venue types, finding and verification steps, red flags, then the shortlist template.

PRO TIP Check your own specialty society's journal first. Many accept case reports.

TRY IT WITH Unusual paediatric neurology case, no publication budget, PubMed indexing preferred.

→ **NEXT #351** From interesting case to CARE-compliant case report

365 Patient-level table for a case series

STANDARD

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT

USE IT For the central table of your case series.

PERSONA	You are a clinical data editor who designs clear patient-level tables.
MY CONTEXT	Variables available for all patients: [list] Data for each patient (de-identified): [PASTE]
TASK	1. Choose the columns that show the key features consistently for every patient. 2. Build the table with one row per patient (Case 1, Case 2...). 3. Add a summary row or column (counts, ranges or medians). 4. Write the table title and footnotes.
RULES	Keep all data de-identified using age bands if needed, and write NA where data is unavailable rather than leaving blanks, because missing cells must be explained.
OUTPUT	The table, then the title and footnotes.

PRO TIP If the table is too wide, split it into clinical features and management/outcome tables.

TRY IT WITH Seven patients with scrub typhus meningoencephalitis.

→ **NEXT #352** Design and write a case series

366 Find previously reported similar cases

STANDARD

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT

USE IT Before writing the discussion of a case report or series.

PERSONA	You are a medical librarian who searches for case reports efficiently.
MY CONTEXT	The key features of my case (diagnosis, unusual presentation, association): [brief]
TASK	1. Build a PubMed search string combining the key concepts with the "Case Reports" publication type. 2. Suggest Google Scholar queries for non-indexed reports. 3. Explain how to screen the results and extract comparison data (age, sex, presentation, treatment, outcome). 4. Give a comparison-table template.
RULES	Don't list case reports from memory; give me the search to run, because any list you produce without live search may contain fabricated cases.
OUTPUT	The search strings, the screening steps, then the table template.

PRO TIP Check the reference lists of the most recent similar case reports. They are often mini-reviews.

TRY IT WITH Case: Castleman disease presenting as a posterior mediastinal mass.

→ **NEXT #353** Case-report discussion grounded in the literature

367 Patient-perspective section

STANDARD

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT

USE IT For the CARE "patient perspective" item, when the patient agrees to contribute.

PERSONA	You are a patient-centred clinician who helps patients describe their experience in their own words.
MY CONTEXT	Brief case summary: [brief] Notes from my conversation with the patient (their own words where possible): [PASTE NOTES] Language the patient spoke in: [language]
TASK	1. Suggest 5 gentle questions to ask the patient about their experience (symptoms, diagnosis journey, treatment and recovery). 2. From my notes, draft a short patient-perspective paragraph in the first person, faithful to their words. 3. Suggest how to have the patient review and approve the text.
RULES	Keep the patient's meaning and words; don't add emotions or statements they did not express, because this section must be genuinely theirs.
OUTPUT	The questions, the draft paragraph, then the approval steps.

PRO TIP Translate carefully and read the translation back to the patient before you use it.

TRY IT WITH A patient who went undiagnosed for 2 years with a rare autoimmune condition.

→ **NEXT #351** From interesting case to CARE-compliant case report

368 Case report → conference poster

STANDARD

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT

USE IT When presenting your case at a conference as a poster or e-poster.

PERSONA	You are a conference poster designer who turns case reports into posters people stop to read.
MY CONTEXT	Case report text or summary: [PASTE] Key images: [list] Poster size and format: [e.g. A0 portrait / e-poster 16:9] Conference word limit, if any: [limit]
TASK	1. Plan the panels: title, introduction (2 lines), case presentation (as a timeline or bullet points), key images, diagnosis, discussion (3 points), learning points, and references. 2. Write the text for each panel within about 400–600 words in total. 3. Suggest the layout and reading flow, with images central. 4. Suggest a one-line "hook" for the top of the poster.
RULES	Use only details from my case report; keep images de-identified and consent in place, because posters are photographed and shared.
OUTPUT	A panel-by-panel plan with text, then layout notes.

PRO TIP People read posters from 2 metres away. If a panel needs a paragraph, it needs cutting.

TRY IT WITH Case report of a rare paediatric pulmonary condition with CT images.

→ **NEXT #382** Research poster from my manuscript or thesis

369 Critique my case report draft

STANDARD

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT

USE IT When your case report is drafted, before submission.

PERSONA	You are a reviewer for a case-reports journal who applies CARE and the journal's standards.
MY CONTEXT	My draft: [PASTE DRAFT] Target journal and word limit: [brief]
TASK	1. Score the draft on novelty, clarity of the timeline, completeness of the diagnostic reasoning, quality of the literature comparison, strength of the learning points, and CARE compliance. 2. Quote the passage behind each score. 3. List the top 5 fixes. 4. Flag privacy risks and missing consent statements.
RULES	Base comments only on my text; quote rather than paraphrase, because vague feedback can't be acted on.
OUTPUT	A scoring table, the top fixes, then the privacy flags.

PRO TIP Run this again after revising. The timeline and diagnostic-reasoning scores usually rise the most.

TRY IT WITH A 1,400-word draft case report of paraquat poisoning with delayed lung fibrosis.

→ **NEXT #351** From interesting case to CARE-compliant case report

370 Adverse drug reaction case report with causality assessment

STANDARD

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT / FACULTY

USE IT When a case involves a suspected adverse drug reaction.

PERSONA	You are a clinical pharmacologist experienced in pharmacovigilance and ADR case reporting.
MY CONTEXT	Suspected drug(s), doses, start and stop dates (relative), reaction, timeline, dechallenge and rechallenge, and other possible causes: [PASTE DE-IDENTIFIED DETAILS]
TASK	1. Apply a causality assessment (e.g. WHO-UMC categories and the Naranjo scale), showing each criterion and its score. 2. Assess severity and seriousness, and whether the reaction is expected (labelled). 3. Draft the ADR case report following CARE and relevant ADR reporting guidance. 4. Remind me to report to the national pharmacovigilance programme, marked [VERIFY CURRENT REPORTING PROCESS].
RULES	Show the reasoning for each causality criterion from my details only, and mark unknown elements as [NOT DOCUMENTED], because causality scores are often challenged.
OUTPUT	The causality table, the severity assessment, the draft, then the reporting reminder.

PRO TIP Report to the pharmacovigilance programme even if you don't publish. Every report counts.

TRY IT WITH Suspected linezolid-induced lactic acidosis in a patient on TB treatment.

→ **NEXT #351** From interesting case to CARE-compliant case report

371 Clinicopathological conference (CPC) write-up

STANDARD

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT / FACULTY

USE IT When presenting or publishing a clinicopathological conference case, where the diagnosis is revealed at the end.

PERSONA	You are a senior physician and pathologist who has run CPCs for years, and knows how to build diagnostic suspense while staying fair to the reader.
MY CONTEXT	De-identified case details in chronological order (history, examination, investigations, course): [PASTE] Final diagnosis and how it was established (histopathology, autopsy or test): [brief] Audience: [residents / faculty / mixed]
TASK	1. Write the case protocol as it would be presented, revealing information in the order the clinicians received it, without giving away the diagnosis. 2. Write the discussant's differential-diagnosis reasoning, narrowing step by step to the most likely diagnosis. 3. Write the pathology or final-diagnosis section with key findings. 4. Add 3 teaching points and 2 questions for the audience.
RULES	Use only the details I gave, and keep the clinical clues fair, because a CPC that hides key data teaches nothing. Keep everything de-identified.
OUTPUT	The case protocol, the discussant's analysis, the final diagnosis, then the teaching points and questions.

PRO TIP Pick a case where the diagnosis could have been reached from the clinical data. Those make the best CPCs.

TRY IT WITH Fever, pancytopenia and splenomegaly in a young man, finally diagnosed as visceral leishmaniasis on marrow.

→ **NEXT #361** Show how differentials were excluded

372 One line on why my case is worth reading

QUICK

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT

Act as a case-reports editor. From my case summary, [PASTE SUMMARY], write one clear sentence of under 30 words that states exactly what the case adds (a rare presentation, association, diagnostic lesson or management lesson), without unverified "first report" claims.

TRY IT WITH A young non-smoker with pulmonary Langerhans cell histiocytosis diagnosed on cryobiopsy.

373 Keywords to search for similar cases

QUICK

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT

Act as a medical librarian. For my case, [KEY FEATURES], give me 8 useful search terms and synonyms (including older names and MeSH terms) and one ready-to-run PubMed search string limited to case reports, to find previously published similar cases.

TRY IT WITH Hypokalaemic periodic paralysis as the presenting feature of thyrotoxicosis.

374 Is my case report properly anonymised?

QUICK

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT

Act as a privacy reviewer for a medical journal. Check this case text and image list, [PASTE TEXT AND IMAGE LIST], for anything that could identify the patient: dates, places, occupation, rare details, faces, tattoos and image metadata. List each risk with a fix.

TRY IT WITH A case mentioning the village name, the exact admission date and a distinctive tattoo.

375 Cut my case report to the word limit

QUICK

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT

Act as a case-reports editor. Cut this case report, [PASTE TEXT], from [CURRENT] to [TARGET] words, keeping the timeline, key investigations, the excluded differentials and the learning points, and moving detail into the timeline table. Show the new word count.

TRY IT WITH A 1,600-word case report for a journal with a 1,000-word limit.

376 Legend for a clinical image

QUICK

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT

Act as a journal image editor. Write a legend for this image, [DESCRIBE IMAGE AND FINDINGS], that states the modality or type, the view, what the arrows or labels point to, and the key finding, in under 50 words, with abbreviations defined.

TRY IT WITH Contrast CT showing a ring-enhancing lesion with a target sign in the right frontal lobe.

377 Turn my case into a teaching caption

QUICK

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT / FACULTY

Act as a medical educator. Turn my case, [BRIEF CASE], into a 60-word teaching caption for a department WhatsApp group or case-of-the-week slide, written for junior residents: the presentation, the key finding, the diagnosis and one learning point, fully de-identified.

TRY IT WITH Elderly man with hiccups for 2 weeks found to have lateral medullary infarct.

378 Explain publication consent to my patient

QUICK

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT

Act as a compassionate clinician. Write a short, simple explanation (under 100 words) that I can say to my patient, [BRIEF CONTEXT], explaining why I want to publish their case, what will be shared, how their identity will be protected, and that they can refuse.

TRY IT WITH A patient who recovered from a rare poisoning, speaking Hindi.

379 Case report or case series?

QUICK

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT

Act as a journal editor. I have [N] patients with [SHARED FEATURE]. Tell me whether to write a single case report, a case series or a clinical image, what each would need, and which journals typically prefer which, in under 120 words.

TRY IT WITH Three patients with the same rare drug reaction in 6 months.

380 The key sentence of my case-report cover letter

QUICK

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT

Act as a senior author. Write the 2 key sentences of a cover letter to the editor for my case report, [TITLE AND ONE-LINE SUMMARY], explaining why the case matters to this journal's readers, [JOURNAL], without exaggerating novelty or making unverified priority claims.

TRY IT WITH Case of tuberculous mastitis mimicking breast carcinoma in a young woman.

SECTION 1.13

1.13

Conference abstracts, posters & oral papers

A conference presentation is often your research's first audience. Write to the exact limit, design for two metres away, and rehearse the questions.

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prompts

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Quick

#381 Conference abstract to the exact word limit

#382 Research poster from my manuscript or thesis

#383 Thesis → 8-minute oral paper, with slides and Q&A

#384 Choose the right conference for my work

#385 Score my abstract like a conference reviewer

#386 Adapt my poster into an e-poster

#387 Layout brief for Canva or PowerPoint

#388 90-second poster walk-through for judges

#389 Slide plan for a short oral presentation

#390 Prepare for questions after my talk

#391 Prepare for an award-paper session

#392 Application for a travel grant or bursary

#393 From conference presentation to published paper

#394 Record a presentation for a virtual or hybrid conference

#395 An accurate, catchy abstract title

#396 25-word summary for the conference programme

#397 One-page handout or QR page for my poster

#398 Will my talk fit the time?

#399 The closing take-home slide

#400 Follow-up email after the conference

Conference abstract to the exact word limit

1.13 CONFERENCE ABSTRACTS, POSTERS & ORAL PAPERS · PG RESIDENT / FACULTY

USE IT

Before a conference abstract deadline, once your results (or interim results) are verified.

WHY

Conference abstracts are judged in minutes by reviewers scoring dozens of submissions. A clear question, a stated design, real numbers and a proportionate conclusion, within the exact limit and format, is what gets accepted, and often what gets you into the award session.

COPY THE PROMPT BELOW

PERSONA	You are a scientific committee member for national and international medical conferences who scores hundreds of abstracts each year. You know what reviewers reward and what gets rejected.
MY CONTEXT	Conference name and its abstract rules (word or character limit, headings, whether the title and authors count, category options): [PASTE RULES] Study design, setting and period: [brief] Sample size and participant flow: [numbers] Primary result with estimate, CI and p: [paste exactly] Key secondary results: [paste] Whether the data is final or interim: [final / interim with n so far]
TASK	1. Draft the abstract in the conference's exact format and headings, within the limit, stating the count used (words or characters, as the conference counts them). 2. Background: the problem and the gap in 2 sentences. Methods: design, setting, participants, key measurements and analysis. Results: the numbers that answer the objective. Conclusion: proportionate, and linked to practice or research. 3. Write 3 title options within any title limit. 4. Recommend the best category, and whether to opt for oral, poster or award consideration. 5. Score the draft against typical reviewer criteria (importance, methods, clarity of results, validity of the conclusion, presentation) and revise the weakest part.
RULES	Use only my numbers, reproduced exactly; label interim data as interim, because presenting interim results as final is misconduct. No references unless the conference allows them. No "results will be presented" placeholders, because reviewers reject abstracts without results.
OUTPUT	The final abstract with its count, the titles, the category advice, then the scoring table.
TWEAK	End with the one sentence a reviewer is most likely to criticise, and a safer rewrite.

How to get more out of it

- › Check how the conference counts: words or characters, and whether spaces, title and authors are included.
- › Submit a day early. Portals slow down or crash near deadlines.
- › Keep the abstract text. The journal version later must not contradict it.
- › → **NEXT #382** Research poster from my manuscript or thesis

Research poster from my manuscript or thesis

1.13 CONFERENCE ABSTRACTS, POSTERS & ORAL PAPERS · PG RESIDENT / FACULTY

USE IT

After abstract acceptance as a poster, 2–3 weeks before the conference.

WHY

Most posters are shrunken papers that nobody reads. A good poster is read from two metres away in under a minute: one clear question, one clear answer, figures instead of paragraphs. This prompt turns your paper into that.

COPY THE PROMPT BELOW

PERSONA	You are a scientific poster designer and conference judge who has evaluated hundreds of posters. You design for visual hierarchy, brevity and a clear take-home message.
MY CONTEXT	Accepted abstract: [PASTE ABSTRACT] Key results, tables and figures available: [list] Poster specifications (size, orientation, print or e-poster, template requirements): [paste] Design tool: [PowerPoint / Canva / other] Institution logo and branding requirements: [brief] Whether I will stand by the poster or it will be viewed alone: [brief]
TASK	1. Write a one-line take-home message to feature prominently. 2. Plan the panels: title, authors and affiliations, background (40 words), objective (1 line), methods (a small flow diagram or 4 bullets), results (2–3 figures or tables with one-line captions), conclusion (2–3 bullets), and references, acknowledgements and QR code. 3. Write the text for every panel, 500–800 words in total. 4. Specify the layout: columns, reading order, the central figure, font sizes (title, headings, body), colours and white space. 5. Convert any results table into a figure where that would be clearer. 6. Give step-by-step build instructions in my tool. 7. Write a 90-second walk-through script for when judges stop by.
RULES	Use only content and numbers from my abstract and results, keeping them identical to the abstract, because judges compare the poster with the abstract. Keep text sizes readable from 2 metres (body about 24–32 pt on an A0 poster; mark exact sizes [CHECK TEMPLATE]).
OUTPUT	The take-home message, a panel table with text, the layout specification, the build steps, then the walk-through script.
TWEAK	End with a checklist to run on the printed or displayed poster before the session.

How to get more out of it

- › If your poster has more than 800 words, it has too many.
- › Put your main result graph in the centre, at eye level.
- › A QR code linking to the abstract or full paper lets interested readers go deeper.
- › → NEXT #388 90-second poster walk-through for judges

Thesis → 8-minute oral paper, with slides and Q&A

1.13 CONFERENCE ABSTRACTS, POSTERS & ORAL PAPERS · PG RESIDENT

USE IT

When your abstract is accepted for oral or award-paper presentation.

WHY

Oral paper slots are short and strictly timed, and chairs cut off speakers who overrun. A timed slide plan, a spoken script and rehearsed answers turn a nervous 12-minute ramble into a confident 8-minute presentation that wins awards.

COPY THE PROMPT BELOW

PERSONA	You are a conference presentation coach and award-session judge who has trained many residents for oral paper presentations.
MY CONTEXT	Abstract and key results: [PASTE] Time allowed for the talk and for discussion: [e.g. 8 + 2 minutes] Figures and tables available: [list] Award session?: [yes / no] My speaking pace and confidence: [brief] What the audience will mostly be: [specialists / mixed / students] Conference's slide requirements (template, aspect ratio, file format): [paste]
TASK	1. Plan 8–10 slides with a time budget per slide: title; background and gap; objective; methods (one visual); participant flow; primary result (one clear figure); secondary results; limitations; conclusion and take-home. 2. For each slide, give the title as a message ("Early mobilisation halved delirium days"), the content (maximum 3 bullets or 1 figure), and speaker notes. 3. Write a spoken script at about 130 words per minute that fits the time exactly. 4. Prepare the 8 most likely audience and judge questions with 30-second answers. 5. Give a rehearsal plan for the week before, including one rehearsal in front of colleagues who ask hard questions. 6. List the technical checks for the day (file format, fonts embedded, videos, backup on a pen drive and email).
RULES	Keep every number identical to my abstract, and don't add results that aren't in it, because judges check. Keep slides visual: no paragraphs, and at least 24 pt text.
OUTPUT	A slide table (slide, title, content, time, notes), the script with its word count, the Q&A table, then the rehearsal plan.
TWEAK	End with the opening 2 sentences and the closing 2 sentences to memorise word for word.

How to get more out of it

- Rehearse with a timer at least 5 times. Most residents overrun by 30–50%.
- Write slide titles as messages, not topics. The audience should get the story from the titles alone.
- Have 2–3 backup slides for predictable questions.
- → NEXT #390 Prepare for questions after my talk

384 Choose the right conference for my work

STANDARD

1.13 CONFERENCE ABSTRACTS, POSTERS & ORAL PAPERS · PG RESIDENT / FACULTY

USE IT When deciding where to submit your abstract.

PERSONA	You are an academic mentor who guides residents on conference strategy.
MY CONTEXT	Specialty and the topic of my work: [brief] Stage of the work: [final results / interim / case report] Budget and travel constraints: [brief] Goals: [award, networking, first presentation, international exposure]
TASK	1. Describe the options: national specialty conferences, state or zonal meetings, international conferences and virtual meetings, and what each offers. 2. Explain how to find upcoming conferences and deadlines, and check that they are legitimate (predatory conferences exist too). 3. Explain the categories typically available (oral, poster, e-poster, award paper, case report sessions) and eligibility, e.g. age or membership requirements for awards. 4. Build a comparison table from the conferences I find.
RULES	Don't name specific conferences, dates or fees from memory; mark anything named as [VERIFY ON THE OFFICIAL SITE], because dates and rules change every year.
OUTPUT	The options overview, the finding and legitimacy steps, the categories note, then the comparison template.

PRO TIP Student and resident award sessions at national specialty conferences are among the best first stages for your work.

TRY IT WITH MD Pulmonary Medicine resident with final thesis results, limited travel budget.

→ **NEXT #381** Conference abstract to the exact word limit

385 Score my abstract like a conference reviewer

STANDARD

1.13 CONFERENCE ABSTRACTS, POSTERS & ORAL PAPERS · PG RESIDENT

USE IT Before you submit a conference abstract.

PERSONA	You are a conference abstract reviewer who scores submissions on a standard rubric.
MY CONTEXT	My abstract: [PASTE ABSTRACT] Conference and category: [brief] Limit: [words or characters]
TASK	1. Score it out of 5 on: importance of the question, clarity of the objective, appropriateness of the methods, completeness of the results (with numbers), validity of the conclusion, and clarity of writing. 2. Quote the text behind each score. 3. Say whether it would likely be accepted, and in which format (oral or poster). 4. Give the 3 most important revisions.
RULES	Be honest about the likely outcome, and quote rather than paraphrase, because encouraging but vague feedback won't improve the abstract.
OUTPUT	A scoring table, the verdict, then the revisions.

PRO TIP Abstracts without numbers in the results are usually rejected. Check yours has them.

TRY IT WITH A 250-word abstract on ultrasound-guided versus landmark-guided central line insertion.

→ **NEXT #381** Conference abstract to the exact word limit

386 Adapt my poster into an e-poster

STANDARD

1.13 CONFERENCE ABSTRACTS, POSTERS & ORAL PAPERS · PG RESIDENT

USE IT When the conference uses screen-displayed e-posters.

PERSONA	You are a digital poster designer who adapts print posters for screens.
MY CONTEXT	My poster content or file description: [PASTE OR DESCRIBE] E-poster specification (aspect ratio, single or multiple slides, resolution, file type, interactivity allowed): [paste]
TASK	1. Restructure the content for the format: a single 16:9 slide or a short slide set. 2. Reduce the text further for on-screen reading distance. 3. Suggest a reading flow and the use of any allowed interactivity (e.g. clickable panels). 4. Specify export settings.
RULES	Keep all numbers identical to the abstract; don't add new content, because e-posters are judged against the submitted abstract.
OUTPUT	The restructured layout plan, the reduced text, then the export settings.

PRO TIP Check your e-poster on a laptop from 1.5 metres before submitting. If you can't read it, neither can judges.

TRY IT WITH An A0 portrait poster converted to one 16:9 e-poster slide.

→ **NEXT #388** 90-second poster walk-through for judges

387 Layout brief for Canva or PowerPoint

STANDARD

1.13 CONFERENCE ABSTRACTS, POSTERS & ORAL PAPERS · PG RESIDENT

USE IT When building the poster yourself.

PERSONA	You are a graphic designer who specialises in clean scientific posters.
MY CONTEXT	Poster size and orientation: [e.g. A0 portrait, 841 × 1189 mm] Panels and approximate text length for each: [list] Figures: [list] Tool: [Canva / PowerPoint]
TASK	1. Give the page set-up for my tool (custom size, margins, grid). 2. Suggest a 3-column grid and the placement of each panel. 3. Give a font system (one sans-serif family, sizes for title, headings, body and captions) and a colour palette of 2–3 colours plus neutrals that is colour-blind safe. 4. Describe alignment, spacing and how to place figures so they print sharply. 5. Give print-export settings.
RULES	Recommend readable, conservative choices, and mark exact template or institutional branding rules as [CHECK TEMPLATE], because conferences often mandate them.
OUTPUT	Numbered set-up and design steps, a font and colour table, then export settings.

PRO TIP Leave about 25% of the poster as white space. Crowded posters are skipped.

TRY IT WITH A0 landscape poster in PowerPoint with 3 figures and 1 table.

→ **NEXT #382** Research poster from my manuscript or thesis

388 90-second poster walk-through for judges

STANDARD

1.13 CONFERENCE ABSTRACTS, POSTERS & ORAL PAPERS · PG RESIDENT

USE IT The week before your poster session.

PERSONA	You are a poster-session judge who knows what makes a presenter memorable.
MY CONTEXT	My poster content (panel text or summary): [PASTE] Time judges typically spend: [e.g. 2–3 minutes]
TASK	1. Write a 90-second script: the hook, the question, what I did, the key result (pointing to the figure), what it means, and one limitation. 2. Write a 30-second version for passers-by. 3. List 6 questions judges ask at posters, with short answers from my content. 4. Give body-language and positioning tips.
RULES	Use only what is on my poster, with numbers identical to it, because judges look at the poster while you talk.
OUTPUT	The two scripts with timing, the Q&A table, then the tips.

PRO TIP Start with the question, not the background. "We asked whether..." gets attention.

TRY IT WITH Poster on the prevalence of vitamin B12 deficiency in metformin-treated diabetics.

→ **NEXT #382** Research poster from my manuscript or thesis

389 Slide plan for a short oral presentation

STANDARD

1.13 CONFERENCE ABSTRACTS, POSTERS & ORAL PAPERS · PG RESIDENT

USE IT When you need slides quickly for a 5–10 minute talk.

PERSONA	You are a presentation designer for medical conferences.
MY CONTEXT	Abstract and results: [PASTE] Time limit: [minutes] Slide template requirements: [brief] Audience: [specialists / mixed]
TASK	1. Recommend the number of slides (about one per minute). 2. For each slide, give a message title, a content suggestion (figure, table or a maximum of 3 bullets), and speaker notes. 3. Suggest the design: font sizes, contrast and one accent colour. 4. List slides to put in the backup section.
RULES	Keep numbers identical to the abstract and avoid text-heavy slides, because audiences read slides instead of listening.
OUTPUT	A slide table, the design notes, then the backup list.

PRO TIP One figure per results slide, with the key number called out in large text.

TRY IT WITH 7-minute oral paper on anaemia prevalence among adolescent girls in a rural block.

→ **NEXT #383** Thesis → 8-minute oral paper, with slides and Q&A

390 Prepare for questions after my talk

STANDARD

1.13 CONFERENCE ABSTRACTS, POSTERS & ORAL PAPERS · PG RESIDENT

USE IT In the days before your oral or poster presentation.

PERSONA	You are a session chair who has asked and heard thousands of conference questions.
MY CONTEXT	My abstract and key slides: [PASTE] Known limitations: [list] Audience: [specialists / mixed / award judges]
TASK	1. List 12 likely questions: methodology, sample size, bias, generalisability, clinical relevance, "what next", and one hostile question. 2. Write a 30-second answer for each, grounded in my study. 3. Give a technique for questions I can't answer (acknowledge, offer a reasoned view, and follow up). 4. Suggest 2 backup slides that answer the most likely questions.
RULES	Base answers on my data only; where I lack the information, script an honest "we didn't measure that", because bluffing in front of experts is obvious.
OUTPUT	A Q&A table, the technique, then the backup-slide suggestions.

PRO TIP Repeat the question before answering. It buys you thinking time and helps the audience hear it.

TRY IT WITH Oral paper on a retrospective cohort of snakebite outcomes.

→ **NEXT #383** Thesis → 8-minute oral paper, with slides and Q&A

391 Prepare for an award-paper session

STANDARD

1.13 CONFERENCE ABSTRACTS, POSTERS & ORAL PAPERS · PG RESIDENT

USE IT When your paper is selected for a best-paper or young-investigator award session.

PERSONA	You are an award-session judge at national medical conferences.
MY CONTEXT	Abstract and presentation plan: [PASTE] Award rules and judging criteria, if published: [paste] Time limits: [brief]
TASK	1. Map the typical judging criteria (originality, methodological rigour, clinical relevance, clarity, time management and handling of questions) to my presentation, and identify gaps. 2. Suggest changes to strengthen my weakest criterion. 3. Prepare answers to the judges' likely tough questions. 4. Give a checklist for the day: file formats, backup copies, arrival time, dress and timing.
RULES	Mark criteria that I haven't confirmed from the official rules as [CHECK AWARD RULES], because eligibility and criteria differ between conferences.
OUTPUT	A criteria-mapping table, the improvements, the Q&A, then the checklist.

PRO TIP Judges remember speakers who finish on time and answer questions honestly.

TRY IT WITH Award session for a prospective cohort on high-flow nasal oxygen in COPD.

→ **NEXT #390** Prepare for questions after my talk

392 Application for a travel grant or bursary

STANDARD

1.13 CONFERENCE ABSTRACTS, POSTERS & ORAL PAPERS · PG RESIDENT

USE IT When applying for conference funding from your institution, a society or a foundation.

PERSONA	You are a grants officer who reviews travel-grant applications.
MY CONTEXT	Conference, dates and the type of presentation accepted: [brief] Estimated costs (registration, travel, stay): [amounts] Funding body and its criteria: [paste] My academic profile (programme, year, publications, presentations): [brief]
TASK	1. Write a concise application letter or statement: what I'm presenting, why it matters, what I'll gain, and how I'll share the learning. 2. Build a cost table with justification. 3. List the documents usually required (acceptance letter, abstract, guide's recommendation, CV).
RULES	Use only facts I provided and mark criteria I haven't confirmed as [CHECK FUNDER CRITERIA], because inaccurate applications are rejected.
OUTPUT	The statement, the cost table, then the document list.

PRO TIP Apply as soon as your abstract is accepted. Travel-grant deadlines are often earlier than you expect.

TRY IT WITH Oral presentation at an international respiratory conference, first-time applicant.

→ **NEXT #384** Choose the right conference for my work

393 From conference presentation to published paper

STANDARD

1.13 CONFERENCE ABSTRACTS, POSTERS & ORAL PAPERS · PG RESIDENT / FACULTY

USE IT After the conference, while feedback is fresh.

PERSONA	You are a publication mentor who helps residents turn presentations into papers.
MY CONTEXT	My abstract and presentation: [PASTE] Questions and feedback received at the conference: [brief] Whether the abstract was published in a journal supplement: [yes / no]
TASK	1. List the feedback that should shape the paper. 2. Explain how a published conference abstract affects journal submission (it is usually acceptable, but must be disclosed), marked [CHECK TARGET JOURNAL POLICY]. 3. Make a plan for writing the full paper within 6 weeks. 4. Draft the disclosure sentence about the prior presentation.
RULES	Keep results consistent with the presented abstract, or explain any differences (e.g. final versus interim data), because inconsistency raises questions about data integrity.
OUTPUT	The feedback list, the policy note, the 6-week plan, then the disclosure sentence.

PRO TIP Conference momentum fades in weeks. Start the paper the week you return.

TRY IT WITH Poster on inhaler technique errors that received strong interest at a national meeting.

→ **NEXT #316** Write the full manuscript for journal submission

394 Record a presentation for a virtual or hybrid conference

STANDARD

1.13 CONFERENCE ABSTRACTS, POSTERS & ORAL PAPERS · PG RESIDENT / FACULTY

USE IT When the conference asks for a pre-recorded talk or you present online.

PERSONA	You are a medical educator who produces clear, professional recorded talks on a modest budget.
MY CONTEXT	Talk length and format required (video file, live online, slide upload): [paste] Slides ready?: [yes / no] Equipment available: [laptop, phone, microphone, quiet room]
TASK	1. Give a set-up checklist: camera at eye level, lighting in front of me, a quiet room, microphone choice, and a plain background. 2. Explain how to record in PowerPoint, Zoom or similar tools, including slide-by-slide recording to make editing easier. 3. Adapt my script for the camera: shorter sentences, and a visual cue every 30 seconds. 4. Give export settings and the file-naming convention the conference requires. 5. Plan for live Q&A online: internet backup, and mute and unmute etiquette.
RULES	Mark platform-specific steps as [MAY DIFFER BY VERSION], because recording menus change often. Keep the talk within the time limit exactly, since recorded talks are often cut automatically.
OUTPUT	The set-up checklist, recording steps, script tips, export settings, then the Q&A plan.

PRO TIP Record one slide at a time. A mistake then costs 40 seconds, not the whole talk.

TRY IT WITH 10-minute pre-recorded oral paper for an international virtual respiratory conference.

→ **NEXT #383** Thesis → 8-minute oral paper, with slides and Q&A

QUICK PROMPTS · 1.13

395 An accurate, catchy abstract title

QUICK

1.13 CONFERENCE ABSTRACTS, POSTERS & ORAL PAPERS · PG RESIDENT

Act as a conference abstract reviewer. Write 4 titles for my abstract, [ONE-LINE SUMMARY WITH DESIGN], under [LIMIT] characters: two descriptive and two that state the finding. None should overclaim or use undefined abbreviations. Mark the one most likely to attract reviewers.

TRY IT WITH Prospective cohort showing frailty predicts delirium after hip fracture surgery.

396 25-word summary for the conference programme

QUICK

1.13 CONFERENCE ABSTRACTS, POSTERS & ORAL PAPERS · PG RESIDENT

Act as a conference programme editor. Summarise my accepted presentation, [PASTE ABSTRACT], in 25 words or fewer for delegates for the printed conference programme or app: what I studied, in whom, where, and the main finding, accurately, clearly and without jargon.

TRY IT WITH Abstract on the diagnostic accuracy of lung ultrasound in paediatric pneumonia.

397 One-page handout or QR page for my poster

QUICK

1.13 CONFERENCE ABSTRACTS, POSTERS & ORAL PAPERS · PG RESIDENT

Act as a scientific communicator. Create the content for a one-page handout or QR-linked page for my poster, [PASTE POSTER TEXT]: title, key figure description, 3 key findings, the take-home message, contact email and the full abstract, in under 300 words.

TRY IT WITH Poster on vitamin B12 deficiency in patients on long-term metformin.

398 Will my talk fit the time?

QUICK

1.13 CONFERENCE ABSTRACTS, POSTERS & ORAL PAPERS · PG RESIDENT

Act as a presentation coach. Check my script, [PASTE SCRIPT], against a [N]-minute slot at about 130 spoken words per minute. Tell me how far over or under I am, and which sentences to cut or merge to fit, without losing the key result.

TRY IT WITH A 1,450-word script for an 8-minute oral paper.

399 The closing take-home slide

QUICK

1.13 CONFERENCE ABSTRACTS, POSTERS & ORAL PAPERS · PG RESIDENT

Act as a conference presentation coach. Write my closing slide for [ONE-LINE FINDING AND DESIGN]: a message title, 2–3 take-home bullets of under 12 words each, in plain language, and one line on what comes next, all proportionate to my design.

TRY IT WITH Retrospective cohort linking delayed antibiotics with higher sepsis mortality.

400 Follow-up email after the conference

QUICK

1.13 CONFERENCE ABSTRACTS, POSTERS & ORAL PAPERS · PG RESIDENT / FACULTY

Act as an academic networking coach. Write a short, professional follow-up email to [NAME, ROLE], whom I met at [CONFERENCE] after my presentation on [TOPIC], thanking them and proposing [collaboration / sharing data / advice on the next study]. Keep it under 120 words.

TRY IT WITH Following up with a senior researcher who asked about multicentre collaboration.

Every prompt at a glance.

Q = Quick · S = Standard · M = Master

1.1 · TOPIC SELECTION & RESEARCH QUESTION

#1	Feasibility-first topic shortlist	M
#2	Five Indian abstracts → six thesis topics	M
#3	Socratic research-question workshop	M
#4	Guide-meeting pitch pack for your top 3 topics	M
#5	India gap finder: 15 topics from scratch	S
#6	Clinical hunch → PICOT question	S
#7	FINER stress-test of my research question	S
#8	Is this topic already done to death?	S
#9	Turn gaps into Consensus yes/no questions	S
#10	Topics I can do from existing hospital records	S
#11	Topics that fit a 12-month data window	S
#12	Topics needing only routine investigations	S
#13	Validate an existing clinical score in my population	S
#14	Quality-improvement and audit thesis topics	S
#15	Questionnaire-based topics without the KAP trap	S
#16	Head-to-head decision matrix for 2–3 topics	S
#17	Thesis title that passes the committee	S
#18	What could kill this topic? A risk register	S
#19	Choose a measurable primary outcome	S
#20	The questions my guide will ask about my topic	S
#21	Topics from the gaps in a clinical guideline	S
#22	Four-week feasibility check before you commit	S
#23	Five original angles on a crowded topic	S
#24	Two-month project for an MBBS student (ICMR-STS style)	S
#25	Ten quick topic seeds in my specialty	Q
#26	Sharpen my research question to one line	Q
#27	PICO or PECO? Frame my question correctly	Q
#28	Check ongoing trials before you commit	Q
#29	The follow-up that surfaces Indian gaps	Q
#30	Identify the variables in my question	Q
#31	My topic is too broad: narrow it three ways	Q
#32	What infrastructure does this topic need?	Q
#33	Does this Western topic apply in India?	Q
#34	Hypothesis-generating or hypothesis-testing?	Q
#35	Summarise my topic on one slide	Q

1.2 · LITERATURE SEARCH STRATEGY

#36	Build a reproducible, multi-database search	M
#37	Search documentation that survives the viva	M
#38	Search rescue: iterate until the results are right	M
#39	MeSH terms and synonyms for one concept	S
#40	PICO → concept blocks table	S
#41	Fix the syntax of my search string	S
#42	Too many or too few hits: fix my search	S
#43	Translate my PubMed string for other databases	S
#44	Use PubMed filters without biasing the review	S

#45	Find studies from Indian centres	S
#46	Grey literature and trial-registry search plan	S
#47	Backward and forward citation chasing plan	S
#48	Map the research landscape in Perplexity	S
#49	Find the systematic reviews before anything else	S
#50	Identify the landmark papers in my field	S
#51	Set up alerts so new papers find you	S
#52	Reference manager set-up and deduplication	S
#53	Write the search-methods paragraph	S
#54	Legal ways to get the full text of a paper	S
#55	Peer review my search strategy (PRESS checklist)	S
#56	Boolean operators explained with my topic	Q
#57	Every synonym for my concept	Q
#58	Explode, major topic, subheadings: explained	Q
#59	Adapt my search for Google Scholar	Q
#60	Which databases for my specialty?	Q
#61	Email an author for their paper or data	Q
#62	Has this paper been retracted or corrected?	Q
#63	Where does this study sit in the evidence hierarchy?	Q
#64	Turn my question into Perplexity and Consensus queries	Q
#65	How far back should my search go?	Q

1.3 · WRITING THE INTRODUCTION

#66	The four-movement thesis introduction	M
#67	Refine my own introduction draft (language only)	M
#68	Examiner-style review of my introduction	M
#69	Where to find sourced burden data	S
#70	Indian-burden paragraph from my sourced figures	S
#71	Paragraph-by-paragraph outline before you write	S
#72	Five opening sentences that are not textbook	S
#73	Write a sharp gap statement	S
#74	"Need for the study" section	S
#75	Every claim that needs a citation	S
#76	Turn a textbook-style intro into a problem-focused one	S
#77	Cut my introduction to the word limit	S
#78	Fix the flow between paragraphs	S
#79	Define key terms by current criteria	S
#80	Clinical and public-health significance	S
#81	Does my introduction lead to my objective?	S
#82	Paraphrase and synthesise sources responsibly	S
#83	Make my draft sound like me, not like AI	S
#84	Consistency check: tense, spelling, abbreviations	S
#85	Short background for a synopsis or protocol	S
#86	Three hook sentences for my topic	Q
#87	Is my tense right?	Q
#88	Turn my bullet notes into a paragraph	Q
#89	Find the textbook sentences	Q
#90	One paragraph on "why now"	Q

#91	Sentence-level clarity edit	Q	#140	Aim versus objective: fix the confusion	S
#92	Does my introduction need sub-headings?	Q	#141	Map each objective to outcome, measurement and analysis	S
#93	Check for causal language in my introduction	Q	#142	Objectives for a descriptive study	S
#94	Build the abbreviation list	Q	#143	Objectives for a diagnostic accuracy study	S
#95	Two-sentence background for my abstract	Q	#144	Too many objectives: trim to what is feasible	S
1.4 · REVIEW OF LITERATURE					
#96	Screen, extract and synthesise: the ROL engine	M	#145	Secondary objectives that add value at no extra cost	S
#97	Thematic ROL draft from my evidence table	M	#146	Fix vague verbs in my objectives	S
#98	Evidence table from uploaded PDFs, with quotes	M	#147	From PICOT question to a full set of objectives	S
#99	Global and Indian studies: the classic thesis ROL	M	#148	Objectives for an interventional study or RCT	S
#100	Examiner-style critique of my ROL chapter	M	#149	Objectives for qualitative or mixed-methods studies	S
#101	Inclusion and exclusion criteria for screening	S	#150	Write my aim in one sentence	Q
#102	Triage a batch of abstracts	S	#151	Rewrite one vague objective	Q
#103	Design my data-extraction sheet	S	#152	Is my hypothesis testable?	Q
#104	Extract one paper into my evidence table	S	#153	One-sided or two-sided test?	Q
#105	Which risk-of-bias tool for which study?	S	#154	Which of my objectives should be primary?	Q
#106	Apply a risk-of-bias tool to one paper	S	#155	Objectives squeezed into a form field	Q
#107	Find the themes in my evidence table	S	1.6 · STUDY DESIGN & METHODOLOGY		
#108	Explain why studies disagree	S	#156	Choose and defend the right study design	M
#109	One study, one paragraph, done properly	S	#157	Write the full methodology chapter	M
#110	Convert a chronological ROL into a thematic one	S	#158	Operational definitions for every variable	M
#111	Build a literature synthesis matrix	S	#159	Case record form (proforma) with a data-entry codebook	M
#112	Heading structure for my ROL chapter	S	#160	Bias and confounding audit of my design	M
#113	The closing paragraph that leads to my study	S	#161	PICO framework and inclusion/exclusion criteria	S
#114	Summarise a paper, grounded in quotes	S	#162	Make eligibility criteria bedside-proof	S
#115	Summary table of key studies	S	#163	Choose and describe my sampling method	S
#116	Check the numbers in my ROL against the sources	S	#164	Randomisation, allocation concealment and blinding	S
#117	Format and check my reference list	S	#165	Choosing and matching controls in a case-control study	S
#118	The questions to ask NotebookLM about my papers	S	#166	Follow-up plan that keeps loss to follow-up low	S
#119	Quick critical appraisal of one paper	S	#167	Tighten a cross-sectional study design	S
#120	Read and summarise a meta-analysis	S	#168	Design a diagnostic accuracy study (STARD)	S
#121	Historical review sub-section without errors	S	#169	Record-review (retrospective) methodology	S
#122	Update my ROL before submission	S	#170	Identify confounders and draw my causal diagram	S
#123	Register a systematic review on PROSPERO	S	#171	Inter-observer agreement and quality-control plan	S
#124	Pull the PICO out of an abstract	Q	#172	Translate and validate a questionnaire	S
#125	One-line takeaway of a paper	Q	#173	Study procedure as a patient journey	S
#126	The limitation that matters most	Q	#174	Blinding when full blinding is impossible	S
#127	Compare two studies in three sentences	Q	#175	Before-after and quasi-experimental studies done right	S
#128	What design is this study really?	Q	#176	Plan a pilot study for my thesis	S
#129	Interpret an effect size and its CI	Q	#177	Data management plan for my thesis	S
#130	Results in plain language	Q	#178	Laboratory sample handling and pre-analytical control	S
#131	Does this reference support my sentence?	Q	#179	Standardised imaging acquisition and reading protocol	S
#132	Format one reference in Vancouver style	Q	#180	Which reporting guideline applies to my study?	S
#133	How much should this study count?	Q	#181	Describe the intervention so others can replicate it (TIDieR)	S
#134	Customise a NotebookLM audio overview	Q	#182	Designing studies in patients who cannot consent	S
#135	Is this ROL paragraph a list or a synthesis?	Q	#183	Convert my protocol into thesis methods	S
1.5 · AIMS, OBJECTIVES & HYPOTHESES					
#136	Aims, objectives and hypotheses, fully aligned	M	#184	Name my study design correctly	Q
#137	Committee review of my objectives	M	#185	Prospective or retrospective: what's the difference for me?	Q
#138	Primary and secondary objectives in SMART form	S	#186	Strengths and weaknesses of my design	Q
#139	Null and alternative hypotheses, correctly stated	S	#187	Operationally define one variable	Q
			#188	Draw a simple random sample in Excel	Q
			#189	Allocation concealment in my trial	Q

#190	Can I apply this criterion at the bedside?	Q	#240	Build Table 1 from my variables	Q
#191	List the likely confounders	Q	1.8 · RESULTS, TABLES & FIGURES		
#192	Cohort or case-control? Decide now	Q	#241	Write the Results chapter from my analysis output	M
#193	Could observer bias affect my results?	Q	#242	Plan every table and figure in my results	M
#194	Write the "study setting" paragraph	Q	#243	Consistency audit: text, tables, figures and abstract	M
#195	My design in one line for the title page	Q	#244	Describe Table 1 in two short paragraphs	S
1.7 · SAMPLE SIZE & STATISTICS			#245	Report the primary outcome properly	S
#196	Pre-specified statistical analysis plan	M	#246	Report secondary outcomes without clutter	S
#197	Sample size with every assumption shown	M	#247	Participant flow diagram and paragraph	S
#198	Run my analysis on the real dataset, safely	M	#248	The best figure for this result	S
#199	Multivariable regression: plan, run and interpret	M	#249	Build the chart step by step in my software	S
#200	Interpret my software output correctly	M	#250	Format my table to journal standard	S
#201	Statistics viva drill for my thesis defence	M	#251	Titles, legends and footnotes for my tables and figures	S
#202	Sample size to estimate a proportion (prevalence)	S	#252	Remove interpretation from my results	S
#203	Sample size to compare two means	S	#253	Stop repeating my tables in the text	S
#204	Sample size to compare two proportions	S	#254	Report a non-significant result honestly	S
#205	Sample size for a diagnostic accuracy study	S	#255	Report diagnostic accuracy results (STARD)	S
#206	Sample size for a correlation study	S	#256	Present regression results clearly	S
#207	Sample size for a case-control study	S	#257	Condense results for an abstract or poster	S
#208	Is my sample big enough for regression?	S	#258	One result, one sentence	Q
#209	Which statistical test? A decision table	S	#259	Check my percentages and denominators	Q
#210	Is my data normally distributed? Decide sensibly	S	#260	How many decimal places?	Q
#211	Choose and report the right non-parametric test	S	#261	Table or figure for this result?	Q
#212	Chi-square, Fisher's exact or something else?	S	#262	Check my chart for common errors	Q
#213	Before-after or matched data: paired analysis	S	#263	Describe a figure in the text	Q
#214	Outcomes measured three or more times	S	#264	Label an exploratory analysis correctly	Q
#215	Correlation or agreement? Methods comparison done right	S	#265	Fix the tense in my results	Q
#216	ROC curve, AUC and the best cut-off	S	1.9 · DISCUSSION, CONCLUSION & LIMITATIONS		
#217	Time-to-event analysis: Kaplan-Meier and Cox	S	#266	Write the discussion: ten paragraphs, no overclaiming	M
#218	Handle missing data honestly	S	#267	Limitations that are honest, ranked and specific	M
#219	Data-cleaning checklist before analysis	S	#268	Hostile-reviewer attack on my discussion	M
#220	Step-by-step SPSS instructions for my analysis	S	#269	Conclusion, recommendations and summary chapter	M
#221	R code for my thesis analysis	S	#270	The first paragraph of the discussion	S
#222	Free point-and-click analysis in Jamovi	S	#271	Compare one finding with the literature	S
#223	Effect sizes and confidence intervals for every result	S	#272	Why my result differs from previous studies	S
#224	Multiple comparisons: correct or not?	S	#273	The mechanism paragraph, clearly labelled as speculation	S
#225	Subgroup and sensitivity analyses that hold up	S	#274	Clinical implications without overreach	S
#226	Report p values and CIs the way journals want	S	#275	Strengths without inflation	S
#227	Quick analysis inside Google Sheets with AI add-ons	S	#276	Future research that someone could actually do	S
#228	Classify my variables correctly	Q	#277	A conclusion that doesn't exceed the data	S
#229	Mean or median for this variable?	Q	#278	Scan my discussion for overclaiming	S
#230	What does my p value actually mean?	Q	#279	Who do my results apply to?	S
#231	Interpret my confidence interval	Q	#280	Discuss a surprising result	S
#232	Odds ratio or relative risk for my study?	Q	#281	Discuss a non-significant primary result	S
#233	Explain statistical power with my study	Q	#282	Check the structure of my discussion	S
#234	My result isn't significant: now what?	Q	#283	The "Summary" chapter of an Indian thesis	S
#235	Should I test baseline differences in my trial?	Q	#284	What my findings mean for Indian practice	S
#236	Interpret my kappa value	Q	#285	Cut my discussion to the word limit	S
#237	Which assumptions does my test need?	Q	#286	Practical recommendations from my thesis	S
#238	Sanity-check an AI-generated sample size	Q	#287	Rewrite "further studies are needed"	Q
#239	Explain any statistics term for the viva	Q			

#288	Calibrate my hedging	Q	#337	Write a letter to the editor about a published paper	S
#289	One take-home message	Q	#338	Check the galley proofs properly	S
#290	Qualify every "significant"	Q	#339	Read and fix a similarity (plagiarism) report honestly	S
#291	Which way does this bias push my result?	Q	#340	Running title under the limit	Q
#292	Swap causal words for correct ones	Q	#341	Word budget for each section	Q
#293	One comparison sentence with a study	Q	#342	Cut my abstract to the limit	Q
#294	Is this really a strength?	Q	#343	Summarise my paper for social media	Q
#295	Do my conclusions match my objectives?	Q	#344	Cut my references to the journal's limit	Q
1.10 · PROTOCOL, ETHICS & SYNOPSIS					
#296	Complete study protocol, IEC-ready	M	#345	Write the acknowledgements	Q
#297	IEC submission pack and pre-review	M	#346	Polite status enquiry to the editor	Q
#298	Participant information sheet and consent form	M	#347	Data-availability statement	Q
#299	The ethics section of my protocol or thesis	S	#348	Plain-language summary of my paper	Q
#300	Respond to IEC queries	S	#349	Reply to one tough reviewer comment	Q
#301	CTRI registration, field by field	S	#350	Is splitting my thesis into two papers acceptable?	Q
#302	University synopsis in the required format	S	1.12 · CASE REPORTS & CASE SERIES		
#303	Month-by-month thesis timeline	S	#351	From interesting case to CARE-compliant case report	M
#304	Realistic thesis budget	S	#352	Design and write a case series	M
#305	Risk-benefit assessment and adverse-event plan	S	#353	Case-report discussion grounded in the literature	M
#306	Confidentiality and data-protection plan	S	#354	Is my case publishable?	S
#307	Request a waiver of consent	S	#355	Build the case timeline table	S
#308	Protocol amendment request to the IEC	S	#356	Consent for publication and de-identification	S
#309	Funding and conflict-of-interest declarations	S	#357	Select and prepare clinical images	S
#310	Rewrite a consent paragraph in plain language	Q	#358	Case report abstract	S
#311	What documents does my IEC need?	Q	#359	Case report titles that follow the CARE convention	S
#312	Does my study involve vulnerable participants?	Q	#360	Write the case presentation from my notes	S
#313	Compensation and insurance: what do I write?	Q	#361	Show how differentials were excluded	S
#314	Assent form for children	Q	#362	Learning points that readers remember	S
#315	Version control for my study documents	Q	#363	"Images in clinical medicine" submission	S
1.11 · THESIS TO MANUSCRIPT & PUBLICATION					
#316	Write the full manuscript for journal submission	M	#364	Journals for case reports: find and verify them	S
#317	Thesis → paper cut plan (and the second paper inside it)	M	#365	Patient-level table for a case series	S
#318	Point-by-point response to peer reviewers	M	#366	Find previously reported similar cases	S
#319	Pre-submission peer review of my manuscript	M	#367	Patient-perspective section	S
#320	Structured abstract within the word limit	S	#368	Case report → conference poster	S
#321	Five manuscript titles for my paper	S	#369	Critique my case report draft	S
#322	Choose the right journal (and avoid predatory ones)	S	#370	Adverse drug reaction case report with causality assessment	S
#323	Check my manuscript against the author guidelines	S	#371	Clinicopathological conference (CPC) write-up	S
#324	Cover letter to the editor	S	#372	One line on why my case is worth reading	Q
#325	Keywords that help my paper get found	S	#373	Keywords to search for similar cases	Q
#326	Authorship order and CRediT contributions	S	#374	Is my case report properly anonymised?	Q
#327	Disclose AI use in my manuscript	S	#375	Cut my case report to the word limit	Q
#328	Ethics, consent, funding, COI and data statements	S	#376	Legend for a clinical image	Q
#329	Compress my thesis methods for a journal	S	#377	Turn my case into a teaching caption	Q
#330	Three-paragraph journal introduction	S	#378	Explain publication consent to my patient	Q
#331	Condense my thesis discussion for a journal	S	#379	Case report or case series?	Q
#332	Rejected: what now?	S	#380	The key sentence of my case-report cover letter	Q
#333	Suggest reviewers properly	S	1.13 · CONFERENCE ABSTRACTS, POSTERS & ORAL PAPERS		
#334	Key points, highlights and "what this study adds"	S	#381	Conference abstract to the exact word limit	M
#335	Brief for a graphical abstract	S	#382	Research poster from my manuscript or thesis	M
#336	Should I post a preprint?	S	#383	Thesis → 8-minute oral paper, with slides and Q&A	M
			#384	Choose the right conference for my work	S

#385	Score my abstract like a conference reviewer	S	#393	From conference presentation to published paper	S
#386	Adapt my poster into an e-poster	S	#394	Record a presentation for a virtual or hybrid conference	S
#387	Layout brief for Canva or PowerPoint	S	#395	An accurate, catchy abstract title	Q
#388	90-second poster walk-through for judges	S	#396	25-word summary for the conference programme	Q
#389	Slide plan for a short oral presentation	S	#397	One-page handout or QR page for my poster	Q
#390	Prepare for questions after my talk	S	#398	Will my talk fit the time?	Q
#391	Prepare for an award-paper session	S	#399	The closing take-home slide	Q
#392	Application for a travel grant or bursary	S	#400	Follow-up email after the conference	Q

Which AI tool for which job.

JOB	BEST TOOL	WHY
Mapping research gaps, current literature	Perplexity (Academic), Consensus	Live search with clickable citations
Reading and comparing your own PDFs	NotebookLM, Claude, ChatGPT (upload)	Answers grounded in the documents you give it
Finding recent Indian papers	Semantic Scholar, PubMed	Filters by year and field; TLDR summaries
Drafting and structuring long text	Claude, ChatGPT, Gemini	Strong long-form structure and formatting control
Running statistics on your data	ChatGPT data analysis, Julius AI, Gemini in Sheets	Reads CSV/XLSX and runs code, but verify every number
Sample size cross-check	OpenEpi, G*Power	Validated calculators to confirm AI arithmetic
Posters and figures	Canva, PowerPoint, ChatGPT images	Layout and custom visuals

THE ETHICS OF AI IN RESEARCH

- › Disclose AI use as your institution and target journal require.
- › Don't submit AI text verbatim. Rewrite it in your own academic voice.
- › Verify every statistic, reference and data point against the primary source.
- › AI cannot replace your guide's approval, your IEC clearance or your clinical judgement.

One last thing.

These prompts will not make you a researcher. They will make you a faster one, and only if the judgement behind them is yours. The clinicians who do well in the next decade will not be the ones who use AI the most. They will be the ones who know exactly when to stop trusting it.

Use them. Check everything. Publish work you can defend.

More AI in medicine every week: Instagram [@abhishekjbenur](#) · [paraboxai.com](#)

This resource is for educational purposes. It does not replace your institutional research guidelines, ethics committee requirements, university thesis regulations or the advice of your supervisor and statistician. Verify all AI-generated content independently before using it in any submitted work. AI cannot be an author; disclose AI use per ICMJE and your target journal's policy.